SPCA110N

POSTGRADUATE COURSE

MASTER OF COMPUTER APPLICATIONS

FIRST YEAR - SECOND SEMESTER

CORE PAPER - VIII

ARTIFICIAL INTELLIGENCE

INSTITUTE OF DISTANCE EDUCATION
UNIVERSITY OF MADRAS

MCA
FIRST YEAR - SECOND SEMESTER

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With best wishes from mind and heart,

DIRECTOR

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SYLLABUS

Objective of the course: The primary objective of this course is to introduce the basic principles, techniques, and applications of Artificial Intelligence. Emphasis will be placed on the teaching of these fundamentals, not on providing a mastery of specific software tools or programming environments.

Course Outcomes: After successful completion of this course, the students should be able to Demonstrate fundamental understanding of the history of artificial intelligence (AI) and its foundations. Apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and learning.

UNIT-I: Introduction: What Is AI? - Foundations of Artificial Intelligence-The History of Artificial Intelligence- The State of the Art- Risks and Benefits of AI. Intelligent Agents: Agents and Environments - The Concept of Rationality - The Nature of Environments- The Structure of Agents.

UNIT-II: Solving problem by Searching: Problem-Solving Agents - Example Problems - Search Algorithms: Best-first search - Search data structures - Redundant paths - Measuring problem-solving performance - Uninformed Search Strategies: BFS-DFS-Depth limited and iterative deepening search. Heuristic Search Strategies: Greedy best-first search - A* search - Search contours - Inadmissible heuristics and weighted A* - Heuristic Functions.

UNIT-III: Local Search and Optimization Problems: Hill-climbing search - Simulated annealing - Local beam search - Local Search in Continuous Spaces - Search with Nondeterministic Actions: The erratic vacuum world - AND—OR search trees. Optimal Decisions in Games: The minimax search algorithm - Optimal decisions in multiplayer games - Alpha—Beta Pruning. Heuristic Alpha—Beta Tree Search: Evaluation

functions - Cutting off search - Forward pruning - Monte Carlo Tree Search - Stochastic Games- Limitations of Game Search Algorithms.

UNIT-IV: Constraint Satisfaction Problems: Defining Constraint Satisfaction Problems - Constraint Propagation: Inference in CSPs - Backtracking Search for CSPs - Local Search for CSPs - The Structure of Problems. Logical agent and Logics: Propositional Logic - Propositional Theorem Proving - Effective Propositional Model Checking - Agents Based on Propositional Logic - First-Order Logic: Syntax and Semantics of First-Order Logic - Using First-Order Logic - Knowledge Engineering in First-Order Logic. Inference in First-Order Logic: Unification and First-Order Inference - Forward Chaining - Backward Chaining – Resolution.

UNIT-V: Knowledge Representation and Reasoning : Ontological Engineering - Categories and Objects - Events - Mental Objects and Modal Logic - Reasoning Systems for Categories - Reasoning with Default Information. Automated Planning: Definition of Classical Planning - Algorithms for Classical Planning - Heuristics for Planning. Quantifying Uncertainty: Acting under Uncertainty - Basic Probability Notation - Inference Using Full Joint Distributions - Independence - Bayes' Rule and Its Use - Naive Bayes Models

Recommended Texts

1) Stuart Russel and Peter Norvig: Artificial Intelligence – A Modern Approach- 4th Edition Pearson Education, 2020.

Reference Books:

1) Elaine Rich and Kevin Knight: Artificial Intelligence- Tata McGraw Hill 2nd Ed, 1991.

2) N.P. padhy: Artificial Intelligence and Intelligent Systems- Oxford Higher Education- Oxford University Press, 2005.

3) George F Luger: Artificial Intelligence- Structures and Strategies for complex Problem Solving- 4 th Ed. Pearson Education, 2002.

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ARTIFICIAL INTELLIGENCE

SCHEME OF LESSONS

LESSON 1
INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Structure

1.1 Introduction

Artificial Intelligence (AI) is one of the newest fields in science and engineering. It attempts not just to understand but also to build intelligent entities. AI currently encompasses a huge variety of subfields, ranging from the general (learning and perception) to the specific, such as playing chess, proving mathematical theorems, writing poetry, driving a car on a crowded street, and diagnosing diseases. AI is relevant to any intellectual task; it is truly a universal field.

1.2 Objectives

- To introduce the basic concepts in Artificial Intelligence.
- To explore the foundations of Artificial Intelligence.
- To know about the components and history of Artificial Intelligence.

1.3 AI Overview

In Figure 1.1, we see eight definitions of AI, laid out along two dimensions. The definitions on top are concerned with thought processes and reasoning, whereas the ones on the bottom address behavior. The definitions on the left measure success in terms of fidelity to human performance, whereas RATIONALITY the ones on the right measure against an ideal performance measure, called rationality. A system is rational if it does the “right thing,” given what it knows.

Thinking Humanly	Thinking Rationally
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Acting Humanly	**Acting Rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998) "AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

Historically, all four approaches to AI have been followed, each by different people with different methods. A human-centered approach must be in part an empirical science, in volving observations and hypotheses about human behavior. A rationalist approach involves a combination of mathematics and engineering. The various group have both disparaged and helped each other. Let us look at the four approaches in more detail.

1.3.1 Acting humanly: The Turing Test approach

The Turing Test, proposed by Alan Turing TURING TEST (1950), was designed to provide a satisfactory operational definition of intelligence. A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer. Chapter 26 discusses the details of the test and whether a computer would really be intelligent if it passed. For now, we note that programming a computer to pass a rigorously applied test provides plenty to work on. The computer would need to possess the following capabilities:

- natural language processing to enable it to communicate successfully in English;
- knowledge representation to store what it knows or hears;
- automated reasoning to use the stored information to answer questions and to draw new conclusions;
- machine learning to adapt to new circumstances and to detect and extrapolate patterns.

Turing's test deliberately avoided direct physical interaction between the interrogator and the computer, because physical simulation of a person is unnecessary for intelligence. However the so-called total Turing TOTAL TURING TEST Test includes a video signal so that the interrogator can test the subject's perceptual abilities, as well as the opportunity for the interrogator to pass physical objects "through the hatch." To pass the total Turing Test, the computer will need

- computer vision to perceive objects, and
- robotics to manipulate objects and move about.

These six disciplines compose most of AI, and Turing deserves credit for designing a test that remains relevant 60 years later. Yet AI researchers have devoted little effort to passing the Turing Test, believing that it is more important to study the underlying principles of intelligence than to duplicate an exemplar.

1.3.2 Thinking humanly: The cognitive modeling approach

If we are going to say that a given program thinks like a human, we must have some way of determining how humans think. We need to get inside the actual workings of human minds. There are three ways to do this: through introspection—trying to catch our own thoughts as they go by; through psychological experiments—observing a person in action; and through brain imaging—observing the brain in action. Once we have a sufficiently precise theory of the mind, it becomes possible to express the theory as a computer program. If the program's input–output behavior matches corresponding human behavior, that is evidence that some of the program's mechanisms could also be operating in humans. The interdisciplinary field of cognitive science brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind. Real cognitive science, however, is necessarily based on experimental investigation of actual humans or animals.

In the early days of AI there was often confusion between the approaches: an author would argue that an algorithm performs well on a task and that it is therefore a good model of human performance, or vice versa. Modern authors separate the two kinds of claims; this distinction has allowed both AI and cognitive science to develop more rapidly. The two fields continue to fertilize each other, most notably in computer vision, which incorporates neurophysiological evidence into computational models.

1.3.3 Thinking rationally: The "laws of thought" approach

The Greek philosopher Aristotle was one of the first to attempt to codify "right thinking," that is, irrefutable reasoning processes. His syllogisms provided patterns for argument structures that always yielded correct conclusions when given correct premises—for example, "Socrates is a man; all men are mortal; therefore, Socrates is mortal." These laws of thought were supposed to govern the operation of the mind; their study initiated the field called logic.

Logicians in the 19th century developed a precise notation for statements about all kinds of objects in the world and the relations among them. (Contrast this with ordinary arithmetic notation, which provides only for statements about numbers.) By 1965, programs existed that could, in principle, solve any solvable problem described in logical notation. (Although if no solution exists, the program might loop forever.) The so-called logicist tradition within artificial intelligence hopes to build on such programs to create intelligent systems.

There are two main obstacles to this approach. First, it is not easy to take informal knowledge and state it in the formal terms required by logical notation, particularly when the knowledge is less than 100% certain. Second, there is a big difference between solving a problem "in principle" and solving it in practice. Even problems with just a few hundred facts can exhaust the computational resources of any computer unless it has some guidance as to which reasoning steps to try first. Although both of these obstacles apply to any attempt to build computational reasoning systems, they appeared first in the logicist tradition.

1.3.4 Acting rationally: The rational agent approach

An agent is just something that acts (agent comes from the Latin agere, to do). Of course, all computer programs do something, but computer agents are expected to do more: operate autonomously, perceive their environment, persist over a prolonged time period, adapt to change, and create and pursue goals. A rational agent is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

In the “laws of thought” approach to AI, the emphasis was on correct inferences. Making correct inferences is sometimes part of being a rational agent, because one way to act rationally is to reason logically to the conclusion that a given action will achieve one’s goals and then to act on that conclusion. On the other hand, correct inference is not all of rationality; in some situations, there is no provably correct thing to do, but something must still be done. There are also ways of acting rationally that cannot be said to involve inference. All the skills needed for the Turing Test also allow an agent to act rationally. Knowledge representation and reasoning enable agents to reach good decisions. We need to be able to generate comprehensible sentences in natural language to get by in a complex society. We need learning not only for erudition, but also because it improves our ability to generate effective behaviour.

The rational-agent approach has two advantages over the other approaches. First, it is more general than the “laws of thought” approach because correct inference is just one of several possible mechanisms for achieving rationality. Second, it is more amenable to scientific development than are approaches based on human behavior or human thought. The standard of rationality is mathematically well defined and completely general, and can be “unpacked” to generate agent designs that provably achieve it. Human behavior, on the other hand, is well adapted for one specific environment and is defined by, well, the sum total of all the things that humans do.

One important point to keep in mind: We will see before too long that achieving perfect rationality—always doing the right thing—is not feasible in complicated environments. The computational demands are just too high. For most of the book, however, we will adopt the working hypothesis that perfect rationality is a good starting point for analysis. It simplifies the problem and provides the appropriate setting for most of the foundational material in the field. Chapters 5 and 17 deal explicitly with the issue of limited rationality—acting appropriately when there is not enough time to do all the computations one might like.

1.4 The Foundations of Artificial Intelligence

In this section, we provide a brief history of the disciplines that contributed ideas, viewpoints, and techniques to AI. Like any history, this one is forced to concentrate on a small number of people, events, and ideas and to ignore others that also were important. We organize the history around a series of questions. We certainly would not wish to give the impression that these questions are the only ones the disciplines address or that the disciplines have all been working toward AI as their ultimate fruition.

1.4.1 Philosophy

- Can formal rules be used to draw valid conclusions?
- How does the mind arise from a physical brain?
- Where does knowledge come from?
- How does knowledge lead to action?

Aristotle (384–322 B.C.) was the first to formulate a precise set of laws governing the rational part of the mind. He developed an informal system of syllogisms for proper reasoning, which in principle allowed one to generate conclusions mechanically, given initial premises. Pascal wrote that “the arithmetical machine produces effects which appear nearer to thought than all the actions of animals.” Gottfried Wilhelm Leibniz (1646–1716) built a mechanical device intended to carry out operations on concepts rather than numbers, but its scope was rather limited. Leibniz did surpass Pascal by building a calculator that could add, subtract, multiply, and take roots, whereas the Pascaline could only add and subtract.

Some speculated that machines might not just do calculations but actually be able to think and act on their own. In his 1651 book Leviathan, Thomas Hobbes suggested the idea of an “artificial animal”. It’s one thing to say that the mind operates, at least in part, according to logical rules, and to build physical systems that emulate some of those rules; it’s another to say that the mind itself is such a physical system. Ren´e Descartes (1596–1650) gave the first clear discussion of the distinction between mind and matter and of the problems that arise. One problem with a purely physical conception of the mind is that it seems to leave little room for free will: if the mind is governed entirely by physical laws, then it has no more free will than a rock “deciding” to fall toward the center of the earth. Descartes was a strong advocate of the power of reasoning in understanding the world, a philosophy now called rationalism, and one that counts Aristotle and Leibnitz as members. But Descartes was also a proponent of dualism.

He held that there is a part of the human mind (or soul or spirit) that is outside of nature, exempt from physical laws. Animals, on the other hand, did not possess this dual quality; they could be treated as machines. An alternative to dualism is materialism, which holds that the brain’s operation according to the laws of physics constitutes the mind. Free will is simply the way that the perception of available choices appears to the choosing entity.

Given a physical mind that manipulates knowledge, the next problem is to establish the source of knowledge. The final element in the philosophical picture of the mind is the connection

between knowledge and action. This question is vital to AI because intelligence requires action as well as reasoning. Moreover, only by understanding how actions are justified can we understand how to build an agent whose actions are justifiable (or rational). Aristotle argued that actions are justified by a logical connection between goals and knowledge of the action's outcome.

Goal-based analysis is useful, but does not say what to do when several actions will achieve the goal or when no action will achieve it completely. Antoine Arnauld (1612–1694) correctly described a quantitative formula for deciding what action to take in cases like this. John Stuart Mill's (1806–1873) book Utilitarianism (Mill, 1863) promoted the idea of rational decision criteria in all spheres of human activity. The more formal theory of decisions is discussed in the following section.

1.4.2 Mathematics

- What are the formal rules to draw valid conclusions?
- What can be computed?
- How do we reason with uncertain information?

Philosophers staked out some of the fundamental ideas of AI, but the leap to a formal science required a level of mathematical formalization in three fundamental areas: logic, computation, and probability. The idea of formal logic can be traced back to the philosophers of ancient Greece, but its mathematical development really began with the work of George Boole (1815–1864), who worked out the details of propositional, or Boolean, logic (Boole, 1847). In 1879, Gottlob Frege (1848–1925) extended Boole's logic to include objects and relations, creating the first order logic that is used today. Alfred Tarski (1902–1983) introduced a theory of reference that shows how to relate the objects in a logic to objects in the real world.

The next step was to determine the limits of what could be done with logic and computation. The integers cannot be represented by nontrivial algorithms – that is, they cannot be computed. This motivated Alan Turing (1912–1954) to try to characterize exactly which functions are computable—capable of being computed. This notion is actually slightly problematic because the notion of a computation or effective procedure really cannot be given a formal definition. However, the Church–Turing thesis, which states that the Turing machine (Turing, 1936) is capable of computing any computable function, is generally accepted as providing a sufficient definition. Turing also showed that there were some functions that no Turing machine can

compute. For example, no machine can tell in general whether a given program will return an answer on a given input or run forever.

Although decidability and computability are important to an understanding of computation, the notion of tractability has had an even greater impact. Roughly speaking, a problem is called intractable if the time required to solve instances of the problem grows exponentially with the size of the instances. The distinction between polynomial and exponential growth in complexity was first emphasized in the mid-1960s. It is important because exponential growth means that even moderately large instances cannot be solved in any reasonable time. Therefore, one should strive to divide the overall problem of generating intelligent behavior into tractable subproblems rather than intractable ones.

How can one recognize an intractable problem? The theory of NP-completeness, pioneered by Steven Cook (1971) and Richard Karp (1972), provides a method. Cook and Karp showed the existence of large classes of canonical combinatorial search and reasoning problems that are NP-complete. Any problem class to which the class of NP-complete problems can be reduced is likely to be intractable. Despite the increasing speed of computers, careful use of resources will characterize intelligent systems.

Besides logic and computation, the third great contribution of mathematics to AI is the theory of probability. The Italian Gerolamo Cardano (1501–1576) first framed the idea of probability, describing it in terms of the possible outcomes of gambling events. Probability quickly became an invaluable part of all the quantitative sciences, helping to deal with uncertain measurements and incomplete theories. James Bernoulli (1654–1705), Pierre Laplace (1749–1827), and others advanced the theory and introduced new statistical methods. Thomas Bayes (1702–1761) proposed a rule for updating probabilities in the light of new evidence. Bayes' rule underlies most modern approaches to uncertain reasoning in AI systems.

1.4.3 Economics

- How should we make decisions so as to maximize payoff?
- How should we do this when others may not go along?
- How should we do this when the payoff may be far in the future?

While the ancient Greeks and others had made contributions to economic thought, Smith was the first to treat it as a science, using the idea that economies can be thought of as consisting

of individual agents maximizing their own economic well-being. Most people think of economics as being about money, but economists will say that they are really studying how people make choices that lead to preferred outcomes. Decision theory, which combines probability theory with utility theory, provides a formal and complete framework for decisions (economic or otherwise) made under uncertainty—that is, in cases where probabilistic descriptions appropriately capture the decision maker's environment. This is suitable for "large" economies where each agent need pay no attention to the actions of other agents as individuals. For "small" economies, the situation is much more like a game: the actions of one player can significantly affect the utility of another (either positively or negatively). Von Neumann and Morgenstern's development of game theory included the surprising result that, for some games, a rational agent should adopt policies that are randomized. Unlike decision theory, game theory does not offer an unambiguous prescription for selecting actions.

For the most part, economists did not address the third question listed above, namely, how to make rational decisions when payoffs from actions are not immediate but instead result from several actions taken in sequence. This topic was pursued in the field of operations research, which emerged in World War II from efforts in Britain to optimize radar installations, and later found civilian applications in complex management decisions. The work of Richard Bellman (1957) formalized a class of sequential decision problems called Markov decision processes. Work in economics and operations research has contributed much to our notion of rational agents, yet for many years AI research developed along entirely separate paths. One reason was the apparent complexity of making rational decisions. The pioneering AI researcher Herbert Simon (1916–2001) won the Nobel Prize in economics in 1978 for his early work showing that models based on satisficing—making decisions that are "good enough," rather than laboriously calculating an optimal decision—gave a better description of actual human behaviour.

1.4.4 Neuroscience

• How do brains process information?

Neuroscience is the study of the nervous system, particularly the brain. Although the exact way in which the brain enables thought is one of the great mysteries of science, the fact that it does enable thought has been appreciated for thousands of years because of the evidence that strong blows to the head can lead to mental incapacitation. It has also long been known that human brains are somehow different; in about 335 B.C. Aristotle wrote, "Of all the animals, man has the largest brain in proportion to his size."5 Still, it was not until the middle of the 18th century that the brain was widely recognized as the seat of consciousness.

Paul Broca showed that speech production was localized to the portion of the left hemisphere now called Broca's area. By that time, it was known that the brain consisted of nerve cells, or neurons and it was a staining technique allowing the observation of individual neurons in the brain (see Figure 1.1). This technique was used by Santiago Ramon y Cajal (1852– 1934) in his pioneering studies of the brain's neuronal structures. Nicolas Rashevsky (1936, 1938) was the first to apply mathematical models to the study of the nervous sytem.

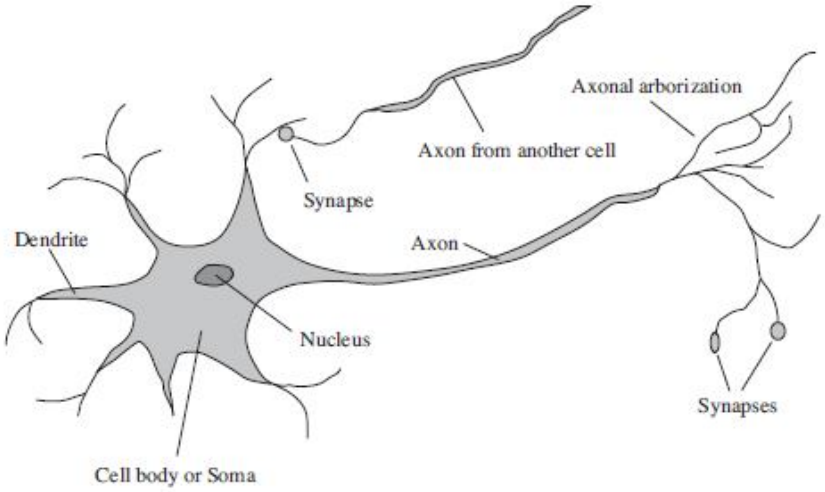


Figure 1.1 The parts of a nerve cell or neuron

We now have some data on the mapping between areas of the brain and the parts of the body that they control or from which they receive sensory input. Such mappings are able to change radically over the course of a few weeks, and some animals seem to have multiple maps. Moreover, we do not fully understand how other areas can take over functions when one area is damaged. There is almost no theory on how an individual memory is stored.

The measurement of intact brain activity began in 1929 with the invention by Hans Berger of the electroencephalograph (EEG). The recent development of functional magnetic resonance imaging (MRI) (Ogawa et al., 1990; Cabeza and Nyberg, 2001) is giving neuroscientists unprecedentedly detailed images of brain activity, enabling measurements that correspond in interesting ways to ongoing cognitive processes. These are augmented by advances in single-cell recording of neuron activity. Individual neurons can be stimulated electrically, chemically, or even optically (Han and Boyden, 2007), allowing neuronal input–output relationships to be mapped. Despite these advances, we are still a long way from understanding how cognitive processes actually work.

The truly amazing conclusion is that a collection of simple cells can lead to thought, action, and consciousness or, in the pithy words of John Searle (1992), brains cause minds. The only real alternative theory is mysticism: that minds operate in some mystical realm that is beyond physical science. Brains and digital computers have somewhat different properties. The brain makes up for that with far more storage and interconnection than even a high-end personal computer, although the largest supercomputers have a capacity that is similar to the brain's. (It should be noted, however, that the brain does not seem to use all of its neurons simultaneously.) Futurists make much of these numbers, pointing to an approaching singularity at which computers reach a superhuman level of performance (Vinge, 1993; Kurzweil, 2005), but the raw comparisons are not especially informative. Even with a computer of virtually unlimited capacity, we still would not know how to achieve the brain's level of intelligence.

1.4.5 Psychology

- How do humans and animals think and act?

The origins of scientific psychology are usually traced to the work of the German physicist Hermann von Helmholtz (1821–1894) and his student Wilhelm Wundt (1832–1920). Helmholtz applied the scientific method to the study of human vision, and his Handbook of Physiological Optics is even now described as "the single most important treatise on the physics and physiology of human vision". In 1879,. Wundt insisted on carefully controlled experiments to perform a perceptual or associative task while introspecting on their thought processes. The careful controls went a long way toward making psychology a science, but the subjective nature of the data made it unlikely that an experimenter would ever disconfirm his or her own theories. Biologists studying animal behavior, on the other hand, lacked introspective data and developed an objective methodology. Applying this viewpoint to humans, the behaviorism movement, led by John Watson (1878–1958), rejected any theory involving mental processes on the grounds that introspection could not provide reliable evidence. Behaviorists insisted on studying only objective measures of the percepts (or stimulus) given to an animal and its resulting actions (or response). Behaviorism discovered a lot about rats and pigeons but had less success at understanding humans.

Cognitive psychology, which views the brain as an information-processing device. Helmholtz also insisted that perception involved a form of unconscious logical inference. Craik specified the three key steps of a knowledge-based agent: (1) the stimulus must be translated into an internal representation, (2) the representation is manipulated by cognitive processes to

derive new internal representations, and (3) these are in turn retranslated back into action. He clearly explained why this was a good design for an agent. Donald Broadbent, whose book Perception and Communication (1958) was one of the first works to model psychological phenomena as information processing. Meanwhile, in the United States, the development of computer modeling led to the creation of the field of cognitive science. At the workshop, George Miller presented The Magic Number Seven, Noam Chomsky presented Three Models of Language, and Allen Newell and Herbert Simon presented The Logic Theory Machine. These three influential papers showed how computer models could be used to address the psychology of memory, language, and logical thinking, respectively. It is now a common view among psychologists that "a cognitive theory should be like a computer program" (Anderson, 1980); that is, it should describe a detailed information processing mechanism whereby some cognitive function might be implemented.

1.4.6 Computer engineering

- How can we build an efficient computer?

For artificial intelligence to succeed, we need two things: intelligence and an artifact. The computer has been the artifact of choice. The modern digital electronic computer was invented independently and almost simultaneously by scientists in three countries embattled in World War II. The first operational computer was the electromechanical Heath Robinson built in 1940 by Alan Turing's team for a single purpose. Since that time, each generation of computer hardware has brought an increase in speed and capacity and a decrease in price. Current expectations are that future increases in power will come from massive parallelism—a curious convergence with the properties of the brain.

Of course, there were calculating devices before the electronic computer. The first programmable machine used punched cards to store instructions for the pattern to be woven. In the mid-19th century, Charles Babbage (1792–1871) designed two machines, neither of which he completed. The Difference Engine was intended to compute mathematical tables for engineering and scientific projects. It was finally built and shown to work in 1991 at the Science Museum in London (Swade, 2000). Babbage's Analytical Engine was far more ambitious: it included addressable memory, stored programs, and conditional jumps and was the first artifact capable of universal computation.

AI also owes a debt to the software side of computer science, which has supplied the operating systems, programming languages, and tools needed to write modern programs Work in AI has pioneered many ideas that have made their way back to mainstream computer science, including time sharing, interactive interpreters, personal computers with windows and mice, rapid development environments, the linked list data type, automatic storage management, and key concepts of symbolic, functional, declarative, and object-oriented programming.

1.4.7 Control theory and cybernetics

The central figure in the creation of what is now called control theory was developed by Norbert Wiener (1894–1964). Meanwhile, a group of researchers in Britain elaborated the idea that intelligence could be created by the use of homeostatic devices containing appropriate feedback loops to achieve stable adaptive behavior. Modern control theory, especially the branch known as stochastic optimal control, has as its goal the design of systems that maximize an objective function over time. Calculus and matrix algebra, the tools of control theory, lend themselves to systems that are describable by fixed sets of continuous variables, whereas AI was founded in part as a way to escape from the these perceived limitations. The tools of logical inference and computation allowed AI researchers to consider problems such as language, vision, and planning that fell completely outside the control theorist's purview.

1.4.8 Linguistics

Chomsky, a researcher pointed out that the behaviorist theory did not address the notion of creativity in language—it did not explain how a child could understand and make up sentences that he or she had never heard before. Chomsky's theory—based on syntactic models going back to the Indian linguist Panini (c. 350 B.C.)—could explain this, and unlike previous theories, it was formal enough that it could in principle be programmed. Modern linguistics and AI, then, were "born" at about the same time, and grew up together, intersecting in a hybrid field called computational linguistics or natural language processing. The problem of understanding language soon turned out to be considerably more complex than it seemed in 1957. Understanding language requires an understanding of the subject matter and context, not just an understanding of the structure of sentences. This might seem obvious, but it was not widely appreciated until the 1960s. Much of the early work in knowledge representation was tied to language and informed by research in linguistics, which was connected in turn to decades of work on the philosophical analysis of language.

1.5 The History of Artificial Intelligence

The origin of artificial intelligence lies in the earliest days of machine computations. During the 1940s and 1950s, AI begins to grow with the emergence of the modern computer. Among the first researchers to attempt to build intelligent programs were Newell and Simon. Their first well known program, logic theorist, was a program that proved statements using the accepted rules of logic and a problem solving program of their own design. By the late fifties, programs existed that could do a passable job of translating technical documents and it was seen as only a matter of extra databases and more computing power to apply the techniques to less formal, more ambiguous texts. Most problem solving work revolved around the work of Newell, Shaw and Simon, on the general problem solver (GPS). Unfortunately the GPS did not fulfill its promise and did not because of some simple lack of computing capacity. In the 1970's the most important concept of AI was developed known as Expert System which exhibits as a set rules the knowledge of an expert. The application area of expert system is very large. The 1980's saw the development of neural networks as a method learning examples.

Prof. Peter Jackson (University of Edinburgh) classified the history of AI into three periods as:

1. Classical
2. Romantic
3. Modern

1. Classical Period:

It was started from 1950. In 1956, the concept of Artificial Intelligence came into existance. During this period, the main research work carried out includes game plying, theorem proving and concept of state space approach for solving a problem.

2. Romantic Period:

It was started from the mid 1960 and continues until the mid 1970. During this period people were interested in making machine understand, that is usually mean the understanding of natural language. During this period the knowledge representation technique "semantic net" was developed.

3. Modern Period:

It was started from 1970 and continues to the present day. This period was developed to solve more complex problems. This period includes the research on both theories and practical aspects of Artificial Intelligence. This period includes the birth of concepts like Expert system, Artificial Neurons, Pattern Recognition etc. The research of the various advanced concepts of Pattern Recognition and Neural Network are still going on.

1.6 The State of the Art

What can AI do today? A concise answer is difficult because there are so many activities in so many subfields. Here we will see a few applications.

Robotic vehicles: A driverless robotic car named STANLEY sped through the rough terrain of the Mojave dessert at 22 mph, finishing the 132-mile course first to win the 2005 DARPA Grand Challenge. STANLEY is a Volkswagen Touareg outfitted with cameras, radar, and laser rangefinders to sense the environment and onboard software to command the steering, braking, and acceleration (Thrun, 2006). The following year CMU's BOSS won the Urban Challenge, safely driving in traffic through the streets of a closed Air Force base, obeying traffic rules and avoiding pedestrians and other vehicles.

Speech recognition: A traveler calling United Airlines to book a flight can have the entire conversation guided by an automated speech recognition and dialog management system.

Autonomous planning and scheduling: A hundred million miles from Earth, NASA's Remote Agent program became the first on-board autonomous planning program to control the scheduling of operations for a spacecraft (Jonsson et al., 2000). REMOTE AGENT generated plans from high-level goals specified from the ground and monitored the execution of those plans—detecting, diagnosing, and recovering from problems as they occurred. Successor program MAPGEN (Al-Chang et al., 2004) plans the daily operations for NASA's Mars Exploration Rovers, and MEXAR2 (Cesta et al., 2007) did mission planning—both logistics and science planning—for the European Space Agency's Mars Express mission in 2008.

Game playing: IBM's DEEP BLUE became the first computer program to defeat the world champion in a chess match when it bested Garry Kasparov by a score of 3.5 to 2.5 in an exhibition match (Goodman and Keene, 1997). Kasparov said that he felt a "new kind of intelligence" across the board from him. Newsweek magazine described the match as "The

brain's last stand." The value of IBM's stock increased by $18 billion. Human champions studied Kasparov's loss and were able to draw a few matches in subsequent years, but the most recent human-computer matches have been won convincingly by the computer.

Spam fighting: Each day, learning algorithms classify over a billion messages as spam, saving the recipient from having to waste time deleting what, for many users, could comprise 80% or 90% of all messages, if not classified away by algorithms. Because the spammers are continually updating their tactics, it is difficult for a static programmed approach to keep up, and learning algorithms work best (Sahami et al., 1998; Goodman and Heckerman, 2004).

Logistics planning: During the Persian Gulf crisis of 1991, U.S. forces deployed a Dynamic Analysis and Replanning Tool, DART (Cross and Walker, 1994), to do automated logistics planning and scheduling for transportation. This involved up to 50,000 vehicles, cargo, and people at a time, and had to account for starting points, destinations, routes, and conflict resolution among all parameters. The AI planning techniques generated in hours a plan that would have taken weeks with older methods. The Defense Advanced Research Project Agency (DARPA) stated that this single application more than paid back DARPA's 30-year investment in AI.

Robotics: The iRobot Corporation has sold over two million Roomba robotic vacuum cleaners for home use. The company also deploys the more rugged PackBot to Iraq and Afghanistan, where it is used to handle hazardous materials, clear explosives, and identify the location of snipers.

Machine Translation: A computer program automatically translates from Arabic to English, allowing an English speaker to see the headline "Ardogan Confirms That Turkey Would Not Accept Any Pressure, Urging Them to Recognize Cyprus." The program uses a statistical model built from examples of Arabic-to-English translations and from examples of English text totaling two trillion words (Brants et al., 2007). None of the computer scientists on the team speak Arabic, but they do understand statistics and machine learning algorithms. These are just a few examples of artificial intelligence systems that exist today. Not magic or science fiction—but rather science, engineering, and mathematics, to which this book provides an introduction.

1.7 Summary

This lesson defines AI and establishes the cultural background against which it has developed. Mathematicians provided the tools to manipulate statements of logical certainty as well as uncertain, probabilistic statements. They also set the groundwork for understanding computation and reasoning about algorithms. Economists formalized the problem of making decisions that maximize the expected outcome to the decision maker. Neuroscientists discovered some facts about how the brain works and the ways in which it is similar to and different from computers. Psychologists adopted the idea that humans and animals can be considered information processing machines. Linguists showed that language use fits into this model. Computer engineers provided the ever-more-powerful machines that make AI applications possible.

Control theory deals with designing devices that act optimally on the basis of feedback from the environment. Initially, the mathematical tools of control theory were quite different from AI, but the fields are coming closer together. The history of AI has had cycles of success, misplaced optimism, and resulting cutbacks in enthusiasm and funding. There have also been cycles of introducing new creative approaches and systematically refining the best ones. AI has advanced more rapidly in the past decade because of greater use of the scientific method in experimenting with and comparing approaches. Recent progress in understanding the theoretical basis for intelligence has gone hand in hand with improvements in the capabilities of real systems. The subfields of AI have become more integrated, and AI has found common ground with other disciplines.

1.7 Model Questions

1. Define: Artificial Intelligence.
2. List the basic components of AI.
3. Describe AI applications in different fields.

LESSON 2
INTELLIGENT AGENTS

Structure

2.1 Introduction

The concept of rationality can be applied to a wide variety of agents operating in any imaginable environment. This concept is used to develop a small set of design principles for building successful agents—systems that can reasonably be called **intelligent**. We begin by examining agents, environments, and the coupling between them. The observation that some agents behave better than others leads naturally to the idea of a rational agent—one that behaves as well as possible. How well an agent can behave depends on the nature of the environment; some environments are more difficult than others. We give a crude categorization of environments and show how properties of an environment influence the design of suitable agents for that environment. We describe a number of basic “skeleton” agent designs.

2.2 Objectives

- To introduce the basic concepts of intelligent agents.
- To explore the structure of different agents.
- To describe the components of agent programs.

2.3 Agents and Environments

An **agent** is anything that can be viewed as perceiving its **environment** through **sensors** and acting upon that environment through **actuators**. This simple idea is illustrated in Figure 2.1. A human agent has eyes, ears, and other organs for sensors and hands, legs, vocal tract, and so on for actuators. A robotic agent might have cameras and infrared range finders for sensors and various motors for actuators. A software agent receives keystrokes, file contents, and network packets as sensory inputs and acts on the environment by displaying on the screen, writing files, and sending network packets.

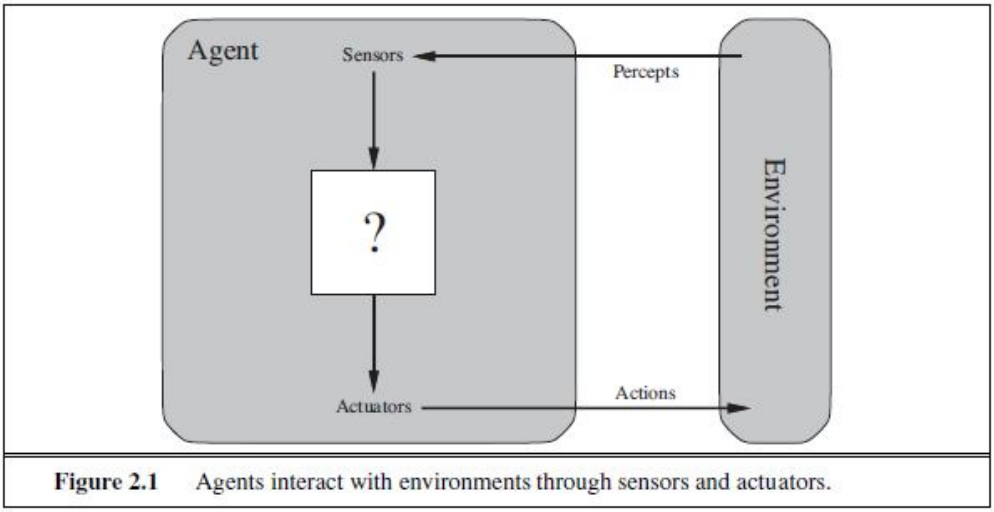


Figure 2.1 Agents interact with environments through sensors and actuators.

The term **percept** to refer to the agent's perceptual inputs at any given instant. An agent's **percept sequence** is the complete history of everything the agent has ever perceived. In general, *an agent's choice of action at any given instant can depend on the entire percept sequence observed to date, but not on anything it hasn't perceived.* Mathematically speaking, an agent's behavior is described by the **agent function** that maps any given percept sequence to an action.

Given an agent to experiment with, in principle, construct this table by trying out all possible percept sequences and recording which actions the agent does in response.1 The table is, of course, an *external* characterization of the agent. *Internally*, the agent function for an artificial agent will be implemented by an **agent program**. It is important to keep these two ideas distinct.

The agent function is an abstract mathematical description; the agent program is a concrete implementation, running within some physical system.

To illustrate these ideas, we use a very simple example—the vacuum-cleaner world shown in Figure 2.2. This world is so simple that we can describe everything that happens; it's also a made-up world, so we can invent many variations. This particular world has just two locations: squares A and B. The vacuum agent perceives which square it is in and whether there is dirt in the square. It can choose to move left, move right, suck up the dirt, or do nothing. One very simple agent function is the following: if the current square is dirty, then suck; otherwise, move to the other square. A partial tabulation of this agent function is shown in Figure 2.3.

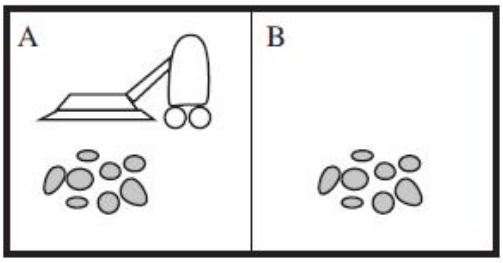


Figure 2.2 A vacuum-cleaner world with just two locations.

Percept sequence	Action
$[A, Clean]$	*Right*
$[A, Dirty]$	*Suck*
$[B, Clean]$	*Left*
$[B, Dirty]$	*Suck*
$[A, Clean], [A, Clean]$	*Right*
$[A, Clean], [A, Dirty]$	*Suck*
⋮	⋮
$[A, Clean], [A, Clean], [A, Clean]$	*Right*
$[A, Clean], [A, Clean], [A, Dirty]$	*Suck*
⋮	⋮

Figure 2.3 Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 2.2.

Looking at Figure 2.3, we see that various vacuum-world agents can be defined simply by filling in the right-hand column in various ways. The obvious question, then, is this: *What is the right way to fill out the table?* In other words, what makes an agent good or bad, intelligent or stupid? We answer these questions in the next section.

2.4 Good Behavior: The Concept of Rationality

A **rational agent** is one that does the right thing—conceptually speaking, every entry in the table for the agent function is filled out correctly. When an agent is plunked down in an environment, it generates a sequence of actions according to the percepts it receives. This sequence of actions causes the environment to go through a sequence of states. If the sequence is desirable, then the agent has performed well. This notion of desirability is captured by a **performance measure** that evaluates any given sequence of environment states. Notice that we said *environment* states, not *agent* states. If we define success in terms of agent's opinion of its own performance, an agent could achieve perfect rationality simply by deluding itself that its performance was perfect. Human agents in particular are notorious for "sour grapes"—believing they did not really want something (e.g., a Nobel Prize) after not getting it.

Obviously, there is not one fixed performance measure for all tasks and agents; typically, a designer will devise one appropriate to the circumstances. This is not as easy as it sounds. Consider, for example, the vacuum-cleaner agent from the preceding section. We might propose to measure performance by the amount of dirt cleaned up in a single eight-hour shift. With a rational agent, of course, what you ask for is what you get. A rational agent can maximize this performance measure by cleaning up the dirt, then dumping it all on the floor, then cleaning it up again, and so on. A more suitable performance measure would reward the agent for having a clean floor. For example, one point could be awarded for each clean square at each time step. As a general rule, it is better to design performance measures according to what one actually wants in the environment, rather than according to how one thinks the agent should behave.

2.4.1 Rationality

What is rational at any given time depends on four things:

• The performance measure that defines the criterion of success.

• The agent's prior knowledge of the environment.

• The actions that the agent can perform.

• The agent's percept sequence to date.

This leads to a **definition of a rational agent**:

For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Consider the simple vacuum-cleaner agent that cleans a square if it is dirty and moves to the other square if not; this is the agent function tabulated in Figure 2.3. Is this a rational agent? That depends! First, we need to say what the performance measure is, what is known about the environment, and what sensors and actuators the agent has. Let us assume the following:

• The performance measure awards one point for each clean square at each time step, over a "lifetime" of 1000 time steps.

• The "geography" of the environment is known *a priori* (Figure 2.2) but the dirt distribution and the initial location of the agent are not. Clean squares stay clean and sucking cleans the current square. The Left and Right actions move the agent left and right except when this would take the agent outside the environment, in which case the agent remains where it is.

• The only available actions are Left , Right, and Suck.

• The agent correctly perceives its location and whether that location contains dirt.

We claim that *under these circumstances* the agent is indeed rational; its expected performance is at least as high as any other agent's. One can see easily that the same agent would be irrational under different circumstances. For example, once all the dirt is cleaned up, the agent will oscillate needlessly back and forth; if the performance measure includes a penalty of one point for each movement left or right, the agent will fare poorly. A better agent for this case would do nothing once it is sure that all the squares are clean. If clean squares can become dirty again, the agent should occasionally check and re-clean them if needed. If the geography of the environment is unknown, the agent will need to explore it rather than stick to squares A and B.

2.5 The Nature of Environments

Now that we have a definition of rationality, we are almost ready to think about building rational agents. First, however, we TASK ENVIRONMENT must think about **task environments**, which are essentially the "problems" to which rational agents are the "solutions." We begin by showing how to specify a task environment, illustrating the process with a number of examples. We then show that task environments come in a variety of flavors. The flavor of the task environment directly affects the appropriate design for the agent program.

2.5.1 Specifying the task environment

In our discussion of the rationality of the simple vacuum-cleaner agent, we had to specify the performance measure, the environment, and the agent's actuators and sensors. We group all these under the heading of the **task environment**. For the acronymically minded, we call this the **PEAS** (**P**erformance, **E**nvironment, **A**ctuators, **S**ensors) description. In designing an agent, the first step must always be to specify the task environment as fully as possible.

The vacuum world was a simple example; let us consider a more complex problem: an automated taxi driver. We should point out, before the reader becomes alarmed, that a fully automated taxi is currently somewhat beyond the capabilities of existing technology. The full driving task is extremely *open-ended*. There is no limit to the novel combinations of circumstances that can arise—another reason we chose it as a focus for discussion. Figure 2.4 summarizes the PEAS description for the taxi's task environment.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Figure 2.4 PEAS description of the task environment for an automated taxi.

First, what is the **performance measure** to which we would like our automated driver to aspire? Desirable qualities include getting to the correct destination; minimizing fuel consumption and wear and tear; minimizing the trip time or cost; minimizing violations of traffic laws and

disturbances to other drivers; maximizing safety and passenger comfort; maximizing profits. Obviously, some of these goals conflict, so trade-offs will be required.

Next, what is the driving **environment** that the taxi will face? Any taxi driver must deal with a variety of roads, ranging from rural lanes and urban alleys to 12-lane freeways. The roads contain other traffic, pedestrians, stray animals, road works, police cars, puddles, and potholes. The taxi must also interact with potential and actual passengers. There are also some optional choices. The taxi might need to operate in Southern California, where snow is seldom a problem, or in Alaska, where it seldom is not. It could always be driving on the right, or we might want it to be flexible enough to drive on the left when in Britain or Japan. Obviously, the more restricted the environment, the easier the design problem.

The **actuators** for an automated taxi include those available to a human driver: control over the engine through the accelerator and control over steering and braking. In addition, it will need output to a display screen or voice synthesizer to talk back to the passengers, and perhaps some way to communicate with other vehicles, politely or otherwise.

The basic **sensors** for the taxi will include one or more controllable video cameras so that it can see the road; it might augment these with infrared or sonar sensors to detect distances to other cars and obstacles. To avoid speeding tickets, the taxi should have a speedometer, and to control the vehicle properly, especially on curves, it should have an accelerometer. To determine the mechanical state of the vehicle, it will need the usual array of engine, fuel, and electrical system sensors. Like many human drivers, it might want a global positioning system (GPS) so that it doesn't get lost. Finally, it will need a keyboard or microphone for the passenger to request a destination.

In Figure 2.5, we have sketched the basic PEAS elements for a number of additional agent types. It may come as a surprise to some readers that our list of agent types includes some programs that operate in the entirely artificial environment defined by keyboard input and character output on a screen. "Surely," one might say, "this is not a real environment, is it?" In fact, what matters is not the distinction between "real" and "artificial" environments, but the complexity of the relationship among the behaviour of the agent, the percept sequence generated by the environment, and the performance measure. Some "real" environments are actually quite simple. For example, a robot designed to inspect parts as they come by on a conveyor belt can make use of a number of simplifying assumptions: that the lighting is always just so, that

the only thing on the conveyor belt will be parts of a kind that it knows about, and that only two actions (accept or reject) are possible.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Figure 2.5 Examples of agent types and their PEAS descriptions.

In contrast, some **software agents** (or software robots or **softbots**) exist in rich, unlimited domains. Imagine a softbot Web site operator designed to scan Internet news sources and show the interesting items to its users, while selling advertising space to generate revenue. To do well, that operator will need some natural language processing abilities, it will need to learn what each user and advertiser is interested in, and it will need to change its plans dynamically—for example, when the connection for one news source goes down or when a new one comes online. The Internet is an environment whose complexity rivals that of the physical world and whose inhabitants include many artificial and human agents.

2.5.2 Properties of Task Environment

The range of task environments that might arise in AI is obviously vast. We can, however, identify a fairly small number of dimensions along which task environments can be categorized. These dimensions determine, to a large extent, the appropriate agent design and the applicability of each of the principal families of techniques for agent implementation. First, we list the dimensions, then we analyze several task environments to illustrate the ideas.

Fully observable vs. partially observable

If an agent's sensors give it access to the complete state of the environment at each point in time, then we say that the task environment is fully observable. A task environment is effectively fully observable if the sensors detect all aspects that are *relevant* to the choice of action; relevance, in turn, depends on the performance measure. Fully observable environments are convenient because the agent need not maintain any internal state to keep track of the world. An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data—for example, a vacuum agent with only a local dirt sensor cannot tell whether there is dirt in other squares, and an automated taxi cannot see what other drivers are thinking. If the agent has no sensors at all then the environment is **unobservable**. One might think that in such cases the agent's plight is hopeless, but, the agent's goals may still be achievable, sometimes with certainty.

Single agent vs. multiagent

The distinction between single-agent and multiagent environments may seem simple enough. For example, an agent solving a crossword puzzle by itself is clearly in a single-agent environment, whereas an agent playing chess is in a two agent environment. There are, however, some subtle issues. First, we have described how an entity *may* be viewed as an agent, but we have not explained which entities *must* be viewed as agents. The key distinction is whether B's behavior is best described as maximizing a performance measure whose value depends on agent A's behavior. For example, in chess, the opponent entity B is trying to maximize its performance measure, which, by the rules of chess, minimizes agent A's performance measure. Thus, chess is a **competitive** multiagent environment. In the taxi-driving environment, on the other hand, avoiding collisions maximizes the performance measure of all agents, so it is a partially **cooperative** multiagent environment. It is also partially competitive because, for example, only one car can occupy a parking space. The agent-design problems in multiagent environments are often quite different from those in single-agent environments; for example, **communication**

often emerges as a rational behavior in multiagent environments; in some competitive environments, **randomized behavior** is rational because it avoids the pitfalls of predictability.

Deterministic vs. stochastic.

If the next state of the environment is completely determined by the current state and the action executed by the agent, then we say the environment is deterministic; otherwise, it is stochastic. In principle, an agent need not worry about uncertainty in a fully observable, deterministic environment. If the environment is partially observable, however, then it could *appear* to be stochastic. Most real situations are so complex that it is impossible to keep track of all the unobserved aspects; for practical purposes, they must be treated as stochastic. Taxi driving is clearly stochastic in this sense, because one can never predict the behavior of traffic exactly; moreover, one's tires blow out and one's engine seizes up without warning. The vacuum world as we described it is deterministic, but variations can include stochastic elements such as randomly appearing dirt and an unreliable suction mechanism. We say an environment is **uncertain** if it is not fully observable or not deterministic. A **nondeterministic** environment is one in which actions are characterized by their *possible* outcomes, but no probabilities are attached to them. Nondeterministic environment descriptions are usually associated with performance measures that require the agent to succeed for *all possible* outcomes of its actions.

Episodic vs. sequential:

In an episodic task environment, the agent's experience is divided into atomic episodes. In each episode the agent receives a percept and then performs a single action. Crucially, the next episode does not depend on the actions taken in previous episodes. Many classification tasks are episodic. For example, an agent that has to spot defective parts on an assembly line bases each decision on the current part, regardless of previous decisions; moreover, the current decision doesn't affect whether the next part is defective. In sequential environments, on the other hand, the current decision could affect all future decisions.3 Chess and taxi driving are sequential: in both cases, short-term actions can have long-term consequences. Episodic environments are much simpler than sequential environments because the agent does not need to think ahead.

Static vs. dynamic

If the environment can change while an agent is deliberating, then we say the environment is dynamic for that agent; otherwise, it is static. Static environments are easy to deal with because the agent need not keep looking at the world while it is deciding on an action, nor need it worry

about the passage of time. Dynamic environments, on the other hand, are continuously asking the agent what it wants to do; if it hasn't decided yet, that counts as deciding to do nothing. If the environment itself does not change with the passage of time but the agent's performance score does, then we say the environment is **semidynamic**. Taxi driving is clearly dynamic: the other cars and the taxi itself keep moving while the driving algorithm is indecisive about what to do next. Chess, when played with a clock, is semidynamic. Crossword puzzles are static.

Discrete vs. continuous:

The discrete/continuous distinction applies to the *state* of the environment, to the way *time* is handled, and to the *percepts* and *actions* of the agent. For example, the chess environment has a finite number of distinct states (excluding the clock). Chess also has a discrete set of percepts and actions. Taxi driving is a continuous-state and continuous-time problem: the speed and location of the taxi and of the other vehicles sweep through a range of continuous values and do so smoothly over time. Taxi-driving actions are also continuous (steering angles, etc.). Input from digital cameras is discrete, strictly speaking, but is typically treated as representing continuously varying intensities and locations.

Known vs. unknown

Strictly speaking, this distinction refers not to the environment itself but to the agent's (or designer's) state of knowledge about the "laws of physics" of the environment. In a known environment, the outcomes (or outcome probabilities if the environment is stochastic) for all actions are given. Obviously, if the environment is unknown, the agent will have to learn how it works in order to make good decisions. Note that the distinction between known and unknown environments is not the same as the one between fully and partially observable environments. It is quite possible for a *known* environment to be *partially* observable—for example, in solitaire card games, I know the rules but am still unable to see the cards that have not yet been turned over. Conversely, an *unknown* environment can be *fully* observable—in a new video game, the screen may show the entire game state but I still don't know what the buttons do until I try them.

As one might expect, the hardest case is *partially observable*, *multiagent*, *stochastic*, *sequential*, *dynamic*, *continuous*, and *unknown*. Taxi driving is hard in all these senses, except that for the most part the driver's environment is known. Driving a rented car in a new country with unfamiliar geography and traffic laws is a lot more exciting.

Figure 2.6 lists the properties of a number of familiar environments. Note that the answers are not always cut and dried. For example, we describe the part-picking robot as episodic, because it normally considers each part in isolation. But if one day there is a large batch of defective parts, the robot should learn from several observations that the distribution of defects has changed, and should modify its behavior for subsequent parts. We have not included a "known/unknown" column because, as explained earlier, this is not strictly a property of the environment. For some environments, such as chess and poker, it is quite easy to supply the agent with full knowledge of the rules, but it is nonetheless interesting to consider how an agent might learn to play these games without such knowledge.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Figure 2.6 Examples of task environments and their characteristics.

Several of the answers in the table depend on how the task environment is defined. We have listed the medical-diagnosis task as single-agent because the disease process in a patient is not profitably modeled as an agent; but a medical-diagnosis system might also have to deal with uncooperative patients and skeptical staff, so the environment could have a multiagent aspect. Furthermore, medical diagnosis is episodic if one conceives of the task as selecting a diagnosis given a list of symptoms; the problem is sequential if the task can include proposing a series of tests, evaluating progress over the course of treatment, and so on. Also, many environments are episodic at higher levels than the agent's individual actions. For example, a chess tournament consists of a sequence of games; each game is an episode because (by and large) the contribution of the moves in one game to the agent's overall performance is not

affected by the moves in its previous game. On the other hand, decision making within a single game is certainly sequential.

2.6 The Structure of Agents

So far we have talked about agents by describing *behavior*—the action that is performed after any given sequence of percepts. The job of AI is to design an **agent program** that implements the agent function—the mapping from percepts to actions. We assume this program will run on some sort of computing device with physical sensors and actuators—we call this the **architecture**:

agent = architecture + program

2.6.1 Agent programs

The agent programs that we design in this book all have the same skeleton: they take the current percept as input from the sensors and return an action to the actuators. Notice the difference between the agent program, which takes the current percept as input, and the agent function, which takes the entire percept history. The agent program takes just the current percept as input because nothing more is available from the environment; if the agent's actions need to depend on the entire percept sequence, the agent will have to remember the percepts.

We describe the agent programs in the simple pseudocode language. For example, Figure 2.7 shows a rather trivial agent program that keeps track of the percept sequence and then uses it to index into a table of actions to decide what to do. To build a rational agent in this way, we as designers must construct a table that contains the appropriate action for every possible percept sequence.

```
function TABLE-DRIVEN-AGENT(percept) returns an action
  persistent: percepts, a sequence, initially empty
              table, a table of actions, indexed by percept sequences, initially fully specified

  append percept to the end of percepts
  action ← LOOKUP(percepts, table)
  return action
```

Figure 2.7 The TABLE-DRIVEN-AGENT program is invoked for each new percept and returns an action each time. It retains the complete percept sequence in memory.

It is instructive to consider why the table-driven approach to agent construction is doomed to failure. Consider the automated taxi: the visual input from a single camera comes in at the rate of roughly 27 megabytes per second (30 frames per second, 640×480 pixels with 24 bits of color information). This gives a lookup table with over 10250,000,000,000 entries for an hour's driving. Even the lookup table for chess—a tiny, well-behaved fragment of the real world—would have at least 10150 entries. The daunting size of these tables (the number of atoms in the observable universe is less than 1080) means that (a) no physical agent in this universe will have the space to store the table, (b) the designer would not have time to create the table, (c) no agent could ever learn all the right table entries from its experience, and (d) even if the environment is simple enough to yield a feasible table size, the designer still has no guidance about how to fill in the table entries.

In the remainder of this section, we outline four basic kinds of agent programs that embody the principles underlying almost all intelligent systems:

• Simple reflex agents;

• Model-based reflex agents;

• Goal-based agents; and

• Utility-based agents.

Each kind of agent program combines particular components in particular ways to generate actions. Section 2.4.6 explains in general terms how to convert all these agents into *learning agents* that can improve the performance of their components so as to generate better actions. Finally, Section 2.4.7 describes the variety of ways in which the components themselves can be represented within the agent.

2.6.2 Simple reflex agents

The simplest kind of agent is the **simple reflex agent**. These agents select actions on the basis of the *current* percept, ignoring the rest of the percept history. For example, the vacuum agent whose agent function is tabulated in Figure 2.3 is a simple reflex agent, because its decision is based only on the current location and on whether that location contains dirt. An agent program for this agent is shown in Figure 2.8.

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action
  if status = Dirty then return Suck
  else if location = A then return Right
  else if location = B then return Left
```

Figure 2.8 The agent program for a simple reflex agent in the two-state vacuum environment. This program implements the agent function tabulated in Figure 2.3.

Notice that the vacuum agent program is very small indeed compared to the corresponding table. The most obvious reduction comes from ignoring the percept history, which cuts down the number of possibilities from 4T to just 4. A further, small reduction comes from the fact that when the current square is dirty, the action does not depend on the location.

Simple reflex behaviors occur even in more complex environments. Imagine yourself as the driver of the automated taxi. If the car in front brakes and its brake lights come on, then you should notice this and initiate braking. In other words, some processing is done on the visual input to establish the condition we call "The car in front is braking." Then, this triggers some established connection in the agent program to the action "initiate braking." We call such a connection a **condition–action rule**, written as

if ***car-in-front-is-braking*** then ***initiate-braking*****.**

Humans also have many such connections, some of which are learned responses (as for driving) and some of which are innate reflexes (such as blinking when something approaches the eye). The program in Figure 2.8 is specific to one particular vacuum environment. A more general and flexible approach is first to build a general-purpose interpreter for condition–action rules and then to create rule sets for specific task environments. Figure 2.9 gives the structure of this general program in schematic form, showing how the condition–action rules allow the agent to make the connection from percept to action. We use rectangles to denote the current internal state of the agent's decision process, and ovals to represent the background information used in the process. The agent program, which is also very simple, is shown in Figure 2.10. The function generates an abstracted description of the current state from the percept, and the function returns the first rule in the set of rules that matches the given state description. Note that the description in terms of "rules" and "matching" is purely conceptual; actual implementations can be as simple as a collection of logic gates implementing a Boolean circuit.

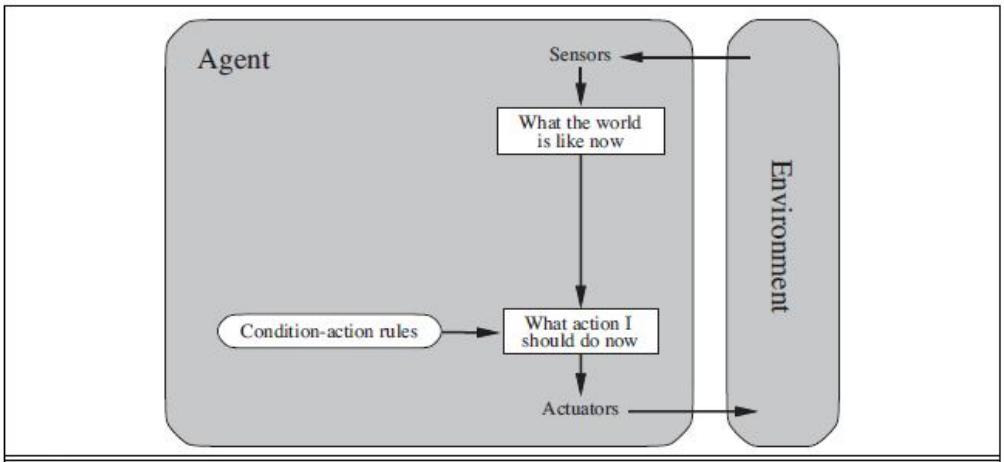


Figure 2.9 Schematic diagram of a simple reflex agent.

```
function SIMPLE-REFLEX-AGENT(percept) returns an action
  persistent: rules, a set of condition–action rules

  state ← INTERPRET-INPUT(percept)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

Figure 2.10 A simple reflex agent. It acts according to a rule whose condition matches the current state, as defined by the percept.

Simple reflex agents have the admirable property of being simple, but they turn out to be of limited intelligence. The agent in Figure 2.10 will work *only if the correct decision can be made on the basis of only the current percept—that is, only if the environment is fully observable.* Even a little bit of unobservability can cause serious trouble. For example, the braking rule given earlier assumes that the condition *car-in-front-is-braking* can be determined from the current percept—a single frame of video. This works if the car in front has a centrally mounted brake light. Unfortunately, older models have different configurations of taillights, brake lights, and turn-signal lights, and it is not always possible to tell from a single image whether the car is braking. A simple reflex agent driving behind such a car would either brake continuously and unnecessarily, or, worse, never brake at all.

We can see a similar problem arising in the vacuum world. Suppose that a simple reflex vacuum agent is deprived of its location sensor and has only a dirt sensor. Such an agent has just two possible percepts: [Dirty] and [Clean]. It can Suck in response to [Dirty]; what should it do in response to [Clean]? Moving Left fails (forever) if it happens to start in square A, and moving Right fails (forever) if it happens to start in square B. Infinite loops are often unavoidable for simple reflex agents operating in partially observable environments.

2.6.3 Model-based reflex agents

The most effective way to handle partial observability is for the agent to *keep track of the part of the world it can't see now*. That is, the agent should maintain some sort of **internal state** that depends on the percept history and thereby reflects at least some of the unobserved aspects of the current state. For the braking problem, the internal state is not too extensive—just the previous frame from the camera, allowing the agent to detect when two red lights at the edge of the vehicle go on or off simultaneously. For other driving tasks such as changing lanes, the agent needs to keep track of where the other cars are if it can't see them all at once. And for any driving to be possible at all, the agent needs to keep track of where its keys are.

Updating this internal state information as time goes by requires two kinds of knowledge to be encoded in the agent program. First, we need some information about how the world evolves independently of the agent—for example, that an overtaking car generally will be closer behind than it was a moment ago. Second, we need some information about how the agent's own actions affect the world—for example, that when the agent turns the steering wheel clockwise, the car turns to the right, or that after driving for five minutes northbound on the freeway, one is usually about five miles north of where one was five minutes ago. This knowledge about "how the world works"—whether implemented in simple Boolean circuits or in complete scientific theories—is called a **model** of the world. An agent that uses such a model is called a **model-based agent**.

Figure 2.11 gives the structure of the model-based reflex agent with internal state, showing how the current percept is combined with the old internal state to generate the updated description of the current state, based on the agent's model of how the world works. The agent program is shown in Figure 2.12. The interesting part is the function, which is responsible for creating the new internal state description. The details of how models and states are represented vary widely depending on the type of environment and the particular technology used in the agent design.

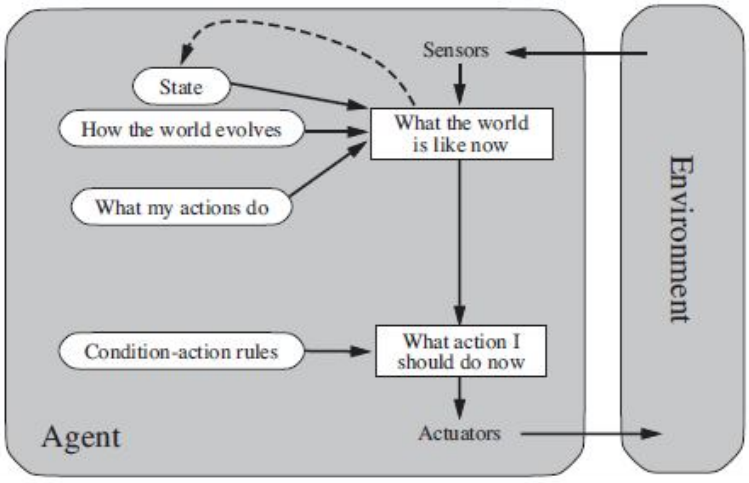


Figure 2.11 A model-based reflex agent.

function MODEL-BASED-REFLEX-AGENT(*percept*) **returns** an action
persistent: *state*, the agent's current conception of the world state
model, a description of how the next state depends on current state and action
rules, a set of condition–action rules
action, the most recent action, initially none

state ← UPDATE-STATE(*state*, *action*, *percept*, *model*)
rule ← RULE-MATCH(*state*, *rules*)
action ← *rule*.ACTION
return *action*

Figure 2.12 A model-based reflex agent. It keeps track of the current state of the world, using an internal model. It then chooses an action in the same way as the reflex agent.

Regardless of the kind of representation used, it is seldom possible for the agent to determine the current state of a partially observable environment *exactly*. Instead, the box labeled "what the world is like now" (Figure 2.11) represents the agent's "best guess". For example, an automated taxi may not be able to see around the large truck that has stopped in front of it and can only guess about what may be causing the hold-up. Thus, uncertainty about the current state may be unavoidable, but the agent still has to make a decision.

2.6.4 Goal-based agents

Knowing something about the current state of the environment is not always enough to decide what to do. For example, at a road junction, the taxi can turn left, turn right, or go straight on. The correct decision depends on where the taxi is trying to get to. In other words, as well as a current state description, the agent needs some sort of **goal** information that describes situations that are desirable—for example, being at the passenger's destination. The agent program can combine this with the model (the same information as was used in the model based reflex agent) to choose actions that achieve the goal. Figure 2.13 shows the goal-based agent's structure.

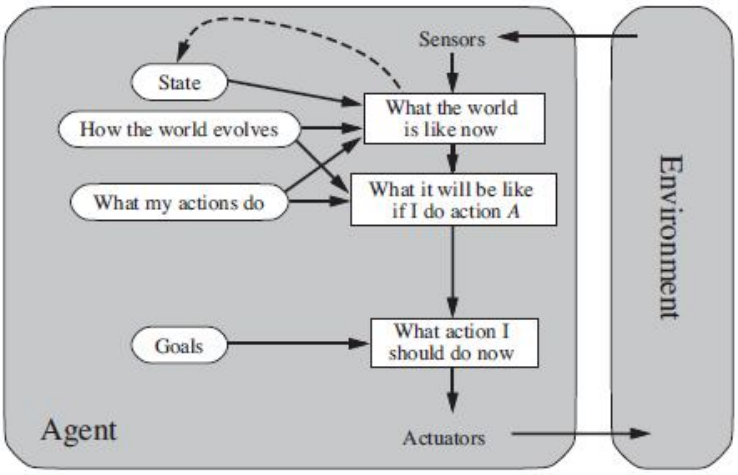


Figure 2.13 A model-based, goal-based agent. It keeps track of the world state as well as a set of goals it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.

Sometimes goal-based action selection is straightforward—for example, when goal satisfaction results immediately from a single action. Sometimes it will be more tricky—for example, when the agent has to consider long sequences of twists and turns in order to find a way to achieve the goal. **Search** and **planning** are the subfields of AI devoted to finding action sequences that achieve the agent's goals.

Notice that decision making of this kind is fundamentally different from the condition–action rules described earlier, in that it involves consideration of the future—both "What will

happen if I do such-and-such?" and "Will that make me happy?" In the reflex agent designs, this information is not explicitly represented, because the built-in rules map directly from percepts to actions. The reflex agent brakes when it sees brake lights. A goal-based agent, in principle, could reason that if the car in front has its brake lights on, it will slow down. Given the way the world usually evolves, the only action that will achieve the goal of not hitting other cars is to brake.

Although the goal-based agent appears less efficient, it is more flexible because the knowledge that supports its decisions is represented explicitly and can be modified. If it starts to rain, the agent can update its knowledge of how effectively its brakes will operate; this will automatically cause all of the relevant behaviors to be altered to suit the new conditions. For the reflex agent, on the other hand, we would have to rewrite many condition–action rules. The goal-based agent's behavior can easily be changed to go to a different destination, simply by specifying that destination as the goal. The reflex agent's rules for when to turn and when to go straight will work only for a single destination; they must all be replaced to go somewhere new.

2.6.5 Utility-based agents

Goals alone are not enough to generate high-quality behavior in most environments. For example, many action sequences will get the taxi to its destination (thereby achieving the goal) but some are quicker, safer, more reliable, or cheaper than others. Goals just provide a crude binary distinction between "happy" and "unhappy" states. A more general performance measure should allow a comparison of different world states according to exactly how happy they would make the agent. Because "happy" does not sound very scientific, economists and computer scientists use the term **utility** instead.

We have already seen that a performance measure assigns a score to any given sequence of environment states, so it can easily distinguish between more and less desirable ways of getting to the taxi's destination. An agent's **utility function** is essentially an internalization of the performance measure. If the internal utility function and the external performance measure are in agreement, then an agent that chooses actions to maximize its utility will be rational according to the external performance measure.

A rational agent program has no idea what its utility function is—but, like goal-based agents, a utility-based agent has many advantages in terms of flexibility and learning. Furthermore, in two kinds of cases, goals are inadequate but a utility-based agent can still make rational decisions.

First, when there are conflicting goals, only some of which can be achieved (for example, speed and safety), the utility function specifies the appropriate tradeoff. Second, when there are several goals that the agent can aim for, none of which can be achieved with certainty, utility provides a way in which the likelihood of success can be weighed against the importance of the goals.

Partial observability and stochasticity are ubiquitous in the real world, and so, therefore, is decision making under uncertainty. Technically speaking, a rational utility-based agent chooses the action that maximizes the **expected utility** of the action outcomes—that is, the utility the agent expects to derive, on average, given the probabilities and utilities of each outcome. Any rational agent must behave *as if* it possesses a utility function whose expected value it tries to maximize. An agent that possesses an *explicit* utility function can make rational decisions with a general-purpose algorithm that does not depend on the specific utility function being maximized. In this way, the "global" definition of rationality—designating as rational those agent functions that have the highest performance—is turned into a "local" constraint on rational-agent designs that can be expressed in a simple program.

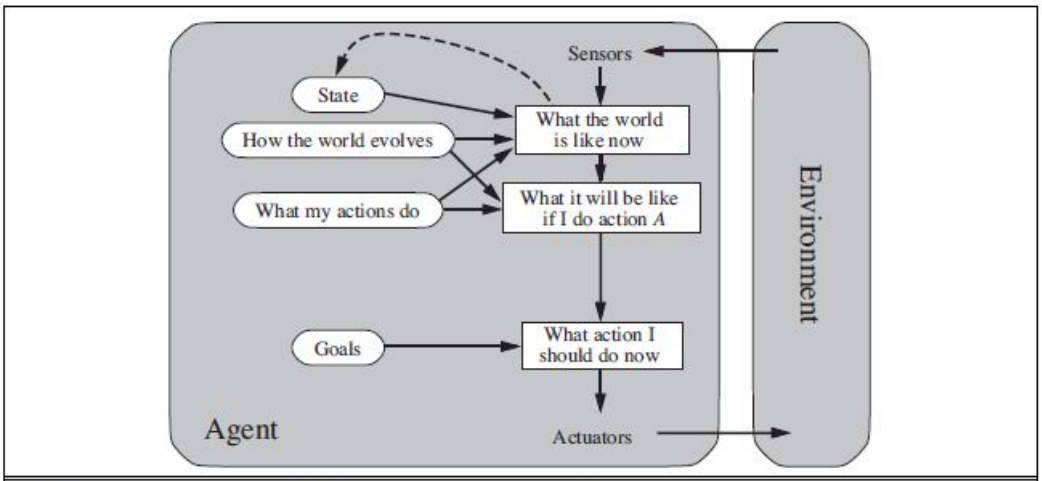


Figure 2.13 A model-based, goal-based agent. It keeps track of the world state as well as a set of goals it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.

The utility-based agent structure appears in Figure 2.14. Utility-based agent programs appear in Part IV, where we design decision-making agents that must handle the uncertainty inherent in stochastic or partially observable environments.

2.6.6 Learning agents

Alan Turing (1950) proposed a method to build learning machines and then to teach them. In many areas of AI, this is now the preferred method for creating state-of-the-art systems. Learning has another advantage like it allows the agent to operate in initially unknown environments and to become more competent than its initial knowledge alone might allow. In this section, we briefly introduce the main ideas of learning agents.

A learning agent can be divided into four conceptual components, as shown in Figure 2.15. The most important distinction is between the **learning element**, which is responsible for making improvements, and the **performance element**, which is responsible for selecting external actions. The performance element is what we have previously considered to be the entire agent: it takes in percepts and decides on actions. The learning element uses feedback from the **critic** on how the agent is doing and determines how the performance element should be modified to do better in the future.

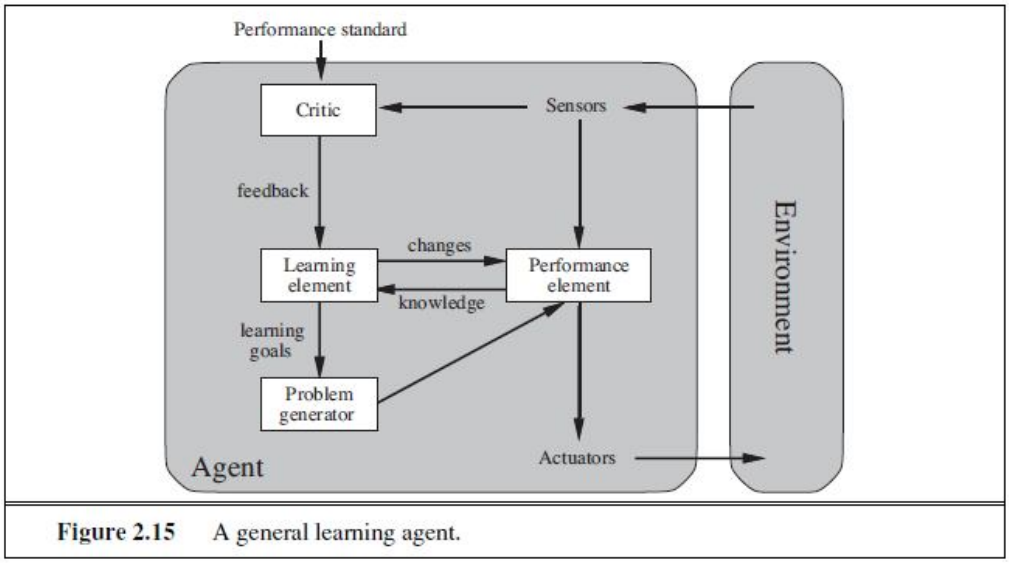


Figure 2.15 A general learning agent.

The design of the learning element depends very much on the design of the performance element. When trying to design an agent that learns a certain capability, the first question is not “How am I going to get it to learn this?” but “What kind of performance element will my agent need to do this once it has learned how?” Given an agent design, learning mechanisms can be constructed to improve every part of the agent.

The critic tells the learning element how well the agent is doing with respect to a fixed performance standard. The critic is necessary because the percepts themselves provide no indication of the agent's success. For example, a chess program could receive a percept indicating that it has checkmated its opponent, but it needs a performance standard to know that this is a good thing; the percept itself does not say so. It is important that the performance standard be fixed. Conceptually, one should think of it as being outside the agent altogether because the agent must not modify it to fit its own behavior.

The last component of the learning agent is the **problem generator**. It is responsible for suggesting actions that will lead to new and informative experiences. The point is that if the performance element had its way, it would keep doing the actions that are best, given what it knows. But if the agent is willing to explore a little and do some perhaps suboptimal actions in the short run, it might discover much better actions for the long run. The problem generator's job is to suggest these exploratory actions. This is what scientists do when they carry out experiments.

To make the overall design more concrete, let us return to the automated taxi example. The performance element consists of whatever collection of knowledge and procedures the taxi has for selecting its driving actions. The taxi goes out on the road and drives, using this performance element. The critic observes the world and passes information along to the learning element. For example, after the taxi makes a quick left turn across three lanes of traffic, the critic observes the shocking language used by other drivers. From this experience, the learning element is able to formulate a rule saying this was a bad action, and the performance element is modified by installation of the new rule. The problem generator might identify certain areas of behavior in need of improvement and suggest experiments, such as trying out the brakes on different road surfaces under different conditions.

The learning element can make changes to any of the "knowledge" components shown in the agent diagrams. The simplest cases involve learning directly from the percept sequence. Observation of pairs of successive states of the environment can allow the agent to learn "How the world evolves," and observation of the results of its actions can allow the agent to learn "What my actions do." For example, if the taxi exerts a certain braking pressure when driving on a wet road, then it will soon find out how much deceleration is actually achieved. Clearly, these two learning tasks are more difficult if the environment is only partially observable.

In summary, agents have a variety of components, and those components can be represented in many ways within the agent program, so there appears to be great variety among learning methods. There is, however, a single unifying theme. Learning in intelligent agents can be summarized as a process of modification of each component of the agent to bring the components into closer agreement with the available feedback information, thereby improving the overall performance of the agent.

2.6.7 How the components of agent programs work

We have described agent programs (in very high-level terms) as consisting of various components. But there are some basic distinctions among the various ways that the components can represent the environment that the agent inhabits. Roughly speaking, we can place the representations along an axis of increasing complexity and expressive power—**atomic**, **factored**, and **structured**. To illustrate these ideas, it helps to consider a particular agent component, such as the one that deals with "What my actions do." This component describes the changes that might occur in the environment as the result of taking an action, and Figure 2.16 provides schematic depictions of how those transitions might be represented.

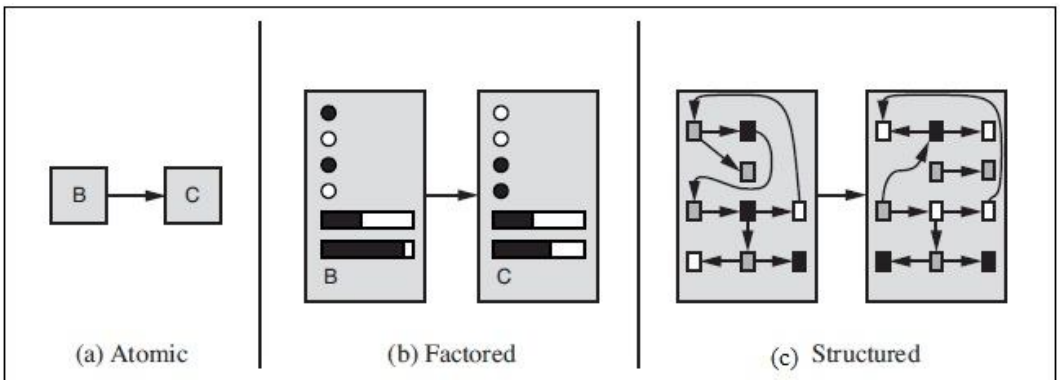


Figure 2.16 Three ways to represent states and the transitions between them. (a) Atomic representation: a state (such as B or C) is a black box with no internal structure; (b) Factored representation: a state consists of a vector of attribute values; values can be Boolean, real-valued, or one of a fixed set of symbols. (c) Structured representation: a state includes objects, each of which may have attributes of its own as well as relationships to other objects.

In an **atomic representation,** each state of the world is indivisible—it has no internal structure. Consider the problem of finding a driving route from one end of a country to the other via some sequence of cities. For the purposes of solving this problem, it may suffice to reduce

the state of world to just the name of the city we are in—a single atom of knowledge; a "black box" whose only discernible property is that of being identical to or different from another black box. The algorithms underlying **search** and **game-playing**, **HiddenMarkov models**, and **Markov decision processes** all work with atomic representations—or, at least, they treat representations *as if* they were atomic.

Now consider a higher-fidelity description for the same problem, where we need to be concerned with more than just atomic location in one city or another; we might need to pay attention to how much gas is in the tank, our current GPS coordinates, whether or not the oil warning light is working, how much spare change we have for toll crossings, what station is on the radio, and so on. A **factored representation** splits up each state into a fixed set of **variables** or **attributes**, each of which can have a **value**. While two different atomic states have nothing in common—they are just different black boxes—two different factored states can share some attributes (such as being at some particular GPS location) and not others (such as having lots of gas or having no gas); this makes it much easier to work out how to turn one state into another. With factored representations, we can also represent *uncertainty*—for example, ignorance about the amount of gas in the tank can be represented by leaving that attribute blank. Many important areas of AI are based on factored representations, including **constraint satisfaction** algorithms, **propositional logic**, **planning**, **Bayesian networks**, and the **machine learning** algorithms.

For many purposes, we need to understand the world as having *things* in it that are *related* to each other, not just variables with values. For example, we might notice that a large truck ahead of us is reversing into the driveway of a dairy farm but a cow has got loose and is blocking the truck's path. A factored representation is unlikely to be pre-equipped with the attribute *TruckAheadBackingIntoDairyFarmDrivewayBlockedByLooseCow* with value true or false. Instead, we would need a **structured representation**, in which objects such as cows and trucks and their various and varying relationships can be described explicitly. (See Figure 2.16(c).) Structured representations underlie **relational databases** and **first-order logic**, **first-order probability models**, **knowledge-based learning** and much of **natural language understanding**. In fact, almost everything that humans express in natural language concerns objects and their relationships.

As we mentioned earlier, the axis along which atomic, factored, and structured representations lie in the axis of increasing **expressiveness**. Roughly speaking, a more

expressive representation can capture, at least as concisely, everything a less expressive one can capture, plus some more. Often, the more expressive language is *much* more concise; for example, rules of chess can be written in a page or two of a structured-representation language such as first-order logic but require thousands of pages when written in a factored-representation language such as propositional logic. On the other hand, reasoning and learning become more complex as the expressive power of the representation increases. To gain the benefits of expressive representations while avoiding their drawbacks, intelligent systems for the real world may need to operate at all points along the axis simultaneously.

2.7 Summary

This chapter has been something of a whirlwind tour of AI, which we have conceived of as the science of agent design. The major points to recall are as follows:

• An **agent** is something that perceives and acts in an environment. The **agent function** for an agent specifies the action taken by the agent in response to any percept sequence.

• The **performance measure** evaluates the behavior of the agent in an environment. A **rational agent** acts so as to maximize the expected value of the performance measure, given the percept sequence it has seen so far.

• A **task environment** specification includes the performance measure, the external environment, the actuators, and the sensors. In designing an agent, the first step must always be to specify the task environment as fully as possible.

• Task environments vary along several significant dimensions. They can be fully or partially observable, single-agent or multiagent, deterministic or stochastic, episodic or sequential, static or dynamic, discrete or continuous, and known or unknown.

• The **agent program** implements the agent function. There exists a variety of basic agent-program designs reflecting the kind of information made explicit and used in the decision process. The designs vary in efficiency, compactness, and flexibility. The appropriate design of the agent program depends on the nature of the environment.

• **Simple reflex agents** respond directly to percepts, whereas **model-based reflex agents** maintain internal state to track aspects of the world that are not evident in the current percept.

Goal-based agents act to achieve their goals, and **utility-based agents** try to maximize their own expected “happiness.”

- All agents can improve their performance through **learning**.

2.8 Model Questions

1. What is an agent?
2. What are Simple Reflex Agents?
3. Explain Goal-based agents with diagram.
4. Differentiate: Single agent and Multiagent.
5. Explain in detail the four basic kinds of agent programs.
6. Describe the components of agent programs.

LESSON 3
SOLVING PROBLEMS BY SEARCHING

Structure

3.1 Introduction

The simplest agents discussed in our previous lesson were the reflex agents, which base their actions on a direct mapping from states to actions. Such agents cannot operate well in environments for which this mapping would be too large to store and would take too long to learn. Goal-based agents, on the other hand, consider future actions and the desirability of their outcomes. This lesson describes one kind of goal-based agent called a **problem-solving agent**. Problem-solving agents use **atomic** representations, that is, states of the world are considered as wholes, with no internal structure visible to the problem solving algorithms. Goal-based agents that use more advanced **factored** or **structured** representations are usually called **planning agents.** Our discussion of problem solving begins with precise definitions of **problems** and their **solutions** and give several examples to illustrate these definitions.

3.2 Objectives

- To introduce the basic concepts of problem-solving agents.
- To describe the example toy and real-world problems.

3.3 Problem-Solving Agents

Intelligent agents are supposed to maximize their performance measure. **Goal formulation**, based on the current situation and the agent's performance measure, is the first step in problem solving. We will consider a goal to be a set of world states—exactly those states in which the

goal is satisfied. The agent's task is to find out how to act, now and in the future, so that it reaches a goal state. Before it can do this, it needs to decide (or we need to decide on its behalf) what sorts of actions and states it should consider. If it were to consider actions at the level of "move the left foot forward an inch" or "turn the steering wheel one degree left," the agent would probably never find its way out of the parking lot, because at that level of detail there is too much uncertainty in the world and there would be too many steps in a solution. **Problem formulation** is the process of deciding what actions and states to consider, given a goal. In general, an agent with several immediate options of unknown value can decide what to do by first examining future actions that eventually lead to states of known value.

To be more specific about what we mean by "examining future actions," we have to be more specific about properties of the environment. For now, we assume that the environment is **observable**, so the agent always knows the current state. For the agent driving in Romania, it's reasonable to suppose that each city on the map has a sign indicating its presence to arriving drivers. We also assume the environment is **discrete**, so at any given state there are only finitely many actions to choose from. This is true for navigating in Romania because each city is connected to a small number of other cities. We will assume the environment is **known**, so the agent knows which states are reached by each action. Finally, we assume that the environment is **deterministic**, so each action has exactly one outcome.

Achieving this is sometimes simplified if the agent can adopt a **goal** and aim at satisfying it. Problem solving is an important aspect of Artificial Intelligence. A problem can be considered to consist of a goal and a set of actions that can be taken to lead to the goal. At any given time, we consider the state of the search space to represent where we have reached as a result of the actions we have applied so far. For example, consider the problem of looking for a contact lens on a football field. The initial state is how we start out, which is to say we know that the lens is somewhere on the field, but we don't know where. If we use the representation where we examine the field in units of one square foot, then our first action might be to examine the square in the top-left corner of the field. If we do not find the lens there, we could consider the state now to be that we have examined the top-left square and have not found the lens. After a number of actions, the state might be that we have examined 500 squares, and we have now just found the lens in the last square we examined. This is a goal state because it satisfies the goal that we had of finding a contact lens.

The process of looking for a sequence of actions that reaches the goal is called **search**. A search algorithm takes a problem as input and returns a **solution** in the form of an action sequence. Once a solution is found, the actions it recommends can be carried out. This is called the **execution** phase. Thus, we have a simple "formulate, search, execute" design for the agent, as shown in Figure 3.1. After formulating a goal and a problem to solve, the agent calls a search procedure to solve it. It then uses the solution to guide its actions, doing whatever the solution recommends as the next thing to do—typically, the first action of the sequence—and then removing that step from the sequence. Once the solution has been executed, the agent will formulate a new goal.

```
function SIMPLE-PROBLEM-SOLVING-AGENT(percept) returns an action
  persistent: seq, an action sequence, initially empty
              state, some description of the current world state
              goal, a goal, initially null
              problem, a problem formulation

  state ← UPDATE-STATE(state, percept)
  if seq is empty then
      goal ← FORMULATE-GOAL(state)
      problem ← FORMULATE-PROBLEM(state, goal)
      seq ← SEARCH(problem)
      if seq = failure then return a null action
  action ← FIRST(seq)
  seq ← REST(seq)
  return action
```

Figure 3.1 A simple problem-solving agent. It first formulates a goal and a problem, searches for a sequence of actions that would solve the problem, and then executes the actions one at a time. When this is complete, it formulates another goal and starts over.

A **problem** can be defined formally by five components:

• The **initial state** that the agent starts in.

• A description of the possible **actions** available to the agent.

• A description of what each action does; the formal name for this is the **transition model**, specified by a function RESULT*(s, a)* that returns the state that results from doing action *a* in state *s*.

- The **goal test**, which determines whether a given state is a goal state.

- A **path cost** function that assigns a numeric cost to each path.

3.4 Example Problems

The problem-solving approach has been applied to a vast array of task environments. We list some of the best known here, distinguishing between *toy* and *real-world* problems. A **toy problem** is intended to illustrate or exercise various problem-solving methods. It can be given a concise, exact description and hence is usable by different researchers to compare the performance of algorithms. A **real-world problem** is one whose solutions people actually care about. Such problems tend not to have a single agreed-upon description, but we can give **the general flavor of their formulations.**

3.4.1 Toy problems

The first example is the **vacuum world** first discussed in the Lesson-2. This can be formulated as a problem as follows:

• **States**: The state is determined by both the agent location and the dirt locations. The agent is in one of two locations, each of which might or might not contain dirt. Thus, there are 2 × 22 = 8 possible world states. A larger environment with n locations has n û0 2n states.

• **Initial state**: Any state can be designated as the initial state.

• **Actions**: In this simple environment, each state has just three actions: *Left*, *Right*, and *Suck*. Larger environments might also include *Up* and *Down*.

• **Transition model**: The actions have their expected effects, except that moving *Left* in the leftmost square, moving *Right* in the rightmost square, and *Suck*ing in a clean square have no effect.

• **Goal test**: This checks whether all the squares are clean.

• **Path cost**: Each step costs 1, so the path cost is the number of steps in the path. Compared with the real world, this toy problem has discrete locations, discrete dirt, reliable cleaning, and it never gets any dirtier. Chapter 4 relaxes some of these assumptions.

The **8-puzzle**, an instance of which is shown in Figure 3.2, consists of a 3×3 board with eight numbered tiles and a blank space. A tile adjacent to the blank space can slide into the space. The object is to reach a specified goal state, such as the one shown on the right of the figure. The standard formulation is as follows:

• **States**: A state description specifies the location of each of the eight tiles and the blank in one of the nine squares.

• **Initial state**: Any state can be designated as the initial state. Note that any given goal can be reached from exactly half of the possible initial states.

• **Actions**: The simplest formulation defines the actions as movements of the blank space *Left*, *Right*, *Up*, or *Down*. Different subsets of these are possible depending on where the blank is.

• **Transition model**: Given a state and action, this returns the resulting state; for example, if we apply *Left* to the start state in Figure 3.2, the resulting state has the 5 and the blank switched.

• **Goal test**: This checks whether the state matches the goal configurations are possible.

• Path cost: **Each step costs 1, so the path cost is the number of steps in the path.**

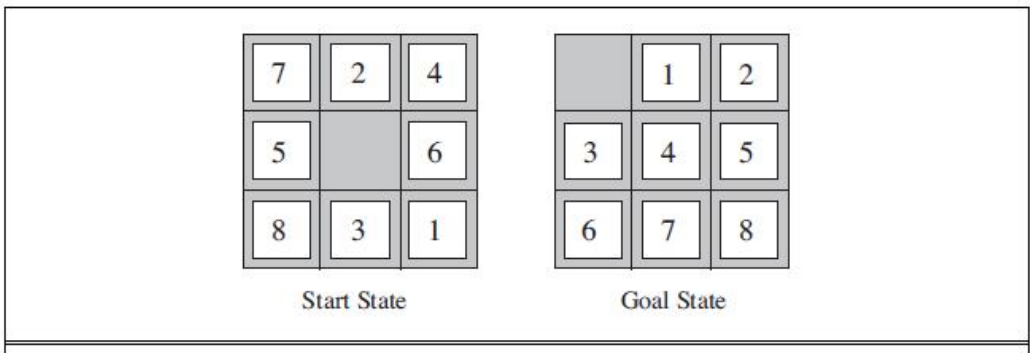


Figure 3.2. A typical instance of the 8-puzzle

The 8-puzzle belongs to the family SLIDING-BLOCK of **sliding-block puzzles**, which are often used as test problems for new search algorithms in AI. This family is known to be NP-complete, so one does not expect to find methods significantly better in the worst case than the search algorithms described in this chapter and the next. The 8-puzzle has 9!/2=181, 440 reachable states and is easily solved. The 15-puzzle (on a 4×4 board) has around 1.3 trillion

states, and random instances can be solved optimally in a few milliseconds by the best search algorithms. The 24-puzzle (on a 5 × 5 board) has around 1025 states, and random instances take several hours to solve optimally.

3.4.2 Real-world problems

We have already seen how the **route-finding problem** ROUTE-FINDING is defined in terms of specified locations and transitions along links between them. Route-finding algorithms are used in a variety of applications. Some, such as Web sites and in-car systems that provide driving directions, are relatively straightforward extensions of the Romania example. Others, such as routing video streams in computer networks, military operations planning, and airline travel-planning systems, involve much more complex specifications. Consider the airline travel problems that must be solved by a travel-planning Web site:

• **States**: Each state obviously includes a location (e.g., an airport) and the current time. Furthermore, because the cost of an action (a flight segment) may depend on previous segments, their fare bases, and their status as domestic or international, the state must record extra information about these "historical" aspects.

• **Initial state**: This is specified by the user's query.

• **Actions**: Take any flight from the current location, in any seat class, leaving after the current time, leaving enough time for within-airport transfer if needed.

• **Transition model**: The state resulting from taking a flight will have the flight's destination as the current location and the flight's arrival time as the current time.

• **Goal test**: Are we at the final destination specified by the user?

• **Path cost**: This depends on monetary cost, waiting time, flight time, customs and immigration procedures, seat quality, time of day, type of airplane, frequent-flyer mileage awards, and so on.

Commercial travel advice systems use a problem formulation of this kind, with many additional complications. **Touring problems** are closely related to route-finding problems. The **Traveling Salesperson Problem** (TSP) is a touring problem in which each city must be visited exactly once. The aim is to find the *shortest* tour. The problem is known to be NP-hard, but an enormous amount of effort has been expended to improve the capabilities of TSP

algorithms. In addition to planning trips for traveling salespersons, these algorithms have been used for tasks such as planning movements of automatic circuit-board drills and of stocking machines on shop floors.

Robot navigation is a generalization of the route-finding problems. Rather than following a discrete set of routes, a robot can move in a continuous space with (in principle) an infinite set of possible actions and states. For a circular robot moving on a flat surface, the space is essentially two-dimensional. When the robot has arms and legs or wheels that must also be controlled, the search space becomes many-dimensional. Advanced techniques are required just to make the search space finite. In addition to the complexity of the problem, real robots must also deal with errors in their sensor readings and motor controls.

Automatic assembly sequencing of complex objects by a robot was first demonstrated by (Michie, 1972). Progress since then has been slow but sure, to the point where the assembly of intricate objects such as electric motors is economically feasible. In assembly problems, the aim is to find an order in which to assemble the parts of some object. If the wrong order is chosen, there will be no way to add some part later in the sequence without undoing some of the work already done. Checking a step in the sequence for feasibility is a difficult geometrical search problem closely related to robot navigation. Thus, the generation of legal actions is the expensive part of assembly sequencing. Any practical algorithm must avoid exploring all but a tiny fraction of the state space. Another important assembly problem is **protein design**, in which the goal is to find a sequence of amino acids that will fold into a three-dimensional protein with the right properties to cure some disease.

3.5 Summary

This chapter has introduced methods that an agent can use to select actions in environments that are deterministic, observable, static, and completely known. In such cases, the agent can construct sequences of actions that achieve its goals; this process is called **search**. Before an agent can start searching for solutions, a **goal** must be identified and a well-defined problem must be formulated. A problem consists of five parts: the **initial state**, a set of **actions**, a **transition model** describing the results of those actions, a **goal test** function, and a **path cost** function. The environment of the problem is represented by a **state space**. A **path** through the state space from the initial state to a goal state is a **solution**.

3.6 Model Questions

1. What is problem-solving agent?
2. Define: Problem formulation.
3. What is a goal?
4. Explain about the five components of a problem.

LESSON 4
UNINFORMED SEARCH ALGORITHMS

Structure

4.1 Introduction

Search is a method that can be used by computers to examine a problem space like this in order to find a goal. Often, we want to find the goal as quickly as possible or without using too many resources. A problem space can also be considered to be a search space because in order to solve the problem, we will search the space for a goal state. We will continue to use the term search space to describe this concept. In this lesson, we will look at a number of methods for examining a search space. These methods are called search methods. There are several general-purpose search algorithms that can be used to solve the problems. In this lesson, we will see several **uninformed** search algorithms—algorithms that are given no information about the problem other than its definition. Although some of these algorithms can solve any solvable problem, none of them can do so efficiently.

4.2 Objectives

- To introduce the different types of search algorithms.
- To explore the various uninformed search algorithms with examples.

4.3 Searching Algorithms

There are various types of searching algorithms. When any type of searching is performed, there may some information about the searching or mayn't be. Also it is possible that the searching procedure may depend upon any constraints or rules. However, generally searching can be

classified into two types i.e. **uninformed (blind search) searching** and **informed (heuristic search) searching**. Also some other classifications of these searches are given below in the Figure 4.1.

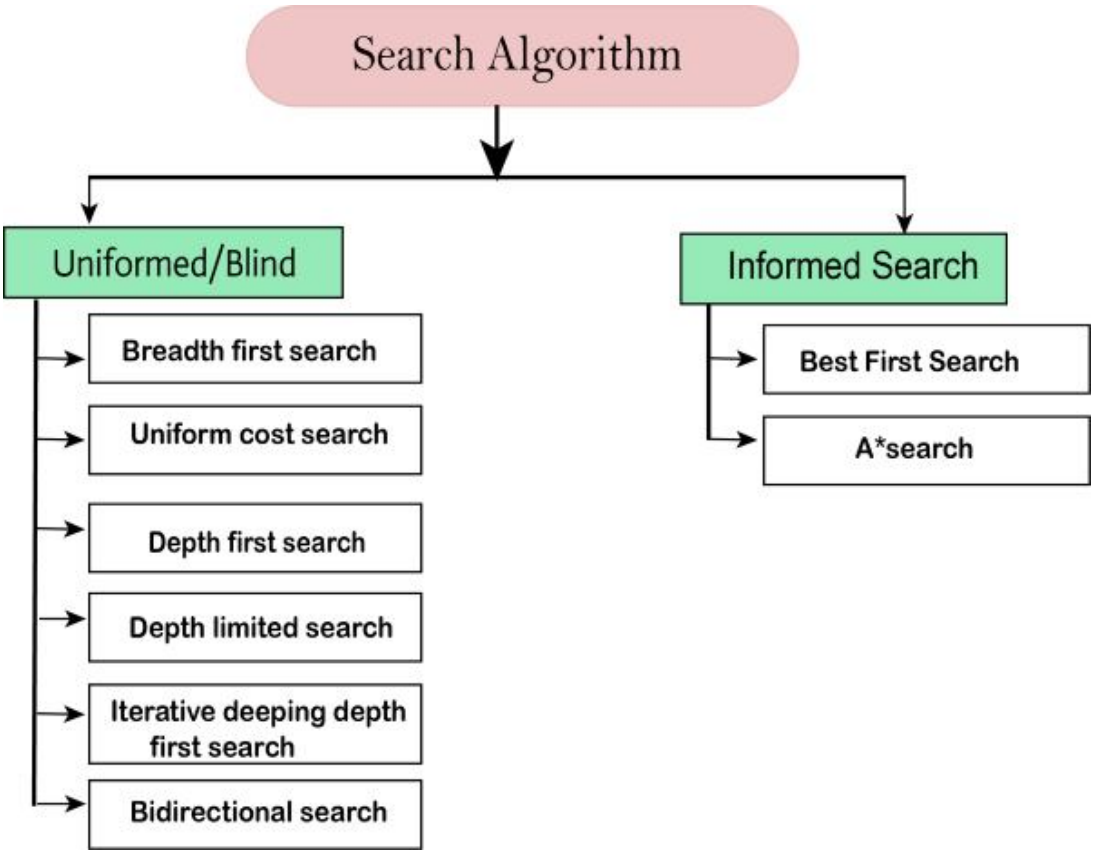


Figure 4.1. Types of Search Algorithms

Following are the four essential properties of search algorithms to compare the efficiency of these algorithms:

1) **Completeness:** A search algorithm is said to be complete if it guarantees to return a solution if at least any solution exists for any random input.

2) **Optimality:** If a solution found for an algorithm is guaranteed to be the best solution (lowest path cost) among all other solutions, then such a solution for is said to be an optimal solution.

3) **Time Complexity:** Time complexity is a measure of time for an algorithm to complete its task.

4) **Space Complexity:** It is the maximum storage space required at any point during the search, as the complexity of the problem.

4.4 Uninformed Search Strategies

This section covers several search strategies that come under the heading of **uninformed search** (also called **blind search**). The uninformed search does not contain any domain knowledge such as closeness, the location of the goal. It operates in a brute-force way as it only includes information about how to traverse the tree and how to identify leaf and goal nodes. Uninformed search applies a way in which search tree is searched without any information about the search space like initial state operators and test for the goal, so it is also called blind search.It examines each node of the tree until it achieves the goal node.

4.4.1 Breadth-First Search

Breadth-first search is a simple strategy in which the root node is expanded first, then all the successors of the root node are expanded next, then *their* successors, and so on. In general, all the nodes are expanded at a given depth in the search tree before any nodes at the next level are expanded.

Breadth first search is a general technique of traversing a graph. Breadth first search may use more memory but will always find the shortest path first. In this type of search the state space is represented in form of a tree. The solution is obtained by traversing through the tree. The nodes of the tree represent the start value or starting state, various intermediate states and the final state. In this search a queue data structure is used and it is level by level traversal. Breadth first search expands nodes in order of their distance from the root. It is a path finding algorithm that is capable of always finding the solution if one exists. The solution which is found is always the optional solution. This task is completed in a very memory intensive manner. Each node in the search tree is expanded in a breadth wise at each level.

Concept:

Step 1: Traverse the root node

Step 2: Traverse all neighbours of root node.

Step 3: Traverse all neighbours of neighbours of the root node.

Step 4: This process will continue until we are getting the goal node.

Algorithm:

Step 1: Place the root node inside the queue.

Step 2: If the queue is empty then stops and return failure.

Step 3: If the FRONT node of the queue is a goal node then stop and return success.

Step 4: Remove the FRONT node from the queue. Process it and find all its neighbours that are in ready state then place them inside the queue in any order.

Step 5: Go to Step 3.

Step 6: Exit.

Implementation:

Let us implement the above algorithm of BFS by taking the following suitable example in Figure 4.2.

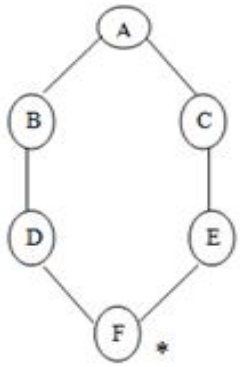


Figure 4.2 Example of BFS

Consider the graph in which let us take A as the starting node and F as the goal node (*)

Step 1:

Place the root node inside the queue i.e. A

Step 2:

Now the queue is not empty and also the FRONT node i.e. A is not our goal node. So move to step 3.

Step 3:

So remove the FRONT node from the queue i.e. A and find the neighbour of A i.e. B and C

B C A

Step 4:

Now B is the FRONT node of the queue .So process B and finds the neighbours of B i.e. D.

C D B

Step 5:

Now find out the neighbours of C i.e. E

D E C

Step 6:

Next find out the neighbours of D as D is the FRONT node of the queue

E F D

Step 7:

Now E is the front node of the queue. So the neighbour of E is F which is our goal node.

F E

Step 8:

Finally F is our goal node which is the FRONT of the queue. So exit.

F

Time Complexity: Time Complexity of BFS algorithm can be obtained by the number of nodes traversed in BFS until the shallowest Node. Where the d= depth of shallowest solution and b is a node at every state.

$$T(b) = 1+b^2+b^3+.....+ b^d= O(b^d)$$

Space Complexity: Space complexity of BFS algorithm is given by the Memory size of frontier which is $O(b^d)$.

Completeness: BFS is complete, which means if the shallowest goal node is at some finite depth, then BFS will find a solution.

Optimality: BFS is optimal if path cost is a non-decreasing function of the depth of the node.

Advantages:

In this procedure at any way it will find the goal. It does not follow a single unfruitful path for a long time. It finds the minimal solution in case of multiple paths.

Disadvantages:

BFS consumes large memory space. Its time complexity is more. It has long pathways, when all paths to a destination are on approximately the same search depth.

4.4.2 Depth First Search

Depth First Search (DFS) is also an important type of uniform search. DFS visits all the vertices in the graph. This type of algorithm always chooses to go deeper into the graph. After DFS visited all the reachable vertices from a particular sources vertices it chooses one of the remaining undiscovered vertices and continues the search. DFS reminds the space limitation

of breath first search by always generating next a child of the deepest unexpanded nodded. The data structure stack or Last In First Out (LIFO) is used for DFS. One interesting property of DFS is that, the discover and finish time of each vertex from a parenthesis structure. If we use one open parenthesis when a vertex is finished then the result is properly nested set of parenthesis.

Concept:

Step 1: Traverse the root node.

Step 2: Traverse any neighbour of the root node.

Step 3: Traverse any neighbour of neighbour of the root node.

Step 4: This process will continue until we are getting the goal node.

Algorithm:

Step 1: PUSH the starting node into the stack.

Step 2: If the stack is empty then stop and return failure.

Step 3: If the top node of the stack is the goal node, then stop and return success.

Step 4: Else POP the top node from the stack and process it. Find all its neighbours that are in ready state and PUSH them into the stack in any order.

Step 5: Go to step 3.

Step 6: Exit.

Implementation:

Let us take an example for implementing the above DFS algorithm.

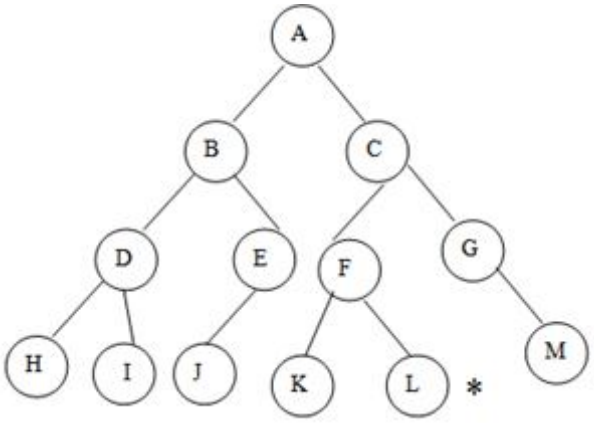


Figure 4.3 Example of DFS

Consider A as the root node and L as the goal node in the graph Figure 4.3.

Step 1: PUSH the starting node into the stack i.e.

Step 2: Now the stack is not empty and A is not our goal node. Hence move to next step.

Step 3: POP the top node from the stack i.e. A and find the neighbours of A i.e. B and C.

B C A

Step 4: Now C is top node of the stack. Find its neighbours i.e. F and G.

B F G C

Step 5: Now G is the top node of the stack. Find its neighbour i.e. M

B F M G

Step 6: Now M is the top node and find its neighbour, but there is no neighbours of M in the graph so POP it from the stack.

B F M

Step 7: Now F is the top node and its neighbours are K and L. so PUSH them on to the stack.

B K L F

Step 8: Now L is the top node of the stack, which is our goal node.

B K L

Also you can traverse the graph starting from the root A and then insert in the order C and B into the stack. Check your answer.

Completeness: DFS search algorithm is complete within finite state space as it will expand every node within a limited search tree.

Time Complexity: Time complexity of DFS will be equivalent to the node traversed by the algorithm. It is given by:

$$T(n)= 1+ n^2+ n^3 +.....+ n^m=O(n^m)$$

Where, m= maximum depth of any node and this can be much larger than d (Shallowest solution depth)

Space Complexity: DFS algorithm needs to store only single path from the root node, hence space complexity of DFS is equivalent to the size of the fringe set, which is **O(bm)**.

Optimal: DFS search algorithm is non-optimal, as it may generate a large number of steps or high cost to reach to the goal node.

Advantages:

DFS consumes very less memory space. It will reach at the goal node in a less time period than BFS if it traverses in a right path. It may find a solution without examining much of search because we may get the desired solution in the very first go.

Disadvantages:

It is possible that may states keep reoccurring. There is no guarantee of finding the goal node. Sometimes the states may also enter into infinite loops.

4.4.3 Uniform Cost Search

Uniform-cost search is a searching algorithm used for traversing a weighted tree or graph. This algorithm comes into play when a different cost is available for each edge. The primary goal of the uniform-cost search is to find a path to the goal node which has the lowest cumulative cost. Uniform-cost search expands nodes according to their path costs form the root node. It can be used to solve any graph/tree where the optimal cost is in demand. A uniform-cost search algorithm is implemented by the priority queue. It gives maximum priority to the lowest cumulative cost. Uniform cost search is equivalent to BFS algorithm if the path cost of all edges is the same. Let us explore Uniform Cost Search algorithm with the following example in Figure 4.4.

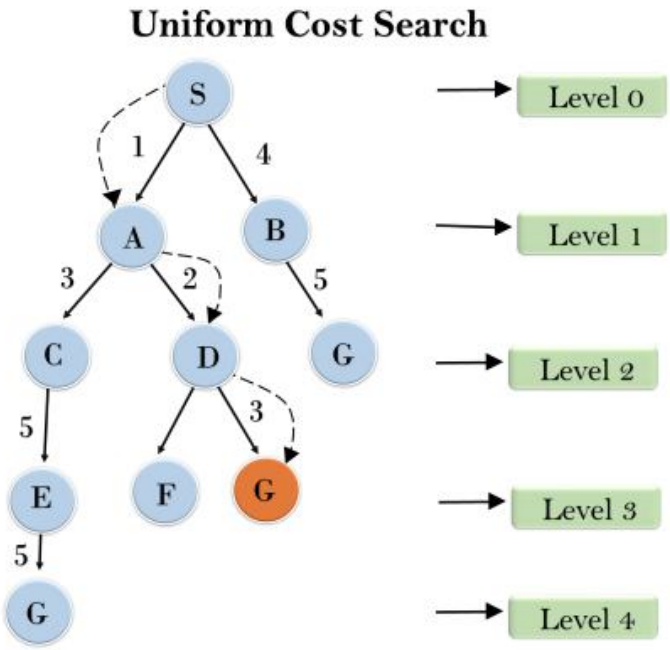


Figure 4.4 Example for Uniform Cost Search

Completeness:

Uniform-cost search is complete, such as if there is a solution, UCS will find it.

Time Complexity:

Let C* **is Cost of the optimal solution**, and **å** is each step to get closer to the goal node. Then the number of steps is = C*/å+1. Here we have taken +1, as we start from state 0 and end to C*/å. Hence, the worst-case time complexity of Uniform-cost search is $\mathbf{O(b^{1 + [C^*/å]})}$.

Space Complexity:

The same logic is for space complexity so, the worst-case space complexity of Uniform-cost search is $\mathbf{O(b^{1 + [C^*/å]})}$.

Optimal:

Uniform-cost search is always optimal as it only selects a path with the lowest path cost.

Advantages:

Uniform cost search is optimal because at every state the path with the least cost is chosen.

Disadvantages:

It does not care about the number of steps involve in searching and only concerned about path cost. Due to which this algorithm may be stuck in an infinite loop.

4.4.4 Depth-Limited Search

A depth-limited search algorithm is similar to depth-first search with a predetermined limit. Depth-limited search can solve the drawback of the infinite path in the Depth-first search. In this algorithm, the node at the depth limit will treat as it has no successor nodes further. Depth-limited search can be terminated with two Conditions of failure: (1) Standard failure value: It indicates that problem does not have any solution and (2) Cutoff failure value: It defines no solution for the problem within a given depth limit. This algorithm is explained in Figure 4.5.

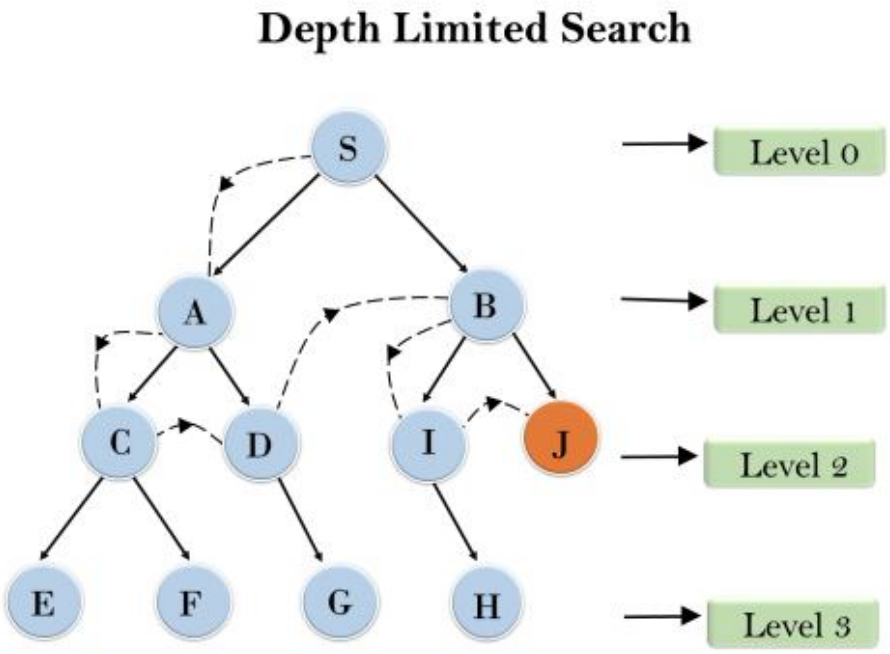


Figure 4.5 Example of Depth-Limited Search

Completeness: DLS search algorithm is complete if the solution is above the depth-limit.

Time Complexity: Time complexity of DLS algorithm is **O(b^{-l})**.

Space Complexity: Space complexity of DLS algorithm is O**(b×–!)**.

Optimal: Depth-limited search can be viewed as a special case of DFS, and it is also not optimal even if –! > d.

Advantages:

Depth-limited search is Memory efficient.

Disadvantages:

Depth-limited search also has a disadvantage of incompleteness. It may not be optimal if the problem has more than one solution.

4.4.5 Iterative Deepening Depth First Search

The iterative deepening algorithm is a combination of DFS and BFS algorithms. This search algorithm finds out the best depth limit and does it by gradually increasing the limit until a goal is found. This algorithm performs depth-first search up to a certain “depth limit”, and it keeps increasing the depth limit after each iteration until the goal node is found. This Search algorithm combines the benefits of Breadth-first search’s fast search and depth-first search’s memory efficiency. The iterative search algorithm is useful uninformed search when search space is large, and depth of goal node is unknown. Figure 4.6 explained about this algorithm with an example. Following tree structure is showing the iterative deepening depth-first search. IDDFS algorithm performs various iterations until it does not find the goal node. The iteration performed by the algorithm is given as:

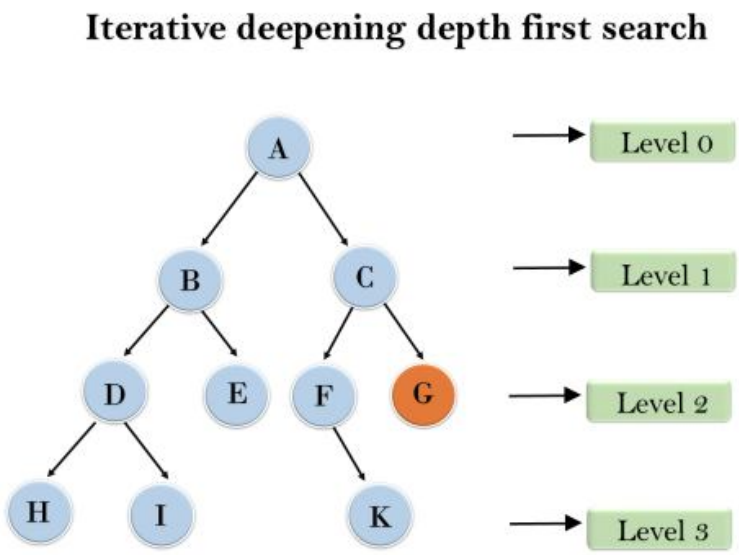


Figure 4.6 Iterative Deepening Depth First Search Example

1’st Iteration——> A

2’nd Iteration——> A, B, C

3’rd Iteration———>A, B, D, E, C, F, G

4’th Iteration———>A, B, D, H, I, E, C, F, K, G

In the fourth iteration, the algorithm will find the goal node.

Completeness:

This algorithm is complete is ifthe branching factor is finite.

Time Complexity:

Let's suppose b is the branching factor and depth is d then the worst-case time complexity is $\mathbf{O(b^d)}$.

Space Complexity:

The space complexity of IDDFS will be **O(bd)**.

Optimal:

IDDFS algorithm is optimal if path cost is a non- decreasing function of the depth of the node.

Advantages:

It combines the benefits of BFS and DFS search algorithm in terms of fast search and memory efficiency.

Disadvantages:

The main drawback of IDDFS is that it repeats all the work of the previous phase.

4.4.6 Bidirectional Search

Bidirectional search algorithm runs two simultaneous searches, one form initial state called as forward-search and other from goal node called as backward-search, to find the goal node. Bidirectional search replaces one single search graph with two small subgraphs in which one starts the search from an initial vertex and other starts from goal vertex. The search stops when these two graphs intersect each other. Bidirectional search can use search techniques such as BFS, DFS, DLS, etc**.** In the below search tree Figure 4.7, bidirectional search algorithm is applied. This algorithm divides one graph/tree into two sub-graphs. It starts traversing from node 1 in the forward direction and starts from goal node 16 in the backward direction. The algorithm terminates at node 9 where two searches meet.

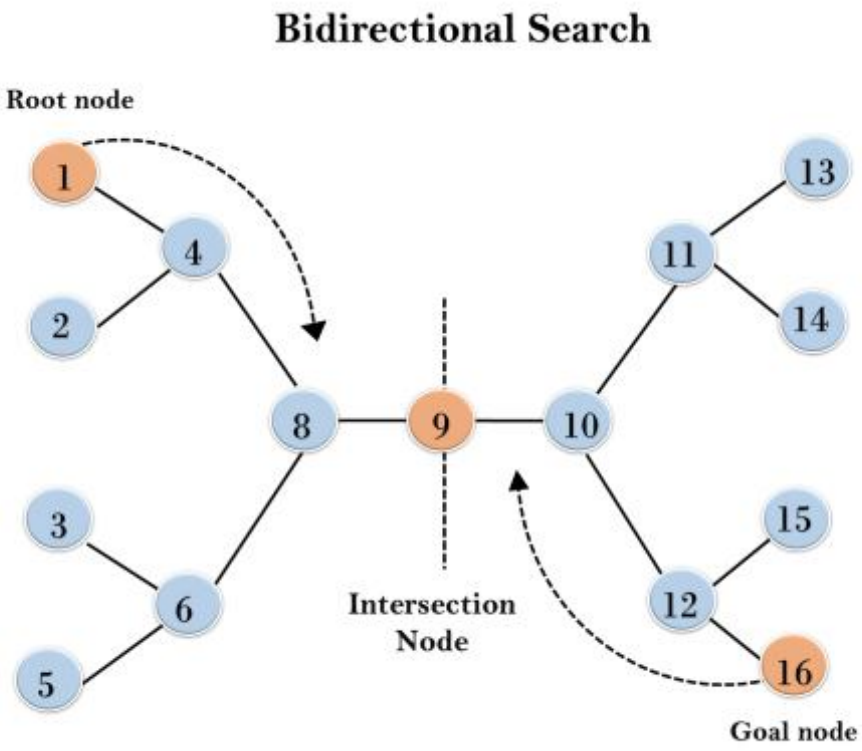


Figure 4.7 Example for Bidirectional Search

Completeness: Bidirectional Search is complete if we use BFS in both searches.

Time Complexity: Time complexity of bidirectional search using BFS is **$O(b^d)$**.

Space Complexity: Space complexity of bidirectional search is **$O(b^d)$**.

Optimal: Bidirectional search is Optimal.

Advantages:

Bidirectional search is fast. Bidirectional search requires less memory

Disadvantages:

Implementation of the bidirectional search tree is difficult. In bidirectional search, one should know the goal state in advance.

4.5 Summary

Uninformed search methods have access only to the problem definition. The basic algorithms are as follows: Breadth-first search expands the shallowest nodes first; it is complete, optimal for unit step costs, but has exponential space complexity. Uniform-cost search expands the node with lowest path cost, g(n), and is optimal for general step costs. Depth-first search expands the deepest unexpanded node first. It is neither complete nor optimal, but has linear space complexity. Depth-limited search adds a depth bound. Iterative deepening search calls depth-first search with increasing depth limits until a goal is found. It is complete, optimal for unit step costs, has time complexity comparable to breadth-first search, and has linear space complexity. Bidirectional search can enormously reduce time complexity, but it is not always applicable and may require too much space.

4.6 Model Questions

1. Mention the classification of search algorithms.
2. Explain in detail with an example: (a) BFS (b) DFS
3. What is the difference between DFS and Depth Limited Search algorithms?
4. Illustrate Bidirectional Search Algorithm with the help of one example.
5. Define: Uniform Cost Search algorithm.

LESSON 5
INFORMED SEARCH ALGORITHMS

Structure

5.1 Introduction

In the previous lesson, we have discussed about the uninformed search algorithms which looked through search space for all possible solutions of the problem without having any additional knowledge about search space. **Informed** search algorithms, on the other hand, can do quite well given some guidance on where to look for solutions. Informed search algorithm contains an array of knowledge such as how far we are from the goal, path cost, how to reach to goal node, etc. This knowledge help agents to explore less to the search space and find more efficiently the goal node.The informed search algorithm is more useful for large search space. Informed search algorithm uses the idea of heuristic, so it is also called **Heuristic search**..

5.2 Objectives

- To introduce the heuristic functions.
- To explore the various informed search algorithms with examples.

5.3 Heuristic Functions

Heuristic is a function which is used in Informed Search, and it finds the most promising path. It takes the current state of the agent as its input and produces the estimation of how close

agent is from the goal. The heuristic method, however, might not always give the best solution, but it guaranteed to find a good solution in reasonable time. Heuristic function estimates how close a state is to the goal. It is represented by h(n), and it calculates the cost of an optimal path between the pair of states. The value of the heuristic function is always positive.

In this section, we look at heuristics for the 8-puzzle, in order to shed light on the nature of heuristics in general. The 8-puzzle was one of the earliest heuristic search problems. As mentioned earlier, the object of the puzzle is to slide the tiles horizontally or vertically into the empty space until the configuration matches the goal configuration (Figure 5.1).

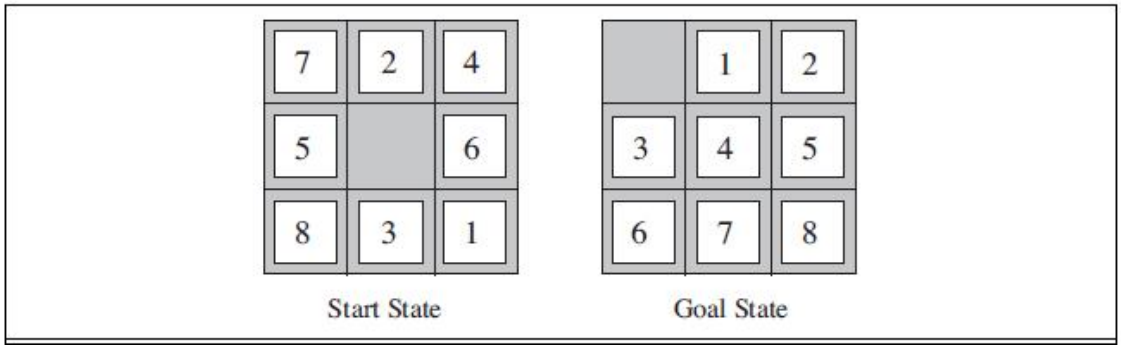


Figure 5.1 A typical instance of the 8-puzzle. The solution is 28 steps long

The average solution cost for a randomly generated 8-puzzle instance is about 22 steps. The branching factor is about 3. (When the empty tile is in the middle, four moves are possible; when it is in a corner, two; and when it is along an edge, three.) This means that an exhaustive tree search to depth 22 would look at about 322 H" 3.1×1010 states. A graph search would cut this down by a factor of about 170,000 because only 9!/2 = 181, 440 distinct states are reachable. This is a manageable number, but the corresponding number for the 15-puzzle is roughly 1013, so the next order of business is to find a good heuristic function. If we want to find the shortest solutions by using A", we need a heuristic function that never overestimates the number of steps to the goal. There is a long history of such heuristics for the 15-puzzle; here are two commonly used candidates:

- h_1 = the number of misplaced tiles. For Figure 5.1, all of the eight tiles are out of position, so the start state would have $h_1 = 8$. h_1 is an admissible heuristic because it is clear that any tile that is out of place must be moved at least once.
- h_2 = the sum of the distances of the tiles from their goal positions. Because tiles cannot move along diagonals, the distance we will count is the sum of the horizontal

and vertical distances. This is sometimes called the **city block distance** or **Manhattan distance**. h_2 is also admissible because all any move can do is move one tile one step closer to the goal. Tiles 1 to 8 in the start state give a Manhattan distance of

$$h_2 = 3+1 + 2 + 2+ 2 + 3+ 3 + 2 = 18 .$$

As expected, neither of these overestimates the true solution cost, which is 26.

5.3.1 Generating admissible heuristics from relaxed problems

We have seen that both h_1 (misplaced tiles) and h_2 (Manhattan distance) are fairly good heuristics for the 8-puzzle and that h_2 is better. How might one have come up with h_2? Is it possible for a computer to invent such a heuristic mechanically?

h_1 and h_2 are estimates of the remaining path length for the 8-puzzle, but they are also perfectly accurate path lengths for *simplified* versions of the puzzle. If the rules of the puzzle were changed so that a tile could move anywhere instead of just to the adjacent empty square, then h_1 would give the exact number of steps in the shortest solution. Similarly, if a tile could move one square in any direction, even onto an occupied square, then h_2 would give the exact number of steps in the shortest solution. A problem with fewer restrictions on the actions is called a **relaxed problem**. The state-space graph of the relaxed problem is a *supergraph* of the original state space because the removal of restrictions creates added edges in the graph.

Because the relaxed problem adds edges to the state space, any optimal solution in the original problem is, by definition, also a solution in the relaxed problem; but the relaxed problem may have *better* solutions if the added edges provide short cuts. Hence, *the cost of an optimal solution to a relaxed problem is an admissible heuristic for the original problem.* Furthermore, because the derived heuristic is an exact cost for the relaxed problem, it must obey the triangle inequality and is therefore **consistent**.

If a problem definition is written down in a formal language, it is possible to construct relaxed problems automatically. For example, if the 8-puzzle actions are described as

A tile can move from square A to square B if

A is horizontally or vertically adjacent to B **and** B is blank,

we can generate three relaxed problems by removing one or both of the conditions:

(a) A tile can move from square A to square B if A is adjacent to B.

(b) A tile can move from square A to square B if B is blank.

(c) A tile can move from square A to square B.

From (a), we can derive h_2 (Manhattan distance). The reasoning is that h_2 would be the proper score if we moved each tile in turn to its destination. We can derive the heuristic derived from (b) and from (c), we can derive h_1 (misplaced tiles) because it would be the proper score if tiles could move to their intended destination in one step. Notice that it is crucial that the relaxed problems generated by this technique can be solved essentially *without search*, because the relaxed rules allow the problem to be decomposed into eight independent subproblems. If the relaxed problem is hard to solve, then the values of the corresponding heuristic will be expensive to obtain.

5.3.2 Generating admissible heuristics from subproblems: Pattern databases

Admissible heuristics can also be derived from the solution cost of a **subproblem** of a given problem. For example, Figure 5.2 shows a subproblem of the 8-puzzle instance in Figure 5.1. The subproblem involves getting tiles 1, 2, 3, 4 into their correct positions. Clearly, the cost of the optimal solution of this subproblem is a lower bound on the cost of the complete problem. It turns out to be more accurate than Manhattan distance in some cases.

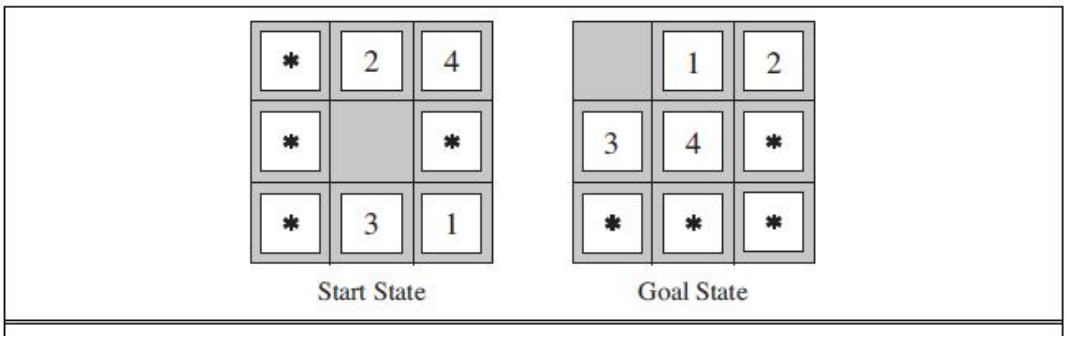


Figure 5.2 A subproblem of the 8-puzzle instrance

The idea behind **pattern databases** is to store these exact solution costs for every possible subproblem instance—in our example, every possible configuration of the four tiles and the blank. (The locations of the other four tiles are irrelevant for the purposes of solving the subproblem, but moves of those tiles do count toward the cost.) Then we compute an admissible heuristic h_{DB} for each complete state encountered during a search simply by looking up the corresponding subproblem configuration in the database. The database itself is constructed by searching back from the goal and recording the cost of each new pattern encountered; the expense of this search is amortized over many subsequent problem instances.

The choice of 1-2-3-4 is fairly arbitrary; we could also construct databases for 5-6-7-8, for 2-4-6-8, and so on. Each database yields an admissible heuristic, and these heuristics can be combined, as explained earlier, by taking the maximum value. A combined heuristic of this kind is much more accurate than the Manhattan distance; the number of nodes generated when solving random 15-puzzles can be reduced by a factor of 1000.

Disjoint pattern databases work for sliding-tile puzzles because the problem can be divided up in such a way that each move affects only one subproblem—because only one tile is moved at a time. For a problem such as Rubik's Cube, this kind of subdivision is difficult because each move affects 8 or 9 of the 26 cubies. More general ways of defining additive, admissible heuristics have been proposed that do apply to Rubik's cube (Yang *et al.*, 2008), but they have not yielded a heuristic better than the best nonadditive heuristic for the problem.

5.4 Pure Heuristic Search

Pure heuristic search is the simplest form of heuristic search algorithms. It expands nodes based on their heuristic value h(n). It maintains two lists, **OPEN** and **CLOSED** list. In the CLOSED list, it places those nodes which have already expanded and in the OPEN list, it places nodes which have yet not been expanded. On each iteration, each node *n* with the lowest heuristic value is expanded and generates all its successors and n is placed to the closed list. The algorithm continues until a goal state is found. In the informed search we will discuss two main algorithms which are given below: (1) Best First Search Algorithm (Greedy search) and (2) A* Search Algorithm.

5.4.1 Best-First Search Algorithm (Greedy Search)

Greedy best-first search algorithm always selects the path which appears best at that moment. It is the combination of depth-first search and breadth-first search algorithms. It uses

the heuristic function and search. Best-first search allows us to take the advantages of both algorithms. With the help of best-first search, at each step, we can choose the most promising node. In the best first search algorithm, we expand the node which is closest to the goal node and the closest cost is estimated by heuristic function, i.e.

$$f(n) = g(n)$$

Where, *h(n)*= estimated cost from node n to the goal. The greedy best first algorithm is implemented by the priority queue.

Algorithm:

· **Step 1:** Place the starting node into the OPEN list.

· **Step 2:** If the OPEN list is empty, Stop and return failure.

· **Step 3:** Remove the node n, from the OPEN list which has the lowest value of h(n), and places it in the CLOSED list.

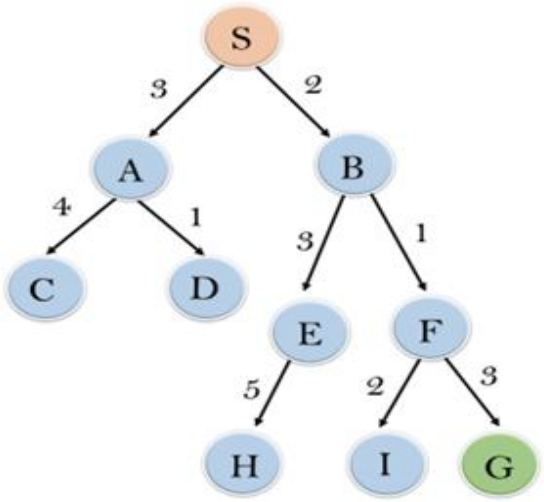
· **Step 4:** Expand the node n, and generate the successors of node n.

· **Step 5:** Check each successor of node n, and find whether any node is a goal node or not. If any successor node is goal node, then return success and terminate the search, else proceed to Step 6.

· **Step 6:** For each successor node, algorithm checks for evaluation function f(n), and then check if the node has been in either OPEN or CLOSED list. If the node has not been in both list, then add it to the OPEN list.

· **Step 7:** Return to Step 2.

Consider the below search problem in Figure 5.3, and we will traverse it using greedy best-first search. At each iteration, each node is expanded using evaluation function *f(n)=h(n),* which is given in the below table.



node	H (n)
A	12
B	4
C	7
D	3
E	8
F	2
H	4
I	9
S	13
G	0

Figure 5.3 Example for Best-First Search Algorithm

In this search example, we are using two lists which are **OPEN** and **CLOSED** Lists. Following Figure 5.4 shows the iteration for traversing the above example.

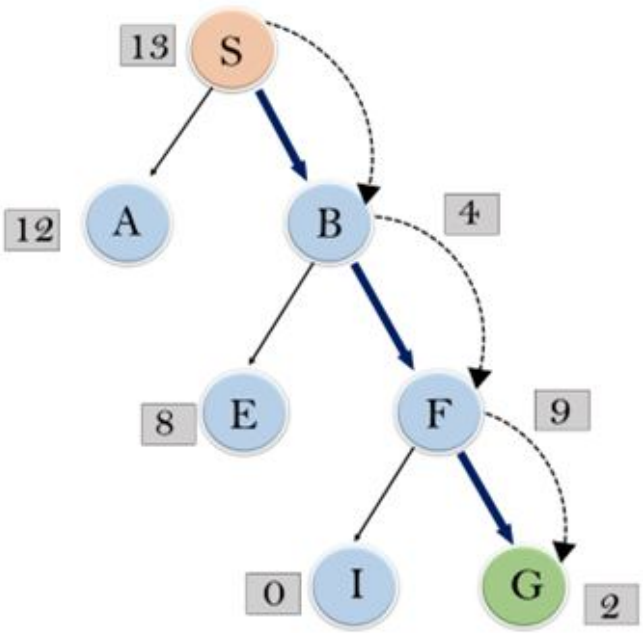


Figure 5.4 Iteration of traversing using Best-First Search Algorithm

Expand the nodes of S and put in the CLOSED list.

Initialization: Open [A, B], Closed [S]

Iteration 1: Open [A], Closed [S, B]

Iteration 2: Open [E, F, A], Closed [S, B]

: Open [E, A], Closed [S, B, F]

Iteration 3: Open [I, G, E, A], Closed [S, B, F]

: Open [I, E, A], Closed [S, B, F, G]

Hence the final solution path will be: **S——> B——>F——> G**

Time Complexity: The worst case time complexity of Greedy best first search is $O(b^m)$.

Space Complexity: The worst case space complexity of Greedy best first search is $O(b^m)$. Where, m is the maximum depth of the search space.

Complete: Greedy best-first search is also incomplete, even if the given state space is finite.

Optimal: Greedy best first search algorithm is not optimal.

Advantages:

Best first search can switch between BFS and DFS by gaining the advantages of both the algorithms. This algorithm is more efficient than BFS and DFS algorithms.

Disadvantages:

It can behave as an unguided depth-first search in the worst case scenario. It can get stuck in a loop as DFS. This algorithm is not optimal.

5.4.2 A* Search Algorithm

A* search is the most commonly known form of best-first search. It uses heuristic function *h(n)*, and cost to reach the node n from the start state *g(n)*. It has combined features of UCS and greedy best-first search, by which it solve the problem efficiently. A* search algorithm finds the

shortest path through the search space using the heuristic function. This search algorithm expands less search tree and provides optimal result faster. A* algorithm is similar to UCS except that it uses *g(n)+h(n)* instead of *g(n)*. In A* search algorithm, we use search heuristic as well as the cost to reach the node. Hence we can combine both costs as following, and this sum is called as a **fitness number** as shown in Figure 5.5.

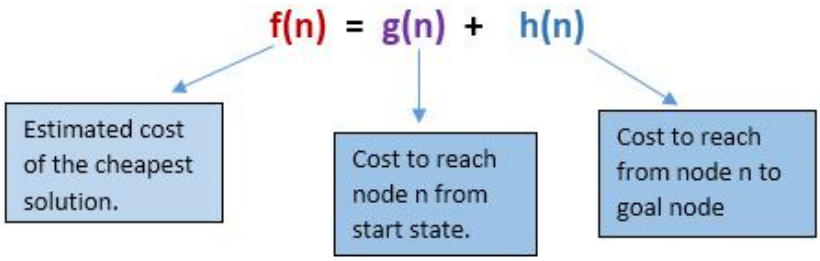


Figure 5.5 Calculation of fitness number in A* Algorithm

Algorithm:

Step1: Place the starting node in the OPEN list.

Step 2: Check if the OPEN list is empty or not, if the list is empty then return failure and stops.

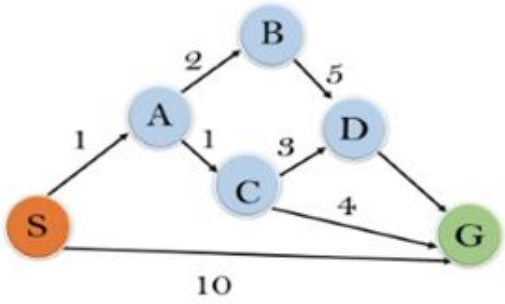
Step 3: Select the node from the OPEN list which has the smallest value of evaluation function *(g+h)*, if node *n* is goal node then return success and stop, otherwise.

Step 4: Expand node *n* and generate all of its successors, and put *n* into the closed list. For each successor *n*', check whether *n*' is already in the OPEN or CLOSED list, if not then compute evaluation function for *n*' and place into Open list.

Step 5: Else if node *n*' is already in OPEN and CLOSED, then it should be attached to the back pointer which reflects the lowest *g(n')* value.

Step 6: Return to **Step 2**.

In the following Figure 5.6, we will traverse the given graph using the A* algorithm. The heuristic value of all states is given in the below table so we will calculate the f(n) of each state using the formula f(n)= g(n) + h(n), where g(n) is the cost to reach any node from start state. Here we will use OPEN and CLOSED list.



State	h(n)
S	5
A	3
B	4
C	2
D	6
G	0

Figure 5.6 Traversing of A* Algorithm

The solution for the above example is given in the Figure 5.7.

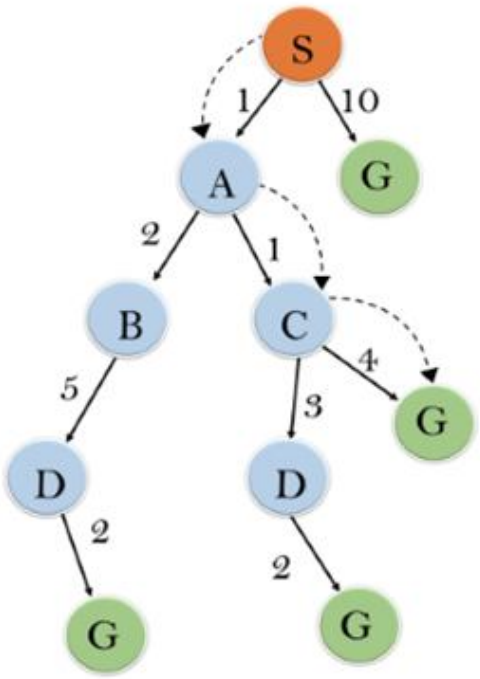


Figure 5.7 Iterative solution for A* algorithm example

Initialization: {(S, 5)}

Iteration1: {(S—> A, 4), (S—>G, 10)}

Iteration2: {(S—> A—>C, 4), (S—> A—>B, 7), (S—>G, 10)}

Iteration3: {(S—> A—>C—>G, 6), (S—> A—>C—>D, 11), (S—> A—>B, 7), (S—>G, 10)}

Iteration 4 will give the final result, as **S—>A—>C—>G** it provides the optimal path with cost 6.

Points to remember:

- A* algorithm returns the path which occurred first, and it does not search for all remaining paths.
- The efficiency of A* algorithm depends on the quality of heuristic.
- A* algorithm expands all nodes which satisfy the condition f(n)<="" li="">

Complete: A* algorithm is complete as long as branching factor is finite and the cost at every action is fixed.

Optimal: A* search algorithm is optimal if it follows below two conditions:

- **Admissible:** the first condition requires for optimality is that h(n) should be an admissible heuristic for A* tree search. An admissible heuristic is optimistic in nature.
- **Consistency:** Second required condition is consistency for only A* graph-search.

If the heuristic function is admissible, then A* tree search will always find the least cost path.

Time Complexity: The time complexity of A* search algorithm depends on heuristic function, and the number of nodes expanded is exponential to the depth of solution d. So the time complexity is O(b^d), where b is the branching factor.

Space Complexity: The space complexity of A* search algorithm is ***O(b^d)*****.**

Advantages:

A* search algorithm is the best algorithm than other search algorithms. A* search algorithm is optimal and complete. This algorithm can solve very complex problems.

Disadvantages:

It does not always produce the shortest path as it mostly based on heuristics and approximation. A* search algorithm has some complexity issues. The main drawback of A* is memory requirement as it keeps all generated nodes in the memory, so it is not practical for various large-scale problems.

5.5 Summary

Informed search methods may have access to a **heuristic** function h(n) that estimates the cost of a solution from n. The generic **best-first search** algorithm selects a node for expansion according to an **evaluation function**. **Greedy best-first search** expands nodes with minimal h(n). It is not optimal but is often efficient. **A* search** expands nodes with minimal f(n) = g(n) + h(n). A* is complete and optimal The performance of heuristic search algorithms depends on the quality of the heuristic function. One can sometimes construct good heuristics by relaxing the problem definition, by storing precomputed solution costs for subproblems in a pattern database, or by learning from experience with the problem class.

5.6 Model Questions

1. What is a heuristic function?
2. What is meant by a relaxed problem?
3. Explain in detail about the Best-First Search algorithm with an example.
4. Discuss the A* algorithm with an example.

LESSON 6
LOCAL SEARCH ALGORITHMS

Structure

6.1 Introduction

In this lesson, we begin with **local search** in the state space, evaluating and modifying one or more current states rather than systematically exploring paths from an initial state. These algorithms are suitable for problems in which all that matters is the solution state, not the path cost to reach it. The family of local search algorithms includes methods inspired by statistical physics (**simulated annealing**) and evolutionary biology. The key idea is that if an agent cannot predict exactly what percept it will receive, then it will need to consider what to do under each **contingency** that its percepts may reveal. With partial observability, the agent will also need to keep track of the states it might be in.

6.2 Objectives

- To introduce the basic concepts in Local search.
- To explore the Hill-Climbing search.
- To discuss the search in continuous spaces.
- To know about the AND-OR search trees.

6.3 Local Search Algorithms

The search algorithms that we have seen so far are designed to explore search spaces systematically. This systematicity is achieved by keeping one or more paths in memory and by recording which alternatives have been explored at each point along the path. When a goal is found, the *path* to that goal also constitutes a *solution* to the problem. In many problems, however, the path to the goal is irrelevant. The same general property holds for many important applications such as integrated-circuit design, factory-floor layout, job-shop scheduling, automatic programming, telecommunications network optimization, vehicle routing, and portfolio management.

If the path to the goal does not matter, we might consider a different class of algorithms, ones that do not worry about paths at all. **Local search** algorithms operate using a single **current node** (rather than multiple paths) and generally move only to neighbors of that node. Typically, the paths followed by the search are not retained. Although local search algorithms are not systematic, they have two key advantages: (1) they use very little memory—usually a constant amount; and (2) they can often find reasonable solutions in large or infinite (continuous) state spaces for which systematic algorithms are unsuitable.

In addition to finding goals, local search algorithms are useful for solving pure **optimization problems**, in which the aim is to find the best state according to an **objective function**. Many optimization problems do not fit the "standard" search models. For example, nature provides an objective function—reproductive fitness—that Darwinian evolution could be seen as attempting to optimize, but there is no "goal test" and no "path cost" for this problem.

To understand local search, we find it useful to consider the **state-space landscape**. A landscape has both "location" (defined by the state) and "elevation" (defined by the value of the heuristic cost function or objective function). If elevation corresponds to cost, then the aim is to find the lowest valley—a **global minimum**; if elevation corresponds to an objective function, then the aim is to find the highest peak—a **global maximum**. (You can convert from one to the other just by inserting a minus sign.) Local search algorithms explore this landscape. A **complete** local search algorithm always finds a goal if one exists; an **optimal** algorithm always finds a global minimum/maximum.

6.3.1 Hill-Climbing Search

The **hill-climbing** search algorithm (**steepest-ascent** version) is shown in Figure 6.1. It is simply a loop that continually moves in the direction of increasing value—that is, uphill. It terminates when it reaches a "peak" where no neighbor has a higher value. The algorithm does not maintain a search tree, so the data structure for the current node need only record the state and the value of the objective function. Hill climbing does not look ahead beyond the immediate neighbors of the current state. This resembles trying to find the top of Mount Everest in a thick fog while suffering from amnesia.

```
function HILL-CLIMBING(problem) returns a state that is a local maximum
  current ← MAKE-NODE(problem.INITIAL-STATE)
  loop do
    neighbor ← a highest-valued successor of current
    if neighbor.VALUE ≤ current.VALUE then return current.STATE
    current ← neighbor
```

Figure 6.1 The Hill-Climbing Algorithm

To illustrate hill climbing, we will use the **8-queens problem.** Local search algorithms typically use a **complete-state formulation**, where each state has 8 queens on the board, one per column. The successors of a state are all possible states generated by moving a single queen to another square in the same column (so each state has 8×7=56 successors). The heuristic cost function *h* is the number of pairs of queens that are attacking each other, either directly or indirectly. The global minimum of this function is zero, which occurs only at perfect solutions. Figure 6.2(a) shows a state with *h=17*. The figure also shows the values of all its successors, with the best successors having *h=12*. Hill-climbing algorithms typically choose randomly among the set of best successors if there is more than one.

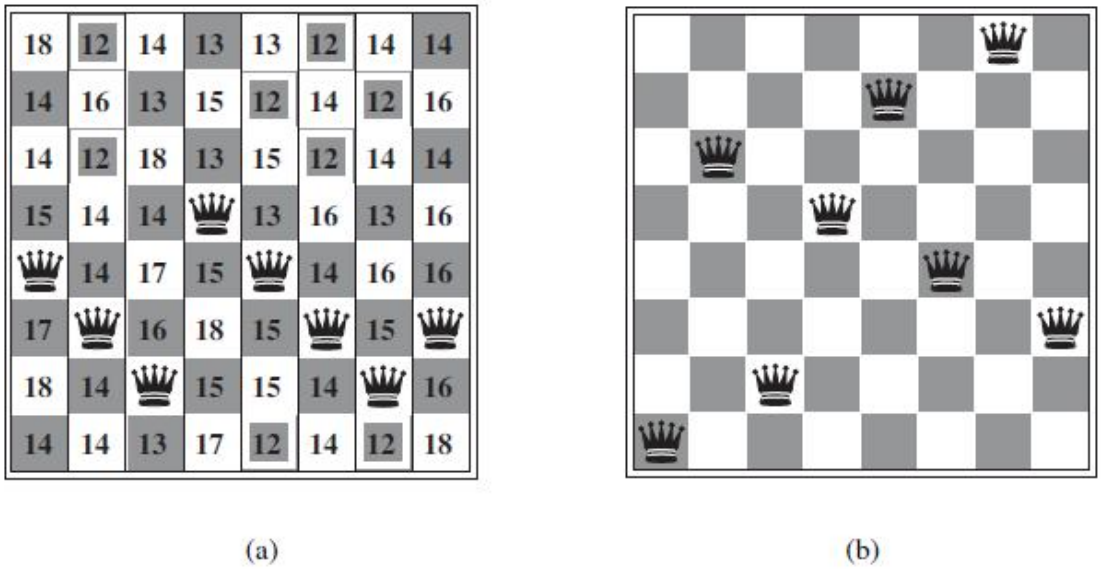


Figure 6.2 (a) An 8-queens state with heuristic cost estimate h=17. (b) A local minimum in the 8-queens state space

Hill climbing is sometimes called **greedy local search** because it grabs a good neighbour state without thinking ahead about where to go next. Although greed is considered one of the seven deadly sins, it turns out that greedy algorithms often perform quite well. Hill climbing often makes rapid progress toward a solution because it is usually quite easy to improve a bad state. For example, from the state in Figure 6.2(a), it takes just five steps to reach the state in Figure 6.2(b), which has *h=1* and is very nearly a solution. Unfortunately, hill climbing often gets stuck for the following reasons:

• **Local maxima**: a local maximum is a peak that is higher than each of its neighboring states but lower than the global maximum. Hill-climbing algorithms that reach the vicinity of a local maximum will be drawn upward toward the peak but will then be stuck with nowhere else to go. More concretely, the state in Figure 6.2(b) is a local maximum (i.e., a local minimum for the cost h); every move of a single queen makes the situation worse.

• **Ridges**: Ridges result in a sequence of local maxima that is very difficult for greedy algorithms to navigate.

• **Plateaux**: a plateau is a flat area of the state-space landscape. It can be a flat local maximum, from which no uphill exit exists, or a **shoulder**, from which progress is possible. A hill-climbing search might get lost on the plateau.

The algorithm in Figure 6.1 halts if it reaches a plateau where the best successor has the same value as the current state. Might it not be a good idea to keep going—to allow a **sideways move** in the hope that the plateau is really a shoulder. The answer is usually yes, but we must take care. If we always allow sideways moves when there are no uphill moves, an infinite loop will occur whenever the algorithm reaches a flat local maximum that is not a shoulder. One common solution is to put a limit on the number of consecutive sideways moves allowed. For example, we could allow up to, say, 100 consecutive sideways moves in the 8-queens problem. This raises the percentage of problem instances solved by hill climbing from 14% to 94%. Success comes at a cost: the algorithm averages roughly 21 steps for each successful instance and 64 for each failure.

Many variants of hill climbing have been invented. **Stochastic hill climbing** chooses at random from among the uphill moves; the probability of selection can vary with the steepness of the uphill move. This usually converges more slowly than steepest ascent, but in some state landscapes, it finds better solutions. **First-choice hill climbing** implements stochastic hill climbing by generating successors randomly until one is generated that is better than the current state. This is a good strategy when a state has many (e.g., thousands) of successors. The success of hill climbing depends very much on the shape of the state-space landscape

6.3.2 Simulated annealing

A hill-climbing algorithm that *never* makes “downhill” moves toward states with lower value (or higher cost) is guaranteed to be incomplete, because it can get stuck on a local maximum. In contrast, a purely random walk—that is, moving to a successor chosen uniformly at random from the set of successors—is complete but extremely inefficient. Therefore, it seems reasonable to try to combine hill climbing with a random walk in some way that yields both efficiency and completeness. **Simulated annealing** is such an algorithm. In metallurgy, **annealing** is the process used to temper or harden metals and glass by heating them to a high temperature and then gradually cooling them, thus allowing the material to reach a low energy crystalline state. To explain simulated annealing, we switch our point of view from hill climbing to **gradient descent** (i.e., minimizing cost) and imagine the task of getting a ping-pong ball into the deepest crevice in a bumpy surface. If we just let the ball roll, it will come to rest at a local minimum. If we shake the

surface, we can bounce the ball out of the local minimum. The trick is to shake just hard enough to bounce the ball out of local minima but not hard enough to dislodge it from the global minimum. The simulated-annealing solution is to start by shaking hard (i.e., at a high temperature) and then gradually reduce the intensity of the shaking (i.e., lower the temperature).

The innermost loop of the simulated-annealing algorithm (Figure 6.3) is quite similar to hill climbing. Instead of picking the *best* move, however, it picks a *random* move. If the move improves the situation, it is always accepted. Otherwise, the algorithm accepts the move with some probability less than 1. The probability decreases exponentially with the "badness" of the move—the amount ÄE by which the evaluation is worsened. The probability also decreases as the "temperature" T goes down: "bad" moves are more likely to be allowed at the start when T is high, and they become more unlikely as T decreases. If the schedule lowers

T slowly enough, the algorithm will find a global optimum with probability approaching 1.

```
function SIMULATED-ANNEALING(problem, schedule) returns a solution state
  inputs: problem, a problem
          schedule, a mapping from time to "temperature"

  current ← MAKE-NODE(problem.INITIAL-STATE)
  for t = 1 to ∞ do
      T ← schedule(t)
      if T = 0 then return current
      next ← a randomly selected successor of current
      ΔE ← next.VALUE – current.VALUE
      if ΔE > 0 then current ← next
      else current ← next only with probability e^(ΔE/T)
```

Figure 6.3 The Simulated Annealing Algorithm

Simulated annealing was first used extensively to solve VLSI layout problems in the early 1980s. It has been applied widely to factory scheduling and other large-scale optimization tasks.

6.3.3 Local beam search

Keeping just one node in memory might seem to be an extreme reaction to the problem of memory limitations. The **local beam search** algorithm keeps track of *k* states rather than just one. It begins with *k* randomly generated states. At each step, all the successors of all *k* states are generated. If any one is a goal, the algorithm halts. Otherwise, it selects the *k* best successors from the complete list and repeats.

At first sight, a local beam search with *k* states might seem to be nothing more than running *k* random restarts in parallel instead of in sequence. In fact, the two algorithms are quite different. In a random-restart search, each search process runs independently of the others. *In a local beam search, useful information is passed among the parallel search threads.* In effect, the states that generate the best successors say to the others, "Come over here, the grass is greener!" The algorithm quickly abandons unfruitful searches and moves its resources to where the most progress is being made.

In its simplest form, local beam search can suffer from a lack of diversity among the *k* states—they can quickly become concentrated in a small region of the state space, making the search little more than an expensive version of hill climbing. A variant called **stochastic beam search**, analogous to stochastic hill climbing, helps alleviate this problem. Instead of choosing the best *k* from the pool of candidate successors, stochastic beam search chooses *k* successors at random, with the probability of choosing a given successor being an increasing function of its value. Stochastic beam search bears some resemblance to the process of natural selection, whereby the "successors" (offspring) of a "state" (organism) populate the next generation according to its "value" (fitness).

6.4 Local Search in Continuous Spaces

This section provides a *very brief* introduction to some local search techniques for finding optimal solutions in continuous spaces. The literature on this topic is vast; many of the basic techniques originated in the 17th century, after the development of calculus by Newton and Leibniz. We find uses for these techniques at several places in the book, including the chapters on learning, vision, and robotics.

We begin with an example. Suppose we want to place three new airports anywhere in Romania, such that the sum of squared distances from each city to its nearest airport is minimized. The state space is then defined by the coordinates of the airports: (x_1, y_1), (x_2, y_2), and (x_3, y_3). This is a *six-dimensional* space; we also say that states are defined by six **variables**. Moving around in this space corresponds to moving one or more of the airports. The objective function $f(x_1, y_1, x_2, y_2, x_3, y_3)$ is relatively easy to compute for any particular state once we compute the closest cities. Let C_i be the set of cities whose closest airport (in the current state) is airport i. Then, *in the neighborhood of the current state*, where the C_{is} remain constant, we have

$$f(x_1, y_1, x_2, y_2, x_3, y_3) = \sum_{i=1}^{3} \left| \sum_{c \in C_i} (x_i - x_c)^2 + (y_i - y_c)^2 \right| .$$

This expression is correct *locally*, but not globally because the sets Ci are (discontinuous) functions of the state.

One way to avoid continuous problems is simply to **discretize** the neighborhood of each state. For example, we can move only one airport at a time in either the x or y direction by a fixed amount plus or minus ä. With 6 variables, this gives 12 possible successors for each state. We can then apply any of the local search algorithms described previously. We could also apply stochastic hill climbing and simulated annealing directly, without discretizing the space. These algorithms choose successors randomly, which can be done by generating random vectors of length ä.

6.5 Searching with Nondeterministic Actions

When the environment is either partially observable or nondeterministic (or both), percepts become useful. In a partially observable environment, every percept helps narrow down the set of possible states the agent might be in, thus making it easier for the agent to achieve its goals. When the environment is nondeterministic, percepts tell the agent which of the possible outcomes of its actions has actually occurred. In both cases, the future percepts cannot be determined in advance and the agent's future actions will depend on those future percepts. So the solution to a problem is not a sequence but a **contingency plan** (also known as a **strategy**) that specifies what to do depending on what percepts are received. In this section, we examine the case of nondeterministic actions in searching.

6.5.1 The erratic vacuum world

As an example, in the vacuum world, the state space has eight states, as shown in Figure 6.4. There are three actions—*Left*, *Right*, and *Suck*—and the goal is to clean up all the dirt (states 7 and 8). If the environment is observable, deterministic, and completely known, then the problem is trivially solvable by any of the algorithms and the solution is an action sequence. For example, if the initial state is 1, then the action sequence [*Suck*,*Right*,*Suck*] will reach a goal state, 8.

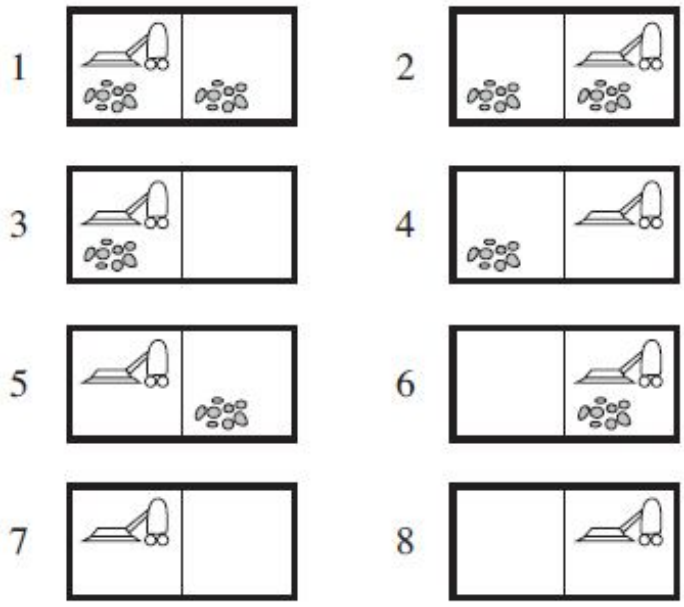


Figure 6.4 The eight possible states of the vacuum world;

States 7 and 8 are goal states

Now suppose that we introduce nondeterminism in the form of a powerful but erratic vacuum cleaner. In the **erratic vacuum world**, the *Suck* action works as follows:

• When applied to a dirty square the action cleans the square and sometimes cleans up dirt in an adjacent square, too.

•·When applied to a clean square the action sometimes deposits dirt on the carpet.

To provide a precise formulation of this problem, we need to generalize the notion of a **transition model.** Instead of defining the transition model by a function that returns a single state, we use a function that returns a *set* of possible outcome states. For example, in the erratic vacuum world, the *Suck* action in state 1 leads to a state in the set {5, 7}—the dirt in the right-hand square may or may not be vacuumed up.

We also need to generalize the notion of a **solution** to the problem. For example, if we start in state 1, there is no single *sequence* of actions that solves the problem. Instead, we need a contingency plan such as the following:

[*Suck*, **if** State =5 **then** [*Right*, *Suck*] **else** [ ]] .

Thus, solutions for nondeterministic problems can contain nested **if–then–else** statements; this means that they are *trees* rather than sequences. This allows the selection of actions based on contingencies arising during execution. Many problems in the real, physical world are contingency problems because exact prediction is impossible. For this reason, many people keep their eyes open while walking around or driving.

6.5.2 AND-OR Search Trees

The next question is how to find contingent solutions to nondeterministic problems. We begin by constructing search trees, but here the trees have a different character. In a deterministic environment, the only branching is introduced by the agent's own choices in each state. We call these nodes **OR nodes**. In the vacuum world, for example, at an OR node the agent chooses *Left or Right or Suck*. In a nondeterministic environment, branching is also introduced by the *environment's* choice of outcome for each action. We call these nodes **AND nodes**. For example, the *Suck* action in state 1 leads to a state in the set {5, 7}, so the agent would need to find a plan for state 5 *and* for state 7. These two kinds of nodes alternate, leading to an AND–OR **tree** as illustrated in Figure 6.5.

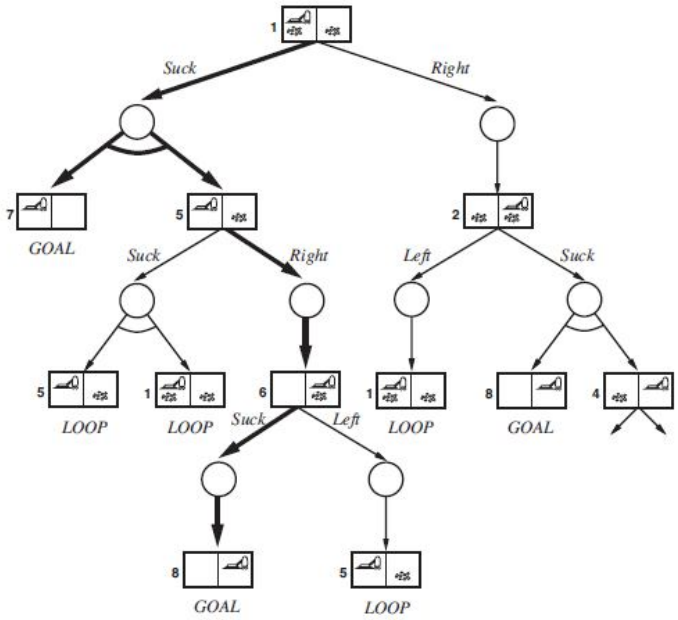


Figure 6.5 AND-OR tree for vacuum world

A solution for an AND–OR search problem is a subtree that (1) has a goal node at every leaf, (2) specifies one action at each of its OR nodes, and (3) includes every outcome branch at each of its AND nodes. The solution is shown in bold lines in the figure; it corresponds to the plan given in the above equation. (The plan uses if–then–else notation to handle the AND branches, but when there are more than two branches at a node, it might be better to use a **case** construct). One may also consider a somewhat different agent design, in which the agent can act *before* it has found a guaranteed plan and deals with some contingencies only as they arise during execution. This type of **interleaving** of search and execution is also useful for exploration problems and for game playing.

AND–OR graphs can also be explored by breadth-first or best-first methods. The concept of a heuristic function must be modified to estimate the cost of a contingent solution rather than a sequence, but the notion of admissibility carries over and there is an analog of the A* algorithm for finding optimal solutions.

6.6 Summary

This lesson has examined search algorithms for problems beyond the “classical” case of finding the shortest path to a goal in an observable, deterministic, discrete environment. *Local search* methods such as **hill climbing** operate on complete-state formulations, keeping only a small number of nodes in memory. Several stochastic algorithms have been developed, including **simulated annealing**, which returns optimal solutions when given an appropriate cooling schedule. In **nondeterministic** environments, agents can apply AND–OR search to generate **contingent** plans that reach the goal regardless of which outcomes occur during execution.

6.7 Model Questions

1. Define: Local Search
2. Illustrate Hill-Climbing search algorithm with an example.
3. Write short notes on Simulated annealing.
4. Discuss about AND-OR Search trees.

LESSON 7
OPTIMAL DECISIONS IN GAMES

Structure

7.1 Introduction

The earlier lesson introduced **multiagent environments**, in which each agent needs to consider the actions of other agents and how they affect its own welfare. The unpredictability of these other agents can introduce **contingencies** into the agent's problem-solving process. In this lesson, we cover **competitive** environments, in which the agents' goals are in conflict, giving rise to **adversarial search** problems—often known as **games**.

7.2 Objectives

- To explore the concepts in games.
- To describe the minimax algorithm and multiplayer games.
- To explain the alpha-beta pruning algorithm.
- To know the evaluation functions for games.

7.3 Games

Mathematical **game theory**, a branch of economics, views any multiagent environment as a game, Games have engaged the intellectual faculties of humans—sometimes to an alarming

degree—for as long as civilization has existed. For AI researchers, the abstract nature of games makes them an appealing subject for study. The state of a game is easy to represent, and agents are usually restricted to a small number of actions whose outcomes are defined by precise rules. Physical games, such as croquet and ice hockey, have much more complicated descriptions, a much larger range of possible actions, and rather imprecise rules defining the legality of actions. With the exception of robot soccer, these physical games have not attracted much interest in the AI community.

Games, unlike most of the toy problems, are interesting *because* they are too hard to solve. For example, chess has an average branching factor of about 35, and games often go to 50 moves by each player, so the search tree has about 35100 or 10154 nodes (although the search graph has "only" about 1040 distinct nodes). Games, like the real world, therefore require the ability to make *some* decision even when calculating the *optimal* decision is infeasible. Games also penalize inefficiency severely. Whereas an implementation of A* search that is half as efficient will simply take twice as long to run to completion, a chess program that is half as efficient in using its available time probably will be beaten into the ground, other things being equal. Game-playing research has therefore spawned a number of interesting ideas on how to make the best possible use of time.

We begin with a definition of the optimal move and an algorithm for finding it. We then look at techniques for choosing a good move when time is limited. **Pruning** allows us to ignore portions of the search tree that make no difference to the final choice, and heuristic **evaluation functions** allow us to approximate the true utility of a state without doing a complete search. Finally, we look at how state-of-the-art game-playing programs fare against human opposition and at directions for future developments.

We first consider games with two players, whom we call MAX and MIN for reasons that will soon become obvious. MAX moves first, and then they take turns moving until the game is over. At the end of the game, points are awarded to the winning player and penalties are given to the loser. A game can be formally defined as a kind of search problem with the following elements:

- S_0: The **initial state**, which specifies how the game is set up at the start.
- PLAYER(s): Defines which player has the move in a state.
- ACTIONS(s): Returns the set of legal moves in a state.

• RESULT(s, a): The **transition model**, which defines the result of a move.

• TERMINAL-TEST(s): A **terminal test**, which is true when the game is over and false otherwise. States where the game has ended are called **terminal states**.

• UTILITY(s, p): A **utility function** (also called an objective function or payoff function), defines the final numeric value for a game that ends in terminal state s for a player p. In chess, the outcome is a win, loss, or draw, with values +1, 0, or 1/2 . Some games have a wider variety of possible outcomes; the payoffs in backgammon range from 0 to +192.

The initial state, ACTIONS function, and RESULT function define the **game tree** for the game—a tree where the nodes are game states and the edges are moves. Figure 7.1 shows part of the game tree for tic-tac-toe (noughts and crosses). From the initial state, MAX has nine possible moves. Play alternates between MAX's placing an X and MIN's placing an O until we reach leaf nodes corresponding to terminal states such that one player has three in a row or all the squares are filled. The number on each leaf node indicates the utility value of the terminal state from the point of view of MAX; high values are assumed to be good for MAX and bad for MIN.

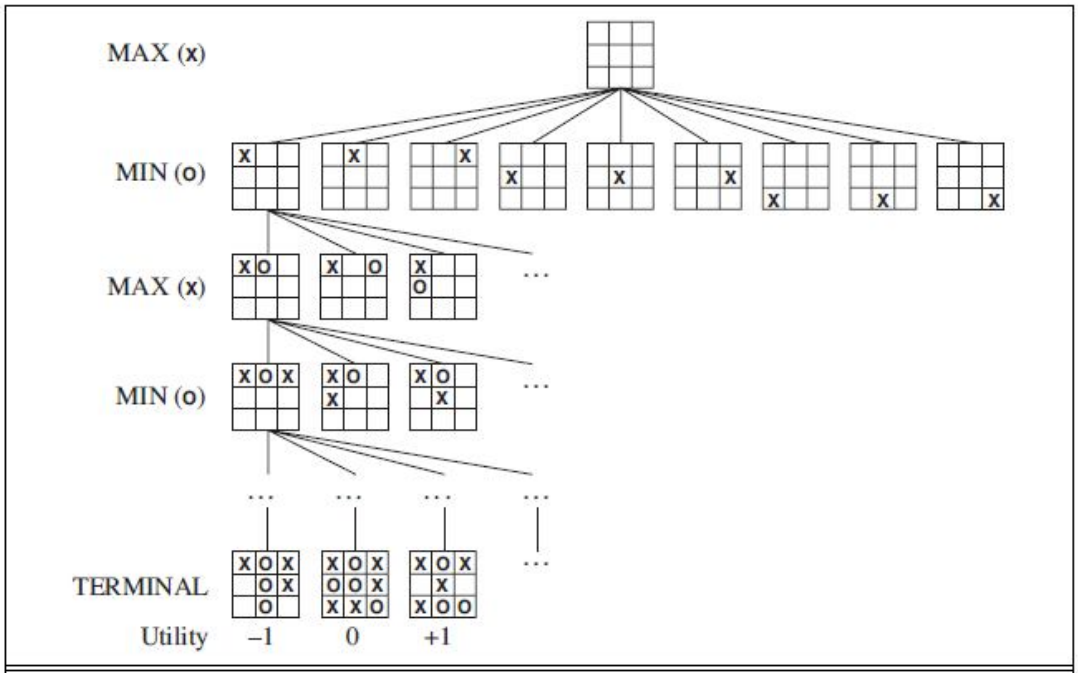


Figure 7.1 A (partial) game tree for the game of tic-tac-toe

For tic-tac-toe the game tree is relatively small—fewer than 9! = 362, 880 terminal nodes. But for chess there are over 10^{40} nodes, so the game tree is best thought of as a theoretical construct that we cannot realize in the physical world. But regardless of the size of the game tree, it is MAX's job to search for a good move. We use the term **search tree** for a tree that is superimposed on the full game tree, and examines enough nodes to allow a player to determine what move to make.

7.4 Optimal Decisions in Games

In a normal search problem, the optimal solution would be a sequence of actions leading to a goal state—a terminal state that is a win. In adversarial search, MIN has something to say about it. MAX therefore must find a contingent **strategy**, which specifies MAX's move in the initial state, then MAX's moves in the states resulting from every possible response by MIN, then MAX's moves in the states resulting from every possible response by MIN to *those* moves, and so on. This is exactly analogous to the AND–OR search algorithm (Figure 7.2) with MAX playing the role of OR and MIN equivalent to AND. Roughly speaking, an optimal strategy leads to outcomes at least as good as any other strategy when one is playing an infallible opponent. We begin by showing how to find this optimal strategy.

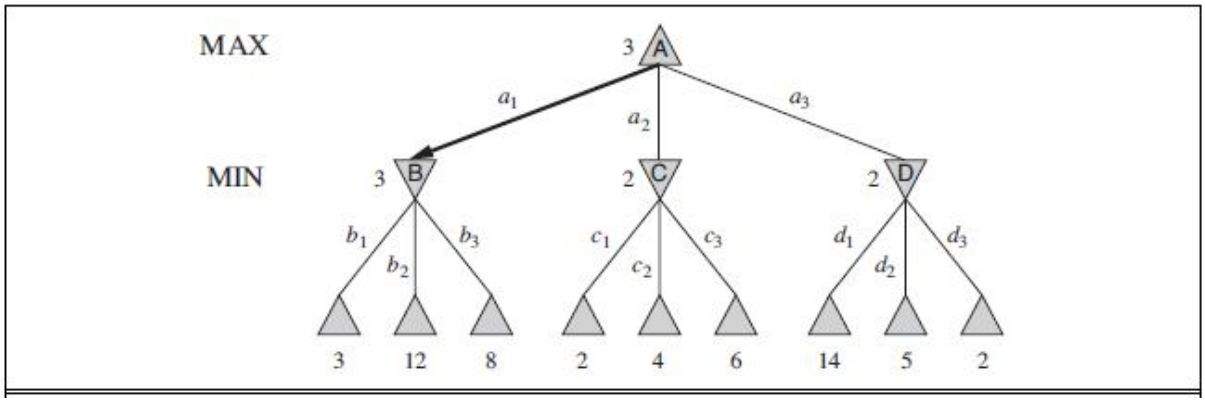


Figure 7.2 A two-ply game tree

Even a simple game like tic-tac-toe is too complex for us to draw the entire game tree on one page, so we will switch to the trivial game in Figure 7.2. The possible moves for MAX at the root node are labeled a_1, a_2, and a_3. The possible replies to a_1 for MIN are b_1, b_2, b_3, and so on. This particular game ends after one move each by MAX and MIN. (In game parlance, we say that this tree is one move deep, consisting of two half-moves, each of which is called a **ply**.) The utilities of PLY the terminal states in this game range from 2 to 14.

Given a game tree, the optimal strategy can be determined from the **minimax value** of each node, which we write as MINIMAX(n). The minimax value of a node is the utility (for MAX) of being in the corresponding state, *assuming that both players play optimally* from there to the end of the game. Obviously, the minimax value of a terminal state is just its utility. Furthermore, given a choice, MAX prefers to move to a state of maximum value, whereas MIN prefers a state of minimum value.

7.4.1 The minimax algorithm

The **minimax algorithm** (Figure 7.3) computes the minimax decision from the current state. It uses a simple recursive computation of the minimax values of each successor state, directly implementing the defining equations. The recursion proceeds all the way down to the leaves of the tree, and then the minimax values are **backed up** through the tree as the recursion unwinds. For example, in Figure 7.2, the algorithm first recurses down to the three bottomleft nodes and uses the UTILITY function on them to discover that their values are 3, 12, and 8, respectively. Then it takes the minimum of these values, 3, and returns it as the backedup value of node B. A similar process gives the backed-up values of 2 for C and 2 for D. Finally, we take the maximum of 3, 2, and 2 to get the backed-up value of 3 for the root node.

```
function MINIMAX-DECISION(state) returns an action
  return arg max_{a ∈ ACTIONS(s)} MIN-VALUE(RESULT(state, a))

function MAX-VALUE(state) returns a utility value
  if TERMINAL-TEST(state) then return UTILITY(state)
  v ← −∞
  for each a in ACTIONS(state) do
    v ← MAX(v, MIN-VALUE(RESULT(s, a)))
  return v

function MIN-VALUE(state) returns a utility value
  if TERMINAL-TEST(state) then return UTILITY(state)
  v ← ∞
  for each a in ACTIONS(state) do
    v ← MIN(v, MAX-VALUE(RESULT(s, a)))
  return v
```

Figure 7.3 An algorithm for calculating Minimax decisions

The minimax algorithm performs a complete depth-first exploration of the game tree. If the maximum depth of the tree is *m* and there are *b* legal moves at each point, then the time complexity of the minimax algorithm is $O(b^m)$. The space complexity is $O(bm)$ for an algorithm that generates all actions at once, or $O(m)$ for an algorithm that generates actions one at a time. For real games, of course, the time cost is totally impractical, but this algorithm serves as the basis for the mathematical analysis of games and for more practical algorithms.

7.4.2 Optimal decisions in multiplayer games

Many popular games allow more than two players. Let us examine how to extend the minimax idea to multiplayer games. This is straightforward from the technical viewpoint, but raises some interesting new conceptual issues. First, we need to replace the single value for each node with a *vector* of values. For example, in a three-player game with players A, B, and C, a vector (v_A, v_B, v_C) is associated with each node. For terminal states, this vector gives the utility of the state from each player's viewpoint. (In two-player, zero-sum games, the two-element vector can be reduced to a single value because the values are always opposite.) The simplest way to implement this is to have the UTILITY function return a vector of utilities.

Now we have to consider nonterminal states. Consider the node marked X in the game tree shown in Figure 7.4. In that state, player C chooses what to do. The two choices lead to terminal states with utility vectors (v_A =1, v_B =2, v_C = 6) and (v_A =4, v_B =2, v_C =3). Since 6 is bigger than 3, C should choose the first move. This means that if state X is reached, subsequent play will lead to a terminal state with utilities (v_A =1, v_B =2, v_C =6). Hence, the backed-up value of X is this vector. The backed-up value of a node n is always the utility vector of the successor state with the highest value for the player choosing at *n*. Anyone who plays multiplayer games, such as Diplomacy, quickly becomes aware that much more is going on than in two-player games.

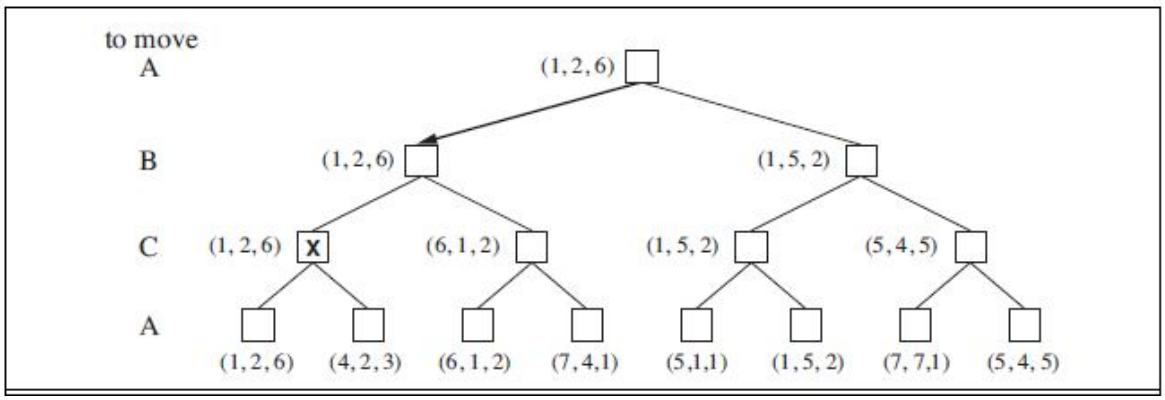


Figure 7.4 The first three plies of a game tree with three players (A, B. C)

Multiplayer games usually involve **alliances**, whether formal or informal, among the players. Alliances are made and broken as the game proceeds. For example, suppose A and B are in weak positions and C is in a stronger position. Then it is often optimal for both A and B to attack C rather than each other, lest C destroy each of them individually. In this way, collaboration emerges from purely selfish behavior. Of course, as soon as C weakens under the joint onslaught, the alliance loses its value, and either A or B could violate the agreement. In some cases, explicit alliances merely make concrete what would have happened anyway. In other cases, a social stigma attaches to breaking an alliance, so players must balance the immediate advantage of breaking an alliance against the long-term disadvantage of being perceived as untrustworthy.

If the game is not zero-sum, then collaboration can also occur with just two players. Suppose, for example, that there is a terminal state with utilities (v_A =1000, v_B =1000) and that 1000 is the highest possible utility for each player. Then the optimal strategy is for both players to do everything possible to reach this state—that is, the players will automatically cooperate to achieve a mutually desirable goal.

7.5 Alpha-Beta Pruning

The problem with minimax search is that the number of game states it has to examine is exponential in the depth of the tree. Unfortunately, we can't eliminate the exponent, but it turns out we can effectively cut it in half. The trick is that it is possible to compute the correct minimax decision without looking at every node in the game tree. That is, we can borrow the idea of **pruning** to eliminate large parts of the tree from consideration. The particular technique we examine is called **alpha–beta pruning**. When applied to a standard minimax tree, it returns the same move as minimax would, but prunes away branches that cannot possibly influence the final decision.

Alpha–beta pruning can be applied to trees of any depth, and it is often possible to prune entire subtrees rather than just leaves. The general principle is this: consider a node *n* somewhere in the tree (see Figure 7.5), such that Player has a choice of moving to that node. If Player has a better choice *m* either at the parent node of *n* or at any choice point further up, then *n will never be reached in actual play.* So once we have found out enough about *n* (by examining some of its descendants) to reach this conclusion, we can prune it.

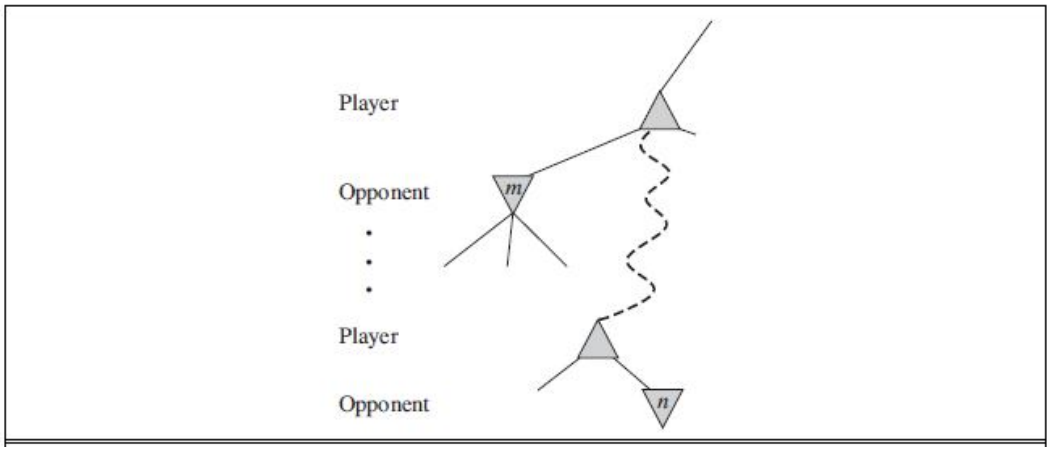


Figure 7.5 The general case of alpha-beta pruning

Remember that minimax search is depth-first, so at any one time we just have to consider the nodes along a single path in the tree. Alpha–beta pruning gets its name from the following two parameters that describe bounds on the backed-up values that appear anywhere along the path:

• α = the value of the best (i.e., highest-value) choice we have found so far at any choice point along the path for MAX.

• β = the value of the best (i.e., lowest-value) choice we have found so far at any choice point along the path for MIN.

Alpha–beta search updates the values of α and β as it goes along and prunes the remaining branches at a node (i.e., terminates the recursive call) as soon as the value of the current node is known to be worse than the current á or â value for MAX or MIN, respectively.

7.6 Evaluation Functions

The minimax algorithm generates the entire game search space, whereas the alpha–beta algorithm allows us to prune large parts of it. However, alpha–beta still has to search all the way to terminal states for at least a portion of the search space. This depth is usually not practical, because moves must be made in a reasonable amount of time—typically a few minutes at most. Claude Shannon (1950) proposed instead that programs should cut off the search earlier and apply a heuristic **evaluation function** to states in the search, effectively turning nonterminal

nodes into terminal leaves. In other words, the suggestion is to alter minimax or alpha–beta in two ways: replace the utility function by a heuristic evaluation function EVAL, which estimates the position's utility, and replace the terminal test by a **cutoff test** that decides when to apply EVAL. That gives us the following for heuristic minimax for state *s* and maximum depth *d*:

$$\text{H-MINIMAX}(s,d) = \begin{cases} \text{EVAL}(s) & \text{if CUTOFF-TEST}(s,d) \\ \max_{a \in Actions(s)} \text{H-MINIMAX}(\text{RESULT}(s,a), d+1) & \text{if PLAYER}(s) = \text{MAX} \\ \min_{a \in Actions(s)} \text{H-MINIMAX}(\text{RESULT}(s,a), d+1) & \text{if PLAYER}(s) = \text{MIN}. \end{cases}$$

An evaluation function returns an *estimate* of the expected utility of the game from a given position, just as the heuristic functions. It should be clear that the performance of a game-playing program depends strongly on the quality of its evaluation function. An inaccurate evaluation function will guide an agent toward positions that turn out to be lost. First, the evaluation function should order the *terminal* states in the same way as the true utility function: states that are wins must evaluate better than draws, which in turn must be better than losses. Otherwise, an agent using the evaluation function might err even if it can see ahead all the way to the end of the game. Second, the computation must not take too long! (The whole point is to search faster.) Third, for nonterminal states, the evaluation function should be strongly correlated with the actual chances of winning.

Given the limited amount of computation that the evaluation function is allowed to do for a given state, the best it can do is make a guess about the final outcome. Most evaluation functions work by calculating various **features** of the state—for example, in chess, we would have features for the number of white pawns, black pawns, white queens, black queens, and so on. The features, taken together, define various *categories* or *equivalence classes* of states: the states in each category have the same values for all the features. Any given category, generally speaking, will contain some states that lead to wins, some that lead to draws, and some that lead to losses. The evaluation function cannot know which states are which, but it can return a single value that reflects the *proportion* of states with each outcome.

Adding up the values of features seems like a reasonable thing to do, but in fact it involves a strong assumption: that the contribution of each feature is *independent* of the values of the other features. For example, assigning the value 3 to a bishop ignores the fact that bishops are more powerful in the endgame, when they have a lot of space to movement. For this reason,

current programs for chess and other games also use *nonlinear* combinations of features. For example, a pair of bishops might be worth slightly more than twice the value of a single bishop, and a bishop is worth more in the endgame (that is, when the *move number* feature is high or the *number of remaining pieces* feature is low). The astute reader will have noticed that the features and weights are *not* part of the rules of chess! They come from centuries of human chess-playing experience. In games where this kind of experience is not available, the weights of the evaluation function can be estimated by the machine learning techniques. Reassuringly, applying these techniques to chess has confirmed that a bishop is indeed worth about three pawns.

7.6.1 Cutting off search

The next step is to modify ALPHA-BETA-SEARCH so that it will call the heuristic EVAL function when it is appropriate to cut off the search.

if CUTOFF-TEST(state, depth) **then return** EVAL(state)

We also must arrange for some bookkeeping so that the current depth is incremented on each recursive call. The most straightforward approach to controlling the amount of search is to set a fixed depth limit so that CUTOFF-TEST(*state, depth*) returns true for all depth greater than some fixed depth d. (It must also return true for all terminal states, just as TERMINAL-TEST did.) The depth d is chosen so that a move is selected within the allocated time. A more robust approach is to apply iterative deepening. When time runs out, the program returns the move selected by the deepest completed search. As a bonus, iterative deepening also helps with move ordering.

These simple approaches can lead to errors due to the approximate nature of the evaluation function. Obviously, a more sophisticated cutoff test is needed. The evaluation function should be applied only to positions that are **quiescent**—that is, unlikely to exhibit wild swings in value in the near future. In chess, for example, positions in which favorable captures can be made are not quiescent for an evaluation function that just counts material. Nonquiescent positions can be expanded further until quiescent positions are reached. This extra search is called a **quiescence search**; sometimes it is restricted to consider only certain types of moves, such as capture moves, that will quickly resolve the uncertainties in the position.

7.6.2 Forward pruning

So far, we have talked about cutting off search at a certain level and about doing alpha–beta pruning that provably has no effect on the result (at least with respect to the heuristic evaluation values). It is also possible to do **forward pruning**, meaning that some moves at a given node are pruned immediately without further consideration. Clearly, most humans playing chess consider only a few moves from each position (at least consciously). One approach to forward pruning is **beam search**: on each ply, consider only a "beam" of the n best moves (according to the evaluation function) rather than considering all possible moves. Unfortunately, this approach is rather dangerous because there is no guarantee that the best move will not be pruned away.

Combining all the techniques described here results in a program that can play creditable chess (or other games). Let us assume we have implemented an evaluation function for chess, a reasonable cutoff test with a quiescence search, and a large transposition table. Let us also assume that, after months of tedious bit-bashing, we can generate and evaluate around a million nodes per second on the latest PC, allowing us to search roughly 200 million nodes per move under standard time controls (three minutes per move). The branching factor for chess is about 35, on average, and 35^5 is about 50 million, so if we used minimax search, we could look ahead only about five plies. Though not incompetent, such a program can be fooled easily by an average human chess player, who can occasionally plan six or eight plies ahead. With alpha–beta search we get to about 10 plies, which results in an expert level of play. To reach grandmaster status we would need an extensively tuned evaluation function and a large database of optimal opening and endgame moves.

7.7 Stochastic Games

In real life, many unpredictable external events can put us into unforeseen situations. Many games mirror this unpredictability by including a random element, such as the throwing of dice. We call these **stochastic games**. Backgammon is a typical game that combines luck and skill. Dice are rolled at the beginning of a player's turn to determine the legal moves. Although White knows what his or her own legal moves are, White does not know what Black is going to roll and thus does not know what Black's legal moves will be. That means White cannot construct a standard game tree of the sort we saw in chess and tic-tac-toe. A game tree in backgammon must include **chance nodes** in addition to MAX and MIN nodes. Chance nodes are shown as circles in Figure 7.6. The branches leading from each chance node denote the possible dice

rolls; each branch is labeled with the roll and its probability. There are 36 ways to roll two dice, each equally likely; but because a 6–5 is the same as a 5–6, there are only 21 distinct rolls. The six doubles (1–1 through 6–6) each have a probability of 1/36, so we say P(1–1) = 1/36. The other 15 distinct rolls each have a 1/18 probability.

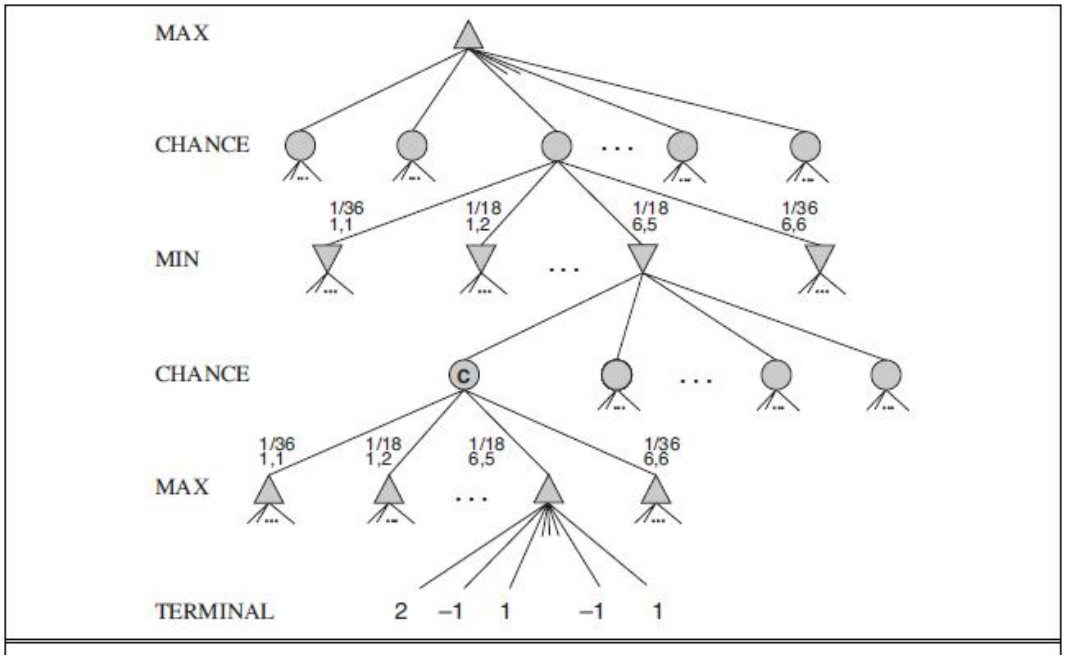


Figure 7.6 Schematic game tree for a back gammon position

The next step is to understand how to make correct decisions. Obviously, we still want to pick the move that leads to the best position. However, positions do not have definite minimax values. Instead, we can only calculate the **expected value** of a position: the average over all possible outcomes of the chance nodes. This leads us to generalize the **minimax value** for deterministic games to an **expectiminimax value** for games with chance nodes. Terminal nodes and MAX and MIN nodes (for which the dice roll is known) work exactly the same way as before. For chance nodes we compute the expected value, which is the sum of the value over all outcomes, weighted by the probability of each chance action:

$$\text{EXPECTIMINIMAX}(s) = \begin{cases} \text{UTILITY}(s) & \text{if TERMINAL-TEST}(s) \\ \max_a \text{EXPECTIMINIMAX}(\text{RESULT}(s,a)) & \text{if PLAYER}(s) = \text{MAX} \\ \min_a \text{EXPECTIMINIMAX}(\text{RESULT}(s,a)) & \text{if PLAYER}(s) = \text{MIN} \\ \sum_r P(r)\text{EXPECTIMINIMAX}(\text{RESULT}(s,r)) & \text{if PLAYER}(s) = \text{CHANCE} \end{cases}$$

where r represents a possible dice roll (or other chance event) and RESULT(s, r) is the same state as s, with the additional fact that the result of the dice roll is r.

As with minimax, the obvious approximation to make with expectiminimax is to cut the search off at some point and apply an evaluation function to each leaf. One might think that evaluation functions for games such as backgammon should be just like evaluation functions for chess—they just need to give higher scores to better positions. But in fact, the presence of chance nodes means that one has to be more careful about what the evaluation values mean.

The advantage of alpha–beta is that it ignores future developments that just are not going to happen, given best play. Thus, it concentrates on likely occurrences. In games with dice, there are *no* likely sequences of moves, because for those moves to take place, the dice would first have to come out the right way to make them legal. This is a general problem whenever uncertainty enters the picture: the possibilities are multiplied enormously, and forming detailed plans of action becomes pointless because the world probably will not play along.

An alternative is to do **Monte Carlo simulation** to evaluate a position. Start with an alpha–beta (or other) search algorithm. From a start position, have the algorithm play thousands of games against itself, using random dice rolls. In the case of backgammon, the resulting win percentage has been shown to be a good approximation of the value of the position, even if the algorithm has an imperfect heuristic and is searching only a few plies (Tesauro, 1995). For games with dice, this type of simulation is called a **rollout**.

Search and evaluation of unnecessary nodes or branches of the **game** tree degrades the overall performance and efficiency of the engine. Both min and max players have lots of choices to decide from. Exploring the entire tree is not possible as there is a restriction of time and space.

7.8 Summary

We have looked at a variety of games to understand what optimal play means and to understand how to play well in practice. The most important ideas are as follows: A game can be defined by the **initial state** (how the board is set up), the legal **actions** in each state, the **result** of each action, a **terminal test** (which says when the game is over), and a **utility function** that applies to terminal states. In two-player zero-sum games with **perfect information**, the **minimax** algorithm can select optimal moves by a depth-first enumeration of the game tree. The **alpha–beta** search algorithm computes the same optimal move as minimax, but achieves much greater efficiency by eliminating subtrees that are provably irrelevant. Usually, it is not feasible to consider the whole game tree (even with alpha–beta), so we need to cut the search off at some point and apply a heuristic **evaluation function** that estimates the utility of a state. Many game programs precompute tables of best moves in the opening and endgame so that they can look up a move rather than search.

7.9 Model Questions

1. Mention the elements of a game problem.
2. Explain in detail: Minimax algorithm.
3. Illustrate Alpha-Beta pruning with an example.
4. Write short notes on Evaluation functions.

LESSON 8
CONSTRAINT SATISFACTION PROBLEMS

Structure

8.1 Introduction

The previous lessons explored the idea that problems can be solved by searching in a space of **states**. These states can be evaluated by domain-specific heuristics and tested to see whether they are goal states. From the point of view of the search algorithm, however, each state is atomic, or indivisible—a black box with no internal structure. This lesson describes a way to solve a wide variety of problems more efficiently. This method uses a **factored representation** for each state: a set of variables, each of which has a value. A problem is solved when each variable has a value that satisfies all the constraints on the variable. A problem described this way is called a **constraint satisfaction problem**, or **CSP**.

8.2 Objectives

- To explore the concepts of Constraint Satisfaction Problems.
- To describe the various inferences in Constraint Satisfaction Problems.

8.3 Defining Constraint Satisfaction Problems

A constraint satisfaction problem consists of three components, X,D, and C:

X is a set of variables, $\{X_1, \ldots, X_n\}$.

D is a set of domains, $\{D_1, \ldots, D_n\}$, one for each variable.

C is a set of constraints that specify allowable combinations of values.

Each domain D_i consists of a set of allowable values, $\{v_1, \ldots, v_k\}$ for variable X_i. Each constraint C_i consists of a pair *(scope, rel)*, where *scope* is a tuple of variables that participate in the constraint and *rel* is a relation that defines the values that those variables can take on. A relation can be represented as an explicit list of all tuples of values that satisfy the constraint, or as an abstract relation that supports two operations: testing if a tuple is a member of the relation and enumerating the members of the relation. For example, if X_1 and X_2 both have the domain {A,B}, then the constraint saying the two variables must have different values can be written as $((X_1,X_2), [(A,B), (B,A)])$ or as $((X_1,X_2),X_1) = X_2)$.

To solve a CSP, we need to define a state space and the notion of a solution. Each state in a CSP is defined by an **assignment** of values ASSIGNMENT to some or all of the variables, $\{X_i$ = CONSISTENT v_i, $X_j = v_j$, . . .}. An assignment that does not violate any constraints is called a **consistent** or legal assignment. A **complete assignment** is one in which every variable is assigned, and a **solution** to a CSP is a consistent, complete assignment. A **partial assignment** is one that assigns values to only some of the variables.

8.3.1 Example problem: Map coloring

Suppose that, having tired of Romania, we are looking at a map of Australia showing each of its states and territories (Figure 8.1(a)). We are given the task of coloring each region either red, green, or blue in such a way that no neighboring regions have the same color. To formulate this as a CSP, we define the variables to be the regions

X = {WA, NT, Q, NSW, V, SA, T} .

The domain of each variable is the set D_i = {red, green, blue}. The constraints require neighboring regions to have distinct colors. Since there are nine places where regions border, there are nine constraints:

C = {SA ‘“ WA, SA ‘“ NT, SA ‘“ Q,SA ‘“ NSW, SA ‘“ V, WA ‘“ NT, NT ‘“ Q,

Q ‘“ NSW, NSW ‘“ V } .

Here we are using abbreviations; SA ‘“ WA is a shortcut for ((SA,WA), SA ‘“ WA), where SA ‘“ WA can be fully enumerated in turn as

{(red , green), (red , blue), (green, red), (green, blue), (blue, red), (blue, green)} .

There are many possible solutions to this problem, such as

{WA = red, NT = green, Q = red, NSW = green, V = red, SA = blue, T = red}.

It can be helpful to visualize a CSP as a **constraint graph**, as shown in Figure 8.1(b). The nodes of the graph correspond to variables of the problem, and a link connects any two variables that participate in a constraint.

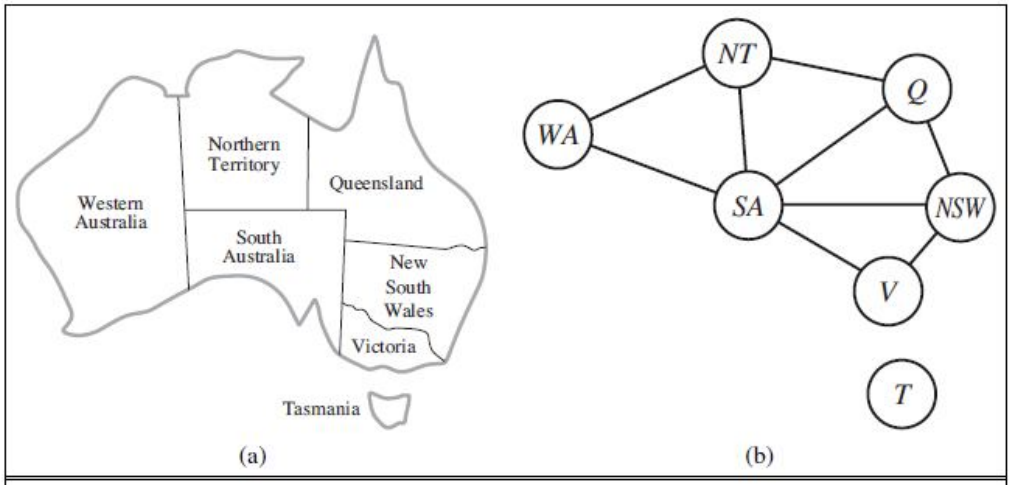


Figure 8.1 (a) The principal state and territories of Australia.
(b) The map-coloring problem represented as a constraint graph

The reason to formulate a problem as a CSP is that the CSPs yield a natural representation for a wide variety of problems; if you already have a CSP-solving system, it is often easier to solve a problem using it than to design a custom solution using another search technique. In addition, CSP solvers can be faster than state-space searchers because the CSP solver can quickly eliminate large swatches of the search space. For example, once we have chosen {SA=blue} in the Australia problem, we can conclude that none of the five neighboring variables can take on the value blue. Without taking advantage of constraint propagation, a search procedure would have to consider 35 =243 assignments for the five neighboring variables; with constraint propagation we never have to consider blue as a value, so we have only 25 =32 assignments to look at, a reduction of 87%.

In regular state-space search we can only ask: is this specific state a goal? With CSPs, once we find out that a partial assignment is not a solution, we can immediately discard further refinements of the partial assignment. Furthermore, we can see *why* the assignment is not a solution—we see which variables violate a constraint—so we can focus attention on the variables that matter. As a result, many problems that are intractable for regular state-space search can be solved quickly when formulated as a CSP.

8.4 Constraint Propagation: Inference in CSPs

In regular state-space search, an algorithm can do only one thing: search. In CSPs there is a choice: an algorithm can search (choose a new variable assignment from several possibilities) or do a specific type of **inference** called **constraint propagation**: using the constraints to reduce the number of legal values for a variable, which in turn can reduce the legal values for another variable, and so on. Constraint propagation may be intertwined with search, or it may be done as a preprocessing step, before search starts. Sometimes this preprocessing can solve the whole problem, so no search is required at all.

The key idea is **local consistency**. If we treat each variable as a node in a graph (see Figure 8.1(b)) and each binary constraint as an arc, then the process of enforcing local consistency in each part of the graph causes inconsistent values to be eliminated throughout the graph. There are different types of local consistency, which we will cover in coming sections.

8.4.1 Node consistency

A single variable (corresponding to a node in the CSP network) is **node-consistent** if all the values in the variable's domain satisfy the variable's unary constraints. For example, in the variant of the Australia map-coloring problem (Figure 8.1) where South Australians dislike green, the variable SA starts with domain {red , green, blue}, and we can make it node consistent by eliminating green, leaving SA with the reduced domain {red , blue}. We say that a network is node-consistent if every variable in the network is node-consistent.

It is always possible to eliminate all the unary constraints in a CSP by running node consistency. It is also possible to transform all *n-ary* constraints into binary ones. Because of this, it is common to define CSP solvers that work with only binary constraints.

8.4.2 Arc consistency

A variable in a CSP is **arc-consistent** if every value in its domain satisfies the variable's binary constraints. More formally, X_i is arc-consistent with respect to another variable X_j if for every value in the current domain D_i there is some value in the domain D_j that satisfies the binary constraint on the arc (X_i, X_j). A network is arc-consistent if every variable is arc consistent with every other variable. For example, consider the constraint $Y = X^2$ where the domain of both X and Y is the set of digits. We can write this constraint explicitly as

((X, Y), {(0, 0), (1, 1), (2, 4), (3, 9))}) .

To make X arc-consistent with respect to Y , we reduce X's domain to {0, 1, 2, 3}. If we also make Y arc-consistent with respect to X, then Y 's domain becomes {0, 1, 4, 9} and the whole CSP is arc-consistent.

On the other hand, arc consistency can do nothing for the Australia map-coloring problem. Consider the following inequality constraint on (SA,WA):

{(red , green), (red , blue), (green, red), (green, blue), (blue, red), (blue, green)} .

No matter what value you choose for SA (or for WA), there is a valid value for the other variable. So applying arc consistency has no effect on the domains of either variable.

The most popular algorithm for arc consistency is called AC-3 (see Figure 8.2). To make every variable arc-consistent, the AC-3 algorithm maintains a queue of arcs to consider. (Actually, the order of consideration is not important, so the data structure is really a set, but tradition calls it a queue.) Initially, the queue contains all the arcs in the CSP. AC-3 then pops off an arbitrary arc (X_i, X_j) from the queue and makes X_i arc-consistent with respect to X_j. If this leaves D_i unchanged, the algorithm just moves on to the next arc. But if this revises D_i (makes the domain smaller), then we add to the queue all arcs (X_k, X_i) where X_k is a neighbor of X_i. We need to do that because the change in D_i might enable further reductions in the domains of D_k, even if we have previously considered X_k. If D_i is revised down to nothing, then we know the whole CSP has no consistent solution, and AC-3 can immediately return failure. Otherwise, we keep checking, trying to remove values from the domains of variables until no more arcs are in the queue. At that point, we are left with a CSP that is equivalent to the original CSP—they both have the same solutions—but the arc-consistent CSP will in most cases be faster to search because its variables have smaller domains.

```
function AC-3( csp) returns false if an inconsistency is found and true otherwise
  inputs: csp, a binary CSP with components (X, D, C)
  local variables: queue, a queue of arcs, initially all the arcs in csp

  while queue is not empty do
    (X_i, X_j) ← REMOVE-FIRST(queue)
    if REVISE(csp, X_i, X_j) then
      if size of D_i = 0 then return false
      for each X_k in X_i.NEIGHBORS - {X_j} do
        add (X_k, X_i) to queue
  return true
```

```
function REVISE( csp, X_i, X_j) returns true iff we revise the domain of X_i
  revised ← false
  for each x in D_i do
    if no value y in D_j allows (x,y) to satisfy the constraint between X_i and X_j then
      delete x from D_i
      revised ← true
  return revised
```

Figure 8.2 The arc-consistency algorithm AC-3

The complexity of AC-3 can be analyzed as follows. Assume a CSP with n variables, each with domain size at most d, and with c binary constraints (arcs). Each arc (X_k ,X_i) can be inserted in the queue only d times because X_i has at most d values to delete. Checking consistency of an arc can be done in $O(d^2)$ time, so we get $O(cd^3)$ total worst-case time.

It is possible to extend the notion of arc consistency to handle n-ary rather than just binary constraints; this is called generalized arc consistency or sometimes hyperarc consistency, depending on the author. A variable X_i is **generalized arc consistent** with respect to an n-ary constraint if for every value v in the domain of X_i there exists a tuple of values that is a member of the constraint, has all its values taken from the domains of the corresponding variables, and has its X_i component equal to v. For example, if all variables have the domain {0, 1, 2, 3}, then to make the variable X consistent with the constraint X < Y < Z, we would have to eliminate 2 and 3 from the domain of X because the constraint cannot be satisfied when X is 2 or 3.

8.4.3 Path consistency

Arc consistency can go a long way toward reducing the domains of variables, sometimes finding a solution (by reducing every domain to size 1) and sometimes finding that the CSP cannot be solved (by reducing some domain to size 0). But for other networks, arc consistency fails to make enough inferences. Consider the map-coloring problem on Australia, but with only two colors allowed, red and blue. Arc consistency can do nothing because every variable is already arc consistent: each can be red with blue at the other end of the arc (or vice versa). But clearly there is no solution to the problem: because Western Australia, Northern Territory and South Australia all touch each other, we need at least three colors for them alone.

Arc consistency tightens down the domains (unary constraints) using the arcs (binary constraints). To make progress on problems like map coloring, we need a stronger notion of consistency. **Path consistency** tightens the binary constraints by using implicit constraints that are inferred by looking at triples of variables.

A two-variable set $\{X_i, X_j\}$ is path-consistent with respect to a third variable X_m if, for every assignment $\{X_i = a, X_j = b\}$ consistent with the constraints on $\{X_i, X_j\}$, there is an assignment to X_m that satisfies the constraints on $\{X_i, X_m\}$ and $\{X_m, X_j\}$. This is called path consistency because one can think of it as looking at a path from X_i to X_j with X_m in the middle.

Let's see how path consistency fares in coloring the Australia map with two colors. We will make the set {WA,SA} path consistent with respect to NT. We start by enumerating the consistent assignments to the set. In this case, there are only two: {WA = red, SA = blue} and {WA = blue, SA = red}. We can see that with both of these assignments NT can be neither red nor blue (because it would conflict with either WA or SA). Because there is no valid choice for NT, we eliminate both assignments, and we end up with no valid assignments for {WA, SA}. Therefore, we know that there can be no solution to this problem. The PC-2 algorithm (Mackworth, 1977) achieves path consistency in much the same way that AC-3 achieves arc consistency.

8.4.4 *K*-consistency

Stronger forms of propagation can be defined with the notion of k**-consistency**. A CSP is k-consistent if, for any set of k " 1 variables and for any consistent assignment to those variables, a consistent value can always be assigned to any kth variable. 1-consistency says that, given the empty set, we can make any set of one variable consistent: this is what we called node

consistency. 2-consistency is the same as arc consistency. For binary constraint networks, 3-consistency is the same as path consistency.

A CSP is **strongly** k**-consistent** if it is k-consistent and is also (k " 1)-consistent, (k " 2)-consistent, all the way down to 1-consistent. Now suppose we have a CSP with n nodes and make it strongly n-consistent (i.e., strongly k-consistent for k = n). We can then solve the problem as follows: First, we choose a consistent value for X_1. We are then guaranteed to be able to choose a value for X_2 because the graph is 2-consistent, for X_3 because it is 3-consistent, and so on. For each variable X_i, we need only search through the d values in the domain to find a value consistent with $X_1, \ldots, X_{i"1}$. We are guaranteed to find a solution in time $O(n^2d)$. Of course, there is no free lunch: any algorithm for establishing n-consistency must take time exponential in n in the worst case. Worse, n-consistency also requires space that is exponential in n. The memory issue is even more severe than the time. In practice, determining the appropriate level of consistency checking is mostly an empirical science. It can be said practitioners commonly compute 2-consistency and less commonly 3-consistency.

8.4.5 Global Constraints

Remember that a **global constraint** is one involving an arbitrary number of variables (but not necessarily all variables). Global constraints occur frequently in real problems and can be handled by special-purpose algorithms that are more efficient than the general-purpose methods described so far. For example, the Alldiff constraint says that all the variables involved must have distinct values (Sudoku puzzle). One simple form of inconsistency detection for Alldiff constraints works as follows: if m variables are involved in the constraint, and if they have n possible distinct values altogether, and m > n, then the constraint cannot be satisfied.

This leads to the following simple algorithm: First, remove any variable in the constraint that has a singleton domain, and delete that variable's value from the domains of the remaining variables. Repeat as long as there are singleton variables. If at any point an empty domain is produced or there are more variables than domain values left, then an inconsistency has been detected.

This method can detect the inconsistency in the assignment {WA = red, NSW = red} for Figure 8.1. Notice that the variables SA, NT, and Q are effectively connected by an Alldiff constraint because each pair must have two different colors. After applying AC-3 with the partial assignment, the domain of each variable is reduced to {green, blue}. That is, we have three variables and

only two colors, so the Alldiff constraint is violated. Thus, a simple consistency procedure for a higher-order constraint is sometimes more effective than applying arc consistency to an equivalent set of binary constraints. There are more complex inference algorithms for Alldiff that propagate more constraints but are more computationally expensive to run.

Another important higher-order RESOURCE constraint is the **resource constraint**, sometimes called the atmost constraint. For example, in a scheduling problem, let $P_1, \ldots, P_4$ denote the numbers of personnel assigned to each of four tasks. The constraint that no more than 10 personnel are assigned in total is written as Atmost $(10, P_1, P_2, P_3, P_4)$. We can detect an inconsistency simply by checking the sum of the minimum values of the current domains; for example, if each variable has the domain {3, 4, 5, 6}, the Atmost constraint cannot be satisfied. We can also enforce consistency by deleting the maximum value of any domain if it is not consistent with the minimum values of the other domains. Thus, if each variable in our example has the domain {2, 3, 4, 5, 6}, the values 5 and 6 can be deleted from each domain.

For large resource-limited problems with integer values—such as logistical problems involving moving thousands of people in hundreds of vehicles—it is usually not possible to represent the domain of each variable as a large set of integers and gradually reduce that set by consistency-checking methods. Instead, domains are represented by upper and lower bounds and are managed by **bounds propagation**. For example, in an airline-scheduling problem, let's suppose there are two flights, F_1 and F_2, for which the planes have capacities 165 and 385, respectively. The initial domains for the numbers of passengers on each flight are then

$$D_1 = [0, 165] \text{ and } D_2 = [0, 385] .$$

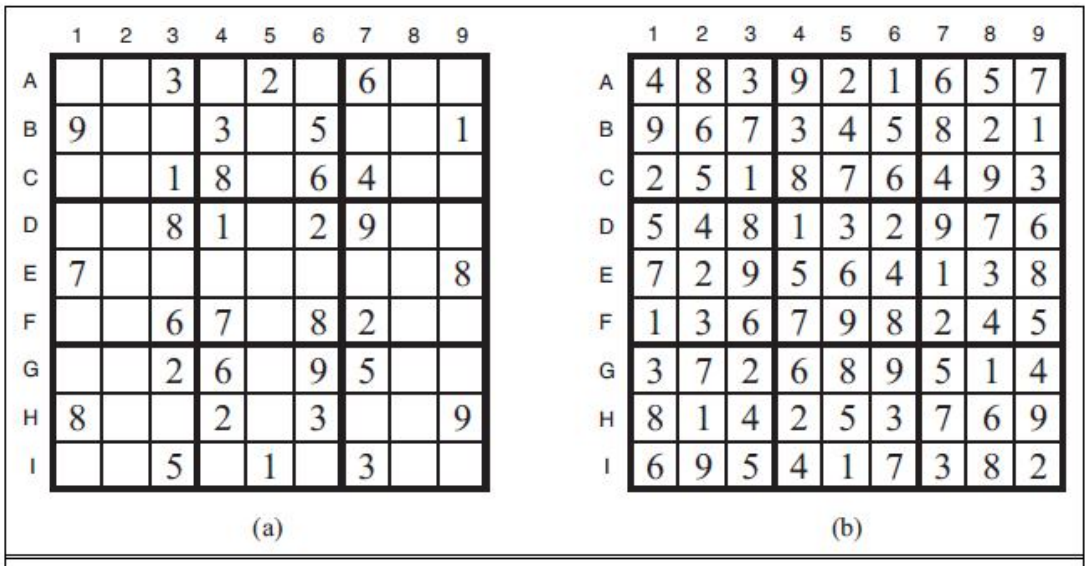
Now suppose we have the additional constraint that the two flights together must carry 420 people: $F_1 + F_2 = 420$. Propagating bounds constraints, we reduce the domains to

$$D_1 = [35, 165] \text{ and } D_2 = [255, 385] .$$

We say that a CSP is **bounds consistent** if for every variable X, and for both the lower-bound and upper-bound values of X, there exists some value of Y that satisfies the constraint between X and Y for every variable Y . This kind of bounds propagation is widely used in practical constraint problems.

8.4.6 Sudoku Example

The popular **Sudoku** puzzle has introduced millions of people to constraint satisfaction problems, although they may not recognize it. A Sudoku board consists of 81 squares, some of which are initially filled with digits from 1 to 9. The puzzle is to fill in all the remaining squares such that no digit appears twice in any row, column, or 3×3 box (see Figure 8.3). A row, column, or box is called a **unit**.



	1	2	3	4	5	6	7	8	9
A			3		2		6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7								8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1		3		

(a)

	1	2	3	4	5	6	7	8	9
A	4	8	3	9	2	1	6	5	7
B	9	6	7	3	4	5	8	2	1
C	2	5	1	8	7	6	4	9	3
D	5	4	8	1	3	2	9	7	6
E	7	2	9	5	6	4	1	3	8
F	1	3	6	7	9	8	2	4	5
G	3	7	2	6	8	9	5	1	4
H	8	1	4	2	5	3	7	6	9
I	6	9	5	4	1	7	3	8	2

(b)

Figure 8.3 (a) A Sudoku puzzle and (b) its solution

The Sudoku puzzles that are printed in newspapers and puzzle books have the property that there is exactly one solution. Although some can be tricky to solve by hand, taking tens of minutes, even the hardest Sudoku problems yield to a CSP solver in less than 0.1 second.

A Sudoku puzzle can be considered a CSP with 81 variables, one for each square. We use the variable names A1 through A9 for the top row (left to right), down to I1 through I9 for the bottom row. The empty squares have the domain {1, 2, 3, 4, 5, 6, 7, 8, 9} and the prefilled squares have a domain consisting of a single value. In addition, there are 27 different Alldiff constraints: one for each row, column, and box of 9 squares.

Alldiff (A1,A2,A3,A4,A5,A6, A7, A8, A9)

Alldiff (B1,B2,B3,B4,B5,B6,B7,B8,B9)

.....

Alldiff (A1,B1,C1,D1,E1, F1,G1,H1, I1)

Alldiff (A2,B2,C2,D2,E2, F2,G2,H2, I2)

.....

Alldiff (A1,A2,A3,B1,B2,B3,C1,C2,C3)

Alldiff (A4,A5,A6,B4,B5,B6,C4,C5,C6)

.....

Let us see how far arc consistency can take us. Assume that the Alldiff constraints have been expanded into binary constraints (such as A1 $\neq$ A2) so that we can apply the AC-3 algorithm directly. Consider variable E6 from Figure 8.3(a)—the empty square between the 2 and the 8 in the middle box. From the constraints in the box, we can remove not only 2 and 8 but also 1 and 7 from E6's domain. From the constraints in its column, we can eliminate 5, 6, 2, 8, 9, and 3. That leaves E6 with a domain of {4}; in other words, we know the answer for E6. Now consider variable I6—the square in the bottom middle box surrounded by 1, 3, and 3. Applying arc consistency in its column, we eliminate 5, 6, 2, 4 (since we now know E6 must be 4), 8, 9, and 3. We eliminate 1 by arc consistency with I5 , and we are left with only the value 7 in the domain of I6 . Now there are 8 known values in column 6, so arc consistency can infer that A6 must be 1. Inference continues along these lines, and eventually, AC-3 can solve the entire puzzle—all the variables have their domains reduced to a single value, as shown in Figure 8.3(b).

Of course, Sudoku would soon lose its appeal if every puzzle could be solved by a mechanical application of AC-3, and indeed AC-3 works only for the easiest Sudoku puzzles. Slightly harder ones can be solved by PC-2, but at a greater computational cost: there are 255, 960 different path constraints to consider in a Sudoku puzzle. To solve the hardest puzzles and to make efficient progress, we will have to be more clever.

Indeed, the appeal of Sudoku puzzles for the human solver is the need to be resourceful in applying more complex inference strategies. Aficionados give them colorful names, such as "naked triples." That strategy works as follows: in any unit (row, column or box), find three squares that each have a domain that contains the same three numbers or a subset of those

numbers. For example, the three domains might be {1, 8}, {3, 8}, and {1, 3, 8}. From that we don't know which square contains 1, 3, or 8, but we do know that the three numbers must be distributed among the three squares. Therefore we can remove 1, 3, and 8 from the domains of every *other* square in the unit.

It is interesting to note how far we can go without saying much that is specific to Sudoku. We do of course have to say that there are 81 variables, that their domains are the digits 1 to 9, and that there are 27 Alldiff constraints. But beyond that, all the strategies—arc consistency, path consistency, etc.—apply generally to all CSPs, not just to Sudoku problems. Even naked triples is really a strategy for enforcing consistency of Alldiff constraints and has nothing to do with Sudoku *per se*. This is the power of the CSP formalism: for each new problem area, we only need to define the problem in terms of constraints; then the general constraint-solving mechanisms can take over.

8.5 Summary

Constraint Satisfaction Problems (CSPs) represent a state with a set of variable/value pairs and represent the conditions for a solution by a set of constraints on the variables. Many important real-world problems can be described as CSPs. A number of inference techniques use the constraints to infer which variable/value pairs are consistent and which are not. These include node, arc, path, and k-consistency.

8.6 Model Questions

1. What do you mean by constraint satisfaction problems?
2. Explain in detail the Map Coloring example of CSP.
3. Mention any two inference techniques of constraint satisfaction problems and explain in detail.
4. Illustrate CSP with Sudoku example.

LESSON 9
SEARCHES FOR CONSTRAINT SATISFACTION PROBLEMS

Structure

9.1 Introduction

CSP search algorithms take advantage of the structure of states and use *general-purpose* rather than *problem-specific* heuristics to enable the solution of complex problems. The main idea is to eliminate large portions of the search space all at once by identifying variable/value combinations that violate the constraints.

9.2 Objectives

- To explain the Backtracking search of CSPs.
- To explore the Local search in CSPs.
- To study the Structure of the problems.

9.3 Backtracking Search for CSPs

Sudoku problems are designed to be solved by inference over constraints. But many other CSPs cannot be solved by inference alone; there comes a time when we must search for a solution. In this section, we will look at backtracking search algorithms that work on partial assignments. We could apply a standard depth-limited search. A state would be a partial assignment, and an action would be adding *var* = *value* to the assignment. But for a CSP with

n variables of domain size *d*, we quickly notice something terrible: the branching factor at the top level is *nd* because any of *d* values can be assigned to any of *n* variables. At the next level, the branching factor is *(n " 1)d*, and so on for *n* levels. We generate a tree with *n! û0 d*^{n} leaves, even though there are only *d*^{n} possible complete assignments!

Our seemingly reasonable but naive formulation ignores crucial property common to all CSPs: **commutativity**. A problem is commutative if the order of application of any given set of actions has no effect on the outcome. CSPs are commutative because when assigning values to variables, we reach the same partial assignment regardless of order. Therefore, we need only consider a *single* variable at each node in the search tree. For example, at the root node of a search tree for coloring the map of Australia, we might make a choice between SA = red, SA = green, and SA = blue, but we would never choose between SA = red and WA = blue. With this restriction, the number of leaves is *d*^{n}, as we would hope.

The term **backtracking search** is used for a depth-first search that chooses values for one variable at a time and backtracks when a variable has no legal values left to assign. The algorithm is shown in Figure 9.1. It repeatedly chooses an unassigned variable, and then tries all values in the domain of that variable in turn, trying to find a solution. If an inconsistency is detected, then BACKTRACK returns failure, causing the previous call to try another value. Because the representation of CSPs is standardized, there is no need to supply BACKTRACKING-SEARCH with a domain-specific initial state, action function, transition model, or goal test.

```
function BACKTRACKING-SEARCH(csp) returns a solution, or failure
  return BACKTRACK({ }, csp)

function BACKTRACK(assignment, csp) returns a solution, or failure
  if assignment is complete then return assignment
  var ← SELECT-UNASSIGNED-VARIABLE(csp)
  for each value in ORDER-DOMAIN-VALUES(var, assignment, csp) do
      if value is consistent with assignment then
          add {var = value} to assignment
          inferences ← INFERENCE(csp, var, value)
          if inferences ≠ failure then
              add inferences to assignment
              result ← BACKTRACK(assignment, csp)
              if result ≠ failure then
                  return result
      remove {var = value} and inferences from assignment
  return failure
```

Figure 9.1 A simple backtracking algorithm for constraint satisfaction problems

The poor performance of uninformed search algorithms are improved by supplying them with domain-specific heuristic functions derived from our knowledge of the problem. It turns out that we can solve CSPs efficiently *without* such domain-specific knowledge.

9.4 Local Search for CSPs

Local search algorithms turn out to be effective in solving many CSPs. They use a complete-state formulation: the initial state assigns a value to every variable, and the search changes the value of one variable at a time. For example, in the 8-queens problem, the initial state might be a random configuration of 8 queens in 8 columns, and each step moves a single queen to a new position in its column. Typically, the initial guess violates several constraints. The point of local search is to eliminate the violated constraints.

In choosing a new value for a variable, the most obvious heuristic is to select the value that results in the minimum number of conflicts with other variables—the **min-conflicts** heuristic. The algorithm is shown in Figure 9.2 and its application to an 8-queens problem is diagrammed in Figure 9.3.

```
function MIN-CONFLICTS(csp, max_steps) returns a solution or failure
  inputs: csp, a constraint satisfaction problem
          max_steps, the number of steps allowed before giving up

  current ← an initial complete assignment for csp
  for i = 1 to max_steps do
      if current is a solution for csp then return current
      var ← a randomly chosen conflicted variable from csp.VARIABLES
      value ← the value v for var that minimizes CONFLICTS(var, v, current, csp)
      set var = value in current
  return failure
```

Figure 9.2 The MIN-CONFLICTS algorithm for solving CSPs by local search

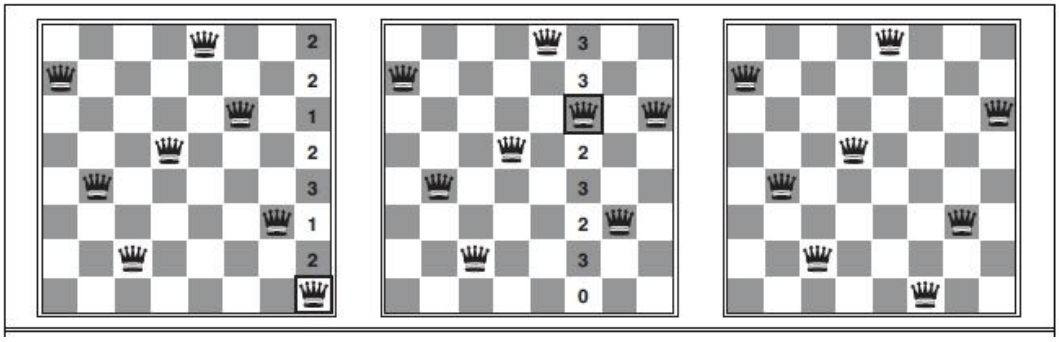


Figure 9.3 A two-step solution using min-conflicts for an 8-queens problem

Min-conflicts is surprisingly effective for many CSPs. Amazingly, on the n-queens problem, if you don't count the initial placement of queens, the run time of min-conflicts is roughly *independent of problem size*. It solves even the *million*-queens problem in an average of 50 steps (after the initial assignment). This remarkable observation was the stimulus leading to a great deal of research in the 1990s on local search and the distinction between easy and hard problems. Roughly speaking, n-queens is easy for local search because solutions are densely distributed throughout the state space. Min-conflicts also works well for hard problems. For example, it has been used to schedule observations for the Hubble Space Telescope, reducing the time taken to schedule a week of observations from three weeks to around 10 minutes.

All the local search techniques are candidates for application to CSPs, and some of those have proved especially effective. The landscape of a CSP under the minconflicts heuristic usually has a series of plateaux. There may be millions of variable assignments that are only one conflict away from a solution. Plateau search—allowing sideways moves to another state with the same score—can help local search find its way off this plateau. This wandering on the plateau can be directed with **tabu search**: keeping a small list of recently visited states and forbidding the algorithm to return to those states. Simulated annealing can also be used to escape from plateaux.

Another technique, called **constraint weighting**, can help concentrate the search on the important constraints. Each constraint is given a numeric weight, W_i, initially all 1. At each step of the search, the algorithm chooses a variable/value pair to change that will result in the lowest total weight of all violated constraints. The weights are then adjusted by incrementing the weight of each constraint that is violated by the current assignment. This has two benefits: it adds topography to plateaux, making sure that it is possible to improve from the current state, and it also, over time, adds weight to the constraints that are proving difficult to solve.

Another advantage of local search is that it can be used in an online setting when the problem changes. This is particularly important in scheduling problems. A week's airline schedule may involve thousands of flights and tens of thousands of personnel assignments, but bad weather at one airport can render the schedule infeasible. We would like to repair the schedule with a minimum number of changes. This can be easily done with a local search algorithm starting from the current schedule. A backtracking search with the new set of constraints usually requires much more time and might find a solution with many changes from the current schedule.

9.5 The Structure of Problems

In this section, we examine ways in which the *structure* of the problem, as represented by the constraint graph, can be used to find solutions quickly. Most of the approaches here also apply to other problems besides CSPs, such as probabilistic reasoning. After all, the only way we can possibly hope to deal with the real world is to decompose it into many subproblems. Looking again at the constraint graph for Australia (Figure 9.4(a)), one fact stands out: Tasmania is not connected to the mainland. Intuitively, it is obvious that coloring Tasmania and coloring the mainland are **independent subproblems**—any solution for the mainland combined with any solution for Tasmania yields a solution for the whole map. Independence can be ascertained simply by finding **connected components** of the constraint graph. Each component corresponds to a subproblem CSP_i. If assignment S_i is a solution of CSP_i, then $U_i S_i$ is a solution of $U_i CSP_i$. Why is this important? Consider the following: suppose each CSP_i has c variables from the total of n variables, where c is a constant. Then there are n/c subproblems, each of which takes at most d^c work to solve, where d is the size of the domain. Hence, the total work is $O(d^c n/c)$, which is *linear* in n; without the decomposition, the total work is $O(d^n)$, which is exponential in n. Let's make this more concrete: dividing a Boolean CSP with 80 variables into four subproblems reduces the worst-case solution time from the lifetime of the universe down to less than a second.

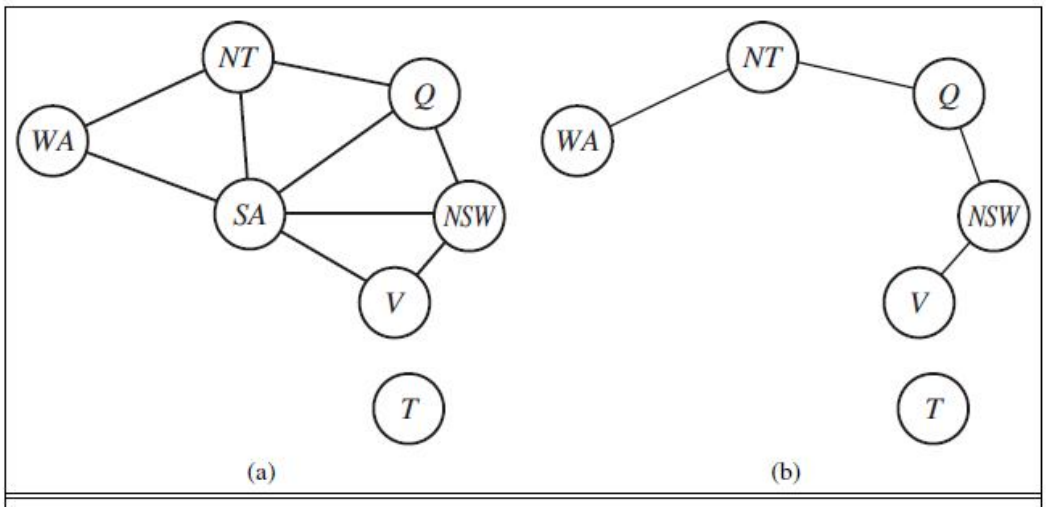


Figure 9.4 (a) The original constraint graph
(b) The constraint graph after the removal of SA

Completely independent subproblems are delicious, then, but rare. Fortunately, some other graph structures are also easy to solve. For example, a constraint graph is a **tree** when any two variables are connected by only one path. We show that *any tree-structured CSP can be solved in time linear in the number of variables.* The key is a new notion of consistency, called **directed arc consistency** or DAC. A CSP is defined to be directed arc-consistent under an ordering of variables $X_1, X_2, \ldots, X_n$ if and only if every X_i is arc-consistent with each X_j for j > i.

To solve a tree-structured CSP, first pick any variable to be the root of the tree, and choose an ordering of the variables such that each variable appears after its parent in the tree. Such an ordering is called a **topological sort**. Figure 9.5(a) shows a sample tree and (b) shows one possible ordering. Any tree with n nodes has *n"1* arcs, so we can make this graph directed arc-consistent in *O(n)* steps, each of which must compare up to *d* possible domain values for two variables, for a total time of $O(nd^2)$. Once we have a directed arc-consistent graph, we can just march down the list of variables and choose any remaining value. Since each link from a parent to its child is arc consistent, we know that for any value we choose for the parent, there will be a valid value left to choose for the child. That means we won't have to backtrack; we can move linearly through the variables. The complete algorithm is shown in Figure 9.6.

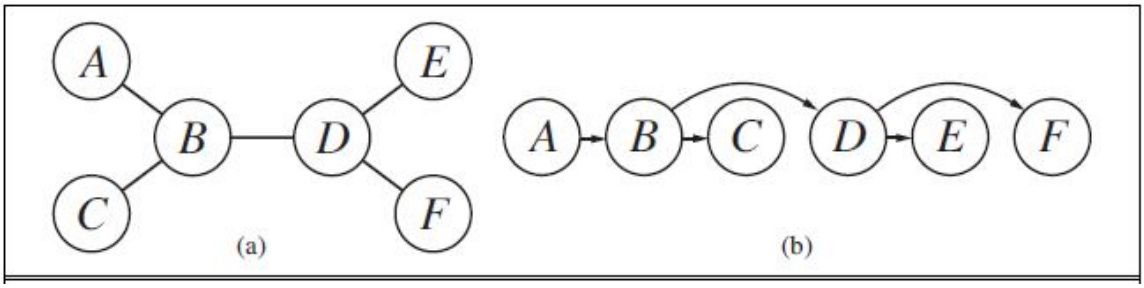


Figure 9.5 (a) The constraint graph of a tree-structured CSP (b) Topological sort of the variables

```
function TREE-CSP-SOLVER(csp) returns a solution, or failure
  inputs: csp, a CSP with components X, D, C

  n ← number of variables in X
  assignment ← an empty assignment
  root ← any variable in X
  X ← TOPOLOGICALSORT(X, root)
  for j = n down to 2 do
    MAKE-ARC-CONSISTENT(PARENT(X_j), X_j)
    if it cannot be made consistent then return failure
  for i = 1 to n do
    assignment[X_i] ← any consistent value from D_i
    if there is no consistent value then return failure
  return assignment
```

Figure 9.6 The TREE-CSP-SOLVER algorithm for solving tree-structured CSPs

Now that we have an efficient algorithm for trees, we can consider whether more general constraint graphs can be *reduced* to trees somehow. There are two primary ways to do this, one based on removing nodes and one based on collapsing nodes together.

The first approach involves assigning values to some variables so that the remaining variables form a tree. Consider the constraint graph for Australia, shown in Figure 9.5(a). If we could delete South Australia, the graph would become a tree, as in (b). Fortunately, we can do this (in the graph, not the continent) by fixing a value for SA and deleting from the domains of the other variables any values that are inconsistent with the value chosen for SA.

Now, any solution for the CSP after SA and its constraints are removed will be consistent with the value chosen for SA. (This works for binary CSPs; the situation is more complicated with higher-order constraints.) Therefore, we can solve the remaining tree with the algorithm given above and thus solve the whole problem. Of course, in the general case (as opposed to map coloring), the value chosen for SA could be the wrong one, so we would need to try each possible value. The general algorithm is as follows:

1. Choose a subset S of the CSP's variables such that the constraint graph becomes a tree after removal of S. S is called a **cycle cutset**.

2. For each possible assignment to the variables in S that satisfies all constraints on S,

(a) remove from the domains of the remaining variables any values that are inconsistent with the assignment for S, and

(b) If the remaining CSP has a solution, return it together with the assignment for S.

If the cycle cutset has size *c*, then the total run time is $O(d^c \hat{u}0 (n " c)d^2)$: we have to try each of the d^c combinations of values for the variables in S, and for each combination we must solve a tree problem of size $n " c$. If the graph is "nearly a tree," then *c* will be small and the savings over straight backtracking will be huge. In the worst case, however, *c* can be as large as $(n " 2)$. Finding the *smallest* cycle cutset is NP-hard, but several efficient approximation algorithms are known. The overall algorithmic approach is called **cutset conditioning**; where it is used for reasoning about probabilities.

The second approach is based on constructing a **tree decomposition** of the constraint graph into a set of connected subproblems. Each subproblem is solved independently, and the resulting solutions are then combined. Like most divide-and-conquer algorithms, this works well if no subproblem is too large. Figure 9.7 shows a tree decomposition of the mapcoloring problem into five subproblems. A tree decomposition must satisfy the following three requirements:

Every variable in the original problem appears in at least one of the subproblems.

• If two variables are connected by a constraint in the original problem, they must appear together (along with the constraint) in at least one of the subproblems.

• If a variable appears in two subproblems in the tree, it must appear in every subproblem along the path connecting those subproblems.

The first two conditions ensure that all the variables and constraints are represented in the decomposition. The third condition seems rather technical, but simply reflects the constraint that any given variable must have the same value in every subproblem in which it appears; the links joining subproblems in the tree enforce this constraint. For example, SA appears in all four of the connected subproblems in Figure 9.7. You can verify from Figure 9.5 that this decomposition makes sense.

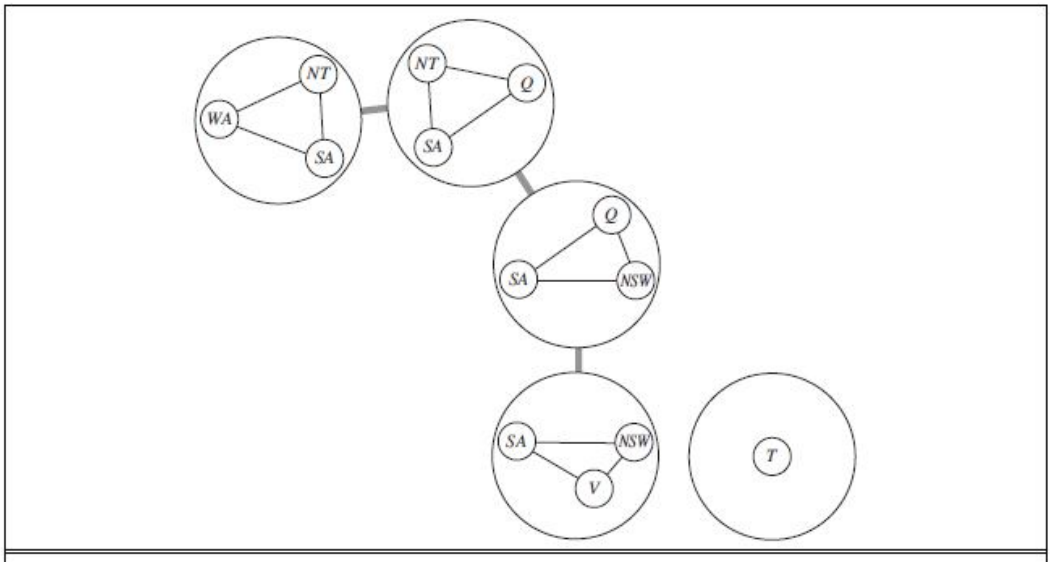


Figure 9.7 A tree decomposition of the constraint graph in Figure 9.5(a)

We solve each subproblem independently; if any one has no solution, we know the entire problem has no solution. If we can solve all the subproblems, then we attempt to construct a global solution as follows. First, we view each subproblem as a "mega-variable" whose domain is the set of all solutions for the subproblem. For example, the leftmost subproblem in Figure 9.7 is a map-coloring problem with three variables and hence has six solutions—one is {WA = red ,SA = blue, NT = green}. Then, we solve the constraints connecting the subproblems, using the efficient algorithm for trees given earlier. The constraints between subproblems simply insist that the subproblem solutions agree on their shared variables. For example, given the solution {WA = red, SA = blue, NT = green} for the first subproblem, the only consistent solution for the next subproblem is {SA = blue, NT = green, Q = red}.

A given constraint graph admits many tree decompositions; in choosing a decompo sition, the aim is to make the subproblems as small as possible. The **tree width** of a tree decomposition of a graph is one less than the size of the largest subproblem; the tree width of the graph itself is defined to be the minimum tree width among all its tree decompositions. If a graph has tree width w and we are given the corresponding tree decomposition, then the problem can be solved in $O(nd^{w+1})$ time. Hence, *CSPs with constraint graphs of bounded tree width are solvable in polynomial time.* Unfortunately, finding the decomposition with minimal tree width is NP-hard, but there are heuristic methods that work well in practice.

So far, we have looked at the structure of the constraint graph. There can be important structure in the *values* of variables as well. Consider the map-coloring problem with *n* colors. For every consistent solution, there is actually a set of n! solutions formed by permuting the color names. For example, on the Australia map we know that *WA, NT*, and *SA* must all have different colors, but there are 3! = 6 ways to assign the three colors to these three regions. This is called **value symmetry**. We would like to reduce the search space by a factor of *n!* by breaking the symmetry. We do this by introducing a **symmetry-breaking constraint**. For our example, we might impose an arbitrary ordering constraint, *NT* < *SA* < *WA*, that requires the three values to be in alphabetical order. This constraint ensures that only one of the *n!* solutions is possible: {*NT* = *blue, SA* = *green, WA* = *red*}.

For map coloring, it was easy to find a constraint that eliminates the symmetry, and in general it is possible to find constraints that eliminate all but one symmetric solution in polynomial time, but it is NP-hard to eliminate all symmetry among intermediate sets of values during search. In practice, breaking value symmetry has proved to be important and effective on a wide range of problems.

9.6 Summary

Backtracking search, a form of depth-first search, is commonly used for solving CSPs. Inference can be interwoven with search. The **minimum-remaining-values** and **degree** heuristics are domain-independent methods for deciding which variable to choose next in a backtracking search. The **least-constraining-value** heuristic helps in deciding which value to try first for a given variable. Backtracking occurs when no legal assignment can be found for a variable. **Conflict-directed backjumping** backtracks directly to the source of the problem. Local search using the **min-conflicts** heuristic has also been applied to constraint satisfaction

problems with great success. The complexity of solving a CSP is strongly related to the structure of its constraint graph. Tree-structured problems can be solved in linear time. **Cutset conditioning** can reduce a general CSP to a tree-structured one and is quite efficient if a small cutset can be found. **Tree decomposition** techniques transform the CSP into a tree of subproblems and are efficient if the **tree width** of the constraint graph is small.

9.7 Model Questions

1. Explain in detail the backtracking search for CSPs.
2. What is local search in CSPs. Explain.
3. Illustrate TREE-CSP-SOLVER algorithm with an example.

LESSON 10
LOGICAL AGENTS

Structure

10.1 Introduction

Humans, it seems, know things; and what they know helps them do things. These are not empty statements. They make strong claims about how the intelligence of humans is achieved—not by purely reflex mechanisms REASONING but by processes of **reasoning** that operate on internal **representations** of knowledge. In AI, this approach to intelligence is embodied in **knowledge-based agents**. In a partially observable environment, an agent's only choice for representing what it knows about the current state is to list all possible concrete states—a hopeless prospect in large environments.

In this lesson, **logic** is developed as a general class of representations to support knowledge-based agents. Such agents can combine and recombine information to suit myriad purposes. Often, this process can be quite far removed from the needs of the moment—as when a mathematician proves a theorem or an astronomer calculates the earth's life expectancy. Knowledge-based agents can accept new tasks in the form of explicitly described goals; they can achieve competence quickly by being told or learning new knowledge about the environment; and they can adapt to changes in the environment by updating the relevant knowledge.

10.2 Objectives

- To explain the concepts about knowledge-based agents.
- To explore the basic concepts of logic.
- To study the syntax and semantics of propositional logic.

10.3 Knowledge-Based Agents

The central component of a knowledge-based agent is its **knowledge base**, or KB. A knowledge base is a set of **sentences**. (Here "sentence" is used as a technical term. It is related but not identical to the sentences of English and other natural languages.) Each sentence is expressed in a language called a **knowledge representation language** and represents some assertion about the world. Sometimes we dignify a sentence with the name **axiom**, when the sentence is taken as given without being derived from other sentences.

There must be a way to add new sentences to the knowledge base and a way to query what is known. The standard names for these operations are TELL and ASK, respectively. Both operations may involve **inference**—that is, deriving new sentences from old. Inference must obey the requirement that when one ASKs a question of the knowledge base, the answer should follow from what has been told (or TELLed) to the knowledge base previously.

Figure 10.1 shows the outline of a knowledge-based agent program. Like all our agents, it takes a percept as input and returns an action. The agent maintains a knowledge base, KB, which may initially contain some **background knowledge**.

```
function KB-AGENT(percept) returns an action
  persistent: KB, a knowledge base
              t, a counter, initially 0, indicating time

  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))
  action ← ASK(KB, MAKE-ACTION-QUERY(t))
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))
  t ← t + 1
  return action
```

Figure 10.1 A generic knowledge-based agent

Each time the agent program is called, it does three things. First, it TELLs the knowledge base what it perceives. Second, it ASKs the knowledge base what action it should perform. In the process of answering this query, extensive reasoning may be done about the current state of the world, about the outcomes of possible action sequences, and so on. Third, the agent program TELLs the knowledge base which action was chosen, and the agent executes the action.

The details of the representation language are hidden inside three functions that implement the interface between the sensors and actuators on one side and the core representation and reasoning system on the other. MAKE-PERCEPT-SENTENCE constructs a sentence asserting that the agent perceived the given percept at the given time. MAKE-ACTION-QUERY constructs a sentence that asks what action should be done at the current time. Finally, MAKE-ACTION-SENTENCE constructs a sentence asserting that the chosen action was executed. The details of the inference mechanisms are hidden inside TELL and ASK.

The agent in Figure 10.1 appears quite similar to the agents with internal states. Because of the definitions of TELL and ASK, however, the knowledge-based agent is not an arbitrary program for calculating actions. It is amenable to a description at the **knowledge level**, where we need specify only what the agent knows and what its goals are, in order to fix its behavior. For example, an automated taxi might have the goal of taking a passenger from San Francisco to Marin County and might know that the Golden Gate Bridge is the only link between the two locations. Then we can expect it to cross the Golden Gate Bridge *because it knows that that will achieve its goal*. Notice that this analysis is independent of how the taxi works at the **implementation level**. It doesn't matter whether its geographical knowledge is implemented as linked lists or pixel maps, or whether it reasons by manipulating strings of symbols stored in registers or by propagating noisy signals in a network of neurons.

A knowledge-based agent can be built simply by TELLing it what it needs to know. Starting with an empty knowledge base, the agent designer can TELL sentences one by one until the agent knows how to operate in its environment. This is called the **declarative** approach to system building. In contrast, the **procedural** approach encodes desired behaviors directly as program code. In the 1970s and 1980s, advocates of the two approaches engaged in heated debates. We now understand that a successful agent often combines both declarative and procedural elements in its design, and that declarative knowledge can often be compiled into more efficient procedural code.

We can also provide a knowledge-based agent with mechanisms that allow it to learn for itself. These mechanisms create general knowledge about the environment from a series of percepts. A learning agent can be fully autonomous.

10.4 Logic

This section summarizes the fundamental concepts of logical representation and reasoning. These beautiful ideas are independent of any of logic's particular forms. We therefore postpone the technical details of those forms until the next section, using instead the familiar example of ordinary arithmetic.

Generally, the knowledge bases consist of sentences. These sentences are expressed according to the **syntax** of the representation language, which specifies all the sentences that are well formed. The notion of syntax is clear enough in ordinary arithmetic: "x + y = 4" is a well-formed sentence, whereas "x4y+ =" is not.

A logic must also define the **semantics** or meaning of sentences. The semantics defines the **truth** of each sentence with respect to each **possible world**. For example, the semantics for arithmetic specifies that the sentence "x + y =4" is true in a world where x is 2 and y is 2, but false in a world where x is 1 and y is 1. In standard logics, every sentence must be either true or false in each possible world—there is no "in between."1.

When we need to be precise, we use the term **model** in place of "possible world." Whereas possible worlds might be thought of as (potentially) real environments that the agent might or might not be in, models are mathematical abstractions, each of which simply fixes the truth or falsehood of every relevant sentence. Informally, we may think of a possible world as, for example, having x men and y women sitting at a table playing bridge, and the sentence x + y =4 is true when there are four people in total. Formally, the possible models are just all possible assignments of real numbers to the variables x and y. Each such assignment fixes the truth of any sentence of arithmetic whose variables are x and y. If a sentence á is true in model *m*, we say that *m* **satisfies** α or sometimes *m* **is a model of** α. We use the notation *M(α)* to mean the set of all models of α.

Now that we have a notion of truth, we are ready to talk about logical reasoning. This involves the relation of logical **entailment** between sentences—the idea that a sentence *follows logically* from another sentence. In mathematical notation, we write

$$\alpha \models \beta$$

to mean that the sentence á entails the sentence β. The formal definition of entailment is this: $\alpha \models \beta$ if and only if, in every model in which á is true, β is also true. Using the notation just introduced, we can write

$$\alpha \models \beta \text{ if and only if } M(\alpha) \subseteq M(\beta) .$$

(Note the direction of the $\subseteq$ here: if $\alpha \models \beta$, then α is a *stronger* assertion than β: it rules out *more* possible worlds.) The relation of entailment is familiar from arithmetic; we are happy with the idea that the sentence x = 0 entails the sentence xy = 0. Obviously, in any model where x is zero, it is the case that xy is zero (regardless of the value of y).

Consider the situation in Figure 10.2(b): the agent has detected nothing in [1,1] and a breeze in [2,1]. These percepts, combined with the agent's knowledge, constitute the KB. The agent is interested (among other things) in whether the adjacent squares [1,2], [2,2], and [3,1] contain pits. Each of the three squares might or might not contain a pit, so (for the purposes of this example) there are $2^3 = 8$ possible models. These eight models are shown in Figure 10.2.

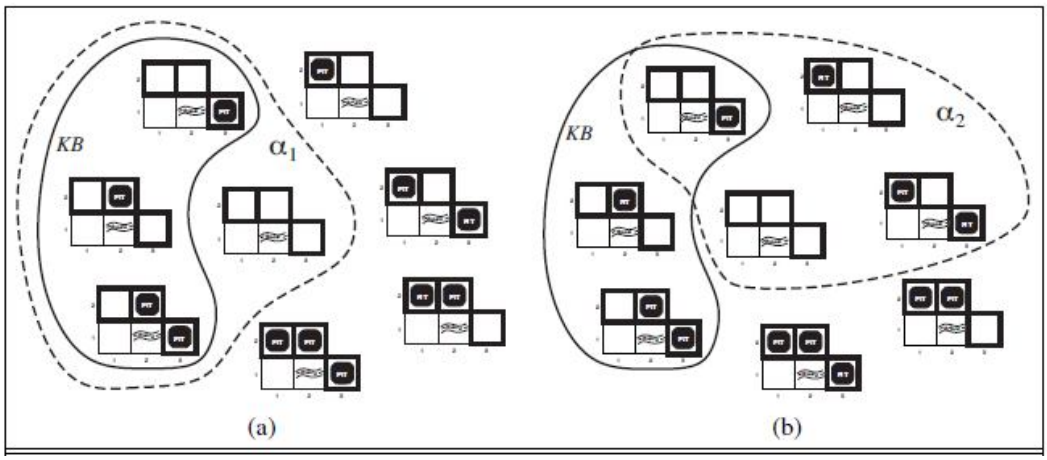


Figure 10.2 Possible models for the presence of pits in squares [1,2], [2,2], and [3,1]

The KB can be thought of as a set of sentences or as a single sentence that asserts all the individual sentences. The KB is false in models that contradict what the agent knows—for example, the KB is false in any model in which [1,2] contains a pit, because there is no breeze in [1,1]. There are in fact just three models in which the KB is true, and these are shown surrounded by a solid line in Figure 10.2. Now let us consider two possible conclusions:

α1 = “There is no pit in [1,2].”

α2 = “There is no pit in [2,2].”

We have surrounded the models of α1 and α2 with dotted lines in Figures 10.2(a) and 10.2(b), respectively. By inspection, we see the following:

in every model in which KB is true, α1 is also true.

Hence, KB |= á1: there is no pit in [1,2]. We can also see that

in some models in which KB is true, α2 is false.

Hence, KB ¥ α2: the agent *cannot* conclude that there is no pit in [2,2]. (Nor can it conclude that there *is* a pit in [2,2].)

The preceding example not only illustrates entailment but also shows how the definition of entailment can be applied to derive LOGICAL INFERENCE conclusions—that is, to carry out **logical inference**. The inference algorithm illustrated in Figure 10.2 is called **model checking**, because it enumerates all possible models to check that á is true in all models in which KB is true, that is, that M(KB) $\subseteq$ M(α).

An inference algorithm that derives only entailed sentences is called **sound** or **truth-preserving**. Soundness is a highly desirable property. An unsound inference procedure essentially makes things up as it goes along—it announces the discovery of nonexistent needles. It is easy to see that model checking, when it is applicable, is a sound procedure.

The property of **completeness** is also desirable: an inference algorithm is complete if it can derive any sentence that is entailed. For real haystacks, which are finite in extent, it seems obvious that a systematic examination can always decide whether the needle is in the haystack. For many knowledge bases, however, the haystack of consequences is infinite, and completeness becomes an important issue. Fortunately, there are complete inference procedures for logics that are sufficiently expressive to handle many knowledge bases.

The final issue to consider is **grounding**—the connection between logical reasoning processes and the real environment in which the agent exists. In particular, *how do we know that* KB *is true in the real world?* (After all, KB is just “syntax” inside the agent’s head.) A simple answer is that the agent’s sensors create the connection. For example, our wumpus-world

agent has a smell sensor. The agent program creates a suitable sentence whenever there is a smell. Then, whenever that sentence is in the knowledge base, it is true in the real world. Thus, the meaning and truth of percept sentences are defined by the processes of sensing and sentence construction that produce them. This is not a direct representation of a single percept, but a general rule—derived, perhaps, from perceptual experience but not identical to a statement of that experience. General rules like this are produced by a sentence construction process called **learning.** Learning is fallible. Thus, KB may not be true in the real world, but with good learning procedures, there is reason for optimism.

10.5 Propositional Logic: A Very Simple Logic

We now present a simple but powerful logic called **propositional logic**. We cover the syntax of propositional logic and its semantics—the way in which the truth of sentences is determined. Then we look at **entailment**—the relation between a sentence and another sentence that follows from it—and see how this leads to a simple algorithm for logical inference.

10.5.1 Syntax

The **syntax** of propositional logic defines the allowable sentences. The **atomic sentences** consist of a single **proposition symbol**. Each such symbol stands for a proposition that can be true or false. We use symbols that start with an uppercase letter and may contain other letters or subscripts, for example*: P, Q, R,* $W_{1,3}$ and North. The names are arbitrary but are often chosen to have some mnemonic value—we use $W_{1,3}$ to stand for the proposition that the wumpus is in [1,3]. (Remember that symbols such as $W_{1,3}$ are *atomic*, i.e., *W, 1,* and *3* are not meaningful parts of the symbol.) There are two proposition symbols with fixed meanings: True is the always-true proposition and False is the always-false proposition. **Complex sentences** are constructed from simpler sentences, using parentheses and **logical connectives**. There are five connectives in common use:

NEGATION ¬ (not). A sentence such as $\neg W_{1,3}$ is called the **negation** of $W_{1,3}$. A **literal** is either an atomic sentence (a **positive literal**) or a negated atomic sentence (a **negative literal**).

CONJUNCTION ∧ (and). A sentence whose main connective is ∧, such as $W_{1,3} \wedge P_{3,1,}$ is called a **conjunction**; its parts are the **conjuncts**. (The ‘" looks like an “A” for “And.”)

DISJUNCTION $\vee$ **(or)**. A sentence using $\vee$, such as $(W_{1,3} \wedge P_{3,1}) \vee W_{2,2,}$ is a **disjunction** of the **disjuncts** $(W_{1,3} \wedge P_{3,1})$ and $W_{2,2}$. (Historically, the $\vee$ comes from the Latin "vel," which means "or." For most people, it is easier to remember $\vee$ as an upside-down $\wedge$.)

IMPLICATION $\Rightarrow$ **(implies)**. A sentence such as $(W_{1,3} \wedge P_{3,1}) \Rightarrow \neg W_{2,2}$ is called an **implication** (or conditional). Its **premise** or **antecedent** is $(W_{1,3} \wedge P_{3,1})$, and its **conclusion** or **consequent** is $\neg W_{2,2}$. Implications are also known as **rules** or **if–then** statements. The implication symbol is sometimes written in other books as $\supset$ or $\rightarrow$.

BICONDITIONAL $\Leftrightarrow$ **(if and only if)**. The sentence $W_{1,3} \Leftrightarrow \neg W_{2,2}$ is a **biconditional**. Some other books write this as $\equiv$.

$$
\begin{array}{rcl}
\mathit{Sentence} & \rightarrow & \mathit{AtomicSentence} \mid \mathit{ComplexSentence} \\
\mathit{AtomicSentence} & \rightarrow & \mathit{True} \mid \mathit{False} \mid P \mid Q \mid R \mid \ldots \\
\mathit{ComplexSentence} & \rightarrow & (\ \mathit{Sentence}\ ) \mid [\ \mathit{Sentence}\ ] \\
 & \mid & \neg\ \mathit{Sentence} \\
 & \mid & \mathit{Sentence} \wedge \mathit{Sentence} \\
 & \mid & \mathit{Sentence} \vee \mathit{Sentence} \\
 & \mid & \mathit{Sentence} \Rightarrow \mathit{Sentence} \\
 & \mid & \mathit{Sentence} \Leftrightarrow \mathit{Sentence} \\
\text{OPERATOR PRECEDENCE} & : & \neg, \wedge, \vee, \Rightarrow, \Leftrightarrow
\end{array}
$$

Figure 10.3 A BNF (Backus-Naur Form) grammar of sentences in propositional logic

Figure 10.3 gives a formal grammar of propositional logic; The BNF(Backus-Naur Form) grammar by itself is ambiguous; a sentence with several operators can be parsed by the grammar in multiple ways. To eliminate the ambiguity we define a precedence for each operator. The "not" operator (¬) has the highest precedence, which means that in the sentence ¬A ∧ B the âÿ binds most tightly, giving us the equivalent of (¬A) ∧ B rather than ¬(A ∧ B). (The notation for ordinary arithmetic is the same: "2+4 is 2, not –6.) When in doubt, use parentheses to make sure of the right interpretation. Square brackets mean the same thing as parentheses; the choice of square brackets or parentheses is solely to make it easier for a human to read a sentence.

10.5.2 Semantics

Having specified the syntax of propositional logic, we now specify its semantics. The semantics defines the rules for determining the truth of a sentence with respect to a particular model. In propositional logic, a model simply fixes the TRUTH VALUE **truth value**—true or false—for every proposition symbol. For example, if the sentences in the knowledge base make use of the proposition symbols $P_{1,2}$, $P_{2,2}$, and $P_{3,1}$, then one possible model is

$$m1 = \{P_{1,2} = false,\ P_{2,2} = false,\ P_{3,1} = true\}\ .$$

With three proposition symbols, there are $2^3 = 8$ possible models—exactly those depicted in Figure 10.2. Notice, however, that the models are purely mathematical objects with no necessary connection to wumpus worlds. $P_{1,2}$ is just a symbol; it might mean "there is a pit in [1,2]" or "I'm in Paris today and tomorrow."

The semantics for propositional logic must specify how to compute the truth value of *any* sentence, given a model. This is done recursively. All sentences are constructed from atomic sentences and the five connectives; therefore, we need to specify how to compute the truth of atomic sentences and how to compute the truth of sentences formed with each of the five connectives. Atomic sentences are easy:

- True is true in every model and False is false in every model.

- The truth value of every other proposition symbol must be specified directly in the model. For example, in the model m1 given earlier, $P_{1,2}$ is false.

For complex sentences, we have five rules, which hold for any subsentences *P* and *Q* in any model m (here "iff" means "if and only if"):

- $\neg$ P is true iff P is false in m.
- P $\wedge$ Q is true iff both P and Q are true in m.
- P $\vee$ Q is true iff either P or Q is true in m.
- P $\Rightarrow$ Q is true unless P is true and Q is false in m.
- P $\Leftrightarrow$ Q is true iff P and Q are both true or both false in m.

The rules can also be expressed with **truth tables** that specify the truth value of a complex sentence for each possible assignment of truth values to its components. Truth tables for the

five connectives are given in Figure 10.4. From these tables, the truth value of any sentences can be computed with respect to any model m by a simple recursive evaluation. For example, the sentence $\neg P_{1,2} \wedge (P_{2,2} \vee P_{3,1})$, evaluated in m_1, gives *true* $\wedge$ *(false* $\vee$ *true) = true* $\wedge$ *true = true*.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
false	*false*	*true*	*false*	*false*	*true*	*true*
false	*true*	*true*	*false*	*true*	*true*	*false*
true	*false*	*false*	*false*	*true*	*false*	*false*
true	*true*	*false*	*true*	*true*	*true*	*true*

Figure 10.4 Truth tables for the five logical connectives

The truth tables for "and," "or," and "not" are in close accord with our intuitions about the English words. The main point of possible confusion is that $P \vee Q$ is true when *P* is true or *Q* is true *or both*. A different connective, called "exclusive or" ("xor" for short), yields false when both disjuncts are true. There is no consensus on the symbol for exclusive or; some choices are $\vee$ or $\neq$ or $\oplus$.

The truth table for $\Rightarrow$ may not quite fit one's intuitive understanding of "*P* implies *Q*" or "if *P* then *Q*." For one thing, propositional logic does not require any relation of *causation* or *relevance* between *P* and *Q*. The sentence "5 is odd implies Tokyo is the capital of Japan" is a true sentence of propositional logic (under the normal interpretation), even though it is a decidedly odd sentence of English. Another point of confusion is that any implication is true whenever its antecedent is false. For example, "5 is even implies Sam is smart" is true, regardless of whether Sam is smart. This seems bizarre, but it makes sense if you think of "$P \Rightarrow Q$" as saying, "If *P* is true, then I am claiming that *Q* is true. Otherwise I am making no claim." The only way for this sentence to be *false* is if *P* is true but *Q* is false.

The biconditional, $P \Leftrightarrow Q$, is true whenever both $P \Rightarrow Q$ and $Q \Rightarrow P$ are true. In English, this is often written as "*P* if and only if *Q*." Many of the rules of the wumpus world are best written using $\Leftrightarrow$. For example, a square is breezy *if* a neighboring square has a pit, and a square is breezy *only if* a neighboring square has a pit. So we need a biconditional,

$$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}),$$

where $B_{1,1}$ means that there is a breeze in [1,1].

10.5.3 A simple knowledge base

Now that we have defined the semantics for propositional logic, we can construct a knowledge base for the wumpus world. We focus first on the *immutable* aspects of the wumpus world, leaving the mutable aspects for a later section. For now, we need the following symbols for each [x, y] location:

$P_{x,y}$ is true if there is a pit in [*x, y*].

$W_{x,y}$ is true if there is a wumpus in [*x, y*], dead or alive.

$B_{x,y}$ is true if the agent perceives a breeze in [*x, y*].

$S_{x,y}$ is true if the agent perceives a stench in [*x, y*].

The sentences we write will suffice to derive $\neg P_{1,2}$ (there is no pit in [1,2]), as was done informally. We label each sentence R_i so that we can refer to them:

· There is no pit in [1,1]:

$$R_1 : \neg P_{1,1} .$$

· A square is breezy if and only if there is a pit in a neighboring square. This has to be stated for each square; for now, we include just the relevant squares:

$$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}) .$$

$$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1}) .$$

· The preceding sentences are true in all wumpus worlds. Now we include the breeze percepts for the first two squares visited in the specific world the agent is in, leading up to the situation.

$$R_4 : \neg B_{1,1} .$$

$$R_5 : B_{2,1} .$$

10.5.4 A simple inference procedure

Our goal now is to decide whether KB |= α for some sentence α. For example, is $\neg P_{1,2}$ entailed by our KB? Our first algorithm for inference is a model-checking approach that is a

direct implementation of the definition of entailment: enumerate the models, and check that á is true in every model in which KB is true. Models are assignments of true or false to every proposition symbol. Returning to our wumpus-world example, the relevant proposition symbols are $B_{1,1}$, $B_{2,1}$, $P_{1,1}$, $P_{1,2}$, $P_{2,1,}$ $P_{2,2}$, and $P_{3,1}$. With seven symbols, there are $2^7 = 128$ possible models; in three of these, KB is true (Figure 10.5). In those three models, ¬ $P_{1,2}$ is true, hence there is no pit in [1,2]. On the other hand, $P_{2,2}$ is true in two of the three models and false in one, so we cannot yet tell whether there is a pit in [2,2].

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
false	*false*	*false*	*false*	*false*	*false*	*false*	*true*	*true*	*true*	*true*	*false*	*false*
false	*false*	*false*	*false*	*false*	*false*	*true*	*true*	*true*	*false*	*true*	*false*	*false*
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
false	*true*	*false*	*false*	*false*	*false*	*false*	*true*	*true*	*false*	*true*	*true*	*false*
false	*true*	*false*	*false*	*false*	*false*	*true*	*true*	*true*	*true*	*true*	*true*	*true*
false	*true*	*false*	*false*	*false*	*true*	*false*	*true*	*true*	*true*	*true*	*true*	*true*
false	*true*	*false*	*false*	*false*	*true*	*true*	*true*	*true*	*true*	*true*	*true*	*true*
false	*true*	*false*	*false*	*true*	*false*	*false*	*true*	*false*	*false*	*true*	*true*	*false*
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
true	*true*	*true*	*true*	*true*	*true*	*true*	*false*	*true*	*true*	*false*	*true*	*false*

Figure 10.5 A truth table constructed for the knowledge base

Figure 10.5 reproduces in a more precise form the reasoning illustrated in Figure 10.2. A general algorithm for deciding entailment in propositional logic is shown in Figure 10.6. Like the BACKTRACKING-SEARCH algorithm, TT-ENTAILS? performs a recursive enumeration of a finite space of assignments to symbols. The algorithm is **sound** because it implements directly the definition of entailment, and **complete** because it works for any KB and á and always terminates—there are only finitely many models to examine.

```
function TT-ENTAILS?(KB, α) returns true or false
  inputs: KB, the knowledge base, a sentence in propositional logic
          α, the query, a sentence in propositional logic

  symbols ← a list of the proposition symbols in KB and α
  return TT-CHECK-ALL(KB, α, symbols, { })

function TT-CHECK-ALL(KB, α, symbols, model) returns true or false
  if EMPTY?(symbols) then
      if PL-TRUE?(KB, model) then return PL-TRUE?(α, model)
      else return true // when KB is false, always return true
  else do
      P ← FIRST(symbols)
      rest ← REST(symbols)
      return (TT-CHECK-ALL(KB, α, rest, model ∪ {P = true})
              and
              TT-CHECK-ALL(KB, α, rest, model ∪ {P = false }))
```

Figure 10.6 A truth-table enumeration algorithm for deciding propositional entailment

Of course, “finitely many” is not always the same as “few.” If KB and á contain n symbols in all, then there are 2^n models. Thus, the time complexity of the algorithm is $O(2^n)$. (The space complexity is only $O(n)$ because the enumeration is depth-first.) Unfortunately, propositional entailment is co-NP-complete. Every known inference algorithm for propositional logic has a worst-case complexity that is exponential in the size of the input.

10.6 Summary

We have introduced knowledge-based agents and have shown how to define a logic with which such agents can reason about the world. The main points are as follows: Intelligent agents need knowledge about the world in order to reach good decisions. Knowledge is contained in agents in the form of **sentences** in a **knowledge representation language** that are stored in a **knowledge base**. A knowledge-based agent is composed of a knowledge base and an inference mechanism. It operates by storing sentences about the world in its knowledge base, using the inference mechanism to infer new sentences, and using these sentences to decide what action to take.

A representation language is defined by its **syntax**, which specifies the structure of sentences, and its **semantics**, which defines the **truth** of each sentence in each **possible**

world or **model**. The relationship of **entailment** between sentences is crucial to our understanding of reasoning. A sentence α entails another sentence β if β is true in all worlds where á is true. Equivalent definitions include the **validity** of the sentence $\alpha \Rightarrow \beta$ and the **unsatisfiability** of the sentence $\alpha \wedge \neg \beta$. Inference is the process of deriving new sentences from old ones. **Sound** inference algorithms derive *only* sentences that are entailed; **complete** algorithms derive *all* sentences that are entailed. **Propositional logic** is a simple language consisting of **proposition symbols** and **logical connectives**. It can handle propositions that are known true, known false, or completely unknown.

10.7 Model Questions

1. Write short notes on: Knowledge-based agents.
2. Explain in detail about the logical representation and reasoning.
3. Discuss the syntax and semantics of propositional logic.

LESSON 11

PROPOSITIONAL THEOREM PROVING

Structure

11.1 Introduction

So far, we have shown how to determine entailment by *model checking*: enumerating models and showing that the sentence must hold in all models. In this section, we show how entailment can be done by **theorem proving**—applying rules of inference directly to the sentences in our knowledge base to construct a proof of the desired sentence without consulting models. If the number of models is large but the length of the proof is short, then theorem proving can be more efficient than model checking.

11.2 Objectives

- To explain the inference rules and proofs of propositional logic.
- To define Conjunctive Normal Form and Resolution algorithm.
- To explore the forward and backward chaining.

11.3 Propositional Theorem Proving

Before we plunge into the details of theorem-proving algorithms, we will need some additional concepts related to entailment. The first concept is **logical equivalence**: two sentences α and β are logically equivalent if they are true in the same set of models. We write this as $\alpha \equiv \beta$. For example, we can easily show (using truth tables) that $P \wedge Q$ and $Q \wedge P$ are logically equivalent; other equivalences are shown in Figure 11.1. These equivalences play much the same role in logic as arithmetic identities do in ordinary mathematics. An alternative definition

of equivalence is as follows: any two sentences α and β are equivalent only if each of them entails the other:

$$\alpha \equiv \beta \text{ if and only if } \alpha \models \beta \text{ and } \beta \models \alpha$$

$$\begin{aligned}
(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\
(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\
((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\
((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\
\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\
(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\
\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{De Morgan} \\
\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{De Morgan} \\
(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\
(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge
\end{aligned}$$

Figure 11.1 Standard logical equivalences

The second concept we will need is **validity**. A sentence is valid if it is true in *all* models. For example, the sentence $P \vee \neg P$ is valid. Valid sentences are also known as **tautologies**—they are *necessarily* true. Because the sentence True is true in all models, every valid sentence is logically equivalent to True. What good are valid sentences? From our definition of entailment, we can derive the **deduction theorem**, which was known to the ancient Greeks:

For any sentences α *and* β, $\alpha \models \beta$ *if and only if the sentence* $(\alpha \Rightarrow \beta)$ *is valid*

Hence, we can decide if $\alpha \models \beta$ by checking that $(\alpha \Rightarrow \beta)$ is true in every model—which is essentially what the inference algorithm does—or by proving that $(\alpha \Rightarrow \beta)$ is equivalent to True. Conversely, the deduction theorem states that every valid implication sentence describes a legitimate inference.

The final concept we will need is **satisfiability**. A sentence is satisfiable if it is true in, or satisfied by, *some* model. For example, the knowledge base given earlier, $(R_1 \wedge R_2 \wedge R_3 \wedge R_4 \wedge R_5)$, is satisfiable because there are three models in which it is true, as shown in Figure 10.5. Satisfiability can be checked by enumerating the possible models until one is found that satisfies the sentence. The problem of determining the satisfiability of sentences in propositional logic—

the **SAT** problem—was the first problem proved to be NP-complete. Many problems in computer science are really satisfiability problems. For example, all the constraint satisfaction problems ask whether the constraints are satisfiable by some assignment.

Validity and satisfiability are of course connected: α is valid iff $\neg\alpha$ is unsatisfiable; contrapositively, α is satisfiable iff $\neg\alpha$ is not valid. We also have the following useful result:

$\alpha \models \beta$ *if and only if the sentence* $(\alpha \wedge \neg \beta)$ *is unsatisfiable.*

Proving β from $\beta\alpha$ by checking the unsatisfiability of $(\alpha \wedge \neg \beta)$ corresponds exactly to the standard mathematical proof technique of *reductio ad absurdum* (literally, “reduction to an absurd thing”). It is also called proof by **refutation** or proof by **contradiction**. One assumes a sentence â to be false and shows that this leads to a contradiction with known axioms á. This contradiction is exactly what is meant by saying that the sentence $(\alpha \wedge \neg \beta)$ is unsatisfiable.

11.3.1 Inference and Proofs

This section covers **inference rules** that can be applied to derive a **proof**—a chain of conclusions that leads to the desired goal. The best-known rule is called **Modus Ponens** (Latin for *mode that affirms*) and is written

$$\alpha \Rightarrow \beta,$$

The notation means that, whenever any sentences of the form $\alpha \Rightarrow \beta$ and α are given, then the sentence β can be inferred. For example, if (WumpusAhead $\wedge$ WumpusAlive) $\Rightarrow$ Shoot and (WumpusAhead $\wedge$ WumpusAlive) are given, then Shoot can be inferred. Another useful inference rule is **And-Elimination**, which says that, from a conjunction, any of the conjuncts can be inferred:

$$\alpha \wedge \beta$$

For example, from (WumpusAhead $\wedge$ WumpusAlive), WumpusAlive can be inferred. By considering the possible truth values of α and β, one can show easily that Modus Ponens and And-Elimination are sound once and for all. These rules can then be used in any particular instances where they apply, generating sound inferences without the need for enumerating models.

All of the logical equivalences in Figure 11.1 can be used as inference rules. For example, the equivalence for biconditional elimination yields the two inference rules

$$\frac{\alpha \Leftrightarrow \beta}{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)} \quad \text{and} \quad \frac{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}{\alpha \Leftrightarrow \beta}.$$

Not all inference rules work in both directions like this. For example, we cannot run Modus Ponens in the opposite direction to obtain á Ò! â and á from â. We can define a proof problem as follows:

• **INITIAL STATE:** the initial knowledge base.

• **ACTIONS:** the set of actions consists of all the inference rules applied to all the sentences that match the top half of the inference rule.

• **RESULT:** the result of an action is to add the sentence in the bottom half of the inference rule.

• **GOAL:** the goal is a state that contains the sentence we are trying to prove.

Thus, searching for proofs is an alternative to enumerating models. In many practical cases finding a proof can be more efficient because the proof can ignore irrelevant propositions, no matter how many of them there are. For example, the proof given earlier leading to $\neg P_{1,2} \wedge \neg P_{2,1}$ does not mention the propositions $B_{2,1}$, $P_{1,1}$, $P_{2,2}$, or $P_{3,1}$. They can be ignored because the goal proposition, $P_{1,2}$, appears only in sentence R_2; the other propositions in R_2 appear only in R_4 and R_2; so R_1, R_3, and *R5* have no bearing on the proof. The same would hold even if we added a million more sentences to the knowledge base; the simple truth-table algorithm, on the other hand, would be overwhelmed by the exponential explosion of models.

One final property of logical systems is **monotonicity**, which says that the set of entailed sentences can only *increase* as information is added to the knowledge base. For any sentences α and β,

$$\text{if KB } \models \alpha \text{ then KB} \wedge \beta = \mid \alpha .$$

For example, suppose the knowledge base contains the additional assertion β stating that there are exactly eight pits in the world. This knowledge might help the agent draw *additional* conclusions, but it cannot invalidate any conclusion α already inferred—such as the conclusion

that there is no pit in [1,2]. Monotonicity means that inference rules can be applied whenever suitable premises are found in the knowledge base—the conclusion of the rule must follow *regardless of what else is in the knowledge base*.

11.3.2 Proof by Resolution

We have argued that the inference rules covered so far are *sound*, but we have not discussed the question of *completeness* for the inference algorithms that use them. Search algorithms such as iterative deepening search are complete in the sense that they will find any reachable goal, but if the available inference rules are inadequate, then the goal is not reachable—no proof exists that uses only those inference rules. For example, if we removed the biconditional elimination rule, the proof in the preceding section would not go through. The current section introduces a single inference rule, **resolution**, that yields a complete inference algorithm when coupled with any complete search algorithm. It turns out that only one particular inference rule is needed to prove any logical entailment (that can be proved): the **resolution rule** in a 2-variable form, the resolution inference rule is this (| here means “or”):

$$\frac{A \mid B, \quad !B}{A} \qquad \frac{A \mid !B, \quad B}{A} \qquad \text{resolution}$$

In English, the first rules says that if A or B is true, and B is not true, then A must be true; the second rule says the same thing, but B and !B are swapped. Note that A | !B is logically equivalent to B -> A, and so the resolution rules can be translated to this:

$$\frac{!B \rightarrow A, \quad !B}{A} \qquad \text{variation of modus ponens}$$

$$\frac{B \rightarrow A, \quad B}{A} \qquad \text{modus ponens}$$

In other words, resolution inference could be viewed as a variation of inference with modus ponens. We can generalize resolution as follows:

A_1 | A_2 | ... | A_i | ... | A_n !A_i
———————————————————————— resolution generalized
A_1 | A_2 | ... | A_i-1 | A_i+1... | A_n

We've written A_i and !A_i, but those could be swapped: the key is that they are complementary literals notice that both sentence above and below the inference line are CNF clauses

• recall that a CNF clause consists of literals (a variable, or the negation of a variable), or-ed together

so resolution can be stated conveniently in terms of clauses, e.g.:

Clause_1 L_i Opposite of literal L_i **is in** Clause_1
————————
Clause_2 Clause_2 **is** Clause_1 **with** opposite of literal L_i removed

Here, L_i is a literal that has the opposite sign of a literal in Clause_1

• i.e. L_i is the complement of a literal in Clause_1

Clause_2 is the same as Clause_1, but with the literal that is the opposite of L_i removed. For example:

(A | !B | !C) !A A and !A are complementary literals
————————————
!B | !C

(A | !B | !C) C C and !C are complementary literals
————————————
A | !B

Full resolution generalizes this even further to two clauses:

Clause_1 Clause_2
———————————————— full resolution
Clause_3

Here, we assume that some literal L app appears in Clause_1, and the complement of L appears in Clause_2 Clause_3 contains all the literals (or-ed together) from both Clause_1 and Clause_2, *except* for L and its opposite — those are not in Clause_3. For example.:

(A | !B | !C) (!A | !C | D) A and !A are complementary literals

———————————————

!B | !C | D

What is surprising is that full resolution is *complete*: it can be used to proved any entailment assuming the entailment can be proven). To do this, resolution requires that all sentences be converted to CNF. Next, recall from above that if A entails B, then A & !B is unsatisfiable. This is a consequence of the deduction theorem. But resolution theorem proves that A entails B by proving that A & !B is unsatisfiable. It follows these steps:

• A & !B is converted to CNF

• Clauses with complementary literals are chosen and resolved (i.e. the resolution rule is applied to them to remove the literal and its complement) and added as a new clause in the knowledge base

• This continues until one of two things happens:

1. no new clause can be added, which means that A does **not** entail B

    • in other words, A & !B is satisfiable

2. two clauses resolve to yield the **empty clause**, which means that A entails B

- the empty clause means there is a contradiction, i.e. if P and !P are clauses then we can apply resolution to them to get the “empty clause”
- clearly, if both P and !P are in the same knowledge base, then it is inconsistent since P & !P is false for all values of P

Example. KB = {P->Q, Q->R}

• does KB entail P->R?

• yes it does, and lets use resolution to prove this

• to prove KB entails P->R, we will prove that KB & !(P->R) is unsatisfiable

• first we convert KB & !(P->R) to CNF:

- !P | Q
- !Q | R
- P
- !R

Next we pick pairs of clauses and resolve them (i.e. apply the resolution inference rule) if we can; we add any clauses produced by this to the collection of clauses:

- !P | Q
- !Q | R
- P
- !R
-
- !P | R // from resolving (!P | Q) with (!Q | R)
- Q // from resolving (!P | Q) with (P)
- !Q // from resolving (!Q | R) with (!R)

- we have Q and !Q, meaning we've reach a contradiction

- this means KB & !(P->R) is unsatisfiable

- which means KB entails P->R

This is a *proof* of entailment, and the various resolutions are the steps of the proof. The exact order in which clauses are resolved could result in shorter or longer proofs, and in practice you usually want short proofs, and so heuristic would be needed to help make decisions.

11.3.3 Conjunctive Normal Form

The resolution rule applies only to clauses (that is, disjunctions of literals), so it would seem to be relevant only to knowledge bases and queries consisting of clauses. How, then, can it lead to a complete inference procedure for all of propositional logic? The answer is that *every sentence of propositional logic is logically equivalent to a conjunction of clauses.* A sentence expressed as a conjunction of clauses is said to be in **conjunctive normal form** or **CNF** (see Figure 11.2). We now describe a procedure for converting to CNF. We illustrate the procedure by converting the sentence $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$ into CNF. The steps are as follows:

1. Eliminate $\Leftrightarrow$, replacing $\alpha \Leftrightarrow \beta$ with $(\alpha \Rightarrow \beta) \wedge (\alpha \Rightarrow \beta)$.

$$(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1}) .$$

2. Eliminate $\Rightarrow$, replacing $\alpha \Rightarrow \beta$ with $\Rightarrow \alpha \vee \beta$:

$$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg(P_{1,2} \vee P_{2,1}) \vee B_{1,1}) .$$

$$\begin{aligned}
CNFSentence &\rightarrow Clause_1 \wedge \cdots \wedge Clause_n \\
Clause &\rightarrow Literal_1 \vee \cdots \vee Literal_m \\
Literal &\rightarrow Symbol \mid \neg Symbol \\
Symbol &\rightarrow P \mid Q \mid R \mid \ldots \\
HornClauseForm &\rightarrow DefiniteClauseForm \mid GoalClauseForm \\
DefiniteClauseForm &\rightarrow (Symbol_1 \wedge \cdots \wedge Symbol_l) \Rightarrow Symbol \\
GoalClauseForm &\rightarrow (Symbol_1 \wedge \cdots \wedge Symbol_l) \Rightarrow False
\end{aligned}$$

Figure 11.2 A grammar for conjunctive normal form

3. CNF requires $\neg$ to appear only in literals, so we "move $\neg$ inwards" by repeated application of the following equivalences from Figure 11.1:

$\neg(\neg\alpha) \equiv \alpha$ (double-negation elimination)

$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta)$ (De Morgan)

$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta)$ (De Morgan)

In the example, we require just one application of the last rule:

$$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge ((\neg P_{1,2} \wedge \neg P_{2,1}) \vee B_{1,1}) .$$

4. Now we have a sentence containing nested $\wedge$ and $\vee$ operators applied to literals. We apply the distributivity law from Figure 11.1, distributing $\vee$ over $\wedge$ wherever possible.

$$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg P_{1,2} \vee B_{1,1}) \wedge (\neg P_{2,1} \vee B_{1,1}) .$$

The original sentence is now in CNF, as a conjunction of three clauses. It is much harder to read, but it can be used as input to a resolution procedure.

11.3.4 A Resolution Algorithm

Inference procedures based on resolution work by using the principle of proof by contradiction introduced to show that KB |= α and to show that (KB $\wedge \neg\alpha$) is unsatisfiable. We do this by proving a contradiction.

A resolution algorithm is shown in Figure 11.3. First, (KB $\wedge \neg\alpha$) is converted into CNF. Then, the resolution rule is applied to the resulting clauses. Each pair that contains complementary literals is resolved to produce a new clause, which is added to the set if it is not already present. The process continues until one of two things happens:

- there are no new clauses that can be added, in which case KB does not entail α; or,
- two clauses resolve to yield the *empty* clause, in which case KB entails α

```
function PL-RESOLUTION(KB, α) returns true or false
  inputs: KB, the knowledge base, a sentence in propositional logic
          α, the query, a sentence in propositional logic

  clauses ← the set of clauses in the CNF representation of KB ∧ ¬α
  new ← { }
  loop do
      for each pair of clauses C_i, C_j in clauses do
          resolvents ← PL-RESOLVE(C_i, C_j)
          if resolvents contains the empty clause then return true
          new ← new ∪ resolvents
      if new ⊆ clauses then return false
      clauses ← clauses ∪ new
```

Figure 11.3 A simple resolution algorithm for propositional logic

The empty clause—a disjunction of no disjuncts—is equivalent to False because a disjunction is true only if at least one of its disjuncts is true. Another way to see that an empty clause represents a contradiction is to observe that it arises only from resolving two complementary unit clauses such as P and ¬P.

We can apply the resolution procedure to a very simple inference in the wumpus world. When the agent is in [1,1], there is no breeze, so there can be no pits in neighboring squares. The relevant knowledge base is

$$KB = R_2 \wedge R_4 = (B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})) \wedge \neg B_{1,1}$$

and we wish to prove á which is, say, $\neg P_{1,2}$. When we convert $(KB \wedge \neg\alpha)$ into CNF, we obtain the clauses shown at the top of Figure 11.4. The second row of the figure shows clauses obtained by resolving pairs in the first row. Then, when $P_{1,2}$ is resolved with $\neg P_{1,2}$, we obtain the empty clause, shown as a small square. Inspection of Figure 11.4 reveals that many resolution steps are pointless. For example, the clause $B_{1,1} \vee \neg B_{1,1} \vee P_{1,2}$ is equivalent to True $\vee P_{1,2}$ which is equivalent to True. Deducing that True is true is not very helpful. Therefore, any clause in which two complementary literals appear can be discarded.

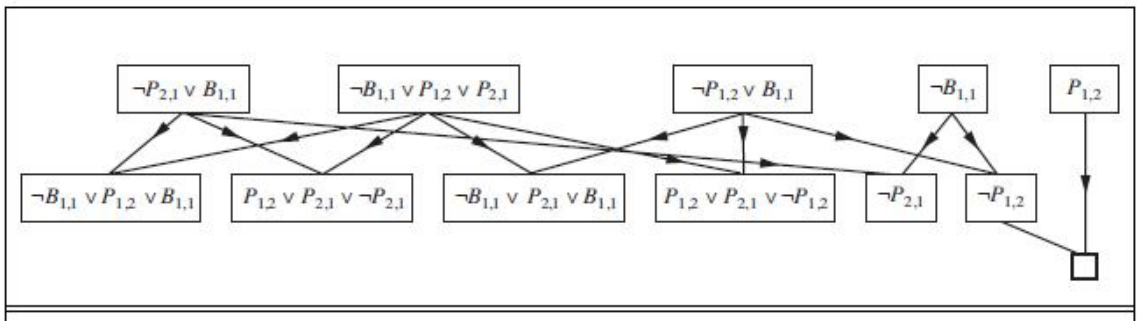


Figure 11.4 Partial application of PL –Resolution to a simple inference

11.3.5 Horn Clauses and Definite Clauses

The completeness of resolution makes it a very important inference method. In many practical situations, however, the full power of resolution is not needed. Some real-world knowledge bases satisfy certain restrictions on the form of sentences they contain, which enables them to use a more restricted and efficient inference algorithm. One such restricted form is the **definite clause**, which is a disjunction of literals of which *exactly one is positive*. For example, the clause $(\neg L_{1,1} \vee \neg$ Breeze $\vee B_{1,1})$ is a definite clause, whereas $(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1})$ is not.

Slightly more general is the **Horn clause**, which is a disjunction of literals of which *at most one is positive*. So all definite clauses are Horn clauses, as are clauses with no positive literals; these are called **goal clauses**. Horn clauses are closed under resolution: if you resolve two Horn clauses, you get back a Horn clause. Knowledge bases containing only definite clauses are interesting for three reasons:

1. Every definite clause can be written as an implication whose premise is a conjunction of positive literals and whose conclusion is a single positive literal. For example, the definite clause ($\neg L_{1,1} \vee \neg$Breeze $\vee B_{1,1}$) can be written as the implication ($L_{1,1} \wedge$ Breeze) $\Rightarrow B_{1,1}$. In the implication form, the sentence is easier to understand: it says that if the agent is in [1,1] and there is a breeze, then [1,1] is breezy. In Horn form, the premise is called the **body** and the conclusion is called the **head**. A sentence consisting of a single positive literal, such as $L_{1,1}$, is called a **fact**. It too can be written in implication form as True $\Rightarrow L_{1,1}$, but it is simpler to write just $L_{1,1}$.
2. Inference with Horn clauses can be done through the **forward-chaining** and **backward-chaining** algorithms. Both of these algorithms are natural, in that the inference steps are obvious and easy for humans to follow. This type of inference is the basis for **logic programming**.
3. Deciding entailment with Horn clauses can be done in time that is *linear* in the size of the knowledge base—a pleasant surprise.

11.3.6 Forward and Backward Chaining

The forward-chaining algorithm PL-FC-ENTAILS?(*KB, q*) determines if a single proposition symbol *q*—the query—is entailed by a knowledge base of definite clauses. It begins from known facts (positive literals) in the knowledge base. If all the premises of an implication are known, then its conclusion is added to the set of known facts. For example, if $L_{1,1}$ and Breeze are known and ($L_{1,1} \wedge$ Breeze) $\Rightarrow B_{1,1}$ is in the knowledge base, then $B_{1,1}$ can be added. This process continues until the query *q* is added or until no further inferences can be made. The detailed algorithm is shown in Figure 11.5; the main point to remember is that it runs in linear time.

```
function PL-FC-ENTAILS?(KB, q) returns true or false
  inputs: KB, the knowledge base, a set of propositional definite clauses
          q, the query, a proposition symbol
  count ← a table, where count[c] is the number of symbols in c's premise
  inferred ← a table, where inferred[s] is initially false for all symbols
  agenda ← a queue of symbols, initially symbols known to be true in KB

  while agenda is not empty do
    p ← POP(agenda)
    if p = q then return true
    if inferred[p] = false then
      inferred[p] ← true
      for each clause c in KB where p is in c.PREMISE do
        decrement count[c]
        if count[c] = 0 then add c.CONCLUSION to agenda
  return false
```

Figure 11.5 The forward-chaining algorithm for propositional logic

The best way to understand the algorithm is through an example and a picture. Figure 11.6(a) shows a simple knowledge base of Horn clauses with A and B as known facts. Figure 11.6(b) shows the same knowledge base drawn as an **AND–OR graph**. In AND–OR graphs, multiple links joined by an arc indicate a conjunction—every link must be proved—while multiple links without an arc indicate a disjunction—any link can be proved. It is easy to see how forward chaining works in the graph. The known leaves (here, A and B) are set, and inference propagates up the graph as far as possible. Wherever a conjunction appears, the propagation waits until all the conjuncts are known before proceeding.

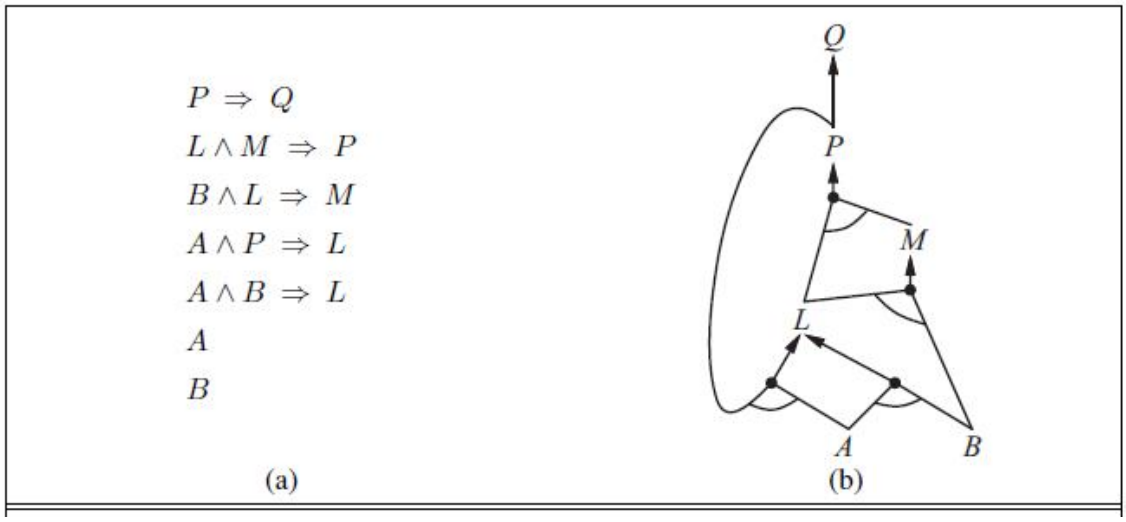


Figure 11.6 (a) A set of Horn clauses (b) The corresponding AND-OR graph

It is easy to see that forward chaining is **sound**: every inference is essentially an application of Modus Ponens. Forward chaining is also **complete**: every entailed atomic sentence will be derived. The easiest way to see this is to consider the final state of the inferred table (after the algorithm reaches a **fixed point** where no new inferences are possible). The table contains true for each symbol inferred during the process, and false for all other symbols. We can view the table as a logical model; moreover, *every definite clause in the original KB is true in this model.* To see this, assume the opposite, namely that some clause $a_1 \wedge . . . \wedge a_k \Rightarrow b$ is false in the model. Then $a_1 \wedge . . . \wedge a_k$ must be true in the model and b must be false in the model. But this contradicts our assumption that the algorithm has reached a fixed point! We can conclude, therefore, that the set of atomic sentences inferred at the fixed point defines a model of the original KB. Furthermore, any atomic sentence *q* that is entailed by the KB must be true in all its

models and in this model in particular. Hence, every entailed atomic sentence *q* must be inferred by the algorithm.

Forward chaining is an example of the general concept of **data-driven** reasoning—that is, reasoning in which the focus of attention starts with the known data. It can be used within an agent to derive conclusions from incoming percepts, often without a specific query in mind. For example, the wumpus agent might TELL its percepts to the knowledge base using an incremental forward-chaining algorithm in which new facts can be added to the agenda to initiate new inferences. In humans, a certain amount of data-driven reasoning occurs as new information arrives. For example, if I am indoors and hear rain starting to fall, it might occur to me that the picnic will be canceled. Yet it will probably not occur to me that the seventeenth petal on the largest rose in my neighbor's garden will get wet; humans keep forward chaining under careful control, lest they be swamped with irrelevant consequences.

The backward-chaining algorithm, as its name suggests, works backward from the query. If the query *q* is known to be true, then no work is needed. Otherwise, the algorithm finds those implications in the knowledge base whose conclusion is *q*. If all the premises of one of those implications can be proved true (by backward chaining), then *q* is true. When applied to the query *Q* in Figure 11.6, it works back down the graph until it reaches a set of known facts, A and B, that forms the basis for a proof. The algorithm is essentially identical to the AND-OR-GRAPH-SEARCH algorithm. As with forward chaining, an efficient implementation runs in linear time. Backward chaining is a form of **goal-directed reasoning**. It is useful for answering specific questions such as "What shall I do now?" and "Where are my keys?" Often, the cost of backward chaining is *much less* than linear in the size of the knowledge base, because the process touches only relevant facts.

11.4 Summary

In propositional logic, the simplest statements are considered as indivisible units, and hence, propositional logic does not study those logical properties and relations that depend upon parts of statements that are not themselves statements on their own, such as the subject and predicate of a statement. The most thoroughly researched branch of propositional logic is classical truth-functional propositional logic, which studies logical operators and connectives that are used to produce complex statements whose truth-value depends entirely on the truth-values of the simpler statements making them up, and in which it is assumed that every statement

is either true or false and not both. Inference rules are patterns of sound inference that can be used to find proofs. The resolution rule yields a complete inference algorithm for knowledge bases that are expressed in conjunctive normal form. Forward chaining and backward chaining are very natural reasoning algorithms for knowledge bases in Horn form.

11.5 Model Questions

1. Discuss in detail about the inference rules of propositional logic.
2. Write short notes on Conjunctive Normal Form (CNF).
3. Define: Resolution algorithm.
4. What do you mean by Horn and Definite clauses?
5. Explain in detail about Forward and Backward chaining.

LESSON 12
PROPOSITIONAL MODEL CHECKING

Structure

12.1 Introduction

In this lesson, we describe two families of efficient algorithms for general propositional inference based on model checking: One approach based on backtracking search, and one on local hill-climbing search. These algorithms are part of the "technology" of propositional logic.

The algorithms we describe are for checking satisfiability: the SAT problem. (As noted earlier, testing entailment, α |= β, can be done by testing *un*satisfiability of $\alpha \wedge \neg \beta$.) We have already noted the connection between finding a satisfying model for a logical sentence and finding a solution for a constraint satisfaction problem. The two families of algorithms closely resemble the backtracking algorithms and the local search algorithms. They are, however, extremely important in their own right because so many combinatorial problems in computer science can be reduced to checking the satisfiability of a propositional sentence. Any improvement in satisfiability algorithms has huge consequences for our ability to handle complexity in general.

12.2 Objectives

- To describe a declarative approach to agent construction.
- To define the backtracking and local search algorithms.
- To explore the agents proposed by propositional logic.

12.3 A Complete Backtracking Algorithm

The first algorithm we consider is often called the **Davis–Putnam algorithm**, after the seminal paper by Martin Davis and Hilary Putnam (1960). The algorithm is in fact the version described by Davis, Logemann, and Loveland (1962), so we will call it DPLL after the initials of all four authors. DPLL takes as input a sentence in conjunctive normal form—a set of clauses. Like BACKTRACKING-SEARCH and TT-ENTAILS?, it is essentially a recursive, depth-first enumeration of possible models. It embodies three improvements over the simple scheme of TT-ENTAILS?:

• *Early termination*: The algorithm detects whether the sentence must be true or false, even with a partially completed model. A clause is true if *any* literal is true, even if the other literals do not yet have truth values; hence, the sentence as a whole could be judged true even before the model is complete. For example, the sentence $(A \vee B) \wedge (A \vee C)$ is true if A is true, regardless of the values of B and C. Similarly, a sentence is false if *any* clause is false, which occurs when each of its literals is false. Again, this can occur long before the model is complete. Early termination avoids examination of entire subtrees in the search space.

• *Pure symbol heuristic*: A **pure symbol** is a symbol that always appears with the same "sign" in all clauses. For example, in the three clauses $(A \vee \neg B)$, $(\neg B \vee \neg C)$, and $(C \vee A)$, the symbol A is pure because only the positive literal appears, B is pure because only the negative literal appears, and C is impure. It is easy to see that if a sentence has a model, then it has a model with the pure symbols assigned so as to make their literals true, because doing so can never make a clause false. Note that, in determining the purity of a symbol, the algorithm can ignore clauses that are already known to be true in the model constructed so far. For example, if the model contains B =false, then the clause $(\neg B \vee \neg C)$ is already true, and in the remaining clauses C appears only as a positive literal; therefore C becomes pure.

• *Unit clause heuristic*: A **unit clause** was defined earlier as a clause with just one literal. In the context of DPLL, it also means clauses in which all literals but one are already assigned false by the model. For example, if the model contains B = true, then $(\neg B \vee \neg C)$ simplifies to âÿC, which is a unit clause. Obviously, for this clause to be true, C must be set to false. The unit clause heuristic assigns all such symbols before branching on the remainder. One important consequence of the heuristic is that any attempt to prove (by refutation) a literal that is already in the knowledge base will succeed immediately. Notice also that assigning one unit clause can create another unit clause—for example, when C is set to false, $(C \vee A)$ becomes a unit clause,

causing true to be assigned to A. This “cascade” of forced assignments is called **unit propagation**. It resembles the process of forward chaining with definite clauses, and indeed, if the CNF expression contains only definite clauses then DPLL essentially replicates forward chaining.

```
function DPLL-SATISFIABLE?(s) returns true or false
  inputs: s, a sentence in propositional logic

  clauses ← the set of clauses in the CNF representation of s
  symbols ← a list of the proposition symbols in s
  return DPLL(clauses, symbols, { })

function DPLL(clauses, symbols, model) returns true or false

  if every clause in clauses is true in model then return true
  if some clause in clauses is false in model then return false
  P, value ← FIND-PURE-SYMBOL(symbols, clauses, model)
  if P is non-null then return DPLL(clauses, symbols – P, model ∪ {P=value})
  P, value ← FIND-UNIT-CLAUSE(clauses, model)
  if P is non-null then return DPLL(clauses, symbols – P, model ∪ {P=value})
  P ← FIRST(symbols); rest ← REST(symbols)
  return DPLL(clauses, rest, model ∪ {P=true}) or
         DPLL(clauses, rest, model ∪ {P=false}))
```

Figure 12.1 The DPLL algorithm

The DPLL algorithm is shown in Figure 12.1, which gives the the essential skeleton of the search process. What Figure 12.1 does not show are the tricks that enable SAT solvers to scale up to large problems. It is interesting that most of these tricks are in fact rather general, and we have seen them before in other guises:

1. **Component analysis**: As DPLL assigns truth values to variables, the set of clauses may become separated into disjoint subsets, called **components**, that share no unassigned variables. Given an efficient way to detect when this occurs, a solver can gain considerable speed by working on each component separately.

2. **Variable and value ordering**: Our simple implementation of DPLL uses an arbitrary variable ordering and always tries the value *true* before *false*. The **degree heuristic** suggests choosing the variable that appears most frequently over all remaining clauses.

3. **Intelligent backtracking:** Many problems that cannot be solved in hours of run time with chronological backtracking can be solved in seconds with intelligent backtracking that backs up all the way to the relevant point of conflict. All SAT solvers that do intelligent backtracking use some form of **conflict clause learning** to record conflicts so that they won't be repeated later in the search. Usually a limited-size set of conflicts is kept, and rarely used ones are dropped.

4. **Random restarts**: Sometimes a run appears not to be making progress. In this case, we can start over from the top of the search tree, rather than trying to continue. After restarting, different random choices (in variable and value selection) are made. Clauses that are learned in the first run are retained after the restart and can help prune the search space. Restarting does not guarantee that a solution will be found faster, but it does reduce the variance on the time to solution.

5. **Clever indexing**: The speedup methods used in DPLL itself, as well as the tricks used in modern solvers, require fast indexing of such things as "the set of clauses in which variable Xi appears as a positive literal." This task is complicated by the fact that the algorithms are interested only in the clauses that have not yet been satisfied by previous assignments to variables, so the indexing structures must be updated dynamically as the computation proceeds.

With these enhancements, modern solvers can handle problems with tens of millions of variables. They have revolutionized areas such as hardware verification and security protocol verification, which previously required laborious, hand-guided proofs.

12.4 Local Search Algorithm

We have seen several local search algorithms, including HILL-CLIMBING and SIMULATED-ANNEALING. These algorithms can be applied directly to satisfiability problems, provided that we choose the right evaluation function. Because the goal is to find an assignment that satisfies every clause, an evaluation function that counts the number of unsatisfied clauses will do the job. In fact, this is exactly the measure used by the MIN-CONFLICTS algorithm for CSPs. All these algorithms take steps in the space of complete assignments, flipping the truth value of one symbol at a time. The space usually contains many local minima, to escape from which various forms of randomness are required. In recent years, there has been a great deal of experimentation to find a good balance between greediness and randomness.

One of the simplest and most effective algorithms to emerge from all this work is called WALKSAT (Figure 12.2). On every iteration, the algorithm picks an unsatisfied clause and picks a symbol in the clause to flip. It chooses randomly between two ways to pick which symbol to flip: (1) a "min-conflicts" step that minimizes the number of unsatisfied clauses in the new state and (2) a "random walk" step that picks the symbol randomly.

```
function WALKSAT(clauses, p, max_flips) returns a satisfying model or failure
  inputs: clauses, a set of clauses in propositional logic
          p, the probability of choosing to do a "random walk" move, typically around 0.5
          max_flips, number of flips allowed before giving up

  model ← a random assignment of true/false to the symbols in clauses
  for i = 1 to max_flips do
      if model satisfies clauses then return model
      clause ← a randomly selected clause from clauses that is false in model
      with probability p flip the value in model of a randomly selected symbol from clause
      else flip whichever symbol in clause maximizes the number of satisfied clauses
  return failure
```

Figure 12.2 The WALKSAT algorithm

When WALKSAT returns a model, the input sentence is indeed satisfiable, but when it returns failure, there are two possible causes: either the sentence is unsatisfiable or we need to give the algorithm more time. If we set *max_flips* = ∞ and *p* > *0*, WALKSAT will eventually return a model (if one exists), because the random-walk steps will eventually hit upon the solution. Alas, if max flips is infinity and the sentence is unsatisfiable, then the algorithm never terminates.

For this reason, WALKSAT is most useful when we expect a solution to exist. On the other hand, WALKSAT cannot always detect *unsatisfiability*, which is required for deciding entailment. For example, an agent cannot *reliably* use WALKSAT to prove that a square is safe in the wumpus world. Instead, it can say, "I thought about it for an hour and couldn't come up with a possible world in which the square *isn't* safe." This may be a good empirical indicator that the square is safe, but it's certainly not a proof.

12.5 Agents Based on Propositional Logic

In this section, we bring together what we have learned so far in order to construct wumpus world agents that use propositional logic. The first step is to enable the agent to deduce, to the

extent possible, the state of the world given its percept history. This requires writing down a complete logical model of the effects of actions. We also show how the agent can keep track of the world efficiently without going back into the percept history for each inference. Finally, we show how the agent can use logical inference to construct plans that are guaranteed to achieve its goals.

12.5.1 The current state of the world

As stated at the beginning of the chapter, a logical agent operates by deducing what to do from a knowledge base of sentences about the world. The knowledge base is composed of axioms—general knowledge about how the world works—and percept sentences obtained from the agent's experience in a particular world. In this section, we focus on the problem of deducing the current state of the wumpus world—where am I, is that square safe, and so on.

We began collecting axioms and the agent knows that the starting square contains no pit ($\neg P_{1,1}$) and no wumpus ($\neg W_{1,1}$). Furthermore, for each square, it knows that the square is breezy if and only if a neighboring square has a pit; and a square is smelly if and only if a neighboring square has a wumpus. Thus, we include a large collection of sentences of the following form:

$$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$
$$S_{1,1} \Leftrightarrow (W_{1,2} \vee W_{2,1})$$

The agent also knows that there is exactly one wumpus. This is expressed in two parts. First,

we have to say that there is *at least one* wumpus:

$$W_{1,1} \vee W_{1,2} \vee \cdots \vee W_{4,3} \vee W_{4,4} .$$

Then, we have to say that there is *at most one* wumpus. For each pair of locations, we add a sentence saying that at least one of them must be wumpus-free:

$$\neg W_{1,1} \vee \neg W_{1,2}$$
$$\neg W_{1,1} \vee \neg W_{1,3}$$
$$\cdots$$
$$\neg W_{4,3} \vee \neg W_{4,4} .$$

So far, so good. Now let's consider the agent's percepts. If there is currently a stench, one might suppose that a proposition Stench should be added to the knowledge base. This is not quite right, however: if there was no stench at the previous time step, then ¬Stench would already be asserted, and the new assertion would simply result in a contradiction. The problem is solved when we realize that a percept asserts something *only about the current time*. Thus, if the time step is 4, then we add *Stench* to the knowledge base, rather than *Stench*—neatly avoiding any contradiction with âÿStench. The same goes for the breeze, bump, glitter, and scream percepts.

The idea of associating propositions with time steps extends to any aspect of the world that changes over time. We use the word **fluent** to refer an aspect of the world that changes. "Fluent" is a synonym for "state variable," in the sense described in the discussion of factored representations. Symbols associated with permanent aspects of the world do not need a time superscript and are sometimes called **atemporal variables**.

12.5.2 A Hybrid Agent

The ability to deduce various aspects of the state of the world can be combined fairly straightforwardly with condition–action rules and with problem-solving algorithms to produce a **hybrid agent** for the wumpus world. Figure 12.3 shows one possible way to do this. The agent program maintains and updates a knowledge base as well as a current plan. The initial knowledge base contains the *atemporal* axioms—those that don't depend on t, such as the axiom relating the breeziness of squares to the presence of pits. At each time step, the new percept sentence is added along with all the axioms that depend on t, such as the successor-state axioms. Then, the agent uses logical inference, by ASKing questions of the knowledge base, to work out which squares are safe and which have yet to be visited.

The main body of the agent program constructs a plan based on a decreasing priority of goals. First, if there is a glitter, the program constructs a plan to grab the gold, follow a route back to the initial location, and climb out of the cave. Otherwise, if there is no current plan, the program plans a route to the closest safe square that it has not visited yet, making sure the route goes through only safe squares. Route planning is done with A* search, not with ASK. If there are no safe squares to explore, the next step—if the agent still has an arrow—is to try to make a safe square by shooting at one of the possible wumpus locations. These are determined by asking where ASK(KB,$\neg W_{x,y}$) is false—that is, where it is *not* known that there is *not* a wumpus. The function PLAN-SHOT (not shown) uses PLAN-ROUTE to plan a sequence of

actions that will line up this shot. If this fails, the program looks for a square to explore that is not provably unsafe—that is, a square for which ASK(KB,$\neg OK^t_{x,y}$) returns false. If there is no such square, then the mission is impossible and the agent retreats to [1, 1] and climbs out of the cave.

```
function HYBRID-WUMPUS-AGENT(percept) returns an action
  inputs: percept, a list, [stench,breeze,glitter,bump,scream]
  persistent: KB, a knowledge base, initially the atemporal "wumpus physics"
              t, a counter, initially 0, indicating time
              plan, an action sequence, initially empty

  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))
  TELL the KB the temporal "physics" sentences for time t
  safe ← {[x, y] : ASK(KB, OK^t_{x,y}) = true}
  if ASK(KB, Glitter^t) = true then
    plan ← [Grab] + PLAN-ROUTE(current, {[1,1]}, safe) + [Climb]
  if plan is empty then
    unvisited ← {[x, y] : ASK(KB, L^{t'}_{x,y}) = false for all t' ≤ t}
    plan ← PLAN-ROUTE(current, unvisited ∩ safe, safe)
  if plan is empty and ASK(KB, HaveArrow^t) = true then
    possible_wumpus ← {[x, y] : ASK(KB, ¬ W_{x,y}) = false}
    plan ← PLAN-SHOT(current, possible_wumpus, safe)
  if plan is empty then  // no choice but to take a risk
    not_unsafe ← {[x, y] : ASK(KB, ¬ OK^t_{x,y}) = false}
    plan ← PLAN-ROUTE(current, unvisited ∩ not_unsafe, safe)
  if plan is empty then
    plan ← PLAN-ROUTE(current, {[1, 1]}, safe) + [Climb]
  action ← POP(plan)
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))
  t ← t + 1
  return action
```

```
function PLAN-ROUTE(current,goals,allowed) returns an action sequence
  inputs: current, the agent's current position
          goals, a set of squares; try to plan a route to one of them
          allowed, a set of squares that can form part of the route

  problem ← ROUTE-PROBLEM(current, goals,allowed)
  return A*-GRAPH-SEARCH(problem)
```

Figure 12.3 A hybrid agent program for the wumpus world

12.5.3 Logical State Estimation

The agent program in Figure 12.3 works quite well, but it has one major weakness: as time goes by, the computational expense involved in the calls to ASK goes up and up. This happens mainly because the required inferences have to go back further and further in time and involve more and more proposition symbols. Obviously, this is unsustainable—we cannot have an agent whose time to process each percept grows in proportion to the length of its life! What we really need is a *constant* update time—that is, independent of t. The obvious answer is to save, or **cache**, the results of inference, so that the inference process at the next time step can build on the results of earlier steps instead of having to start again from scratch.

As we saw earlier, the past history of percepts and all their ramifications can be replaced by the **belief state**—that is, some representation of the set of all possible current states of the world. The process of updating the belief state as new percepts arrive is called **state estimation**. Here we can use a logical sentence involving the proposition symbols associated with the current time step, as well as the atemporal symbols. For example, the logical sentence

$$WumpusAlive^1 \wedge L^1_{2,1} \wedge B_{2,1} \wedge (P_{3,1} \vee P_{2,2})$$

represents the set of all states at time 1 in which the wumpus is alive, the agent is at [2, 1], that square is breezy, and there is a pit in [3, 1] or [2, 2] or both.

Maintaining an exact belief state as a logical formula turns out not to be easy. If there are n fluent symbols for time t, then there are 2^n possible states—that is, assignments of truth values to those symbols. Now, the set of belief states is the powerset (set of all subsets) of the set of physical states. There are 2^n physical states, hence 2^{2n} belief states. Even if we used the most compact possible encoding of logical formulas, with each belief state represented by a unique binary number, we would need numbers with $log_2(2^{2n})=2^n$ bits to label the current belief state. That is, exact state estimation may require logical formulas whose size is exponential in the number of symbols.

One very common and natural scheme for *approximate* state estimation is to represent belief states as conjunctions of literals, that is, 1-CNF formulas. To do this, the agent program simply tries to prove X^t and $\neg X^t$ for each symbol X^t (as well as each atemporal symbol whose truth value is not yet known), given the belief state at t " 1. The conjunction of provable literals becomes the new belief state, and the previous belief state is discarded.

It is important to understand that this scheme may lose some information as time goes along. On the other hand, because every literal in the 1-CNF belief state is proved from the previous belief state, and the initial belief state is a true assertion, we know that entire 1-CNF belief state must be true. Thus, *the set of possible states represented by the 1-CNF belief state includes all states that are in fact possible given the full percept history.* The 1-CNF belief state acts as a simple outer envelope, or **conservative approximation**, around the exact belief state. We see this idea of conservative approximations to complicated sets as a recurring theme in many areas of AI.

12.5.4 Making Plans by Propositional Inference

The agent in Figure 12.3 uses logical inference to determine which squares are safe, but uses

A* search to make plans. In this section, we show how to make plans by logical inference. The basic idea is very simple:

1. Construct a sentence that includes

   (a) $Init^0$, a collection of assertions about the initial state;

   (b) $Transition^1, \ldots, Transition^t$, the successor-state axioms for all possible actions at each time up to some maximum time t;

   (c) the assertion that the goal is achieved at time t: $HaveGold^t \wedge ClimbedOut^t$.

2. Present the whole sentence to a SAT solver. If the solver finds a satisfying model, then the goal is achievable; if the sentence is unsatisfiable, then the planning problem is impossible.

3. Assuming a model is found, extract from the model those variables that represent actions and are assigned true. Together they represent a plan to achieve the goals.

A propositional planning procedure, SATPLAN, is shown in Figure 12.4. It implements the basic idea just given, with one twist. Because the agent does not know how many steps it will take to reach the goal, the algorithm tries each possible number of steps t, up to some maximum conceivable plan length T_{max}. In this way, it is guaranteed to find the shortest plan if one exists. Because of the way SATPLAN searches for a solution, this approach cannot be used in a partially observable environment; SATPLAN would just set the unobservable variables to the values it needs to create a solution.

```
function SATPLAN(init, transition, goal, T_max) returns solution or failure
  inputs: init, transition, goal, constitute a description of the problem
          T_max, an upper limit for plan length

  for t = 0 to T_max do
    cnf ← TRANSLATE-TO-SAT(init, transition, goal, t)
    model ← SAT-SOLVER(cnf)
    if model is not null then
      return EXTRACT-SOLUTION(model)
  return failure
```

Figure 12.4 The SATPLAN algorithm

The key step in using SATPLAN is the construction of the knowledge base. It might seem, on casual inspection, that the wumpus world axioms suffice for steps 1(a) and 1(b) above. There is, however, a significant difference between the requirements for entailment (as tested by ASK) and those for satisfiability.

SATPLAN has more surprises in store, however. The first is that it finds models with impossible actions, such as shooting with no arrow. To understand why, we need to look more carefully at what the successor-state axioms say about actions whose preconditions are not satisfied. The axioms *do* predict correctly that nothing will happen when such an action is executed, but they do *not* say that the action cannot be executed! To avoid generating plans with illegal actions, we must add **precondition axioms** stating that an action occurrence requires the preconditions to be satisfied. SATPLAN's second surprise is the creation of plans with multiple simultaneous actions.

To summarize, SATPLAN finds models for a sentence containing the initial state, the goal, the successor-state axioms, the precondition axioms, and the action exclusion axioms. It can be shown that this collection of axioms is sufficient, in the sense that there are no longer any spurious "solutions." Any model satisfying the propositional sentence will be a valid plan for the original problem. Modern SAT-solving technology makes the approach quite practical. For example, a DPLL-style solver has no difficulty in generating the 11-step solution for the wumpus world instance.

12.6 Summary

The set of possible models, given a fixed propositional vocabulary, is finite, so entailment can be checked by enumerating models. Efficient **model-checking** inference algorithms for propositional logic include backtracking and local search methods and can often solve large problems quickly. **Local search** methods such as WALKSAT can be used to find solutions. Such algorithms are sound but not complete. Logical **state estimation** involves maintaining a logical sentence that describes the set of possible states consistent with the observation history. Each update step requires inference using the transition model of the environment, which is built from **successorstate axioms** that specify how each **fluent** changes.

Decisions within a logical agent can be made by SAT solving: finding possible models specifying future action sequences that reach the goal. This approach works only for fully observable or sensorless environments. Propositional logic does not scale to environments of unbounded size because it lacks the expressive power to deal concisely with time, space, and universal patterns of relationships among objects.

12.7 Model Questions

1. What is meant by propositional model checking?
2. Explain in detail the complete backtracking algorithm.
3. Discuss the local search algorithm in propositional logic.
4. Write short notes on: Hybrid agents.

LESSON 13
FIRST ORDER LOGIC

Structure

13.1 Introduction

In previous lessons, we used propositional logic as our representation language because it sufficed to illustrate the basic concepts of logic and knowledge-based agents. Unfortunately, propositional logic is too puny a language to represent knowledge of complex environments in a concise way. In this lesson, we examine **first-order logic**, which is sufficiently expressive to represent a good deal of our commonsense knowledge. It also either subsumes or forms the foundation of many other representation languages and has been studied intensively for many decades.

13.2 Objectives

- To introduce the concept of First-Order logic.
- To explore the syntax and semantics of first-order logic.
- To study the terms, sentences and quantifiers in first-order logic.

13.3 Representation Revisited

In this section, we discuss the nature of representation languages. Our discussion motivates the development of first-order logic, a much more expressive language than the propositional logic. We look at propositional logic and at other kinds of languages to understand what works and what fails.

Programming languages (such as C++ or Java or Lisp) are by far the largest class of formal languages in common use. Programs themselves represent, in a direct sense, only computational processes. Data structures within programs can represent facts; for example, a program could use a 4 × 4 array to represent the contents of the wumpus world. Thus, the programming language statement World [2,2] $\leftarrow$ Pit is a fairly natural way to assert that there is a pit in square [2,2]. What programming languages lack is any general mechanism for deriving facts from other facts; each update to a data structure is done by a domain-specific procedure whose details are derived by the programmer from his or her own knowledge of the domain. This procedural approach can be contrasted with the **declarative** nature of propositional logic, in which knowledge and inference are separate, and inference is entirely domain independent.

A second drawback of data structures in programs (and of databases, for that matter) is the lack of any easy way to say, for example, "There is a pit in [2,2] or [3,1]" or "If the wumpus is in [1,1] then he is not in [2,2]." Programs can store a single value for each variable, and some systems allow the value to be "unknown," but they lack the expressiveness required to handle partial information.

Propositional logic is a declarative language because its semantics is based on a truth relation between sentences and possible worlds. It also has sufficient expressive power to deal with partial information, using disjunction and negation. Propositional logic has a third property that is desirable in representation languages, namely, **compositionality**. In a compositional language, the meaning of a sentence is a function of the meaning of its parts. For example, the meaning of "$S_{1,4} \wedge S_{1,2}$" is related to the meanings of "$S_{1,4}$" and "$S_{1,2}$." Clearly, noncompositionality makes life much more difficult for the reasoning system.

However, propositional logic lacks the expressive power to *concisely* describe an environment with many objects. For example, we were forced to write a separate rule about breezes and pits for each square, such as

$$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}) \ .$$

In English, on the other hand, it seems easy enough to say, once and for all, "Squares adjacent to pits are breezy." The syntax and semantics of English somehow make it possible to describe the environment concisely.

13.3.1 Combining the best of formal and natural languages

We can adopt the foundation of propositional logic—a declarative, compositional semantics that is context-independent and unambiguous—and build a more expressive logic on that foundation, borrowing representational ideas from natural language while avoiding its drawbacks. When we look at the syntax of natural language, the most obvious elements are nouns and noun phrases that refer to **objects** (squares, pits, wumpuses) and verbs and verb phrases that refer to **relations** among objects. Some of these relations are **functions**—relations in which there is only one "value" for a given "input." It is easy to start listing examples of objects, relations, and functions:

- **Objects:** people, houses, numbers, theories, Ronald McDonald, colors, baseball games, wars, centuries . . .
- **Relations:** these can be unary relations or **properties** such as red, round, bogus, prime, multistoried . . ., or more general *n-ary* relations such as brother of, bigger than, inside, part of, has color, occurred after, owns, comes between, . . .
- **Functions:** father of, best friend, third inning of, one more than, beginning of . . .

Indeed, almost any assertion can be thought of as referring to objects and properties or relations. Some examples follow:

- "One plus two equals three."

Objects: one, two, three, one plus two; Relation: equals; Function: plus. ("One plus two" is a name for the object that is obtained by applying the function "plus" to the objects "one" and "two." "Three" is another name for this object.)

- "Squares neighboring the wumpus are smelly."

Objects: wumpus, squares; Property: smelly; Relation: neighboring.

- "Evil King John ruled England in 1200."

Objects: John, England, 1200; Relation: ruled; Properties: evil, king.

The language of **first-order logic**, whose syntax and semantics is built around objects and relations. It has been so important to mathematics, philosophy, and artificial intelligence precisely because those fields—and indeed, much of everyday human existence—can be usefully thought of as dealing with objects and the relations among them. First-order logic can also

express facts about *some* or *all* of the objects in the universe. This enables one to represent general laws or rules, such as the statement “Squares neighboring the wumpus are smelly.”

The primary difference between propositional and first-order logic lies in the **ontological commitment** made by each language—that is, what it assumes about the nature of *reality*. Mathematically, this commitment is expressed through the nature of the formal **models** with respect to which the truth of sentences is defined. For example, propositional logic assumes that there are facts that either hold or do not hold in the world. Each fact can be in one of two states: true or false, and each model assigns true or false to each proposition symbol. First-order logic assumes more; namely, that the world consists of objects with certain relations among them that do or do not hold. The formal models are correspondingly more complicated than those for propositional logic. Special-purpose logics make still further ontological commitments; for example, **temporal logic** assumes that facts hold at particular *times* and that those times are ordered. Thus, special-purpose logics give certain kinds of objects “first class” status within the logic, rather than simply defining them within the knowledge base.

Higher-order logic views the relations and functions referred to by first-order logic as objects in themselves. This allows one to make assertions about *all* relations—for example, one could wish to define what it means for a relation to be transitive. Unlike most special-purpose logics, higher-order logic is strictly more expressive than first-order logic, in the sense that some sentences of higher-order logic cannot be expressed by any finite number of first-order logic sentences.

A logic can also be characterized by its **epistemological commitments**—the possible states of knowledge that it allows with respect to each fact. In both propositional and firstorder logic, a sentence represents a fact and the agent either believes the sentence to be true, believes it to be false, or has no opinion. These logics therefore have three possible states of knowledge regarding any sentence. Systems using **probability theory**, on the other hand, can have any *degree of belief*, ranging from 0 (total disbelief) to 1 (total belief). For example, a probabilistic wumpus-world agent might believe that the wumpus is in [1,3] with probability 0.75. The ontological and epistemological commitments of five different logics are summarized in Figure 13.1.

Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

Figure 13.1 Formal languages and their ontological and epistemological commitments

In the next section, we will launch into the details of first-order logic. Just as a student of physics requires some familiarity with mathematics, a student of AI must develop a talent for working with logical notation. On the other hand, it is also important *not* to get too concerned with the *specifics* of logical notation—after all, there are dozens of different versions. The main things to keep hold of are how the language facilitates concise representations and how its semantics leads to sound reasoning procedures.

13.4 Syntax and Semantics of First-Order Logic

We begin this section by specifying more precisely the way in which the possible worlds of first-order logic reflect the ontological commitment to objects and relations. Then we introduce the various elements of the language, explaining their semantics as we go along.

13.4.1 Models for first-order logic

The models of a logical language are the formal structures that constitute the possible worlds under consideration. Each model links the vocabulary of the logical sentences to elements of the possible world, so that the truth of any sentence can be determined. Thus, models for propositional logic link proposition symbols to predefined truth values. Models for first-order logic are much more interesting. First, they have objects in them! The **domain** of a model is the set of objects or **domain elements** it contains. The domain is required to be *nonempty*—every possible world must contain at least one object. Mathematically speaking, it doesn't matter *what* these objects are—all that matters is *how many* there are in each particular model—but for pedagogical purposes we'll use a concrete example. Figure 13.2 shows a model with five objects: Richard the Lionheart, King of England from 1189 to 1199; his younger brother, the evil King John, who ruled from 1199 to 1215; the left legs of Richard and John; and a crown.

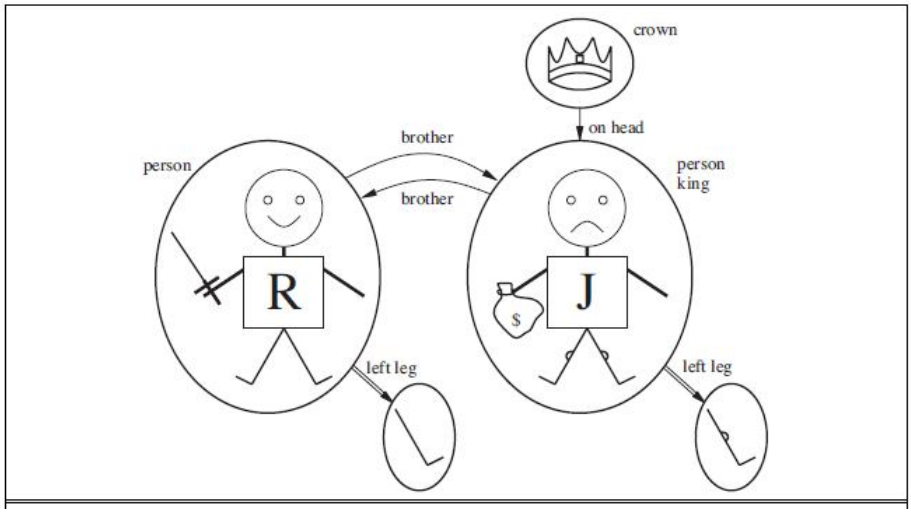


Figure 13.2 A model containing five objects

The objects in the model may be *related* in various ways. In the figure, Richard and John are brothers. Formally speaking, a relation is just the set of **tuples** of objects that are related. (A tuple is a collection of objects arranged in a fixed order and is written with angle brackets surrounding the objects.) Thus, the brotherhood relation in this model is the set

{ _Richard the Lionheart, King John_, _King John, Richard the Lionheart_ } . (13.1)

(Here we have named the objects in English, but you may, if you wish, mentally substitute the pictures for the names.) The crown is on King John's head, so the "on head" relation contains just one tuple, _the crown, King John_. The "brother" and "on head" relations are binary relations—that is, they relate pairs of objects. The model also contains unary relations, or properties: the "person" property is true of both Richard and John; the "king" property is true only of John (presumably because Richard is dead at this point); and the "crown" property is true only of the crown.

Certain kinds of relationships are best considered as functions, in that a given object must be related to exactly one object in this way. For example, each person has one left leg, so the model has a unary "left leg" function that includes the following mappings:

(Richard the Lionheart) '! Richard's left leg
(King John) '! John's left leg (13.2)

Strictly speaking, models in first-order logic require **total functions**, that is, there must be a value for every input tuple. Thus, the crown must have a left leg and so must each of the left legs. There is a technical solution to this awkward problem involving an additional "invisible" object that is the left leg of everything that has no left leg, including itself. Fortunately, as long as one makes no assertions about the left legs of things that have no left legs, these technicalities are of no import.

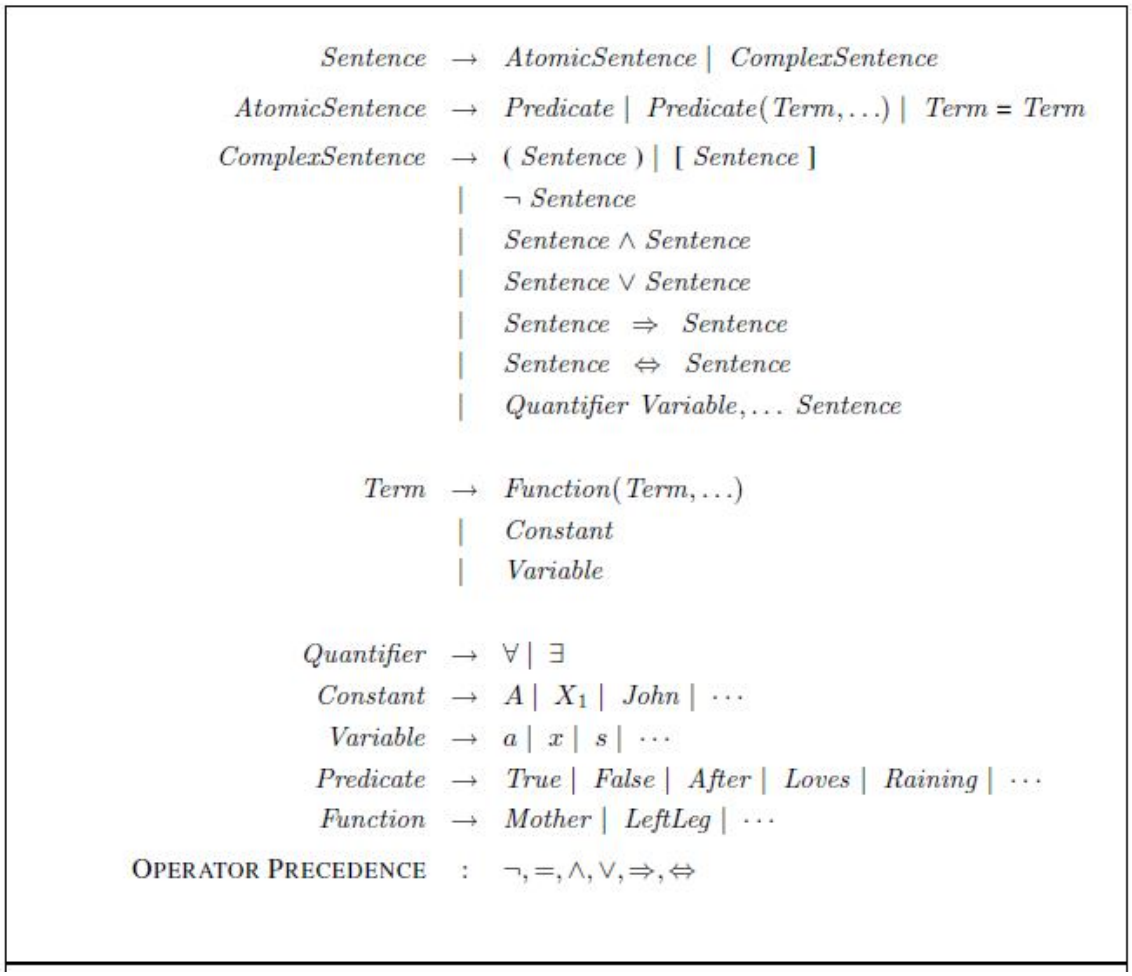
So far, we have described the elements that populate models for first-order logic. The other essential part of a model is the link between those elements and the vocabulary of the logical sentences, which we explain next.

13.4.2 Symbols and interpretations

A complete description of syntax of first-order logic is obtained from the formal grammar in Figure 13.3. The basic syntactic elements of first-order logic are the symbols that stand for objects, relations, and functions. The symbols, therefore, come in three kinds: **constant symbols**, which stand for objects; **predicate symbols**, which stand for relations; and **function symbols**, which stand for functions. We adopt the convention that these symbols will begin with uppercase letters. For example, we might use the constant symbols *Richard* and *John*; the predicate symbols *Brother*, *OnHead, Person, King*, and *Crown*; and the function symbol *LeftLeg*. As with proposition symbols, the choice of names is entirely up to the user. Each predicate and function symbol comes with an **arity** that fixes the number of arguments.

As in propositional logic, every model must provide the information required to determine if any given sentence is true or false. Thus, in addition to its objects, relations, and functions, each model includes an **interpretation** that specifies exactly which objects, relations and functions are referred to by the constant, predicate, and function symbols. One possible interpretation for our example—which a logician would call the **intended interpretation**—is as follows:

- Richard refers to Richard the Lionheart and John refers to the evil King John.
- Brother refers to the brotherhood relation, that is, the set of tuples of objects given in Equation (13.1); OnHead refers to the "on head" relation that holds between the crown and King John; Person, King, and Crown refer to the sets of objects that are persons, kings, and crowns.
- LeftLeg refers to the "left leg" function, that is, the mapping given in Equation (13.2).



$$
\begin{aligned}
\textit{Sentence} &\rightarrow \textit{AtomicSentence} \mid \textit{ComplexSentence} \\
\textit{AtomicSentence} &\rightarrow \textit{Predicate} \mid \textit{Predicate}(\textit{Term}, \ldots) \mid \textit{Term} = \textit{Term} \\
\textit{ComplexSentence} &\rightarrow (\textit{Sentence}) \mid [\textit{Sentence}] \\
&\mid \neg \textit{Sentence} \\
&\mid \textit{Sentence} \wedge \textit{Sentence} \\
&\mid \textit{Sentence} \vee \textit{Sentence} \\
&\mid \textit{Sentence} \Rightarrow \textit{Sentence} \\
&\mid \textit{Sentence} \Leftrightarrow \textit{Sentence} \\
&\mid \textit{Quantifier}\ \textit{Variable}, \ldots\ \textit{Sentence} \\
\textit{Term} &\rightarrow \textit{Function}(\textit{Term}, \ldots) \\
&\mid \textit{Constant} \\
&\mid \textit{Variable} \\
\textit{Quantifier} &\rightarrow \forall \mid \exists \\
\textit{Constant} &\rightarrow A \mid X_1 \mid \textit{John} \mid \cdots \\
\textit{Variable} &\rightarrow a \mid x \mid s \mid \cdots \\
\textit{Predicate} &\rightarrow \textit{True} \mid \textit{False} \mid \textit{After} \mid \textit{Loves} \mid \textit{Raining} \mid \cdots \\
\textit{Function} &\rightarrow \textit{Mother} \mid \textit{LeftLeg} \mid \cdots \\
\text{OPERATOR PRECEDENCE} &: \neg, =, \wedge, \vee, \Rightarrow, \Leftrightarrow
\end{aligned}
$$

Figure 13.3 The syntax of first-order logic

There are many other possible interpretations, of course. For example, one interpretation maps Richard to the crown and John to King John's left leg. There are five objects in the model, so there are 25 possible interpretations just for the constant symbols Richard and John. Notice that not all the objects need have a name—for example, the intended interpretation does not name the crown or the legs. It is also possible for an object to have several names; there is an interpretation under which both Richard and John refer to the crown. If you find this possibility confusing, remember that, in propositional logic, it is perfectly possible to have a model in which Cloudy and Sunny are both true; it is the job of the knowledge base to rule out models that are inconsistent with our knowledge.

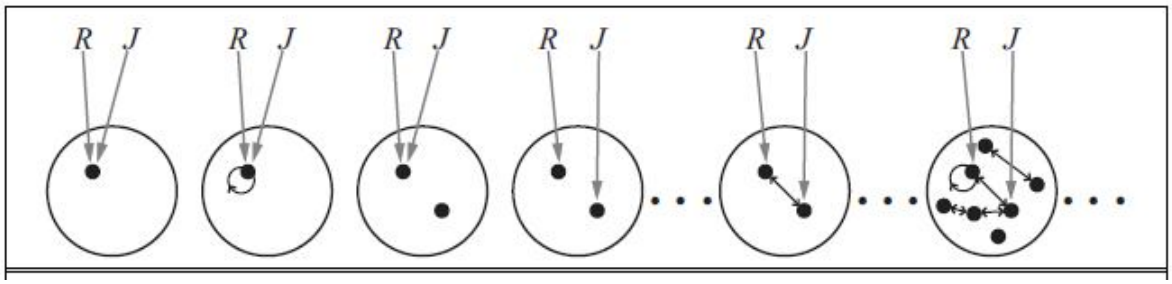


Figure 13.4 Some members of the set of all models for a language

In summary, a model in first-order logic consists of a set of objects and an interpretation that maps constant symbols to objects, predicate symbols to relations on those objects, and function symbols to functions on those objects. Just as with propositional logic, entailment, validity, and so on are defined in terms of *all possible models*. To get an idea of what the set of all possible models looks like, see Figure 13.4. It shows that models vary in how many objects they contain—from one up to infinity—and in the way the constant symbols map to objects. If there are two constant symbols and one object, then both symbols must refer to the same object; but this can still happen even with more objects. When there are more objects than constant symbols, some of the objects will have no names. Because the number of possible models is unbounded, checking entailment by the enumeration of all possible models is not feasible for first-order logic (unlike propositional logic). Even if the number of objects is restricted, the number of combinations can be very large. For the example in Figure 13.4, there are 137,506,194,466 models with six or fewer objects.

13.4.3 Terms

A **term** is a logical expression that refers to an object. Constant symbols are therefore terms, but it is not always convenient to have a distinct symbol to name every object. For example, in English we might use the expression “King John’s left leg” rather than giving a name to his leg. This is what function symbols are for: instead of using a constant symbol, we use LeftLeg (John). In the general case, a complex term is formed by a function symbol followed by a parenthesized list of terms as arguments to the function symbol. It is important to remember that a complex term is just a complicated kind of name. It is not a “subroutine call” that “returns a value.” There is no LeftLeg subroutine that takes a person as input and returns a leg. We can reason about left legs (e.g., stating the general rule that everyone has one and then deducing that John must have one) without ever providing a definition of LeftLeg.

This is something that cannot be done with subroutines in programming languages.

The formal semantics of terms is straightforward. Consider a term $f(t_1, \ldots, t_n)$. The function symbol f refers to some function in the model (call it *F*); the argument terms refer to objects in the domain (call them $d_1, \ldots, d_n$); and the term as a whole refers to the object that is the value of the function *F* applied to $d_1, \ldots, d_n$. For example, suppose the LeftLeg function symbol refers to the function shown in Equation (13.2) and John refers to King John, then LeftLeg(John) refers to King John's left leg. In this way, the interpretation fixes the referent of every term.

13.4.4 Atomic Sentences

Now that we have both terms for referring to objects and predicate symbols for referring to relations, we can put them together to make **atomic sentences** that state facts. An **atomic sentence** (or **atom** for short) is formed from a predicate symbol optionally followed by a parenthesized list of terms, such as

Brother (Richard , John).

This states, under the intended interpretation given earlier, that Richard the Lionheart is the brother of King John.6 Atomic sentences can have complex terms as arguments. Thus,

Married(Father (Richard),Mother (John))

states that Richard the Lionheart's father is married to King John's mother (again, under a suitable interpretation). An atomic sentence is **true** in a given model if the relation referred to by the predicate symbol holds among the objects referred to by the arguments.

13.4.5 Complex Sentences

We can use **logical connectives** to construct more complex sentences, with the same syntax and semantics as in propositional calculus. Here are four sentences that are true in the model of Figure 13.2 under our intended interpretation:

¬Brother (LeftLeg(Richard), John)

Brother (Richard , John) "" Brother (John,Richard)

King(Richard) (" King(John)

¬King(Richard) Ò! King(John) .

13.4.6 Quantifiers

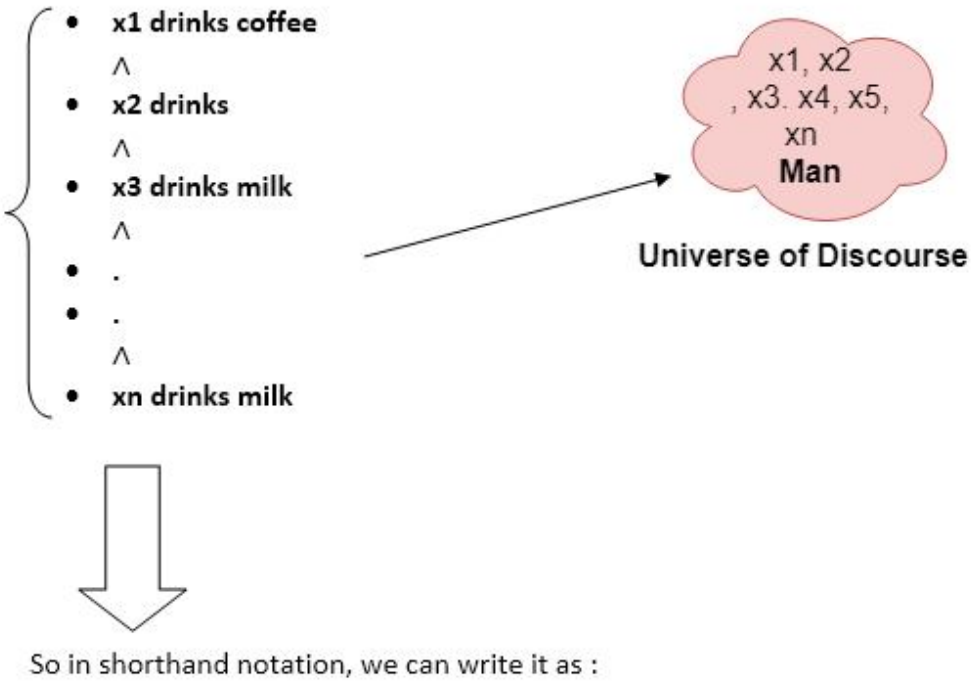
Once we have a logic that allows objects, it is only natural to want to express properties of entire collections of objects, instead of enumerating the objects by name. **Quantifiers** let us do this. First-order logic contains two standard quantifiers, called *universal* and *existential*.

Universal quantifier ()

Universal quantifier is a symbol of logical representation, which specifies that the statement within its range is true for everything or every instance of a particular thing. The Universal quantifier is represented by a symbol ", which resembles an inverted A. If x is a variable, then "x is read as:

- *For all x*
- *For each x*
- *For every x.*

For example, consider the sentence " All man drink coffee". Let a variable *x* which refers to a cat so all *x* can be represented in UOD as below:



“x man(x) ’! drink (x, coffee).

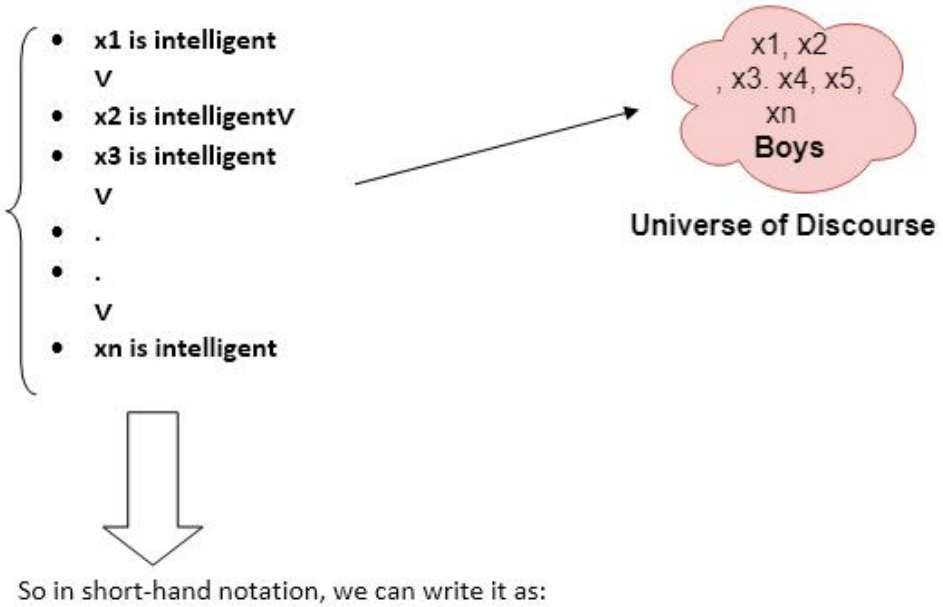
It will be read as: There are all x where x is a man who drink coffee.

Existential Quantifier

Existential quantifiers are the type of quantifiers, which express that the statement within its scope is true for at least one instance of something. It is denoted by the logical operator ∃, which resembles as inverted E. When it is used with a predicate variable then it is called as an existential quantifier. If x is a variable, then existential quantifier will be ∃x or ∃(x). And it will be read as:

- There exists a ‘x.’
- For some ‘x.’
- For at least one ‘x.’

For example, consider the sentence **“Some boys are intelligent”.**



∃x: boys(x) ∧ intelligent(x)

It will be read as: There are some x where x is a boy who is intelligent. The main connective for universal quantifier **”** is implication →. The main connective for existential quantifier ∃ is and ∧.

13.4.7 Properties of Quantifiers

- In universal quantifier, “x”y is similar to “y”x.
- In Existential quantifier, ∃x∃y is similar to ∃y∃x.
- ∃x”y is not similar to “y∃x.

Some Examples of FOL using quantifier:

1. All birds fly.

In this question the predicate is “**fly(bird)**” and since there are all birds who fly so it will be represented as follows **“x bird(x) → fly(x)**.

2. Every man respects his parent.

In this question, the predicate is “**respect(x, y)**,” **where x = man, and y= parent**. Since there is every man so will use “, and it will be represented as **“x man(x) → respects (x, parent)**.

3. Some boys play cricket.

In this question, the predicate is “**play(x, y)**,” where x = boys, and y= game. Since there are some boys so we will use **”, and it will be represented as: ”x boys(x) → play(x, cricket)**.

4. Not all students like both Mathematics and Science.

In this question, the predicate is “**like(x, y),” where x= student, and y= subject**. Since there are not all students, so we will use **” with negation, so** following representation for this: **¬” (x) [student(x) → like(x, Mathematics) ∧ like(x, Science)].**

5. Only one student failed in Mathematics.

In this question, the predicate is “**failed(x, y),” where x= student, and y= subject**.

Since there is only one student who failed in Mathematics, so we will use following representation for this: **“(x) [student(x) ’! failed (x, Mathematics) ∧” (y) [¬(x==y) ∧ student(y) → ¬failed (x, Mathematics)]**.

13.5 Summary

This lesson has introduced **first-order logic**, a representation language that is far more powerful than propositional logic. The important points are as follows: Knowledge representation languages should be declarative, compositional, expressive, context independent, and unambiguous. Logics differ in their ontological commitments and epistemological commitments. While propositional logic commits only to the existence of facts, first-order logic commits to the existence of objects and relations and thereby gains expressive power. The syntax of first-order logic builds on that of propositional logic. It adds terms to represent objects, and has universal and existential quantifiers to construct assertions about all or some of the possible values of the quantified variables.

13.6 Model Questions

1. Define: Domain and Domain elements.
2. Write short notes on Terms in First-Order logic.
3. Discuss in detail about the quantifiers in First-Order logic.
4. Illustrate atomic and complex sentences in first-order logic with examples.

LESSON 14
KNOWLEDGE REPRESENTATION & ENGINEERING IN FIRST ORDER LOGIC

Structure

14.1 Introduction

In the previous lesson, we have defined an expressive logical language, it is time to learn how to use it. The best way to do this is through examples. We have seen some simple sentences illustrating the various aspects of logical syntax. In knowledge representation, a domain is just some part of the world about which we wish to express some knowledge. We begin with a brief description of the TELL/ASK interface for first-order knowledge bases. Then we look at the domains of family relationships, numbers, sets, and lists, and at the wumpus world.

The process of constructing a knowledge-base in first-order logic is called as knowledge-engineering. In **knowledge-engineering**, someone who investigates a particular domain, learns important concept of that domain, and generates a formal representation of the objects, is known as **knowledge engineer**.

14.2 Objectives

- To introduce the concept of knowledge representation.
- To introduce the concept of knowledge engineering and it's process with an example.

14.3 Knowledge Representation

14.3.1 Assertions and queries in first-order logic

Sentences are added to a knowledge base using TELL, exactly as in propositional logic. Such sentences are called **assertions**. For example, we can assert that John is a king, Richard is a person, and all kings are persons:

TELL(KB, King(John)) .

TELL(KB, Person(Richard)) .

TELL(KB, “ x King(x) Þ Person(x)) .

We can ask questions of the knowledge base using ASK. For example,

ASK(KB, King(John))

returns true. Questions asked with ASK are called **queries** or **goals**. Generally speaking, any query that is logically entailed by the knowledge base should be answered affirmatively. For example, given the two preceding assertions, the query

ASK(KB, Person(John))

should also return true. We can ask quantified queries, such as

ASK(KB, ∃ x Person(x)) .

The answer is true, but this is perhaps not as helpful as we would like. It is rather like answering “Can you tell me the time?” with “Yes.” If we want to know what value of *x* makes the sentence true, we will need a different function, ASKVARS, which we call with

ASKVARS(KB, Person(x))

and which yields a stream of answers. In this case there will be two answers*: {x/John}* and *{x/Richard}*. Such an answer is called a **substitution** or **binding list**. ASKVARS is usually ST reserved for knowledge bases consisting solely of Horn clauses, because in such knowledge bases every way of making the query true will bind the variables to specific values. That is not the case with first-order logic; if KB has been told *King(John)* (“ *King(Richard)*, then there is no binding to *x* for the query ∃ *x King(x)*, even though the query is true.

14.3.2 The Kinship Domain

The first example we consider is the domain of family relationships, or kinship. This domain includes facts such as "*Elizabeth is the mother of Charles*" and "*Charles is the father of William*" and rules such as "*One's grandmother is the mother of one's parent.*"

Clearly, the objects in our domain are people. We have two unary predicates, *Male* and *Female*. Kinship relations—parenthood, brotherhood, marriage, and so on—are represented by binary predicates: *Parent, Sibling, Brother, Sister, Child, Daughter, Son, Spouse, Wife, Husband, Grandparent, Grandchild, Cousin, Aunt*, and *Uncle*. We use functions for *Mother* and *Father*, because every person has exactly one of each of these (at least according to nature's design). We can go through each function and predicate, writing down what we know in terms of the other symbols. For example, one's mother is one's female parent:

$$\forall\, m, c\ Mother(c) = m \Leftrightarrow Female(m) \wedge Parent(m, c) .$$

One's husband is one's male spouse:

$$\forall\, w, h\ Husband(h,w) \Leftrightarrow Male(h) \wedge Spouse(h,w) .$$

Male and female are disjoint categories:

$$\forall\, x\ Male(x) \Leftrightarrow \neg Female(x) .$$

Parent and child are inverse relations:

$$\forall\, p, c\ Parent(p, c) \Leftrightarrow Child(c, p) .$$

A grandparent is a parent of one's parent:

$$\forall\, g, c\ Grandparent(g, c) \Leftrightarrow \exists p\ Parent(g, p) \wedge Parent(p, c) .$$

A sibling is another child of one's parents:

$$\forall\, x, y\ Sibling(x, y) \Leftrightarrow x \_= y \wedge \exists p\ Parent(p, x) \wedge Parent(p, y) .$$

Each of these sentences can be viewed as an **axiom** of the kinship domain. Axioms are commonly associated with purely mathematical domains—we will see some axioms for numbers shortly—but they are needed in all domains. They provide the basic factual information from

which useful conclusions can be derived. Our kinship axioms are also **definitions**; they have the form $\forall x, y\ P(x, y) \Leftrightarrow \ldots$. The axioms define the *Mother* function and the *Husband, Male, Parent, Grandparent*, and *Sibling* predicates in terms of other predicates. Our definitions "bottom out" at a basic set of predicates (*Child, Spouse*, and *Female*) in terms of which the others are ultimately defined. This is a natural way in which to build up the representation of a domain, and it is analogous to the way in which software packages are built up by successive definitions of subroutines from primitive library functions. Notice that there is not necessarily a unique set of primitive predicates; we could equally well have used *Parent, Spouse,* and *Male*. In some domains, as we show, there is no clearly identifiable basic set.

From a purely logical point of view, a knowledge base need contain only axioms and no theorems, because the theorems do not increase the set of conclusions that follow from the knowledge base. From a practical point of view, theorems are essential to reduce the computational cost of deriving new sentences. Without them, a reasoning system has to start from first principles every time, rather like a physicist having to rederive the rules of calculus for every new problem.

14.3.3 Numbers, Sets and Lists

Numbers are perhaps the most vivid example of how a large theory can be built up from a tiny kernel of axioms. We describe here the theory of **natural numbers** or non-negative integers. We need a predicate NatNum that will be true of natural numbers; we need one constant symbol, 0; and we need one function symbol, *S* (successor). The **Peano axioms** define natural numbers and addition. Natural numbers are defined recursively:

$$NatNum(0) .$$

$$\forall n\ NatNum(n) \Rightarrow NatNum(S(n)) .$$

That is, 0 is a natural number, and for every object n, if n is a natural number, then S(n) is a natural number. So the natural numbers are 0, S(0), S(S(0)), and so on. (After reading Section 8.2.8, you will notice that these axioms allow for other natural numbers besides the usual ones; see Exercise 8.12.) We also need axioms to constrain the successor function:

$$\forall n\ 0 \neq S(n) .$$

$$\forall m, n\ m \neq n \Rightarrow S(m) \neq S(n) .$$

Now we can define addition in terms of the successor function:

$$\forall m\ NatNum(m) \Rightarrow +(0,m) = m .$$

$$\forall m, n\ NatNum(m) \wedge NatNum(n) \Rightarrow +(S(m), n) = S(+(m, n)) .$$

The first of these axioms says that adding 0 to any natural number m gives m itself. Notice the use of the binary function symbol "+" in the term +(m, 0); in ordinary mathematics, the term would be written m + 0 using **infix** notation. (The notation we have used for first-order logic is called **prefix**.) To make our sentences about numbers easier to read, we allow the use of infix notation. We can also write S(n) as n+ 1, so the second axiom becomes

$$\forall m, n\ NatNum(m) \wedge NatNum(n) \Rightarrow (m + 1) + n = (m + n) + 1 .$$

This axiom reduces addition to repeated application of the successor function.

The use of infix notation is an example of **syntactic sugar**, that is, an extension to or abbreviation of the standard syntax that does not change the semantics. Any sentence that uses sugar can be "desugared" to produce an equivalent sentence in ordinary first-order logic.

Once we have addition, it is straightforward to define multiplication as repeated addition, exponentiation as repeated multiplication, integer division and remainders, prime numbers, and so on. Thus, the whole of number theory (including cryptography) can be built up from one constant, one function, one predicate and four axioms.

The domain of **sets** is also fundamental to mathematics as well as to commonsense reasoning. (In fact, it is possible to define number theory in terms of set theory.) We want to be able to represent individual sets, including the empty set. We need a way to build up sets by adding an element to a set or taking the union or intersection of two sets. We will want to know whether an element is a member of a set and we will want to distinguish sets from objects that are not sets.

We will use the normal vocabulary of set theory as syntactic sugar. The empty set is a constant written as {}. There is one unary predicate, Set, which is true of sets. The binary predicates are $x \in s$ (x is a member of set s) and $s1 \subseteq s2$ (set s1 is a subset, not necessarily proper, of set s2). The binary functions are s1 $\cap$ s2 (the intersection of two sets), s1 $\cup$ s2 (the union of two sets), and $\{x|s\}$ (the set resulting from adjoining element x to set s). One possible set of axioms is as follows:

1. The only sets are the empty set and those made by adjoining something to a set:

$$\forall\ s\ Set(s) \Leftrightarrow (s=\{\}) \vee (,\ s2\ Set(s2) \wedge s=\{x|s2\})\ .$$

2. The empty set has no elements adjoined into it. In other words, there is no way to decompose {} into a smaller set and an element:0

$$\neg\exists\ x,\ s\ \{x|s\}=\{\}\ .$$

3. Adjoining an element already in the set has no effect:

$$\forall\ x,\ s\ x\ s \Leftrightarrow s=\{x|s\}\ .$$

4. The only members of a set are the elements that were adjoined into it. We express this recursively, saying that x is a member of s if and only if s is equal to some set s2 adjoined with some element y, where either y is the same as x or x is a member of s2:

$$\forall\ x,\ s\ x\in s \Leftrightarrow \exists y,\ s2\ (s=\{y|s2\} \wedge (x=y \vee x\in\ s2))\ .$$

5. A set is a subset of another set if and only if all of the first set's members are members of the second set:

$$\forall\ s1,\ s2\ s1\ \dagger" s2 \Leftrightarrow (\forall x\ x\in\ s1 \Leftrightarrow x\in\ s2)\ .$$

6. Two sets are equal if and only if each is a subset of the other:

$$\forall\ s1,\ s2\ (s1\ =s2) \Leftrightarrow (s1 \subseteq s2 \wedge\ \ s2 \subseteq s1)\ .$$

7. An object is in the intersection of two sets if and only if it is a member of both sets:

$$\forall\ x,\ s1,\ s2\ x\in(s1 \cap s2) \Leftrightarrow (x\in\ s1 \vee x\in s2)\ .$$

8. An object is in the union of two sets if and only if it is a member of either set:

$$\forall\ x,\ s1,\ s2\ x\in(s1 \cup s2) \Leftrightarrow (x\in s1 \vee x\in s2)\ .$$

Lists are similar to sets. The differences are that lists are ordered and the same element can appear more than once in a list. We can use the vocabulary of Lisp for lists: *Nil* is the constant list with no elements; *Cons, Append, First*, and *Rest* are functions; and Find is the predicate that does for lists what Member does for sets. *List?* is a predicate that is true only of

lists. As with sets, it is common to use syntactic sugar in logical sentences involving lists. The empty list is [ ]. The term *Cons(x, y)*, where *y* is a nonempty list, is written *[x|y]*. The term *Cons(x, Nil)* (i.e., the list containing the element *x*) is written as *[x]*. A list of several elements, such as *[A,B,C]*, corresponds to the nested term *Cons(A, Cons(B, Cons(C, Nil)))*.

14.3.4 The Wumpus World

In this section, we will see some propositional logic axioms for the wumpus world. The firstorder axioms in this section are much more concise, capturing in a natural way exactly what we want to say. Recall that the wumpus agent receives a percept vector with five elements. The corresponding first-order sentence stored in the knowledge base must include both the percept and the time at which it occurred; otherwise, the agent will get confused about when it saw what. We use integers for time steps. A typical percept sentence would be

$$Percept([Stench, Breeze, Glitter , None, None], 5) .$$

Here, *Percept* is a binary predicate, and *Stench* and so on are constants placed in a list. The actions in the wumpus world can be represented by logical terms:

$$Turn(Right), Turn(Left), Forward , Shoot , Grab, Climb .$$

To determine which is best, the agent program executes the query

$$ASKVARS(“ a BestAction(a, 5)) ,$$

which returns a binding list such as {a/Grab}. The agent program can then return Grab as the action to take. The raw percept data implies certain facts about the current state. For example:

$$\forall\ t, s, g, m, c\ Percept([s, Breeze, g,m, c], t) \Rightarrow Breeze(t) ,$$

$$\forall\ t, s, b, m, c\ Percept([s, b, Glitter,m, c], t) \Rightarrow Glitter (t) ,$$

and so on. These rules exhibit a trivial form of the reasoning process called **perception.** Notice the quantification over time *t*. In propositional logic, we would need copies of each sentence for each time step.

14.4 Knowledge Engineering

The preceding section illustrated the use of first-order logic to represent knowledge in three simple domains. This section describes the general process of knowledge-base construction—a process called **knowledge engineering**. A knowledge engineer is someone who investigates a particular domain, learns what concepts are important in that domain, and creates a formal representation of the objects and relations in the domain. *General-purpose* knowledge bases, which cover a broad range of human knowledge are intended to support tasks such as natural language understanding.

14.4.1 The knowledge-engineering process

Knowledge engineering projects vary widely in content, scope, and difficulty, but all such projects include the following steps:

1. ***Identify the task.*** The knowledge engineer must delineate the range of questions that the knowledge base will support and the kinds of facts that will be available for each specific problem instance. For example, does the wumpus knowledge base need to be able to choose actions or is it required to answer questions only about the contents of the environment? Will the sensor facts include the current location? The task will determine what knowledge must be represented in order to connect problem instances to answers. This step is analogous to the process for designing agents.

2. ***Assemble the relevant knowledge***. The knowledge engineer might already be an expert in the domain, or might need to work with real experts to extract what they know—a process called **knowledge acquisition**. At this stage, the knowledge is not represented formally. The idea is to understand the scope of the knowledge base, as determined by the task, and to understand how the domain actually works.

For the wumpus world, which is defined by an artificial set of rules, the relevant knowledge is easy to identify. (Notice, however, that the definition of adjacency was not supplied explicitly in the wumpus-world rules.) For real domains, the issue of relevance can be quite difficult—for example, a system for simulating VLSI designs might or might not need to take into account stray capacitances and skin effects

3. ***Decide on a vocabulary of predicates, functions, and constants***. That is, translate the important domain-level concepts into logic-level names. This involves many

questions of knowledge-engineering *style*. Like programming style, this can have a significant impact on the eventual success of the project. For example, should pits be represented by objects or by a unary predicate on squares? Should the agent's orientation be a function or a predicate? Should the wumpus's location depend on time? Once the choices have been made, the result is a vocabulary that is known as the **ontology** of the domain. The word *ontology* means a particular theory of the nature of being or existence. The ontology determines what kinds of things exist, but does not determine their specific properties and interrelationships.

4. ***Encode general knowledge about the domain.*** The knowledge engineer writes down the axioms for all the vocabulary terms. This pins down (to the extent possible) the meaning of the terms, enabling the expert to check the content. Often, this step reveals misconceptions or gaps in the vocabulary that must be fixed by returning to step 3 and iterating through the process.

5. ***Encode a description of the specific problem instance.*** If the ontology is well thought out, this step will be easy. It will involve writing simple atomic sentences about instances of concepts that are already part of the ontology. For a logical agent, problem instances are supplied by the sensors, whereas a "disembodied" knowledge base is supplied with additional sentences in the same way that traditional programs are supplied with input data.

6. ***Pose queries to the inference procedure and get answers.*** This is where the reward is: we can let the inference procedure operate on the axioms and problem-specific facts to derive the facts we are interested in knowing. Thus, we avoid the need for writing an application-specific solution algorithm.

7. ***Debug the knowledge base.*** Alas, the answers to queries will seldom be correct on the first try. More precisely, the answers will be correct *for the knowledge base as written*, assuming that the inference procedure is sound, but they will not be the ones that the user is expecting. For example, if an axiom is missing, some queries will not be answerable from the knowledge base. A considerable debugging process could ensue. Missing axioms or axioms that are too weak can be easily identified by noticing places where the chain of reasoning stops unexpectedly. For example, if the knowledge base includes a diagnostic rule (see Exercise 8.13) for finding the wumpus,

$$\forall s \; Smelly(s) \Rightarrow Adjacent(Home(Wumpus), s) ,$$

instead of the biconditional, then the agent will never be able to prove the *absence* of wumpuses. Incorrect axioms can be identified because they are false statements about the world. For example, the sentence

$$\forall x \; NumOfLegs(x, 4) \Rightarrow Mammal(x)$$

is false for reptiles, amphibians, and, more importantly, tables. *The falsehood of this sentence can be determined independently of the rest of the knowledge base.* In contrast, a typical error in a program looks like this:

$$offset = position + 1.$$

It is impossible to tell whether this statement is correct without looking at the rest of the program to see whether, for example, offset is used to refer to the current position, or to one beyond the current position, or whether the value of position is changed by another statement and so offset should also be changed again.

To understand this seven-step process better, we now apply it to an extended example—the domain of electronic circuits.

14.4.2 The Electronic Circuits Domain

We will develop an ontology and knowledge base that allow us to reason about digital circuits of the kind shown in Figure 14.1. We follow the seven-step process for knowledge engineering.

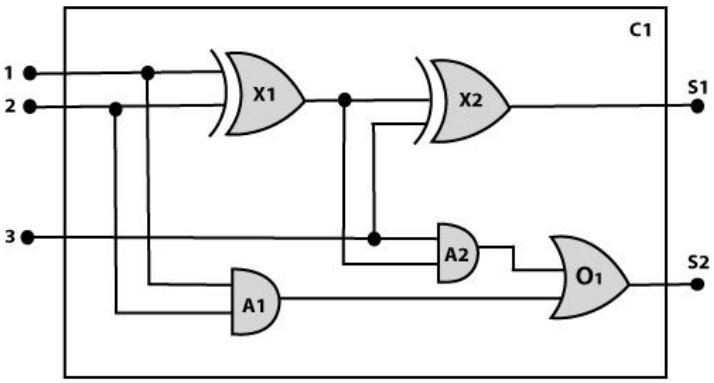


Figure 14.1 One-bit Full Adder

1. Identify the task:

The first step of the process is to identify the task, and for the digital circuit, there are various reasoning tasks. At the first level or highest level, we will examine the functionality of the circuit:

- **Does the circuit add properly?**
- **What will be the output of gate A2, if all the inputs are high?**

At the second level, we will examine the circuit structure details such as:

- **Which gate is connected to the first input terminal?**
- **Does the circuit have feedback loops?**

2. Assemble the relevant knowledge:

In the second step, we will assemble the relevant knowledge which is required for digital circuits. So for digital circuits, we have the following required knowledge:

- Logic circuits are made up of wires and gates.
- Signal flows through wires to the input terminal of the gate, and each gate produces the corresponding output which flows further.
- In this logic circuit, there are four types of gates used: **AND, OR, XOR, and NOT**.
- All these gates have one output terminal and two input terminals (except NOT gate, it has one input terminal).

3. Decide on vocabulary:

The next step of the process is to select functions, predicate, and constants to represent the circuits, terminals, signals, and gates. Firstly we will distinguish the gates from each other and from other objects. Each gate is represented as an object which is named by a constant, such as, **Gate(X1)**. The functionality of each gate is determined by its type, which is taken as constants such as **AND, OR, XOR, or NOT**. Circuits will be identified by a predicate: **Circuit (C1)**.

For the terminal, we will use predicate: **Terminal(x)**.

For gate input, we will use the function **In(1, X1)** for denoting the first input terminal of the gate, and for output terminal we will use **Out (1, X1)**.

The function **Arity(c, i, j)** is used to denote that circuit c has i input, j output.

The connectivity between gates can be represented by predicate **Connect(Out(1, X1), In(1, X1))**.

We use a unary predicate **On (t)**, which is true if the signal at a terminal is on.

4. Encode general knowledge about the domain:

To encode the general knowledge about the logic circuit, we need some following rules:

- If two terminals are connected then they have the same input signal, it can be represented as:

$\forall$ *t1, t2 Terminal (t1)* $\wedge$ *Terminal (t2)* $\wedge$ *Connect (t1, t2)* $\rightarrow$ *Signal (t1) = Signal (2).*

- Signal at every terminal will have either value 0 or 1, it will be represented as:

$\forall$ *t Terminal (t)* $\rightarrow$ *Signal (t) = 1* $\vee$ *Signal (t) = 0.*

- Connect predicates are commutative:

$\forall$ *t1, t2 Connect(t1, t2)* $\rightarrow$ *Connect (t2, t1).*

- Representation of types of gates:

$\forall$ *g Gate(g)* $\wedge$ *r = Type(g)* $\rightarrow$ *r = OR* $\vee$ *r = AND* $\vee$ *r = XOR* $\vee$ *r = NOT.*

- Output of AND gate will be zero if and only if any of its input is zero.

$\forall$ *g Gate(g)* $\vee$ *Type(g) = AND* $\rightarrow$ *Signal (Out(1, g))= 0* $\Leftrightarrow$ $\exists$*n Signal (In(n, g))= 0.*

- Output of OR gate is 1 if and only if any of its input is 1:

$\forall$ *g Gate(g)* $\wedge$ *Type(g) = OR* $\rightarrow$ *Signal (Out(1, g))= 1* $\Leftrightarrow$ $\exists$*n Signal (In(n, g))= 1*

- Output of XOR gate is 1 if and only if its inputs are different:

$\forall$ *g Gate(g)* $\vee$*Type(g) = XOR* $\rightarrow$ *Signal (Out(1, g)) = 1* $\Leftrightarrow$ *Signal (In(1, g))* $\neq$ *Signal (In(2, g))*

- Output of NOT gate is invert of its input:

$\forall$ *g Gate(g)* $\vee$ *Type(g) = NOT* $\rightarrow$ *Signal (In(1, g))* $\neq$ *Signal (Out(1, g)).*

- All the gates in the above circuit have two inputs and one output (except NOT gate).

$$\forall g\ Gate(g) \vee Type(g) = NOT \rightarrow Arity(g, 1, 1)$$

$$\forall g\ Gate(g) \vee r = Type(g) \vee (r = AND \vee r = OR \vee r = XOR) \rightarrow Arity(g, 2, 1).$$

- All gates are logic circuits:

$$\forall g\ Gate(g) \rightarrow Circuit(g).$$

5. Encode a description of the problem instance:

Now we encode problem of circuit C1, firstly we categorize the circuit and its gate components. This step is easy if ontology about the problem is already thought. This step involves the writing simple atomics sentences of instances of concepts, which is known as ontology. For the given circuit C1, we can encode the problem instance in atomic sentences as below: Since in the circuit there are two XOR, two AND, and one OR gate so atomic sentences for these gates will be:

1. *For XOR gate: Type(x1)= XOR, Type(X2) = XOR*
2. *For AND gate: Type(A1) = AND, Type(A2)= AND*
3. *For OR gate: Type (O1) = OR.*

And then represent the connections between all the gates.

6. Pose queries to the inference procedure and get answers:

In this step, we will find all the possible set of values of all the terminal for the adder circuit. The first query will be:

What should be the combination of input which would generate the first output of circuit C1, as 0 and a second output to be 1?

1. $\exists i1, i2, i3\ Signal(In(1, C1)) = i1 \wedge Signal(In(2, C1)) = i2 \wedge Signal(In(3, C1)) = i3$
2. $\wedge Signal(Out(1, C1)) = 0 \wedge Signal(Out(2, C1)) = 1$

7. Debug the knowledge base

Now we will debug the knowledge base, and this is the last step of the complete process. In this step, we will try to debug the issues of knowledge base. In the knowledge base, we may have omitted assertions like $1 \neq 0$.

14.5 Summary

A **possible world**, or **model**, for first-order logic includes a set of objects and an **interpretation** that maps constant symbols to objects, predicate symbols to relations among objects, and function symbols to functions on objects. An atomic sentence is true just when the relation named by the predicate holds between the objects named by the terms. **Extended interpretations**, which map quantifier variables to objects in the model, define the truth of quantified sentences. Developing a knowledge base in first-order logic requires a careful process of analysing the domain, choosing a vocabulary, and encoding the axioms required to support the desired inferences.

14.6 Model Questions

1. What is meant by knowledge representation.
2. Define: Knowledge engineering.
3. Write short notes on kinship domain.
4. Discuss about numbers, sets and lists in knowledge representation.
5. Illustrate knowledge representation of wumpus world with an example.
6. Explain the steps involved in knowledge engineering process.

LESSON 15
INFERENCE IN FIRST ORDER LOGIC

Structure

15.1 Introduction

This lesson introduces inference rules for quantifiers and shows how to reduce first-order inference to propositional inference. Also, this lesson describes the idea of **unification**, showing how it can be used to construct inference rules that work directly with first-order sentences. We then discuss three major families of first-order inference algorithms. **Forward chaining, backward chaining** and resolution systems. Forward and backward chaining can be very efficient, but are applicable only to knowledge bases that can be expressed as sets of Horn clauses. General first-order sentences require resolution-based **theorem proving**.

15.2 Objectives

- To introduce the concept of first-order inference rules and Unification.
- To explore the forward and backward chaining algorithms.
- To study the concept of resolution for first order logic.

15.3 Propositional vs First-Order Inference

This section and the next introduce the ideas underlying modern logical inference systems. We begin with some simple inference rules that can be applied to sentences with quantifiers to obtain sentences without quantifiers. These rules lead naturally to the idea that *first-order* inference can be done by converting the knowledge base to *propositional* logic and using *propositional* inference, which we already know how to do. The next section points out an obvious shortcut, leading to inference methods that manipulate first-order sentences directly.

15.3.1 Inference rules for quantifiers

Let us begin with universal quantifiers. Suppose our knowledge base contains the standard folkloric axiom stating that all greedy kings are evil:

$$\forall x \; King(x) \wedge Greedy(x) \Rightarrow Evil(x) .$$

Then it seems quite permissible to infer any of the following sentences:

$King(John) \wedge Greedy(John) \Rightarrow Evil(John)$

$King(Richard) \wedge Greedy(Richard) \Rightarrow Evil(Richard)$

$King(Father (John)) \wedge Greedy(Father (John)) \Rightarrow Evil(Father (John)) .$

...

The rule of **Universal Instantiation** (**UI** for short) says that we can infer any sentence obtained by substituting a **ground term** (a term without variables) for the variable. To write out the inference rule formally, we use the notion of **substitutions.** Let $SUBST(\theta,\alpha)$ denote the result of applying the substitution θ to the sentence α. Then the rule is written

$$\frac{\forall v \; \alpha}{SUBST(\{v/g\}, \; \alpha)}$$

for any variable *v* and ground term *g*. For example, the three sentences given earlier are obtained with the substitutions *{x/John}, {x/Richard }*, and *{x/Father (John)}*. In the rule for **Existential Instantiation**, the variable is replaced by a single *new constant symbol*. The formal statement is as follows: for any sentence α, variable *v*, and constant symbol *k* that does not appear elsewhere in the knowledge base,

$$\exists v\ \alpha$$

$$\text{SUBST}(\{v/k\}, \alpha)$$

For example, from the sentence

$$\exists x\ Crown(x) \wedge OnHead(x, John)$$

we can infer the sentence

$$Crown(C1) \wedge OnHead(C1, John)$$

as long as C1 does not appear elsewhere in the knowledge base. Basically, the existential sentence says there is some object satisfying a condition, and applying the existential instantiation rule just gives a name to that object. Of course, that name must not already belong to another object. Mathematics provides a nice example: suppose we discover that there is a number that is a little bigger than 2.71828 and that satisfies the equation $d(x^y)/dy = x^y$ for x. We can give this number a name, such as e, but it would be a mistake to give it the name of an existing object, such as π. In logic, the new name is called a **Skolem constant**. Existential Instantiation is a special case of a more general process called **skolemization.**

Whereas Universal Instantiation can be applied many times to produce many different consequences, Existential Instantiation can be applied once, and then the existentially quantified sentence can be discarded. For example, we no longer need $\exists x\ Kill(x, Victim)$ once we have added the sentence *Kill (Murderer, Victim)*. Strictly speaking, the new knowledge base is not logically equivalent to the old, but it can be shown to be **inferentially equivalent** in the sense that it is satisfiable exactly when the original knowledge base is satisfiable.

15.3.2 Reduction to propositional inference

Once we have rules for inferring nonquantified sentences from quantified sentences, it becomes possible to reduce first-order inference to propositional inference. In this section we give the main ideas. The first idea is that, just as an existentially quantified sentence can be replaced by one instantiation, a universally quantified sentence can be replaced by the set of *all possible* instantiations. For example, suppose our knowledge base contains just the sentences

$$\forall x\ King(x) \wedge Greedy(x) \Rightarrow Evil(x)$$

$$King(John)$$

Greedy(John)

Brother (Richard, John) (15.1)

Then we apply UI to the first sentence using all possible ground-term substitutions from the vocabulary of the knowledge base—in this case, *{x/John}* and *{x/Richard }*. We obtain

$King(John) \wedge Greedy(John) \Rightarrow Evil(John)$

$King(Richard) \wedge Greedy(Richard) \Rightarrow Evil(Richard)$,

and we discard the universally quantified sentence. Now, the knowledge base is essentially propositional if we view the ground atomic sentences—*King(John), Greedy(John)*, and so on—as proposition symbols. Therefore, we can apply any of the complete propositional algorithms to obtain conclusions such as *Evil(John)*.

This technique of **propositionalization** can be made completely general; that is, every first-order knowledge base and query can be propositionalized in such a way that entailment is preserved. Thus, we have a complete decision procedure for entailment . . . or perhaps not. There is a problem: when the knowledge base includes a function symbol, the set of possible ground-term substitutions is infinite! For example, if the knowledge base mentions the *Father* symbol, then infinitely many nested terms such as *Father(Father(Father(John)))* can be constructed. Our propositional algorithms will have difficulty with an infinitely large set of sentences.

Fortunately, there is a famous theorem due to Jacques Herbrand (1930) to the effect that if a sentence is entailed by the original, first-order knowledge base, then there is a proof involving just a *finite* subset of the propositionalized knowledge base. Since any such subset has a maximum depth of nesting among its ground terms, we can find the subset by first generating all the instantiations with constant symbols (*Richard* and *John*), then all terms of depth 1 *(Father(Richard)* and *Father(John))*, then all terms of depth 2, and so on, until we are able to construct a propositional proof of the entailed sentence.

We have sketched an approach to first-order inference via propositionalization that is **complete**—that is, any entailed sentence can be proved. This is a major achievement, given that the space of possible models is infinite. On the other hand, we do not know until the proof is done that the sentence *is* entailed! What happens when the sentence is *not* entailed? Can we

tell? Well, for first-order logic, it turns out that we cannot. Our proof procedure can go on and on, generating more and more deeply nested terms, but we will not know whether it is stuck in a hopeless loop or whether the proof is just about to pop out. This is very much like the halting problem for Turing machines. Alan Turing (1936) and Alonzo Church (1936) both proved, in rather different ways, the inevitability of this state of affairs. The question of entailment for first-order logic is semidecidable—that is, algorithms exist that say yes to every entailed sentence, but no algorithm exists that also says no to every nonentailed sentence.

15.4 Unification

Unification is a process of making two different logical atomic expressions identical by finding a substitution. Unification depends on the substitution process. It takes two literals as input and makes them identical using substitution. Let ψ_1 and ψ_2 be two atomic sentences and σ be a unifier such that, $\psi_1\sigma = \psi_2\sigma$, then it can be expressed as **UNIFY(ψ_1, ψ_2)**.

Example: Find the MGU (Most General Unifier) for Unify{King(x), King(John)}

Let ψ_1 = King(x), ψ_2 = King(John),

Substitution θ = {John/x} is a unifier for these atoms and applying this substitution, and both expressions will be identical. The UNIFY algorithm is used for unification, which takes two atomic sentences and returns a unifier for those sentences (If any exist). Unification is a key component of all first-order inference algorithms. It returns fail if the expressions do not match with each other. The substitution variables are called Most General Unifier or MGU.

E.g. Let's say there are two different expressions, **P(x, y), and P(a, f(z))**.

In this example, we need to make both above statements identical to each other. For this, we will perform the substitution.

P(x, y)..... (i)

P(a, f(z))..... (ii)

Substitute *x* with *a*, and *y* with *f(z)* in the first expression, and it will be represented as ***a/x*** and *f(z)/y*. With both the substitutions, the first expression will be identical to the second expression and the substitution set will be: ***[a/x, f(z)/y]***.

15.4.1 Conditions for Unification and Unification Algorithm

Following are some basic conditions for unification. (1) Predicate symbol must be same, atoms or expression with different predicate symbol can never be unified. (2) Number of Arguments in both expressions must be identical. (3) Unification will fail if there are two similar variables present in the same expression.

Algorithm: Unify(Ψ_1, Ψ_2)

Step. 1: If ψ_1 or ψ_2 is a variable or constant, then:

a) If ψ_1 or ψ_2 are identical, then return NIL.

b) Else if $\psi_1$is a variable,

a. then if ψ_1 occurs in ψ_2, then return FAILURE

b. Else return { (ψ_2/ ψ_1)}.

c) Else if ψ_2 is a variable,

a. If ψ_2 occurs in ψ_1 then return FAILURE,

b. Else return {(ψ_1/ ψ_2)}.

d) Else return FAILURE.

Step.2: If the initial Predicate symbol in ψ_1 and ψ_2 are not same, then return FAILURE.

Step. 3: IF ψ_1 and ψ_2 have a different number of arguments, then return FAILURE.

Step. 4: Set Substitution set(SUBST) to NIL.

Step. 5: For i=1 to the number of elements in ψ_1.

a) Call Unify function with the ith element of ψ_1 and ith element of ψ_2, and put the result into S.

b) If S = failure then returns Failure

c) If S $\neq$ NIL then do,

a. Apply S to the remainder of both L1 and L2.

b. SUBST= APPEND(S, SUBST).

Step.6: Return SUBST.

15.4.2 Implementation of the Algorithm

Step.1: Initialize the substitution set to be empty.

Step.2: Recursively unify atomic sentences:

a) Check for Identical expression match.

b) If one expression is a variable v_i, and the other is a term t_i which does not contain variable v_i, then:

(i) Substitute t_i / v_i in the existing substitutions

(ii) Add t_i /v_i to the substitution setlist.

(iii) If both the expressions are functions, then function name must be similar, and the number of arguments must be the same in both the expression.

For each pair of the following atomic sentences find the most general unifier (If exist).

1. Find the MGU of {p(f(a), g(Y)) and p(X, X)}

Sol: S_0 => Here, ψ_1 = p(f(a), g(Y)), and ψ_2 = p(X, X)

SUBST θ= {f(a) / X}

S1 => ψ_1 = p(f(a), g(Y)), and ψ_2 = p(f(a), f(a))

SUBST θ= {f(a) / g(y)}, **Unification failed**.

Unification is not possible for these expressions.

2. Find the MGU of {p(b, X, f(g(Z))) and p(Z, f(Y), f(Y))}

Here, ψ_1 = p(b, X, f(g(Z))) , and ψ_2 = p(Z, f(Y), f(Y))

S_0 => { p(b, X, f(g(Z))); p(Z, f(Y), f(Y))}
SUBST θ={b/Z}

S_1 => { p(b, X, f(g(b))); p(b, f(Y), f(Y))}
SUBST θ={f(Y) /X}

S_2 => { p(b, f(Y), f(g(b))); p(b, f(Y), f(Y))}
SUBST θ= {g(b) /Y}

S_2 => { p(b, f(g(b)), f(g(b)); p(b, f(g(b)), f(g(b))} **Unified Successfully.**
And Unifier = { b/Z, f(Y) /X , g(b) /Y}.

3. Find the MGU of {p (X, X), and p (Z, f(Z))}

Here, ψ_1 = {p (X, X), and ψ_2 = p (Z, f(Z))

S_0 => {p (X, X), p (Z, f(Z))}
SUBST θ= {X/Z}

S1 => {p (Z, Z), p (Z, f(Z))}
SUBST θ= {f(Z) / Z}, **Unification Failed**.

Hence, unification is not possible for these expressions.

4. Find the MGU of UNIFY(prime (11), prime(y))

Here, ψ_1 = {prime(11) , and ψ_2 = prime(y)}

S_0 => {prime(11) , prime(y)}
SUBST θ= {11/y}

S_1 => {prime(11) , prime(11)} , **Successfully unified.**

Unifier: {11/y}.

5. Find the MGU of Q(a, g(x, a), f(y)), Q(a, g(f(b), a), x)}

Here, ψ_1 = Q(a, g(x, a), f(y)), and ψ_2 = Q(a, g(f(b), a), x)

S_0 => {Q(a, g(x, a), f(y)); Q(a, g(f(b), a), x)}

SUBST θ= {f(b)/x}

S_1 => {Q(a, g(f(b), a), f(y)); Q(a, g(f(b), a), f(b))}

SUBST θ= {b/y}

S_1 => {Q(a, g(f(b), a), f(b)); Q(a, g(f(b), a), f(b))}, **Successfully Unified.**

Unifier: [a/a, f(b)/x, b/y].

6. UNIFY(knows(Richard, x), knows(Richard, John))

Here, ψ_1 = knows(Richard, x), and ψ_2 = knows(Richard, John)

S_0 => { knows(Richard, x); knows(Richard, John)}

SUBST θ= {John/x}

S_1 => { knows(Richard, John); knows(Richard, John)}, **Successfully Unified.**

Unifier: {John/x}.

15.5 Forward Chaining

A forward-chaining algorithm for propositional definite clauses is discussed in this section. The idea is simple: start with the atomic sentences in the knowledge base and apply Modus Ponens in the forward direction, adding new atomic sentences, until no further inferences can be made. Here, we explain how the algorithm is applied to first-order definite clauses. Definite clauses such as *Situation* $\Rightarrow$ *Response* are especially useful for systems that make inferences in response to newly arrived information. Many systems can be defined this way, and forward chaining can be implemented very efficiently. The inference engine is the component of the intelligent system in artificial intelligence, which applies logical rules to the knowledge base to infer new information from known facts. The first inference engine was part of the expert system. Inference engine commonly proceeds in two modes, which are: (a) **Forward chaining and (b) Backward chaining**

Horn clause and definite clause are the forms of sentences, which enables knowledge base to use a more restricted and efficient inference algorithm. Logical inference algorithms use forward and backward chaining approaches, which require KB in the form of the **first-**

order definite clause. A clause which is a disjunction of literals with **exactly one positive literal** is known as a definite clause or strict horn clause. A clause which is a disjunction of literals with **at most one positive literal** is known as horn clause. Hence all the definite clauses are horn clauses. For example, ***(¬ p* V *¬ q* V *k)***. It has only one positive literal k. It is equivalent to $p \wedge q \rightarrow k$.

Forward chaining is also known as a **forward deduction** or **forward reasoning** method when using an inference engine. Forward chaining is a form of reasoning which start with atomic sentences in the knowledge base and applies inference rules (Modus Ponens) in the forward direction to extract more data until a goal is reached. The Forward-chaining algorithm starts from known facts, triggers all rules whose premises are satisfied, and add their conclusion to the known facts. This process repeats until the problem is solved.

15.5.1 Properties of Forward-Chaining

It is a down-up approach, as it moves from bottom to top. It is a process of making a conclusion based on known facts or data, by starting from the initial state and reaches the goal state. Forward-chaining approach is also called as data-driven as we reach to the goal using available data. Forward -chaining approach is commonly used in the expert system, such as CLIPS, business, and production rule systems. Consider the following famous example which we will use in both approaches. Consider the following example.

"As per the law, it is a crime for an American to sell weapons to hostile nations. Country A, an enemy of America, has some missiles, and all the missiles were sold to it by Robert, who is an American citizen."

Prove that **"Robert is criminal."**

To solve the above problem, first, we will convert all the above facts into first-order definite clauses, and then we will use a forward-chaining algorithm to reach the goal.

Facts conversion into FOL:

- It is a crime for an American to sell weapons to hostile nations. (Let's say p, q, and r are variables)

  American (p) ^ weapon(q) ^ sells (p, q, r) ^ hostile(r) $\rightarrow$ Criminal(p) ...(1)

- Country A has some missiles. **?p Owns(A, p) ^ Missile(p)**. It can be written in two definite clauses by using Existential Instantiation, introducing new Constant T1.

  Owns(A, T1) .....(2)

  Missile(T1) .....(3)

- All of the missiles were sold to country A by Robert.

  ?p Missiles(p) ^ Owns (A, p) → Sells (Robert, p, A) .....(4)

- Missiles are weapons.

  Missile(p) → Weapons (p) .....(5)

- Enemy of America is known as hostile.

  Enemy(p, America) → Hostile(p) .....(6)

- Country A is an enemy of America.

  Enemy (A, America) .....(7)

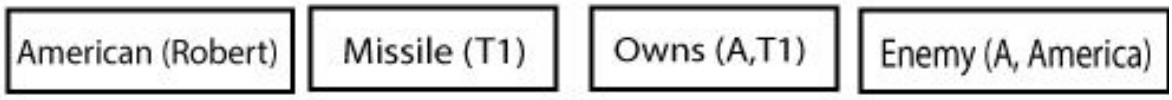
- Robert is American

  American(Robert). .....(8)

Forward chaining proof:

Step-1:

In the first step we will start with the known facts and will choose the sentences which do not have implications, such as: ***American(Robert), Enemy(A, America), Owns(A, T1),*** **and** ***Missile(T1)***. All these facts will be represented as below.



Step-2:

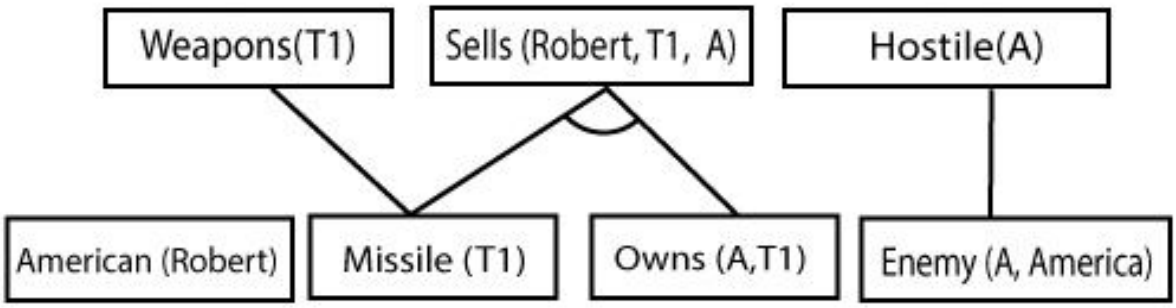
At the second step, we will see those facts which infer from available facts and with satisfied premises.

Rule-(1) does not satisfy premises, so it will not be added in the first iteration.

Rule-(2) and (3) are already added.

Rule-(4) satisfy with the substitution *{p/T1},* **so Sells (*Robert, T1, A*)** is added, which infers from the conjunction of Rule (2) and (3).

Rule-(6) is satisfied with the substitution *(p/A),* so *Hostile(A)* is added and which infers from Rule-(7).



Step-3:

At step-3, as we can check Rule-(1) is satisfied with the substitution ***{p/Robert, q/T1, r/ A}*, so we can add *Criminal(Robert)*** which infers all the available facts. And hence we reached our goal statement (See Figure 15.1).

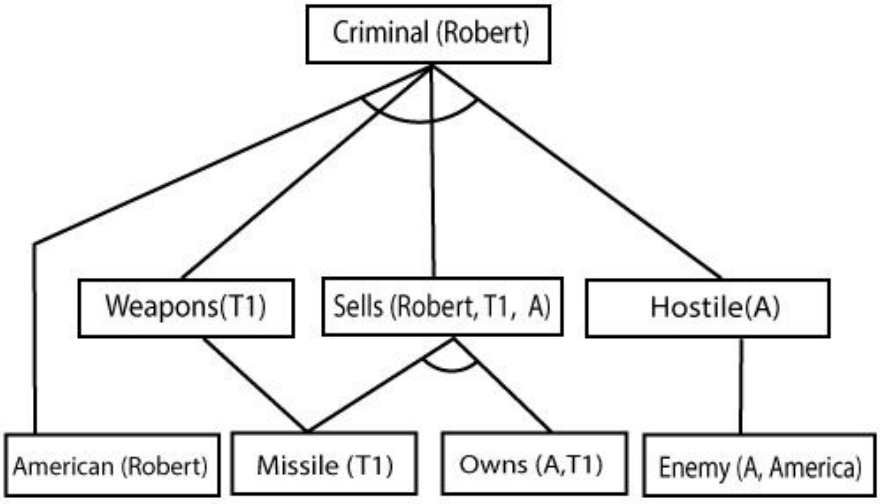


Figure 15.1 The proof tree generated by forward chaining

Hence it is proved that Robert is Criminal using forward chaining approach.

15.6 Backward Chaining

The second major family of logical inference algorithms uses the **backward chaining** approach for definite clauses. These algorithms work backward from the goal, chaining through rules to find known facts that support the proof. We describe the basic algorithm, and then we describe how it is used in **logic programming**, which is the most widely used form of automated reasoning. We also see that backward chaining has some disadvantages compared with forward chaining, and we look at ways to overcome them. Finally, we look at the close connection between logic programming and constraint satisfaction problems.

Backward-chaining is also known as a backward deduction or backward reasoning method when using an inference engine. A backward chaining algorithm is a form of reasoning, which starts with the goal and works backward, chaining through rules to find known facts that support the goal.

Properties of backward chaining:

It is known as a top-down approach. Backward-chaining is based on modus ponens inference rule. In backward chaining, the goal is broken into sub-goal or sub-goals to prove the facts true. It is called a goal-driven approach, as a list of goals decides which rules are selected and used. Backward -chaining algorithm is used in game theory, automated theorem proving tools, inference engines, proof assistants, and various AI applications. The backward-chaining method mostly used a **depth-first search** strategy for proof. Consider the following example. In backward-chaining, we will use the same above example, and will rewrite all the rules.

- **American (p) ^ weapon(q) ^ sells (p, q, r) ^ hostile(r) → Criminal(p) ...(1)**

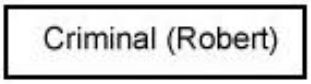
  Owns(A, T1) .....(2)
- **Missile(T1)**
- **?p Missiles(p) ^ Owns (A, p) → Sells (Robert, p, A) .....(4)**
- **Missile(p) → Weapons (p) .....(5)**
- **Enemy(p, America) → Hostile(p) .....(6)**
- **Enemy (A, America) .....(7)**
- **American(Robert). .....(8)**

15.6.1 Backward-Chaining Proof

In Backward chaining, we will start with our goal predicate, which is ***Criminal(Robert)***, and then infer further rules.

Step-1:

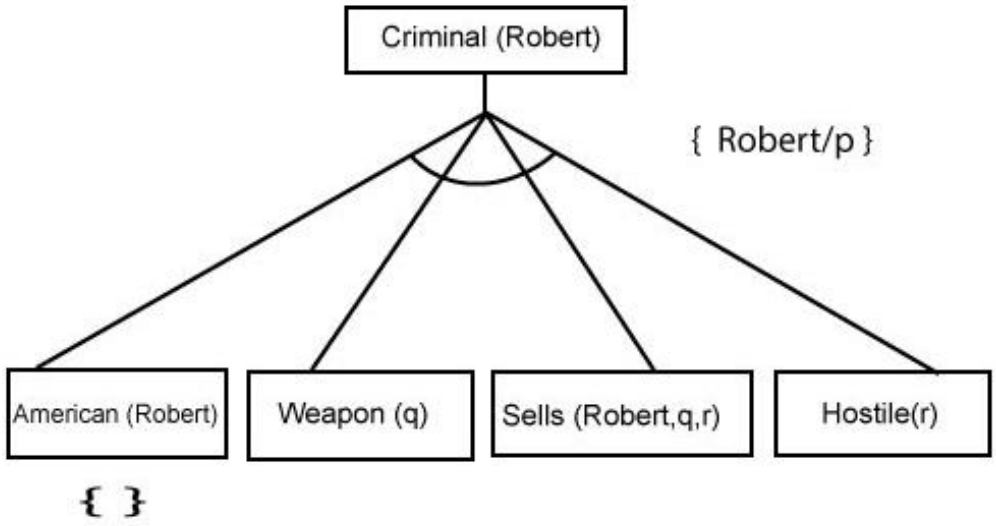
At the first step, we will take the goal fact. And from the goal fact, we will infer other facts, and at last, we will prove those facts true. So our goal fact is “Robert is Criminal,” so following is the predicate of it.



Step-2:

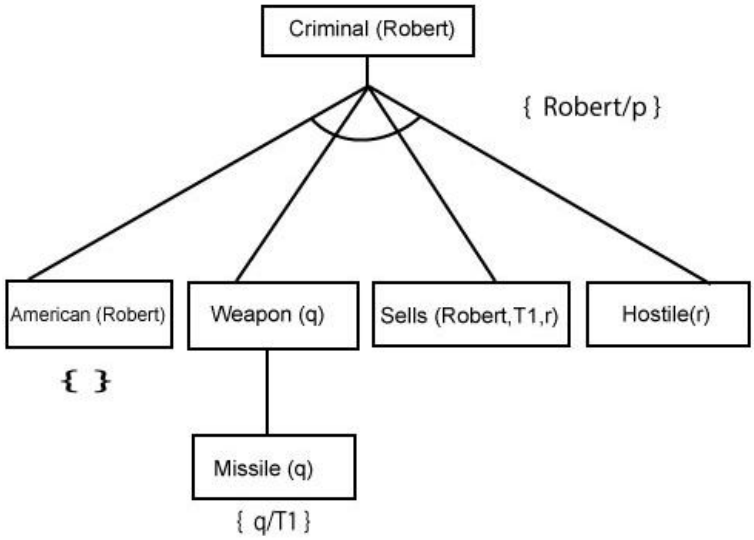
At the second step, we will infer other facts form goal fact which satisfies the rules. So as we can see in Rule-1, the goal predicate *Criminal (Robert)* is present with substitution *{Robert/P}*. So we will add all the conjunctive facts below the first level and will replace *p* with *Robert*.

Here we can see *American (Robert)* is a fact, so it is proved here.



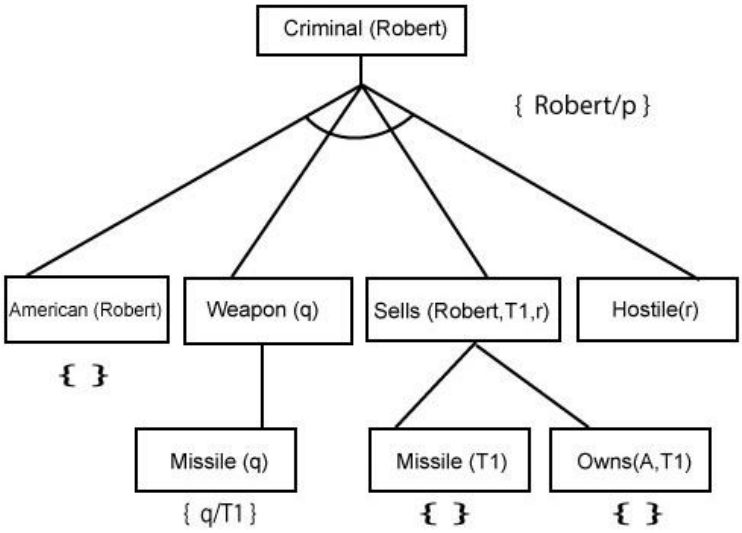
Step-3:

At step-3, we will extract further fact *Missile(q)* which infer from *Weapon(q)*, as it satisfies Rule-(5). *Weapon (q)* is also true with the substitution of a constant *T1* at *q*.



Step-4:

At step-4, we can infer facts *Missile(T1)* and *Owns(A, T1)* form *Sells(Robert, T1, r)* which satisfies the **Rule- 4**, with the substitution of *A* in place of *r*. So these two statements are proved here.



Step-5:

At step-5, we can infer the fact ***Enemy(A, America)*** from ***Hostile(A)*** which satisfies Rule-6. And hence all the statements are proved true using backward chaining.

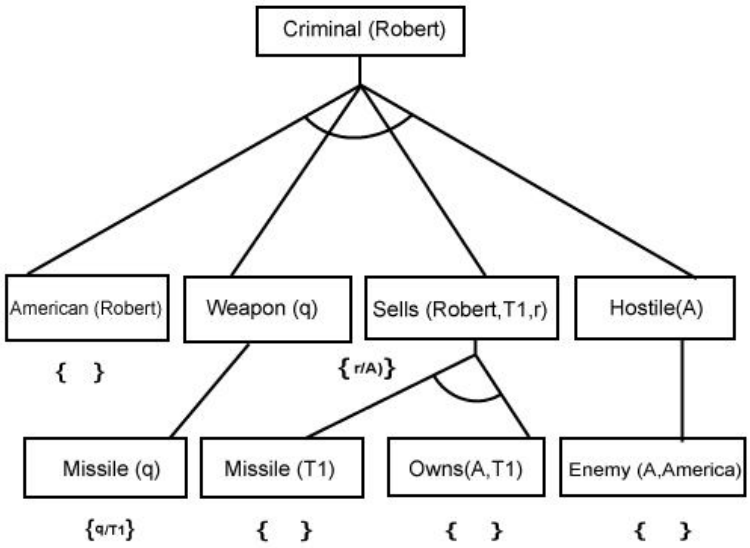


Figure 15.2 Proof tree constructed by backward chaining

15.7 Differences between Forward and Backward Chaining

Forward chaining as the name suggests, start from the known facts and move forward by applying inference rules to extract more data, and it continues until it reaches to the goal, whereas backward chaining starts from the goal, move backward by using inference rules to determine the facts that satisfy the goal. Forward chaining is called a **data-driven** inference technique, whereas backward chaining is called a **goal-driven** inference technique. Forward chaining is known as the **down-up** approach, whereas backward chaining is known as a **top-down** approach. Forward chaining uses **breadth-first search** strategy, whereas backward chaining uses **depth-first search** strategy. Forward and backward chaining both applies **Modus ponens** inference rule. Forward chaining can be used for tasks such as planning, design process monitoring, diagnosis, and classification, whereas backward chaining can be used for classification and diagnosis tasks. Forward chaining can be like an exhaustive search, whereas backward chaining tries to avoid the unnecessary path of reasoning. In forward-chaining

there can be various ASK questions from the knowledge base, whereas in backward chaining there can be fewer ASK questions. Forward chaining is slow as it checks for all the rules, whereas backward chaining is fast as it checks few required rules only.

15.8 Resolution in First Order Logic

Resolution is a theorem proving technique that proceeds by building refutation proofs, i.e., proofs by contradictions. It was invented by a Mathematician John Alan Robinson in the year 1965. Resolution is used, if there are various statements are given, and we need to prove a conclusion of those statements. Unification is a key concept in proofs by resolutions. Resolution is a single inference rule which can efficiently operate on the **conjunctive normal form or clausal form**. Disjunction of literals (an atomic sentence) is called a **clause**. It is also known as a unit clause. A sentence represented as a conjunction of clauses is said to be **conjunctive normal form** or **CNF**.

15.8.1 The resolution inference rule

The resolution rule for first-order logic is simply a lifted version of the propositional rule. Resolution can resolve two clauses if they contain complementary literals, which are assumed to be standardized apart so that they share no variables.

$$\frac{l_1 \vee \ldots\ldots \vee l_k, \qquad m_1 \vee \ldots\ldots \vee m_n}{\text{SUBST}(\theta, l_1 \vee \ldots\ldots \vee l_{i-1} \vee l_{i+1} \vee \ldots \vee l_k \vee m_1 \vee \ldots\ldots \vee m_{j-1} \vee m_{j+1} \vee \ldots\ldots \vee m_n)}$$

Where $\mathbf{l_i}$ and $\mathbf{m_j}$ are complementary literals. This rule is also called the **binary resolution rule** because it only resolves exactly two literals. We can resolve two clauses which are given below:

[Animal (g(x) V Loves (f(x), x)] **and** ***[¬ Loves(a, b) V ¬Kills(a, b)]***

Where two complimentary literals are: **Loves (f(x), x) and ¬ Loves (a, b).** These literals can be unified with unifier **θ= [a/f(x), and b/x]** , and it will generate a resolvent clause:

[Animal (g(x) V ¬ Kills(f(x), x)].

15.8.2 Steps for resolution

1. Conversion of facts into first-order logic.
2. Convert FOL statements into CNF
3. Negate the statement which needs to prove (proof by contradiction)
4. Draw resolution graph (unification).

To better understand all the above steps, we will take an example in which we will apply resolution.

Example:

a. John likes all kind of food.

b. Apple and vegetable are food

c. Anything anyone eats and not killed is food.

d. Anil eats peanuts and still alive

e. Harry eats everything that Anil eats.

Prove by resolution that:

f. John likes peanuts.

Step-1: Conversion of Facts into FOL

In the first step we will convert all the given statements into its first order logic.

a. $\forall x: food(x) \rightarrow likes(John, x)$

b. $food(Apple) \wedge food(vegetables)$

c. $\forall x \forall y: eats(x, y) \wedge \neg killed(x) \rightarrow food(y)$

d. $eats(Anil, Peanuts) \wedge alive(Anil)$.

e. $\forall x : eats(Anil, x) \rightarrow eats(Harry, x)$

f. $\forall x: \neg killed(x) \rightarrow alive(x)$ } **added predicates.**

g. $\forall x: alive(x) \rightarrow \neg killed(x)$ }

h. $likes(John, Peanuts)$

Step-2: Conversion of FOL into CNF

In First order logic resolution, it is required to convert the FOL into CNF as CNF form makes easier for resolution proofs.

- **Eliminate all implication (®) and rewrite**
  a. ∀x ¬ food(x) V likes(John, x)
  b. food(Apple) ∨ food(vegetables)
  c. ∀x ∀y ¬ [eats(x, y) ∨ ¬ killed(x)] V food(y)
  d. eats (Anil, Peanuts) ∨ alive(Anil)
  e. ∀x ¬ eats(Anil, x) V eats(Harry, x)
  f. ∀x¬ [¬ killed(x)] V alive(x)
  g. ∀x ¬ alive(x) V ¬ killed(x)
  h. likes(John, Peanuts).
- **Move negation (¬)inwards and rewrite**
  a. ∀x ¬ food(x) V likes(John, x)
  b. food(Apple) ∨ food(vegetables)
  c. ∀x ∀y ¬ eats(x, y) V killed(x) V food(y)
  d. eats (Anil, Peanuts) ∨ alive(Anil)
  e. ∀x ¬ eats(Anil, x) V eats(Harry, x)
  f. ∀x ¬killed(x)] V alive(x)
  g. ∀x ¬ alive(x) V ¬ killed(x)
  h. likes(John, Peanuts).
- **Rename variables or standardize variables**
  a. x ¬ food(x) V likes(John, x)
  b. food(Apple) ∨ food(vegetables)
  c. ∀y ∀z ¬ eats(y, z) V killed(y) V food(z)
  d. eats (Anil, Peanuts) ∨ alive(Anil)

e. ∀w¬ eats(Anil, w) V eats(Harry, w)

f. ∀g ¬killed(g)] V alive(g)

g. ∀k ¬ alive(k) V ¬ killed(k)

h. likes(John, Peanuts).

- **Eliminate existential instantiation quantifier by elimination.**

In this step, we will eliminate existential quantifier ∃, and this process is known as **Skolemization**. But in this example problem since there is no existential quantifier so all the statements will remain same in this step.

- **Drop Universal quantifiers.**

In this step we will drop all universal quantifier since all the statements are not implicitly quantified so we don't need it.

a. ¬ food(x) V likes(John, x)

b. food(Apple)

c. food(vegetables)

d. ¬ eats(y, z) V killed(y) V food(z)

e. eats (Anil, Peanuts)

f. alive(Anil)

g. ¬ eats(Anil, w) V eats(Harry, w)

h. killed(g) V alive(g)

i. ¬ alive(k) V ¬ killed(k)

j. likes(John, Peanuts).

- **Distribute conjunction ∧ over disjunction ¬.**

This step will not make any change in this problem.

Step-3: Negate the statement to be proved

In this statement, we will apply negation to the conclusion statements, which will be written as ¬likes(John, Peanuts)

Step-4: Draw Resolution graph:

Now in this step, we will solve the problem by resolution tree using substitution. For the above problem, resolution tree is given in the Figure 15.3.

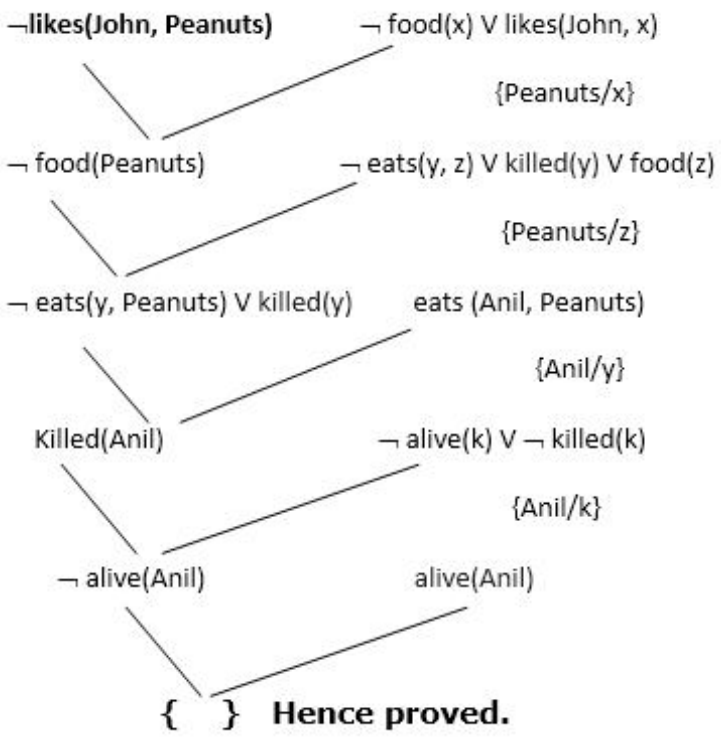


Figure 15.3 Resolution Tree or Graph

Hence the negation of the conclusion has been proved as a complete contradiction with the given set of statements.

Explanation of Resolution Graph

In the first step of resolution graph, ***¬likes(John, Peanuts)***, and ***likes(John, x)*** get resolved(canceled) by substitution of ***{Peanuts/x}***, and we are left with ***¬ food(Peanuts)*.** In the second step of the resolution graph, ***¬food(Peanuts)***, and ***food(z)*** get resolved (canceled) by substitution of ***{Peanuts/z}***, and we are left with ***¬eats(y, Peanuts)* V *killed(y)*.** In the third step of the resolution graph, ***¬eats(y, Peanuts)*** and ***eats(Anil, Peanuts)*** get resolved by substitution ***{Anil/y}***, and we are left with ***Killed(Anil)*.** In the fourth step of the resolution

graph, ***Killed(Anil)*** and ***¬killed(k)*** get resolve by substitution ***{Anil/k}***, and we are left with ***¬alive(Anil).*** In the last step of the resolution graph ***¬alive(Anil)*** and ***alive(Anil)*** get resolved.

15.9 Summary

We have presented an analysis of logical inference in first-order logic and a number of algorithms for doing it. A first approach uses inference rules (**universal instantiation** and **existential instantiation**) to **propositionalize** the inference problem. Typically, this approach is slow, unless the domain is small. The use of **unification** to identify appropriate substitutions for variables eliminates the instantiation step in first-order proofs, making the process more efficient in many cases. A lifted version of **Modus Ponens** uses unification to provide a natural and powerful inference rule, **generalized Modus Ponens**. The **forward-chaining** and **backward-chaining** algorithms apply this rule to sets of definite clauses. The generalized **resolution** inference rule provides a complete proof system for firstorder logic, using knowledge bases in conjunctive normal form.

15.10 Model Questions

1. Illustrate Unification with an example.
2. Write short notes on: (a) Forward Chaining (b) Backward Chaining
3. Mention any five differences between Forward and Backward Chaining.
4. What is meant by knowledge representation.
5. Explain in detail about the Resolution in First Order Logic with an example.

LESSON 16
REASONING SYSTEMS

Structure

16.1 Introduction

The previous chapters described the technology for knowledge-based agents: the syntax, semantics, and proof theory of propositional and first-order logic, and the implementation of agents that use these logics. In this chapter we address the question of what *content* to put into such an agent's knowledge base—how to represent facts about the world. The first section introduces the idea of a general ontology, which organizes everything in the world into a hierarchy of categories. Next section covers the basic categories of objects, substances, and measures. Reasoning systems designed for efficient inference with categories and reasoning with default information are also discussed.

16.2 Objectives

- To introduce the concept of ontological engineering.
- To study the concepts of categories, objects and events.
- To explore the reasoning for categories and default information.

16.3 Ontological Engineering

In “toy” domains, the choice of representation is not that important; many choices will work. Complex domains such as shopping on the Internet or driving a car in traffic require more general and flexible representations. This lesson shows how to create these representations, concentrating on general concepts—such as *Events, Time, Physical Objects*, and *Beliefs*—that occur in many different domains. Representing these abstract concepts is sometimes called **ontological engineering**.

The prospect of representing *everything* in the world is daunting. Of course, we won’t actually write a complete description of everything—that would be far too much for even a 1000-page textbook—but we will leave placeholders where new knowledge for any domain can fit in. For example, we will define what it means to be a physical object, and the details of different types of objects—robots, televisions, books, or whatever—can be filled in later. This is analogous to the way that designers of an object-oriented programming framework (such as the Java Swing graphical framework) define general concepts like *Window*, expecting users to use these to define more specific concepts like *SpreadsheetWindow*. The general framework of concepts is called an **upper ontology** because of the convention of drawing graphs with the general concepts at the top and the more specific concepts below them, as in Figure 16.1.

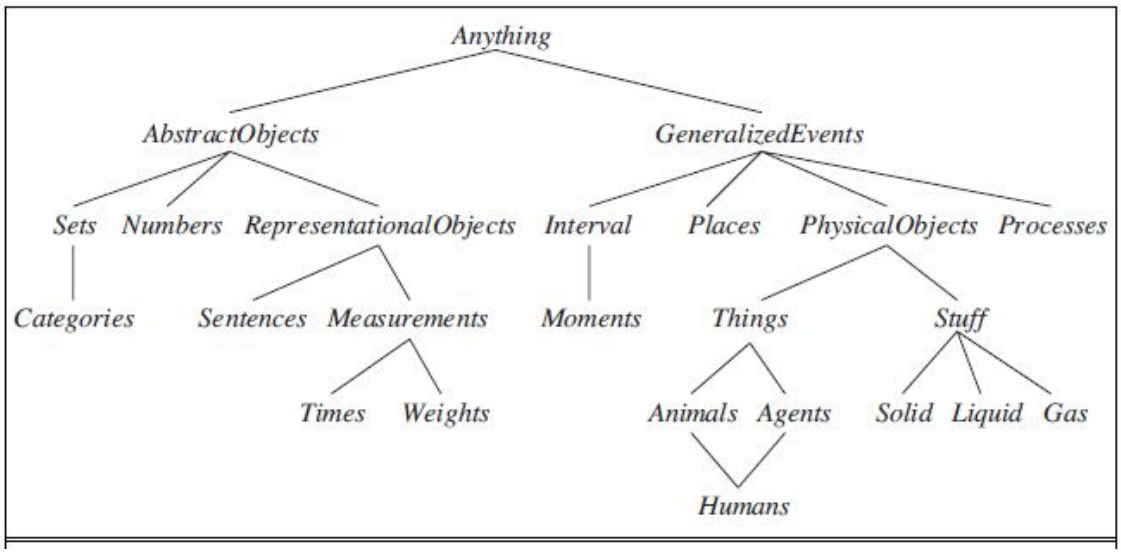


Figure 16.1 The upper ontology of the world

Before considering the ontology further, we should state one important caveat. We have elected to use first-order logic to discuss the content and organization of knowledge although certain aspects of the real world are hard to capture in FOL. The principal difficulty is that most generalizations have exceptions or hold only to a degree. For example, although "tomatoes are red" is a useful rule, some tomatoes are green, yellow, or orange. Similar exceptions can be found to almost all the rules in this chapter. The ability to handle exceptions and uncertainty is extremely important, but is orthogonal to the task of understanding the general ontology.

If we look at the wumpus world, similar considerations apply. Although we do represent time, it has a simple structure: Nothing happens except when the agent acts, and all changes are instantaneous. A more general ontology, better suited for the real world, would allow for simultaneous changes extended over time. We also used a Pit predicate to say which squares have pits. We could have allowed for different kinds of pits by having several individuals belonging to the class of pits, each having different properties. Similarly, we might want to allow for other animals besides wumpuses. It might not be possible to pin down the exact species from the available percepts, so we would need to build up a biological taxonomy to help the agent predict the behavior of cave-dwellers from scanty clues.

For any special-purpose ontology, it is possible to make changes like these to move toward greater generality. An obvious question then arises: do all these ontologies converge on a general-purpose ontology? After centuries of philosophical and computational investigation, the answer is "Maybe." In this section, we present one general-purpose ontology that synthesizes ideas from those centuries. Two major characteristics of general-purpose ontologies distinguish them from collections of special-purpose ontologies:

- A general-purpose ontology should be applicable in more or less any special-purpose domain (with the addition of domain-specific axioms). This means that no representational issue can be finessed or brushed under the carpet.
- In any sufficiently demanding domain, different areas of knowledge must be *unified*, because reasoning and problem solving could involve several areas simultaneously. A robot circuit-repair system, for instance, needs to reason about circuits in terms of electrical connectivity and physical layout, and about time, both for circuit timing analysis and estimating labor costs. The sentences describing time therefore must be capable of being combined with those describing spatial layout and must work equally well for nanoseconds and minutes and for angstroms and meters.

16.4 Categories and Objects

The organization of objects into **categories** is a vital part of knowledge representation. Although interaction with the world takes place at the level of individual objects, *much reasoning takes place at the level of categories.* For example, a shopper would normally have the goal of buying a basketball, rather than a *particular* basketball such as BB9. Categories also serve to make predictions about objects once they are classified. One infers the presence of certain objects from perceptual input, infers category membership from the perceived properties of the objects, and then uses category information to make predictions about the objects. For example, from its green and yellow mottled skin, one-foot diameter, ovoid shape, red flesh, black seeds, and presence in the fruit aisle, one can infer that an object is a watermelon; from this, one infers that it would be useful for fruit salad.

There are two choices for representing categories in first-order logic: **predicates** and **objects**. That is, we can use the predicate *Basketball (b)*, or we can **reify** the category as an object, *Basketballs*. We could then say *Member(b, Basketballs)*, which we will abbreviate as $b \in Basketballs$, to say that b is a member of the category of basketballs. We say *Subset(Basketballs, Balls)*, abbreviated as $Basketballs \subset Balls$, to say that Basketballs is a **subcategory** of Balls. We will use subcategory, subclass, and subset interchangeably.

Categories serve to organize and simplify the knowledge base through **inheritance**. If we say that all instances of the category *Food* are edible, and if we assert that *Fruit* is a subclass of *Food* and *Apples* is a subclass of *Fruit*, then we can infer that every apple is edible. We say that the individual apples **inherit** the property of edibility, in this case from their membership in the *Food* category.

Subclass relations organize categories into a **taxonomy**, or **taxonomic hierarchy**. Taxonomies have been used explicitly for centuries in technical fields. The largest such taxonomy organizes about 10 million living and extinct species, many of them beetles, into a single hierarchy; library science has developed a taxonomy of all fields of knowledge, encoded as the Dewey Decimal system; and tax authorities and other government departments have developed extensive taxonomies of occupations and commercial products. Taxonomies are also an important aspect of general commonsense knowledge.

First-order logic makes it easy to state facts about categories, either by relating objects to categories or by quantifying over their members. Here are some types of facts, with examples of each:

- An object is a member of a category.

$$BB9 \in Basketballs$$

- A category is a subclass of another category.

$$Basketballs , \subset Balls$$

- All members of a category have some properties.

$$(x \in Basketballs) \Rightarrow Spherical\ (x)$$

- Members of a category can be recognized by some properties.

$$Orange(x) \wedge Round\ (x) \wedge Diameter(x)=9.5\_\_ \wedge x \in Balls \Rightarrow x \in Basketballs$$

- A category as a whole has some properties.

$$Dogs \in Domesticated\ Species$$

Notice that because Dogs is a category and is a member of *DomesticatedSpecies*, the latter must be a category of categories. Of course there are exceptions to many of the above rules (punctured basketballs are not spherical); we deal with these exceptions later.

Although subclass and member relations are the most important ones for categories, we also want to be able to state relations between categories that are not subclasses of each other. For example, if we just say that Males and Females are subclasses of Animals, then we have not said that a male cannot be a female. We say that two or more categories are **disjoint** if they have no members in common. And even if we know that males and females are disjoint, we will not know that an animal that is not a male must be a female, unless we say that males and females constitute an **exhaustive decomposition** of the animals. A disjoint exhaustive decomposition is known as a **partition**. The following examples illustrate these three concepts:

Disjoint({Animals,Vegetables})

ExhaustiveDecomposition({Americans,Canadians, Mexicans},

NorthAmericans)

Partition({Males, Females}, Animals).

(Note that the Exhaustive Decomposition of North Americans is not a Partition, because some people have dual citizenship.) The three predicates are defined as follows:

$Disjoint(s) \Leftrightarrow (\forall c1, c2\ c1 \in s \wedge c2 \in s \wedge c1 \_= c2 \Rightarrow Intersection(c1, c2)=\{\})$

$ExhaustiveDecomposition(s, c) \Leftrightarrow (\forall i\ i \in c \Leftrightarrow \exists c2\ c2 \in s \wedge i \in c2)$

$Partition(s, c) \Leftrightarrow Disjoint(s) \wedge Exhaustive\ Decomposition(s, c)$.

Categories can also be *defined* by providing necessary and sufficient conditions for membership. For example, a bachelor is an unmarried adult male:

$$x \in Bachelors \Leftrightarrow Unmarried(x) \wedge x \in Adults \wedge x \in Males.$$

16.4.1 Physical composition

The idea that one object can be part of another is a familiar one. One's nose is part of one's head, Romania is part of Europe, and this chapter is part of this book. We use the general PartOf relation to say that one thing is part of another. Objects can be grouped into part of hierarchies, reminiscent of the Subset hierarchy:

$PartOf\ (Bucharest\ ,\ Romania)$
$PartOf\ (Romania,\ EasternEurope)$
$PartOf\ (EasternEurope,\ Europe)$
$PartOf\ (Europe,\ Earth)$.

The *PartOf* relation is transitive and reflexive; that is,

$PartOf\ (x, y) \wedge PartOf\ (y, z) \Rightarrow PartOf\ (x, z)$.
$PartOf\ (x, x)$.

Therefore, we can conclude *PartOf(Bucharest, Earth)*. Categories of **composite objects** are often characterized by structural relations among parts. For example, a biped has two legs attached to a body:

$$\begin{aligned} Biped(a) \Rightarrow \quad & \exists l1, l2, b\ Leg(l1) \wedge Leg(l2) \wedge Body(b) \wedge \\ & PartOf\ (l1, a) \wedge PartOf\ (l2, a) \wedge PartOf\ (b, a) \wedge \\ & Attached(l1, b) \wedge Attached(l2, b) \wedge \\ & l1 \neq l2 \wedge [\forall l3\ Leg(l3) \wedge PartOf\ (l3, a) \Rightarrow (l3 = l1 \vee l3 = l2)]\ . \end{aligned}$$

The notation for "exactly two" is a little awkward; we are forced to say that there are two legs, that they are not the same, and that if anyone proposes a third leg, it must be the same as

one of the other two. We can define a *PartPartition* relation analogous to the Partition relation for categories. An object is composed of the parts in its *PartPartition* and can be viewed as deriving some properties from those parts. For example, the mass of a composite object is the sum of the masses of the parts. Notice that this is not the case with categories, which have no mass, even though their elements might.

It is also useful to define composite objects with definite parts but no particular structure. For example, we might want to say "The apples in this bag weigh two pounds." The temptation would be to ascribe this weight to the *set* of apples in the bag, but this would be a mistake because the set is an abstract mathematical concept that has elements but does not BUNCH have weight. Instead, we need a new concept, which we will call a **bunch**. For example, if the apples are Apple1, Apple2, and Apple3, then

$$BunchOf(\{Apple1, Apple2, Apple3\})$$

denotes the composite object with the three apples as parts (not elements). We can then use the bunch as a normal, albeit unstructured, object. Notice that $BunchOf(\{x\}) = x$. Furthermore, $BunchOf(Apples)$ is the composite object consisting of all apples—not to be confused with Apples, the category or set of all apples. These axioms are an example of a general technique called **logical minimization**, which means defining an object as the smallest one satisfying certain conditions.

16.4.2 Measurements

In both scientific and commonsense theories of the world, objects have height, mass, cost, and so on. The values that we assign for these MEASURE properties are called **measures**. Ordinary quantitative measures are quite easy to represent. We imagine that the universe includes abstract "measure objects," such as the *length* that is the length of a line segment. We can call this length 1.5 inches or 3.81 centimeters. Thus, the same length has different names in our language.We represent the length with a **units function** that takes a number as argument. If the line segment is called L1, we can write

$$Length(L1) = Inches(1.5) = Centimeters(3.81) .$$

Conversion between units is done by equating multiples of one unit to another:

$$Centimeters(2.54 \times d) = Inches(d) .$$

Similar axioms can be written for pounds and kilograms, seconds and days, and dollars and cents. Measures can be used to describe objects as follows:

$$Diameter(Basketball\ 12) = Inches(9.5) .$$

$$ListPrice(Basketball\ 12) = \$(19) .$$

$$d \in Days \Rightarrow Duration(d) = Hours(24) .$$

Note that $(1) is *not* a dollar bill! One can have two dollar bills, but there is only one object named $(1). Note also that, while *Inches(0)* and *Centimeters(0)* refer to the same zero length, they are not identical to other zero measures, such as *Seconds(0)*.

Simple, quantitative measures are easy to represent. Other measures present more of a problem, because they have no agreed scale of values. Exercises have difficulty, desserts have deliciousness, and poems have beauty, yet numbers cannot be assigned to these qualities. One might, in a moment of pure accountancy, dismiss such properties as useless for the purpose of logical reasoning; or, still worse, attempt to impose a numerical scale on beauty. This would be a grave mistake, because it is unnecessary. The most important aspect of measures is not the particular numerical values, but the fact that measures can be *ordered*.

Although measures are not numbers, we can still compare them, using an ordering symbol such as >. These sorts of monotonic relationships among measures form the basis for the field of **qualitative physics**, a subfield of AI that investigates how to reason about physical systems without plunging into detailed equations and numerical simulations.

16.5 Events

Situation calculus is limited in its applicability: it was designed to describe a world in which actions are discrete, instantaneous, and happen one at a time. Consider a continuous action, such as filling a bathtub. Situation calculus can say that the tub is empty before the action and full when the action is done, but it can't talk about what happens *during* the action. It also can't describe two actions happening at the same time—such as brushing one's teeth while waiting for the tub to fill. To handle such cases we introduce an alternative formalism known as **event calculus**, which is based on points of time rather than on situations.

Event calculus reifies fluents and events. The fluent *At(Shankar, Berkeley)* is an object that refers to the fact of Shankar being in Berkeley, but does not by itself say anything about whether it is true. To assert that a fluent is actually true at some point in time we use the predicate *T*, as in *T(At(Shankar, Berkeley), t)*. Events are described as instances of event categories. The event E1 of Shankar flying from San Francisco to Washington, D.C. is described as

$$E1 \in Flyings \wedge Flyer(E1, Shankar) \wedge Origin(E1, SF) \wedge Destination(E1, DC) .$$

We then use *Happens(E1, i)* to say that the event *E1* took place over the time interval *i*, and we say the same thing in functional form with *Extent(E1) = i*. We represent time intervals by a (start, end) pair of times; that is, *i = (t1, t2)* is the time interval that starts at *t1* and ends at *t2*. The complete set of predicates for one version of the event calculus is

T(f, t) Fluent f is true at time t

Happens(e, i) Event e happens over the time interval i

Initiates(e, f, t) Event e causes fluent f to start to hold at time t

Terminates(e, f, t) Event e causes fluent f to cease to hold at time t

Clipped(f, i) Fluent f ceases to be true at some point during time interval i

Restored (f, i) Fluent f becomes true sometime during time interval i

We assume a distinguished event, Start , that describes the initial state by saying which fluents are initiated or terminated at the start time. We define *T* by saying that a fluent holds at a point in time if the fluent was initiated by an event at some time in the past and was not made false (clipped) by an intervening event. A fluent does not hold if it was terminated by an event and not made true (restored) by another event. Formally, the axioms are:

$$Happens(e, (t1, t2)) \wedge Initiates(e, f, t1) \wedge \neg Clipped(f, (t1, t)) \wedge t1 < t \Rightarrow T(f, t)$$

$$Happens(e, (t1, t2)) \wedge Terminates(e, f, t1) \wedge \neg Restored(f, (t1, t)) \wedge t1 < t \Rightarrow \neg T(f, t)$$

We can extend event calculus to make it possible to represent simultaneous events (such as two people being necessary to ride a seesaw), exogenous events (such as the wind blowing and changing the location of an object), continuous events (such as the level of water in the bathtub continuously rising) and other complications.

16.5.1 Processes

The events we have seen so far are what we call **discrete events**—they have a definite structure. Shankar's trip has a beginning, middle, and end. If interrupted halfway, the event would be something different—it would not be a trip from San Francisco to Washington, but instead a trip from San Francisco to somewhere over Kansas. On the other hand, the category of events denoted by *Flyings* has a different quality. If we take a small interval of Shankar's flight, say, the third 20-minute segment (while he waits anxiously for a bag of peanuts), that event is still a member of Flyings. In fact, this is true for any subinterval. Categories of events with this property are called **process** categories or **liquid event** categories. Any process e that happens over an interval also happens over any subinterval:

$$(e \in Processes) \wedge Happens(e, (t1, t4)) \wedge (t1 < t2 < t3 < t4) \Rightarrow Happens(e, (t2, t3)) .$$

The distinction between liquid and nonliquid events is exactly analogous to the difference between substances, or *stuff*, and individual objects, or *things*. In fact, some have called liquid events **temporal substances**, whereas substances like butter are **spatial substances**.

16.5.2 Time intervals

Event calculus opens us up to the possibility of talking about time, and time intervals. We will consider two kinds of time intervals: moments and extended intervals. The distinction is that only moments have zero duration:

$$Partition(\{Moments, ExtendedIntervals\}, Intervals)$$

$$i \in Moments \Leftrightarrow Duration(i) = Seconds(0) .$$

Next we invent a time scale and associate points on that scale with moments, giving us absolute times. The time scale is arbitrary; we measure it in seconds and say that the moment at midnight (GMT) on January 1, 1900, has time 0. The functions Begin and End pick out the earliest and latest moments in an interval, and the function Time delivers the point on the time scale for a moment. The function Duration gives the difference between the end time and the start time.

$$Interval(i) \Rightarrow Duration(i) = (Time(End(i)) - Time(Begin(i))) .$$

$$Time(Begin(AD1900)) = Seconds(0) .$$

$$Time(Begin(AD2001)) = Seconds(3187324800) .$$

Time(End(AD2001))=Seconds(3218860800) .

Duration(AD2001)=Seconds(31536000) .

To make these numbers easier to read, we also introduce a function Date, which takes six arguments (hours, minutes, seconds, day, month, and year) and returns a time point:

Time(Begin(AD2001))=Date(0, 0, 0, 1, Jan, 2001)

Date(0, 20, 21, 24, 1, 1995)=Seconds(3000000000) .

Two intervals Meet if the end time of the first equals the start time of the second. The complete set of interval relations, as proposed by Allen (1983), is shown graphically in Figure 16.2 and logically below:

Meet(i, j) ⇔ End(i)=Begin(j)

Before(i, j) ⇔ End(i) < Begin(j)

After (j, i) ⇔ Before(i, j)

During(i, j) ⇔ Begin(j) < Begin(i) < End(i) < End(j)

Overlap(i, j) ⇔ Begin(i) < Begin(j) < End(i) < End(j)

Begins(i, j) ⇔ Begin(i) = Begin(j)

Finishes(i, j) ⇔ End(i) = End(j)

Equals(i, j) ⇔ Begin(i) = Begin(j) ∧ End(i) = End(j)

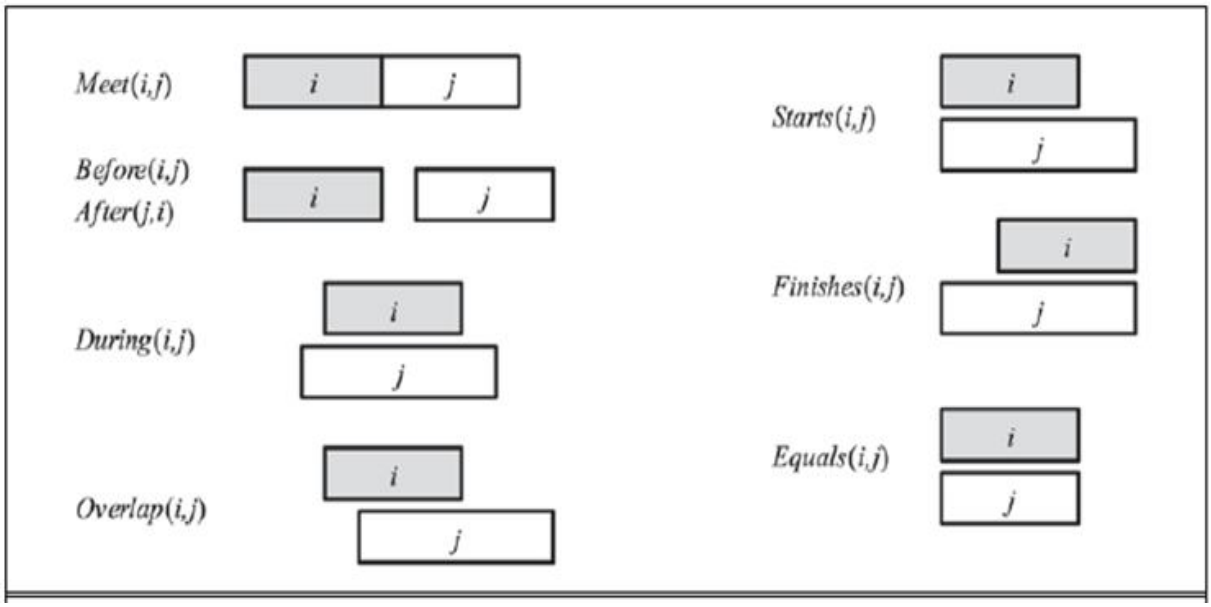


Figure 16.2 Predicates on time intervals

These all have their intuitive meaning, with the exception of *Overlap*: we tend to think of overlap as symmetric (if *i* overlaps *j* then *j* overlaps *i*), but in this definition, *Overlap(i, j)* only holds if *i* begins before *j*. To say that the reign of Elizabeth II immediately followed that of George VI, and the reign of Elvis overlapped with the 1950s, we can write the following:

Meets(ReignOf (GeorgeVI), ReignOf (ElizabethII)) .

Overlap(Fifties, ReignOf (Elvis)) .

Begin(Fifties)=Begin(AD1950) .

End(Fifties)=End(AD1959) .

16.6 Mental Events and Mental Objects

The agents we have constructed so far have beliefs and can deduce new beliefs. Yet none of them has any knowledge *about* beliefs or *about* deduction. Knowledge about one's own knowledge and reasoning processes is useful for controlling inference. For example, suppose Alice asks "what is the square root of 1764" and Bob replies "I don't know." If Alice insists "think harder," Bob should realize that with some more thought, this question can in fact be answered. On the other hand, if the question were "Is your mother sitting down right now?" then Bob should realize that thinking harder is unlikely to help. Knowledge about the knowledge of other agents is also important; Bob should realize that his mother knows whether she is sitting or not, and that asking her would be a way to find out.

What we need is a model of the mental objects that are in someone's head (or something's knowledge base) and of the mental processes that manipulate those mental objects. The model does not have to be detailed. We do not have to be able to predict how many milliseconds it will take for a particular agent to make a deduction. We will be happy just to be able to conclude that mother knows whether or not she is sitting.

We begin with the **propositional attitudes** that an agent can have toward mental objects: attitudes such as Believes, Knows, Wants, Intends, and Informs. The difficulty is that these attitudes do not behave like "normal" predicates. For example, suppose we try to assert that Lois knows that Superman can fly:

Knows(Lois, CanFly(Superman)) .

One minor issue with this is that we normally think of *CanFly(Superman)* as a sentence, but here it appears as a term. That issue can be patched up just be reifying *CanFly(Superman)*; making it a fluent. A more serious problem is that, if it is true that Superman is Clark Kent, then we must conclude that Lois knows that Clark can fly:

$$(Superman = Clark) \wedge Knows(Lois\ ,\ CanFly(Superman))$$

$$\models Knows(Lois,\ CanFly(Clark\ ))\ .$$

This is a consequence of the fact that equality reasoning is built into logic. Normally that is a good thing; if our agent knows that 2 + 2 = 4 and 4 < 5, then we want our agent to know that 2 + 2 < 5. This property is called **referential transparency**—it doesn't matter what term a logic uses to refer to an object, what matters is the object that the term names. But for propositional attitudes like *believes* and *knows*, we would like to have referential opacity—the terms used *do* matter, because not all agents know which terms are co-referential.

Modal logic is designed to address this problem. Regular logic is concerned with a single modality, the modality of truth, allowing us to express "*P* is true." Modal logic includes special modal operators that take sentences (rather than terms) as arguments. For example, "*A* knows *P*" is represented with the notation **K**AP, where **K** is the modal operator for knowledge. It takes two arguments, an agent (written as the subscript) and a sentence. The syntax of modal logic is the same as first-order logic, except that sentences can also be formed with modal operators.

Modal logic solves some tricky issues with the interplay of quantifiers and knowledge. The English sentence "Bond knows that someone is a spy" is ambiguous. The first reading is that there is a particular someone who Bond knows is a spy; we can write this as

$$\exists\ x\ \boldsymbol{K}BondSpy(x),$$

which in modal logic means that there is an x that, in all accessible worlds, Bond knows to be a spy. The second reading is that Bond just knows that there is at least one spy:

$$\boldsymbol{K}Bond\ \exists\ x\ Spy(x)\ .$$

The modal logic interpretation is that in each accessible world there is an x that is a spy, but it need not be the same x in each world.

However, one problem with the modal logic approach is that it assumes **logical omniscience** on the part of agents. That is, if an agent knows a set of axioms, then it knows all consequences of those axioms. This is on shaky ground even for the somewhat abstract notion of knowledge, but it seems even worse for belief, because belief has more connotation of referring to things that are physically represented in the agent, not just potentially derivable. There have been attempts to define a form of limited rationality for agents; to say that agents believe those assertions that can be derived with the application of no more than *k* reasoning steps, or no more than *s* seconds of computation. These attempts have been generally unsatisfactory.

16.7 Reasoning Systems for Categories

Categories are the primary building blocks of large-scale knowledge representation schemes. This section describes systems specially designed for organizing and reasoning with categories. There are two closely related families of systems: **semantic networks** provide graphical aids for visualizing a knowledge base and efficient algorithms for inferring properties of an object on the basis of its category membership; and **description logics** provide a formal language for constructing and combining category definitions and efficient algorithms for deciding subset and superset relationships between categories.

16.7.1 Semantic Networks

Semantic networks provides graphical aids for visualizing a knowledge base in patterns of interconnected nodes and arcs. It has an efficient algorithms for inferring of and object on the basis of its category membership. In 1909 Charles Peirce proposed a graphical notation of nodes and arcs called existential graphs that is a visual notation for logical expressions. A typical graphical notation displays objects or categories names in oval or boxes and connects them with labeled arcs. For example, a *MemberOf* link between Mary and female persons corresponds to the logical assertion *Mary E (element)* of female person. We can connect categories using *SubsetOf* of links. A single box used to assert properties of every member of a category.

Figure 16.3 has a *MemberOf* link between *Mary* and *FemalePersons*, corresponding to the logical assertion *Mary* $\in$ *FemalePersons*; similarly, the *SisterOf* link between *Mary* and *John* corresponds to the assertion *SisterOf (Mary, John)*. We can connect categories using *SubsetOf* links, and so on. It is such fun drawing bubbles and arrows that one can get carried away. For example, we know that persons have female persons as mothers, so can we draw a *HasMother* link from *Persons* to *FemalePersons*? The answer is no, because *HasMother* is a

relation between a person and his or her mother, and categories do not have mothers. For this reason, we have used a special notation—the double-boxed link—in Figure 16.3.

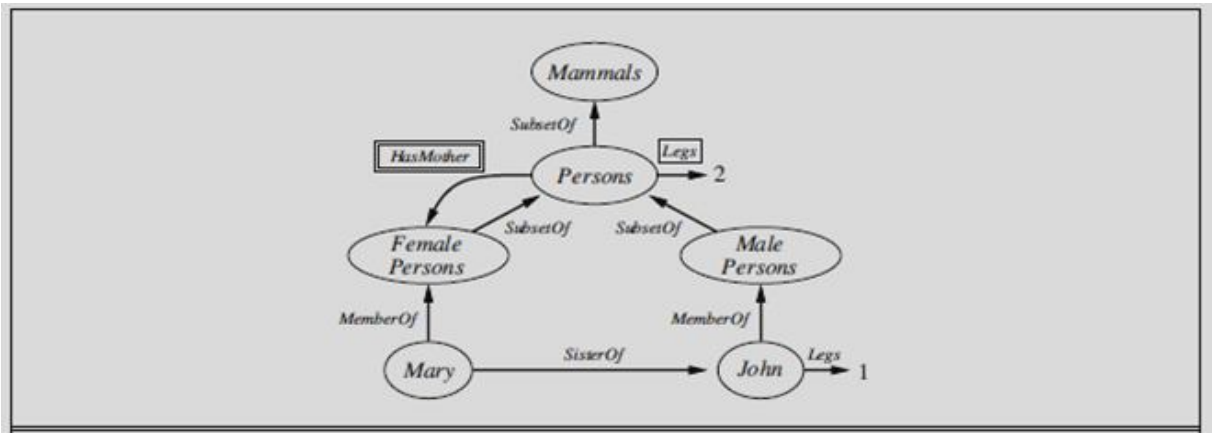


Figure 16.3 A semantic network with four objects (John, Mary, 1, and 2)

Semantic networks Semantic network notation makes it very convenient to perform inheritance reasoning. For example by virtue of being a person marry inherits the property of having two legs. The simplicity and efficiency of this inference mechanism, compared with logical theorem has been one of the main attractions of semantic networks. Inheritance becomes complicated when an object belong to more than one category. And also when a category can be a subset of more than one other category which is called multiple inheritance.

Semantic networks multiple inheritance can cause the algorithm to find two or more conflicting values answering a query. For this reason multiple inheritance is banned in some object oriented programming like Java. Another form of inference is the use of inverse links. For example *HasSister* is the inverse of *SisterOf*. Its ok to have inverse links as long as they are made into objects in their own right. Semantic networks provide direct indexing only for objects, categories and the links emanating from them. The drawback is that links between bubbles represent only one binary relation.

One of the most important aspects of semantic networks is their ability to represent **default values** for categories. Examining Figure 16.3 carefully, one notices that John has one leg, despite the fact that he is a person and all persons have two legs. In a strictly logical KB, this would be a contradiction, but in a semantic network, the assertion that all persons have two legs has only default status; that is, a person is assumed to have two legs unless this is contradicted by more specific information. The default semantics is enforced naturally by the inheritance algorithm, because it follows links upwards from the object itself (John in this case) and stops

as soon as it finds a value. We say that the default is **overridden** by the more specific value. Notice that we could also override the default number of legs by creating a category of *OneLeggedPersons,* a subset of *Persons* of which *John* is a member.

16.7.2 Description Logics

The syntax of first-order logic is designed to make it easy to say things about objects. **Description logics** are notations that are designed to make it easier to describe definitions and properties of categories. Description logic systems evolved from semantic networks in response to pressure to formalize what the networks mean while retaining the emphasis on taxonomic structure as an organizing principle.

The principal inference tasks for description logics are **subsumption** (checking if one category is a subset of another by comparing their definitions) and **classification** (checking whether an object belongs to a category). Some systems also include **consistency** of a category definition—whether the membership criteria are logically satisfiable. The CLASSIC language (Borgida *et al.*, 1989) is a typical description logic. The syntax of CLASSIC descriptions is shown in Figure 16.4. For example, to say that bachelors are unmarried adult males we would write

$$Bachelor = And(Unmarried, Adult, Male) .$$

The equivalent in first-order logic would be

$$Bachelor(x) \Leftrightarrow Unmarried(x) \wedge Adult(x) \wedge Male(x) .$$

$$
\begin{array}{rcl}
Concept & \rightarrow & \textbf{Thing} \mid ConceptName \\
& \mid & \textbf{And}(Concept, \ldots) \\
& \mid & \textbf{All}(RoleName, Concept) \\
& \mid & \textbf{AtLeast}(Integer, RoleName) \\
& \mid & \textbf{AtMost}(Integer, RoleName) \\
& \mid & \textbf{Fills}(RoleName, IndividualName, \ldots) \\
& \mid & \textbf{SameAs}(Path, Path) \\
& \mid & \textbf{OneOf}(IndividualName, \ldots) \\
Path & \rightarrow & [RoleName, \ldots]
\end{array}
$$

Figure 16.4 The syntax of descriptions in a subset of the CLASSIC language

Notice that the description logic has an an algebra of operations on predicates, which of course we can't do in first-order logic. Any description in CLASSIC can be translated into an equivalent first-order sentence, but some descriptions are more straightforward in CLASSIC. Perhaps the most important aspect of description logics is their emphasis on tractability of inference. A problem instance is solved by describing it and then asking if it is subsumed by one of several possible solution categories. In standard first-order logic systems, predicting the solution time is often impossible. It is frequently left to the user to engineer the representation to detour around sets of sentences that seem to be causing the system to take several weeks to solve a problem. The thrust in description logics, on the other hand, is to ensure that subsumption-testing can be solved in time polynomial in the size of the descriptions.

16.8 Reasoning with Default Information

In the preceding section, we saw a simple example of an assertion with default status: people have two legs. This default can be overridden by more specific information, such as that Long John Silver has one leg. We saw that the inheritance mechanism in semantic networks implements the overriding of defaults in a simple and natural way. In this section, we study defaults more generally, with a view toward understanding the *semantics* of defaults rather than just providing a procedural mechanism.

16.8.1 Circumscription and default logic

In the previous section, we saw that a property inherited by all members of a category in a semantic network could be overridden by more specific information for a subcategory. Simple introspection suggests that these failures of monotonicity are widespread in commonsense reasoning. It seems that humans often "jump to conclusions." For example, when one sees a car parked on the street, one is normally willing to believe that it has four wheels even though only three are visible. Now, probability theory can certainly provide a conclusion that the fourth wheel exists with high probability, yet, for most people, the possibility of the car's not having four wheels *does not arise unless some new evidence presents itself*. Thus, it seems that the four-wheel conclusion is reached *by default*, in the absence of any reason to doubt it. If new evidence arrives—for example, if one sees the owner carrying a wheel and notices that the car is jacked up—then the conclusion can be retracted. This kind of reasoning is said to exhibit **nonmonotonicity**, because the set of beliefs does not grow monotonically over time as new evidence arrives. **Nonmonotonic logics** have been devised with modified notions of truth and

entailment in order to capture such behavior. We will look at two such logics that have been studied extensively: **circumscription** and **default logic**.

Circumscription can be seen as a more powerful and precise version of the closedworld assumption. The idea is to specify particular predicates that are assumed to be "as false as possible"—that is, false for every object except those for which they are known to be true. For example, suppose we want to assert the default rule that birds fly. We would introduce a predicate, say *Abnormal1(x),* and write

$$Bird(x) \land \neg Abnormal1(x) \Rightarrow Flies(x) .$$

If we say that Abnormal1 is to be **circumscribed**, a circumscriptive reasoner is entitled to assume $\neg Abnormal1$(x) unless *Abnormal 1(x)* is known to be true. This allows the conclusion

Flies(Tweety) to be drawn from the premise *Bird(Tweety)*, but the conclusion no longer holds if *Abnormal1(Tweety)* is asserted.

Circumscription can be viewed as an example of a **model preference** logic. In such logics, a sentence is entailed (with default status) if it is true in all *preferred* models of the KB, as opposed to the requirement of truth in *all* models in classical logic. For circumscription, one model is preferred to another if it has fewer abnormal objects.

Default logic is a formalism in which **default rules** can be written to generate contingent, nonmonotonic conclusions. A default rule looks like this:

$$Bird(x) : Flies(x)/Flies(x).$$

This rule means that if Bird(x) is true, and if Flies(x) is consistent with the knowledge base, then Flies(x) may be concluded by default. In general, a default rule has the form

$$P : J_1, \ldots, J_n/C$$

where *P* is called the prerequisite, *C* is the conclusion, and Ji are the justifications—if any one of them can be proven false, then the conclusion cannot be drawn. Any variable that appears in J_i or *C* must also appear in *P*. The Nixon-diamond example can be represented in default logic with one fact and two default rules:

Republican(Nixon) $\wedge$ *Quaker(Nixon)* .

Republican(x) : $\neg$*Pacifist (x)*/$\neg$*Pacifist (x)* .

Quaker(x) : *Pacifist (x)*/*Pacifist (x)* .

16.8.2 Truth Maintenance Systems

We have seen that many of the inferences drawn by a knowledge representation system will have only default status, rather than being absolutely certain. Inevitably, some of these inferred facts will turn out to be wrong and will have to be retracted in the face of new information. This process is called **belief revision**. Suppose that a knowledge base KB contains a sentence P—perhaps a default conclusion recorded by a forward-chaining algorithm, or perhaps just an incorrect assertion—and we want to execute *TELL(KB,* $\neg$*P).* To avoid creating a contradiction, we must first execute *RETRACT(KB, P)*. This sounds easy enough.

Problems arise, however, if any *additional* sentences were inferred from P and asserted in the KB. For example, the implication P $\Rightarrow$ Q might have been used to add Q. The obvious "solution"—retracting all sentences inferred from P—fails because such sentences may have other justifications besides P. For example, if R and R $\Rightarrow$ Q are also in the KB, then Q does not have to be removed after all. **Truth maintenance systems**, or TMSs, are designed to handle exactly these kinds of complications.

One simple approach to truth maintenance is to keep track of the order in which sentences are told to the knowledge base by numbering them from P1 to Pn. When the call *RETRACT(KB, Pi)* is made, the system reverts to the state just before Pi was added, thereby removing both Pi and any inferences that were derived from Pi. The sentences Pi+1 through Pn can then be added again. This is simple, and it guarantees that the knowledge base will be consistent, but retracting Pi requires retracting and reasserting *n " i* sentences as well as undoing and redoing all the inferences drawn from those sentences. For systems to which many facts are being added—such as large commercial databases—this is impractical.

A more efficient approach JTMS is the **justification-based truth maintenance system**, or **JTMS**. In a JTMS, each sentence in the knowledge base is annotated with a **justification** consisting of the set of sentences from which it was inferred. For example, if the knowledge base already contains P $\Rightarrow$ Q, then *TELL(P)* will cause Q to be added with the justification {P, P $\Rightarrow$ Q}. In general, a sentence can have any number of justifications. Justifications make retraction efficient. Given the call *RETRACT(P)*, the JTMS will delete exactly those sentences

for which P is a member of every justification. So, if a sentence Q had the single justification {P, P $\Rightarrow$ Q}, it would be removed; if it had the additional justification {P, P V R $\Rightarrow$ Q}, it would still be removed; but if it also had the justification {R, P V R $\Rightarrow$ Q}, then it would be spared. In this way, the time required for retraction of P depends only on the number of sentences derived from P rather than on the number of other sentences added since P entered the knowledge base.

The JTMS assumes that sentences that are considered once will probably be considered again, so rather than deleting a sentence from the knowledge base entirely when it loses all justifications, we merely mark the sentence as being *out* of the knowledge base. If a subsequent assertion restores one of the justifications, then we mark the sentence as being back *in*. In this way, the JTMS retains all the inference chains that it uses and need not rederive sentences when a justification becomes valid again.

An **assumption-based truth maintenance system**, or **ATMS**, makes this type of context switching between hypothetical worlds particularly efficient. In a JTMS, the maintenance of justifications allows you to move quickly from one state to another by making a few retractions and assertions, but at any time only one state is represented. An ATMS represents *all* the states that have ever been considered at the same time. Whereas a JTMS simply labels each sentence as being *in* or *out*, an ATMS keeps track, for each sentence, of which assumptions would cause the sentence to be true. In other words, each sentence has a label that consists of a set of assumption sets. The sentence holds just in those cases in which all the assumptions in one of the assumption sets hold.

16.9 Summary

Large-scale knowledge representation requires a general-purpose ontology to organize and tie together the various specific domains of knowledge. A general-purpose ontology needs to cover a wide variety of knowledge and should be capable, in principle, of handling any domain. Building a large, general-purpose ontology is a significant challenge that has yet to be fully realized, although current frameworks seem to be quite robust. We presented an **upper ontology** based on categories and the event calculus. We covered categories, subcategories, parts, structured objects, measurements, substances, events, time and space, change, and beliefs. Natural kinds cannot be defined completely in logic, but properties of natural kinds can be represented.

Actions, events, and time can be represented either in situation calculus or in more expressive representations such as event calculus. Such representations enable an agent to construct plans by logical inference. We presented a detailed analysis of the Internet shopping domain, exercising the general ontology and showing how the domain knowledge can be used by a shopping agent. Special-purpose representation systems, such as **semantic networks** and **description logics**, have been devised to help in organizing a hierarchy of categories. **Inheritance** is an important form of inference, allowing the properties of objects to be deduced from their membership in categories. **Nonmonotonic logics**, such as **circumscription** and **default logic**, are intended to capture default reasoning in general. **Truth maintenance systems** handle knowledge updates and revisions efficiently.

16.10 Model Questions

1. Discuss Ontological engineering.
2. Write short notes on (i) Categories (ii) Objects (iii) Events.
3. What is meant by knowledge representation.
4. Explain about Semantic Networks.
5. What are Truth Maintenance Systems? Explain the types of TMS.

LESSON 17
CLASSICAL PLANNING

Structure

17.1 Introduction

We have defined AI as the study of rational action, which means that **planning**—devising a plan of action to achieve one's goals—is a critical part of AI. In this lesson we introduce a representation for planning problems that scales up to problems that could not be handled by those earlier approaches. AI develops an expressive yet carefully constrained language for representing planning problems. This lesson also shows how forward and backward search algorithms can take advantage of this representation, primarily through accurate heuristics that can be derived automatically from the structure of the representation.

17.2 Objectives

- To study the concepts behind classical planning.
- To discuss the algorithms for classical planning.
- To study the heuristics for planning.

17.3 Definition of Classical Planning

Actions are described by a set of action schemas that implicitly define the ACTIONS(s) and RESULT(s, a) functions needed to do a problem-solving search. Any system for action description needs to solve the frame problem—to say what changes and what stays the same as the result of the action. **Classical planning** concentrates on problems where most actions leave most things unchanged. Think of a world consisting of a bunch of objects on a flat surface.

The action of nudging an object causes that object to change its location by a vector Ä. A concise description of the action should mention only Ä; it shouldn't have to mention all the objects that stay in place. PDDL (Planning Domain Definition Language) does that by specifying the result of an action in terms of what changes; everything that stays the same is left unmentioned.

A set of ground (variable-free) actions can be represented by a single **action schema**. The schema is a **lifted** representation—it lifts the level of reasoning from propositional logic to a restricted subset of first-order logic. For example, here is an action schema for flying a plane from one location to another:

Action(Fly(p, from, to),
PRECOND:At(p, from) ∧ Plane(p) ∧ Airport (from) ∧ Airport (to)
EFFECT:¬At(p, from) ∧ At(p, to))

The schema consists of the action name, a list of all the variables used in the schema, a **precondition** and an **effect**. Although we haven't said yet how the action schema converts into logical sentences, think of the variables as being universally quantified. We are free to choose whatever values we want to instantiate the variables. For example, here is one ground action that results from substituting values for all the variables:

Action(Fly(P1, SFO, JFK),
PRECOND:At(P1, SFO) ∧ Plane(P1) ∧ Airport (SFO) ∧ Airport (JFK)
EFFECT:¬At(P1, SFO) ∧ At(P1, JFK))

The precondition and effect of an action are each conjunctions of literals (positive or negated atomic sentences). The precondition defines the states in which the action can be executed, and the effect defines the result of executing the action.

A set of action schemas serves as a definition of a planning *domain*. A specific *problem* within the domain is defined with the addition of an initial state and a goal. The **initial state** is a conjunction of ground atoms. The **goal** is just like a precondition: a conjunction of literals (positive or negative) that may contain variables, such as *At(p, SFO) ∧ Plane(p)*. Any variables are treated as existentially quantified, so this goal is to have *any* plane at SFO. The problem is solved when we can find a sequence of actions that end in a state s that entails the goal. For example, the state *Rich ∧ Famous ∧ Miserable* entails the goal *Rich ∧ Famous*, and the state *Plane(Plane1) ∧ At(Plane1, SFO)* entails the goal *At(p, SFO) ∧ Plane(p)*. Now we have

defined planning as a search problem: we have an initial state, an ACTIONS function, a RESULT function, and a goal test. We'll look at some example problems before investigating efficient search algorithms.

17.3.1 Example: Air cargo transport

Figure 17.1 shows an air cargo transport problem involving loading and unloading cargo and flying it from place to place. The problem can be defined with three actions: *Load , Unload,* and *Fly*. The actions affect two predicates: *In(c, p)* means that cargo c is inside plane p, and *At(x, a)* means that object x (either plane or cargo) is at airport a. Note that some care must be taken to make sure the *At* predicates are maintained properly. When a plane flies from one airport to another, all the cargo inside the plane goes with it. In first-order logic it would be easy to quantify over all objects that are inside the plane. But basic PDDL does not have a universal quantifier, so we need a different solution. The approach we use is to say that a piece of cargo ceases to be *At* anywhere when it is In a plane; the cargo only becomes At the new airport when it is unloaded. So *At* really means "available for use at a given location." The following plan is a solution to the problem:

[Load (C1, P1, SFO), Fly(P1, SFO, JFK),Unload(C1, P1, JFK),

Load (C2, P2, JFK), Fly(P2, JFK, SFO),Unload(C2, P2, SFO)] .

Finally, there is the problem of spurious actions such as *Fly(P1, JFK, JFK)*, which should be a no-op, but which has contradictory effects (according to the definition, the effect would include $At(P1, JFK) \wedge \neg At(P1, JFK)$). It is common to ignore such problems, because they seldom cause incorrect plans to be produced. The correct approach is to add inequality preconditions saying that the from and to airports must be different.

$Init(At(C_1, SFO) \wedge At(C_2, JFK) \wedge At(P_1, SFO) \wedge At(P_2, JFK)$
$\wedge Cargo(C_1) \wedge Cargo(C_2) \wedge Plane(P_1) \wedge Plane(P_2)$
$\wedge Airport(JFK) \wedge Airport(SFO))$
$Goal(At(C_1, JFK) \wedge At(C_2, SFO))$
$Action(Load(c, p, a),$
PRECOND: $At(c, a) \wedge At(p, a) \wedge Cargo(c) \wedge Plane(p) \wedge Airport(a)$
EFFECT: $\neg At(c, a) \wedge In(c, p))$
$Action(Unload(c, p, a),$
PRECOND: $In(c, p) \wedge At(p, a) \wedge Cargo(c) \wedge Plane(p) \wedge Airport(a)$
EFFECT: $At(c, a) \wedge \neg In(c, p))$
$Action(Fly(p, from, to),$
PRECOND: $At(p, from) \wedge Plane(p) \wedge Airport(from) \wedge Airport(to)$
EFFECT: $\neg At(p, from) \wedge At(p, to))$

Figure 17.1 A PDDL description of an air cargo transportation planning problem

17.3.2 The complexity of classical planning

In this section, we consider the theoretical complexity of planning and distinguish two decision problems. **PlanSAT** is the question of whether there exists any plan that solves a planning problem. **Bounded PlanSAT** asks whether there is a solution of length *k* or less; this can be used to find an optimal plan.

The first result is that both decision problems are decidable for classical planning. The proof follows from the fact that the number of states is finite. But if we add function symbols to the language, then the number of states becomes infinite, and PlanSAT becomes only semidecidable: an algorithm exists that will terminate with the correct answer for any solvable problem, but may not terminate on unsolvable problems. The Bounded PlanSAT problem remains decidable even in the presence of function symbols. For proofs of the assertions in this section, see Ghallab *et al.* (2004).

Both PlanSAT and Bounded PlanSAT are in the complexity class PSPACE, a class that is arger (and hence more difficult) than NP and refers to problems that can be solved by a deterministic Turing machine with a polynomial amount of space. Even if we make some rather severe restrictions, the problems remain quite difficult. For example, if we disallow negative effects, both problems are still NP-hard. However, if we also disallow negative preconditions, PlanSAT reduces to the class *P*.

These worst-case results may seem discouraging. We can take solace in the fact that agents are usually not asked to find plans for arbitrary worst-case problem instances, but rather are asked for plans in specific domains (such as blocks-world problems with n blocks), which can be much easier than the theoretical worst case. For many domains (including the blocks world and the air cargo world), Bounded PlanSAT is NP-complete while PlanSAT is in *P*; in other words, optimal planning is usually hard, but sub-optimal planning is sometimes easy. To do well on easier-than-worst-case problems, we will need good search heuristics. That's the true advantage of the classical planning formalism: it has facilitated the development of very accurate domain-independent heuristics, whereas systems based on successor-state axioms in first-order logic have had less success in coming up with good heuristics.

17.4 Algorithms for Classical Planning

Now we turn our attention to planning algorithms. We saw how the description of a planning problem defines a search problem: we can search from the initial state through the space of

states, looking for a goal. One of the nice advantages of the declarative representation of action schemas is that we can also search backward from the goal, looking for the initial state. Figure 10.5 compares forward and backward searches.

17.4.1 Forward (progression) state-space search

Now that we have shown how a planning problem maps into a search problem, we can solve planning problems with any of the heuristic search algorithms or a local search algorithm. From the earliest days of planning research (around 1961) until around 1998 it was assumed that forward state-space search was too inefficient to be practical. It is not hard to come up with reasons why.

First, forward search is prone to exploring irrelevant actions. Consider the noble task of buying a copy of *AI: A Modern Approach* from an online bookseller. Suppose there is an action schema *Buy(isbn)* with effect *Own(isbn)*. ISBNs are 10 digits, so this action schema represents 10 billion ground actions. An uninformed forward-search algorithm would have to start enumerating these 10 billion actions to find one that leads to the goal.

Second, planning problems often have large state spaces. Consider an air cargo problem with 10 airports, where each airport has 5 planes and 20 pieces of cargo. The goal is to move all the cargo at airport A to airport B. There is a simple solution to the problem: load the 20 pieces of cargo into one of the planes at A, fly the plane to B, and unload the cargo. Finding the solution can be difficult because the average branching factor is huge: each of the 50 planes can fly to 9 other airports, and each of the 200 packages can be either unloaded (if it is loaded) or loaded into any plane at its airport (if it is unloaded). So in any state there is a minimum of 450 actions (when all the packages are at airports with no planes) and a maximum of 10,450 (when all packages and planes are at the same airport). On average, let's say there are about 2000 possible actions per state, so the search graph up to the depth of the obvious solution has about 200041 nodes.

Clearly, even this relatively small problem instance is hopeless without an accurate heuristic. Although many real-world applications of planning have relied on domain-specific heuristics, it turns out that strong domain-independent heuristics can be derived automatically; that is what makes forward search feasible.

17.4.2 Backward (regression) relevant-states search

In regression search we start at the goal and apply the actions backward until we find a sequence of steps that reaches the initial state. It is called **relevant-states** search because we only consider actions that are relevant to the goal (or current state). As in belief-state search, there is a *set* of relevant states to consider at each step, not just a single state.

We start with the goal, which is a conjunction of literals forming a description of a set of states—for example, the goal $\neg Poor \wedge Famous$ describes those states in which *Poor* is false, *Famous* is true, and any other fluent can have any value. If there are *n* ground fluents in a domain, then there are 2^n ground states (each fluent can be true or false), but 3^n descriptions of sets of goal states (each fluent can be positive, negative, or not mentioned).

In general, backward search works only when we know how to regress from a state description to the predecessor state description. For example, it is hard to search backwards for a solution to the n-queens problem because there is no easy way to describe the states that are one move away from the goal. Happily, the PDDL representation was designed to make it easy to regress actions—if a domain can be expressed in PDDL, then we can do regression search on it. Given a ground goal description g and a ground action a, the regression from g over a gives us a state description g' defined by

$$g' = (g - ADD(a)) \cup Precond(a).$$

That is, the effects that were added by the action need not have been true before, and also the preconditions must have held before, or else the action could not have been executed. Note that *DEL(a)* does not appear in the formula; that's because while we know the fluents in *DEL(a)* are no longer true after the action, we don't know whether or not they were true before, so there's nothing to be said about them.

To get the full advantage of backward search, we need to deal with partially uninstantiated actions and states, not just ground ones. For example, suppose the goal is to deliver a specific piece of cargo to SFO: *At(C2, SFO)*. That suggests the action *Unload(C2, p', SFO)*:

Action(Unload(C2, p', SFO),
PRECOND:In(C2, p') ∧ At(p', SFO) ∧ Cargo(C2) ∧ Plane(p') ∧ Airport(SFO)
EFFECT:At(C2, SFO)"" ¬ In(C2, p') .

This represents unloading the package from an *unspecified* plane at SFO; any plane will do, but we need not say which one now. We can take advantage of the power of first-order representations: a single description summarizes the possibility of using *any* of the planes by implicitly quantifying over p'. The regressed state description is

$$g' = In(C2, p') \land At(p', SFO) \land Cargo(C2) \land Plane(p') \land Airport(SFO) .$$

The final issue is deciding which actions are candidates to regress over. In the forward direction we chose actions that were **applicable**—those actions that could be the next step in the plan. In backward search we want actions that are **relevant**—those actions that could be the *last* step in a plan leading up to the current goal state.

Backward search keeps the branching factor lower than forward search, for most problem domains. However, the fact that backward search uses state sets rather than individual states makes it harder to come up with good heuristics. That is the main reason why the majority of current systems favor forward search.

17.4.3 Heuristics for planning

Neither forward nor backward search is efficient without a good heuristic function. A heuristic function *h(s)* estimates the distance from a state s to the goal and that if we can derive an **admissible** heuristic for this distance—one that does not overestimate—then we can use A* search to find optimal solutions. An admissible heuristic can be derived by defining a **relaxed problem** that is easier to solve. The exact cost of a solution to this easier problem then becomes the heuristic for the original problem.

By definition, there is no way to analyze an atomic state, and thus it requires some ingenuity by a human analyst to define good domain-specific heuristics for search problems with atomic states. Planning uses a factored representation for states and action schemas. That makes it possible to define good domain-independent heuristics and for programs to automatically apply a good domain-independent heuristic for a given problem.

Think of a search problem as a graph where the nodes are states and the edges are actions. The problem is to find a path connecting the initial state to a goal state. There are two ways we can relax this problem to make it easier: by adding more edges to the graph, making it strictly easier to find a path, or by grouping multiple nodes together, forming an abstraction of the state space that has fewer states, and thus is easier to search.

We look first at heuristics that add edges to the graph. For example, the **ignore preconditions heuristic** drops all preconditions from actions. Every action becomes applicable in every state, and any single goal fluent can be achieved in one step (if there is an applicable action—if not, the problem is impossible). This almost implies that the number of steps required to solve the relaxed problem is the number of unsatisfied goals—almost but not quite, because (1) some action may achieve multiple goals and (2) some actions may undo the effects of others. For many problems an accurate heuristic is obtained by considering (1) and ignoring (2). First, we relax the actions by removing all preconditions and all effects except those that are literals in the goal. Then, we count the minimum number of actions required such that the union of those actions' effects satisfies the goal. This is an instance of the **set-cover problem**. There is one minor irritation: the set-cover problem is NP-hard. Fortunately a simple greedy algorithm is guaranteed to return a set covering whose size is within a factor of log n of the true minimum covering, where n is the number of literals in the goal. Unfortunately, the greedy algorithm loses the guarantee of admissibility.

It is also possible to ignore only *selected* preconditions of actions. Consider the slidingblock puzzle (8-puzzle or 15-puzzle). We could encode this as a planning problem involving tiles with a single schema *Slide*:

Action(Slide(t, s1, s2),
PRECOND:On(t, s1) ∧ Tile(t) ∧ Blank (s2) ∧ Adjacent (s1, s2)
EFFECT:On(t, s2) ∧ Blank(s1)'' ¬On(t, s1) ∧ ¬Blank(s2))

If we remove the preconditions Blank (s2) ∧ Adjacent (s1, s2) then any tile can move in one action to any space and we get the number-of-misplaced-tiles heuristic. If we remove Blank(s2) then we get the Manhattan-distance heuristic. It is easy to see how these heuristics could be derived automatically from the action schema description. The ease of manipulating the schemas is the great advantage of the factored representation of planning problems, as compared with the atomic representation of search problems.

Another possibility is the **ignore delete lists** heuristic. Assume for a moment that all goals and preconditions contain only positive literals We want to create a relaxed version of the original problem that will be easier to solve, and where the length of the solution will serve as a good heuristic. We can do that by removing the delete lists from all actions (i.e., removing all negative literals from effects). That makes it possible to make monotonic progress towards the

goal—no action will ever undo progress made by another action. It turns out it is still NP-hard to find the optimal solution to this relaxed problem, but an approximate solution can be found in polynomial time by hill-climbing.

A key idea in defining heuristics is **decomposition**: dividing a problem into parts, solving each part independently, and then combining the parts. The **subgoal independence** assumption is that the cost of solving a conjunction of subgoals is approximated by the sum of the costs of solving each subgoal *independently*. The subgoal independence assumption can be optimistic or pessimistic. It is optimistic when there are negative interactions between the subplans for each subgoal—for example, when an action in one subplan deletes a goal achieved by another subplan. It is pessimistic, and therefore inadmissible, when subplans contain redundant actions—for instance, two actions that could be replaced by a single action in the merged plan.

An example of a system that makes use of effective heuristics is FF, or FASTFORWARD (Hoffmann, 2005), a forward state-space searcher that uses the ignore-delete-lists heuristic, estimating the heuristic with the help of a planning graph. FF then uses hill-climbing search (modified to keep track of the plan) with the heuristic to find a solution. When it hits a plateau or local maximum—when no action leads to a state with better heuristic score—then FF uses iterative deepening search until it finds a state that is better, or it gives up and restarts hill-climbing.

17.5 Summary

In this lesson, we defined the problem of planning in deterministic, fully observable, static environments. We described the PDDL representation for planning problems and several algorithmic approaches for solving them. The points to remember: Planning systems are problem-solving algorithms that operate on explicit propositional or relational representations of states and actions. These representations make possible the derivation of effective heuristics and the development of powerful and flexible algorithms for solving problems. PDDL, the Planning Domain Definition Language, describes the initial and goal states as conjunctions of literals, and actions in terms of their preconditions and effects. State-space search can operate in the forward direction (**progression**) or the backward direction (**regression**). Effective heuristics can be derived by subgoal independence assumptions and by various relaxations of the planning problem.

17.6 Model Questions

1. Define: Classical Planning with an example.
2. State the algorithms for Classical Planning and explain.
3. Write short notes on the complexities in classical planning.
4. Discuss bout the heuristics for planning.

LESSON 18
UNCERTAIN KNOWLEDGE & REASONING

Structure

18.1 Introduction

Till now, we have learned knowledge representation using first-order logic and propositional logic with certainty, which means we were sure about the predicates. With this knowledge representation, we might write A→B, which means if A is true then B is true, but consider a situation where we are not sure about whether A is true or not then we cannot express this statement, this situation is called **uncertainty**. Agents may need to handle uncertainty, whether due to partial observability, nondeterminism, or a combination of the two. An agent may never know for certain what state it's in or where it will end up after a sequence of actions. We have seen problem-solving agents and logical agents designed to handle uncertainty by keeping track of a **belief state**—a representation of the set of all possible world states that it might be in—and generating a contingency plan that handles every possible eventuality that its sensors may report during execution. Despite its many virtues, however, this approach has significant drawbacks. So to represent uncertain knowledge, where we are not sure about the predicates, we need uncertain reasoning or probabilistic reasoning.

18.2 Objectives

- To introduce uncertainty and it's principles.
- To study the probability notations.
- To explore the Bayes' theorem and it's use.

18.3 Acting Under Uncertainty

18.3.1 Summarizing uncertainty

Let's consider an example of uncertain reasoning: diagnosing a dental patient's toothache. Diagnosis—whether for medicine, automobile repair, or whatever—almost always involves uncertainty. Let us try to write rules for dental diagnosis using propositional logic, so that we can see how the logical approach breaks down. Consider the following simple rule:

$$Toothache \Rightarrow Cavity .$$

The problem is that this rule is wrong. Not all patients with toothaches have cavities; some of them have gum disease, an abscess, or one of several other problems:

$$Toothache \Rightarrow Cavity \ V \ GumProblem \ V \ Abscess \ . \ . \ .$$

Unfortunately, in order to make the rule true, we have to add an almost unlimited list of possile problems. We could try turning the rule into a causal rule:

$$Cavity \Rightarrow Toothache .$$

But this rule is not right either; not all cavities cause pain. The only way to fix the rule is to make it logically exhaustive: to augment the left-hand side with all the qualifications required for a cavity to cause a toothache. Trying to use logic to cope with a domain like medical diagnosis thus fails for three main reasons:

- **Laziness**: It is too much work to list the complete set of antecedents or consequents needed to ensure an exceptionless rule and too hard to use such rules.
- **Theoretical ignorance**: Medical science has no complete theory for the domain.
- **Practical ignorance**: Even if we know all the rules, we might be uncertain about a particular patient because not all the necessary tests have been or can be run.

The connection between toothaches and cavities is just not a logical consequence in either direction. This is typical of the medical domain, as well as most other judgmental domains: law, business, design, automobile repair, gardening, dating, and so on. The agent's knowledge can at best provide only a **degree of belief** in the relevant sentences. Our main tool for dealing with degrees of belief is **probability theory**. In the terminology, **ontological commitments** of logic and probability theory are the same—that the world is composed of facts that do or do not

hold in any particular case—but the **epistemological commitments** are different: a logical agent believes each sentence to be true or false or has no opinion, whereas a probabilistic agent may have a numerical degree of belief between 0 (for sentences that are certainly false) and 1 (certainly true).

Probability provides a way of **summarizing** *the uncertainty that comes from our laziness and ignorance,* thereby solving the qualification problem. We might not know for sure what afflicts a particular patient, but we believe that there is, say, an 80% chance—that is, a probability of 0.8—that the patient who has a toothache has a cavity. That is, we expect that out of all the situations that are indistinguishable from the current situation as far as our knowledge goes, the patient will have a cavity in 80% of them. This belief could be derived from statistical data—80% of the toothache patients seen so far have had cavities—or from some general dental knowledge, or from a combination of evidence sources.

18.3.2 Probabilistic Reasoning

Probabilistic reasoning is a way of knowledge representation where we apply the concept of probability to indicate the uncertainty in knowledge. In probabilistic reasoning, we combine probability theory with logic to handle the uncertainty. We use probability in probabilistic reasoning because it provides a way to handle the uncertainty that is the result of someone's laziness and ignorance. In the real world, there are lots of scenarios, where the certainty of something is not confirmed, such as "It will rain today," "behavior of someone for some situations," "A match between two teams or two players." These are probable sentences for which we can assume that it will happen but not sure about it, so here we use probabilistic reasoning. Approximately, there are three reasons for using probabilistic in Artificial Intelligence. They are: (i) When there are unpredictable outcomes. (ii) When specifications or possibilities of predicates becomes too large to handle and (iii) When an unknown error occurs during an experiment. In probabilistic reasoning, there are two ways to solve problems with uncertain knowledge: (i) Bayes' rule and (ii) Bayesian Statistics. As probabilistic reasoning uses probability and related terms, so before understanding probabilistic reasoning, let's understand some common terms:

Probability: Probability can be defined as a chance that an uncertain event will occur. It is the numerical measure of the likelihood that an event will occur. The value of probability always remains between 0 and 1 that represent ideal uncertainties.

$$0 \leq P(A) \leq 1, \quad \textit{where } P(A) \textit{ is the probability of an event } A.$$

P(A) = 0, indicates total uncertainty in an event A.

P(A) =1, indicates total certainty in an event A.

We can find the probability of an uncertain event by using the below formula.

$$\textbf{Probability of occurrence} = \frac{\text{Number of desired outcomes}}{\text{Total number of outcomes}}$$

- P(¬A) = probability of a not happening event.
- P(¬A) + P(A) = 1.

Event: Each possible outcome of a variable is called an event.

Sample space: The collection of all possible events is called sample space.

Random variables: Random variables are used to represent the events and objects in the real world.

Prior probability: The prior probability of an event is probability computed before observing new information.

Posterior Probability: The probability that is calculated after all evidence or information has taken into account. It is a combination of prior probability and new information.

18.3.3 Conditional Probability

Conditional probability is a probability of occurring an event when another event has already happened. Let's suppose, we want to calculate the event A when event B has already occurred, "the probability of A under the conditions of B", it can be written as:

$$P(A \mid B) = \frac{P(A \wedge B)}{P(B)}$$

Where P($A \wedge B$) = Joint probability of a and B and P(B)= Marginal probability of B. If the probability of A is given and we need to find the probability of B, then it will be given as:

$$P(B \mid A) = \frac{P(A \wedge B)}{P(A)}$$

It can be explained by using the below Venn diagram (Figure 18.1), where B is occurred event, so sample space will be reduced to set B, and now we can only calculate event A when event B is already occurred by dividing the probability of **P(A ∧ *B*) by P(B)**.

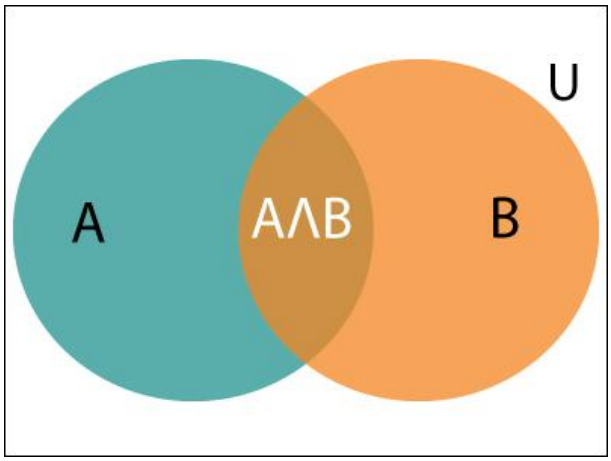


Figure 18.1 Conditional probability

Example:

In a class, there are 70% of the students who like English and 40% of the students who likes English and mathematics, and then what is the percent of students those who like English also like mathematics?

Solution:

Let, A is an event that a student likes Mathematics and B is an event that a student likes English.

$$P(A|B) = \frac{P(A \wedge B)}{P(B)} = \frac{0.4}{0.7} = 57\%$$

Hence, 57% are the students who like English also like Mathematics.

18.4 Inference Using Full Joint Distributions

In this section we describe a simple method for PROBABILISTIC **probabilistic inference**—that is, the computation of posterior probabilities for query propositions given

observed evidence. We use the full joint distribution as the “knowledge base” from which answers to all questions may be derived. Along the way we also introduce several useful techniques for manipulating equations involving probabilities.

We begin with a simple example: a domain consisting of just the three Boolean variables Toothache, Cavity, and Catch (the dentist’s nasty steel probe catches in my tooth). The full joint distribution is a 2 × 2 × 2 table as shown in Figure 18.2.

	toothache		*¬toothache*	
	catch	*¬catch*	*catch*	*¬catch*
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

Figure 18.2 A full joint distribution

Notice that the probabilities in the joint distribution sum to 1, as required by the axioms of probability. There is a direct way to calculate the probability of any proposition, simple or complex: simply identify those possible worlds in which the proposition is true and add up their probabilities. For example, there are six possible worlds in which cavity (“ toothache holds:

$$P(cavity \vee toothache) = 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064 = 0.28 .$$

One particularly common task is to extract the distribution over some subset of variables or a single variable. For example, adding the entries in the first row gives the unconditional or **marginal probability** of cavity:

$$P(cavity) = 0.108 + 0.012 + 0.072 + 0.008 = 0.2$$

This process is called **marginalization**, or **summing out**—because we sum up the probabilities for each possible value of the other variables, thereby taking them out of the equation. A variant of this rule involves conditional probabilities instead of joint probabilities, using the product rule:

$$\mathbf{P}(\mathbf{Y}) = \sum_{\mathbf{z}} \mathbf{P}(\mathbf{Y} \mid \mathbf{z}) P(\mathbf{z}) .$$

This rule is called **conditioning**. Marginalization and conditioning turn out to be useful rules for all kinds of derivations involving probability expressions. In most cases, we are interested

in computing *conditional* probabilities of some variables, given evidence about others. For example, we can compute the probability of a cavity, given evidence of a toothache, as follows:

$$\begin{aligned} P(cavity \mid toothache) &= \frac{P(cavity \wedge toothache)}{P(toothache)} \\ &= \frac{0.108 + 0.012}{0.108 + 0.012 + 0.016 + 0.064} = 0.6 \, . \end{aligned}$$

Just to check, we can also compute the probability that there is no cavity, given a toothache:

$$\begin{aligned} P(\neg cavity \mid toothache) &= \frac{P(\neg cavity \wedge toothache)}{P(toothache)} \\ &= \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} = 0.4 \, . \end{aligned}$$

The two values sum to 1.0, as they should. Notice that in these two calculations the term *1/P(toothache)* remains constant, no matter which value of Cavity we calculate. In fact, it can be viewed as a **normalization** constant for the distribution ***P****(Cavity* | *toothache*), ensuring that it adds up to 1.

The full joint distribution can answer probabilistic queries for discrete variables. It does not scale well, however: for a domain described by *n* Boolean variables, it requires an input table of size $O(2^n)$ and takes $O(2^n)$ time to process the table. In a realistic problem we could easily have $n > 100$, making $O(2^n)$ impractical. The full joint distribution in tabular form is just not a practical tool for building reasoning systems. Instead, it should be viewed as the theoretical foundation on which more effective approaches may be built, just as truth tables formed a theoretical foundation for more practical algorithms like DPLL.

18.5 Independence

Let us expand the full joint distribution in Figure 18.2 by adding a fourth variable, Weather. The full joint distribution then becomes ***P****(Toothache, Catch, Cavity,Weather)*, which has 2 × 2 × 2 × 4 = 32 entries. It contains four “editions” of the table shown in Figure 18.2, one for each kind of weather. What relationship do these editions have to each other and to the original three-variable table? For example, how are *P(toothache, catch, cavity, cloudy)* and *P(toothache, catch, cavity)* related? We can use the product rule:

P(toothache, catch, cavity, cloudy)

$$= P(cloudy \mid toothache, catch, cavity)P(toothache, catch, cavity) .$$

Now, unless one is in the deity business, one should not imagine that one's dental problems influence the weather. And for indoor dentistry, at least, it seems safe to say that the weather does not influence the dental variables. Therefore, the following assertion seems reasonable:

$$P(cloudy \mid toothache, catch, cavity) = P(cloudy) \qquad (18.1)$$

From this, we can deduce

$$P(toothache, catch, cavity, cloudy) = P(cloudy)P(toothache, catch, cavity) .$$

A similar equation exists for *every entry* in $\boldsymbol{P}(Toothache, Catch, Cavity, Weather)$. In fact, we can write the general equation

$$\boldsymbol{P}(Toothache, Catch, Cavity, Weather) = \boldsymbol{P}(Toothache, Catch, Cavity)\boldsymbol{P}(Weather) .$$

Thus, the 32-element table for four variables can be constructed from one 8-element table and one 4-element table. This decomposition is illustrated schematically in Figure 18.3(a). The property we used in Equation (18.1) is called **independence** (also **marginal independence** and **absolute independence**). In particular, the weather is independent of one's dental problems. Independence between propositions *a* and *b* can be written as

$$P(a \mid b) = P(a) \text{ or } P(b \mid a) = P(b) \text{ or } P(a \wedge b) = P(a)P(b)$$

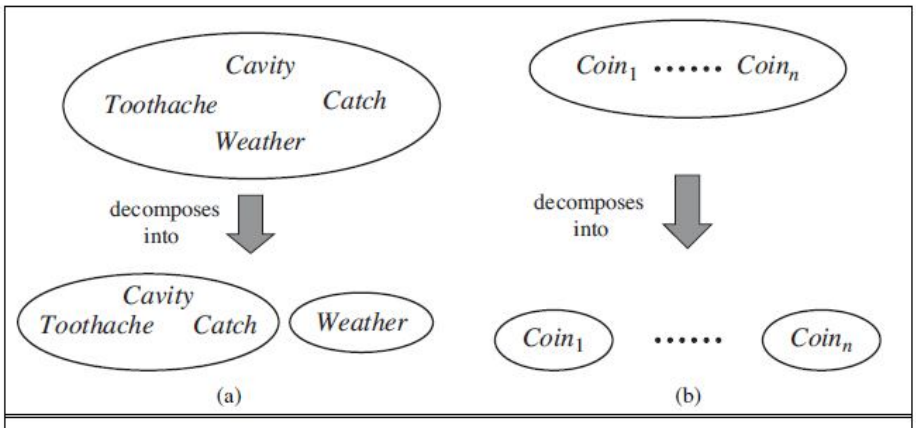


Figure 18.3 Two examples of factoring a large joint distribution into smaller distribution

All these forms are equivalent. Independence between variables X and Y can be written as follows (again, these are all equivalent):

$$\boldsymbol{P}(X \mid Y) = \boldsymbol{P}(X) \text{ or } \boldsymbol{P}(Y \mid X) = \boldsymbol{P}(Y) \text{ or } \boldsymbol{P}(X, Y) = \boldsymbol{P}(X)\boldsymbol{P}(Y) .$$

Independence assertions are usually based on knowledge of the domain. As the toothache–weather example illustrates, they can dramatically reduce the amount of information necessary to specify the full joint distribution. If the complete set of variables can be divided into independent subsets, then the full joint distribution can be *factored* into separate joint distributions on those subsets. For example, the full joint distribution on the outcome of n independent coin flips, $\boldsymbol{P}(C_1, \ldots, C_n)$, has 2^n entries, but it can be represented as the product of n single-variable distributions $\boldsymbol{P}(C_i)$. In a more practical vein, the independence of dentistry and meteorology is a good thing, because otherwise the practice of dentistry might require intimate knowledge of meteorology, and vice versa.

When they are available, then, independence assertions can help in reducing the size of the domain representation and the complexity of the inference problem. Unfortunately, clean separation of entire sets of variables by independence is quite rare. Whenever a connection, however indirect, exists between two variables, independence will fail to hold. Moreover, even independent subsets can be quite large—for example, dentistry might involve dozens of diseases and hundreds of symptoms, all of which are interrelated. To handle such problems, we need more subtle methods than the straightforward concept of independence.

18.6 Bayes' Theorem

Bayes' theorem is also known as **Bayes' rule, Bayes' law**, or **Bayesian reasoning**, which determines the probability of an event with uncertain knowledge. In probability theory, it relates the conditional probability and marginal probabilities of two random events. Bayes' theorem was named after the British mathematician **Thomas Bayes**. The **Bayesian inference** is an application of Bayes' theorem, which is fundamental to Bayesian statistics. It is a way to calculate the value of $P(B|A)$ with the knowledge of $P(A|B)$. Bayes' theorem allows updating the probability prediction of an event by observing new information of the real world.

Example: If cancer corresponds to one's age then by using Bayes' theorem, we can determine the probability of cancer more accurately with the help of age. Bayes' theorem can be derived using product rule and conditional probability of event A with known event B: As from product rule we can write:

1. P(A À” B)= P(A|B) P(B) or

Similarly, the probability of event B with known event A:

1. P(A À” B)= P(B|A) P(A)

Equating right hand side of both the equations, we will get:

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)} \quad \text{....(a)}$$

The above equation (a) is called as **Bayes' rule** or **Bayes' theorem**. This equation is basic of most modern AI systems for **probabilistic inference**. It shows the simple relationship between joint and conditional probabilities. Here, P(A|B) is known as **posterior**, which we need to calculate, and it will be read as Probability of hypothesis A when we have occurred an evidence B. P(B|A) is called the likelihood, in which we consider that hypothesis is true, then we calculate the probability of evidence. P(A) is called the **prior probability**, probability of hypothesis before considering the evidence. P(B) is called **marginal probability**, pure probability of an evidence. In the equation (a), in general, we can write P (B) = P(A)*P(B|A_i), hence the Bayes' rule can be written as:

$$P(A_i|B) = \frac{P(A_i)*P(B|A_i)}{\sum_{i=1}^{k} P(A_i)*P(B|A_i)}$$

Where A_1, A_2, A_3,....., A_n is a set of mutually exclusive and exhaustive events.

18.6.1 Applying Baye's Rule

Bayes' rule allows us to compute the single term P(B|A) in terms of P(A|B), P(*B*), and P(A). This is very useful in cases where we have a good probability of these three terms and want to determine the fourth one. Suppose we want to perceive the effect of some unknown cause, and want to compute that cause, then the Bayes' rule becomes:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})\,P(\text{cause})}{P(\text{effect})}$$

Example-1:

Question: What is the probability that a patient has diseases meningitis with a stiff neck?

Given Data:

A doctor is aware that disease meningitis causes a patient to have a stiff neck, and it occurs 80% of the time. He is also aware of some more facts, which are given as follows:

- The Known probability that a patient has meningitis disease is 1/30,000.
- The Known probability that a patient has a stiff neck is 2%.

Let a be the proposition that patient has stiff neck and b be the proposition that patient has meningitis. , so we can calculate the following as:

$$P(a|b) = 0.8$$

$$P(b) = 1/30000$$

$$P(a) = .02$$

$$\mathbf{P(b|a)} = \frac{P(a|b)P(b)}{P(a)} = \frac{0.8*(\frac{1}{30000})}{0.02} = 0.001333333.$$

Hence, we can assume that 1 patient out of 750 patients has meningitis disease with a stiff neck.

Example-2:

Question: From a standard deck of playing cards, a single card is drawn. The probability that the card is king is 4/52, then calculate posterior probability P(King|Face), which means the drawn face card is a king card.

Solution:

$$\mathbf{P(king|face)} = \frac{P(\text{Face}|\text{king})*P(\text{King})}{P(\text{Face})} \quad .....(i)$$

P(king): probability that the card is King= 4/52= 1/13

P(face): probability that a card is a face card= 3/13

P(Face|King): probability of face card when we assume it is a king = 1

Putting all values in equation (i) we will get:

$$P(\text{king}|\text{face}) = \frac{1 * (\frac{1}{13})}{(\frac{3}{13})} = 1/3,$$ **it is a probability that a face card is a king card.**

Some applications of Bayes' theorem are: (i) It is used to calculate the next step of the robot when the already executed step is given. (ii) Bayes' theorem is helpful in weather forecasting and (iii) It can solve the Monty Hall problem.

18.7 Summary

This lesson has suggested probability theory as a suitable foundation for uncertain reasoning and provided a gentle introduction to its use. Uncertainty arises because of both laziness and ignorance. It is inescapable in complex, nondeterministic, or partially observable environments. Probabilities express the agent's inability to reach a definite decision regarding the truth of a sentence. Probabilities summarize the agent's beliefs relative to the evidence. Decision theory combines the agent's beliefs and desires, defining the best action as the one that maximizes expected utility. Basic probability statements include **prior probabilities** and **conditional probabilities** over simple and complex propositions.

The **full joint probability distribution** specifies the probability of each complete assignment of values to random variables. It is usually too large to create or use in its explicit form, but when it is available it can be used to answer queries simply by adding up entries for the possible worlds corresponding to the query propositions. **Bayes' rule** allows unknown probabilities to be computed from known conditional probabilities, usually in the causal direction. Applying Bayes' rule with many pieces of evidence runs into the same scaling problems as does the full joint distribution. **Conditional independence** brought about by direct causal relationships in the domain might allow the full joint distribution to be factored into smaller, conditional distributions. The **naive Bayes** model assumes the conditional independence of all effect variables, given a single cause variable, and grows linearly with the number of effects.

18.8 Model Questions

1. Define: Uncertainty with an example.
2. What do you mean by prior and posterior probability?
3. Write short notes on: Conditional probability with an example.
4. What is Full joint distribution?
5. Illustrate Bayes' theorem with an example.

MODEL QUESTION PAPER

MASTER OF COMPUTER APPLICATIONS

FIRST YEAR – SECOND SEMESTER

CORE PAPER – VIII

ARTIFICIAL INTELLIGENCE

Time: 3 Hours Maximum: 80 Marks

PART – A (10 x 2 = 20 Marks)

Answer any TEN Questions.

1. Define Artificial Intelligence.
2. What is Rational Agent?
3. What is the structure of Intelligent agent?
4. List the steps involved in simple problem solving technique.
5. Define: Path
6. What is Backtracking search?
7. What is informed search?
8. Define: Annealing.
9. What is meant by Alpha Beta pruning?
10. What is First Order Logic.
11. What is Full joint distribution?
12. What are the three levels in describing knowledge based agent?

PART – B (5 x 6 = 30 Marks)

Answer any FIVE Questions.

13. Discuss Hill Climbing technique.

14. What are the advantages and disadvantages of declarative knowledge?

15. Write short notes on Knowledge refining.

16. What is understanding? Explain.

17. Compare: Expert systems and Knowledge based systems.

18. Discuss about the heuristics for planning.

19. Discuss Ontological engineering.

PART – C (3 x 10 = 30 Marks)

Answer any THREE Questions.

20. Elaborate the advantages of heuristic search.

21. Explain structural representation of knowledge.

22. Explain in detail about Semantic networks.

23. State the algorithms for classical planning and explain.

24. Illustrate Baye's theorem with an example.